# cSlope

## Verification Manual

Limit equilibrium slope stability analysis — comparison with published results

This document presents factors of safety computed by cSlope for 172 verification problems from the geotechnical literature and from the verification manuals of other slope stability programs, next to the values the sources publish and the range of other programs' results.

| | |
|---|---|
| Program version | cSlope 0.1.0-beta.1 (commit 7e37a28) |
| Report generated | 2026-11-03 |
| Problems | 172 |
| Values published by the sources | 200 |
| cSlope analyses | 337 |

*Generated automatically by packages/desktop/scripts/verification-report.ts from the engine's verification test cases.*

# 1 Introduction

Each problem below is defined once in the engine's verification data (packages/engine/src/verification/problems.ts). The same definitions are run by the automated test suite on every change, so the values in this report are the values the program computes.

## Analysis settings

| Setting | Value |
|---|---|
| Slip surfaces | Circular, or the non-circular surface the source gives. Fixed surfaces where the source gives one, otherwise a critical-circle search (with non-circular optimisation where stated). |
| Number of slices | 50 unless the problem states otherwise |
| Search | 4000 entry/exit combinations plus 2000 refined around the best surfaces |
| Convergence tolerance | 0.001 on the factor of safety; 0.001 with non-circular optimisation |
| Methods | Bishop simplified, Janbu simplified (uncorrected unless stated), Morgenstern-Price with half-sine interslice function; Spencer = Morgenstern-Price with a constant interslice function |
| Pore pressure | Piezometric line; head = vertical distance below the line (default, the usual convention). Checks marked "phreatic correction" use vertical distance × $\cos^2\theta$. |
| Unit weight of water | 9.81 kN/m³ |

## Units

Problems published in imperial units without water are entered in their own units (ft, psf, pcf); the factor of safety is dimensionless. Problems with water are converted to SI because the program uses $\gamma w$ = 9.81 kN/m³ (62.4 pcf = 9.80 kN/m³, < 0.1 % difference).

## Source of the input data

The input data were transcribed from the publications and from the verification manuals of other slope stability programs, most of them commercial, and cross-checked against an independent open-source transcription of the same problems. For each problem the report lists the values the source publishes, with the method where the source names one; some are hand calculations, charts, trial wedges or finite elements rather than the method of slices. For each analysis cSlope runs, it gives the range of results that other programs publish in their verification manuals, without naming the programs, and cSlope's result. The report presents the results; it does not mark them as passed or failed. Problem ids are cSlope's own: a number is one slope, a letter a variant of it.

# 2 Summary of results

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| V001 | Arai & Tagyo (1985) example 1 | 1.451; 1.265; 1.357 | Bishop simplified | 1.409–1.417 | 1.404 |
| | | | Morgenstern-Price (half-sine) | 1.414 | 1.401 |
| | | | Spencer, non-circular | 1.386 | 1.384 |
| | | | Janbu simplified, non-circular | – | 1.252 |
| | | | Janbu corrected, non-circular | – | 1.345 |
| V002 | Arai & Tagyo (1985) example 2, three layers | 0.417; 0.405; 0.43; 0.43; 0.44; 0.40–0.45; 0.39 | Bishop simplified | 0.420 | 0.419 |
| | | | Spencer, non-circular | 0.414 | 0.411 |
| | | | Janbu simplified, non-circular | – | 0.393 |
| | | | Janbu corrected, non-circular | – | 0.413 |
| V003 | Arai & Tagyo (1985) example 3, water table | 1.138; 0.995; 1.071 | Bishop simplified | 1.112–1.118 | 1.113 |
| | | | Spencer | 1.118 | 1.114 |
| | | | Janbu corrected | 1.131 | 1.129 |
| | | | Janbu simplified, non-circular | – | 0.966 |
| | | | Janbu corrected, non-circular | – | 1.047 |
| V004a | 4:1 slope, Mohr-Coulomb fit | 2.66 | Spencer | 2.794 | 2.802 |
| | | | Janbu simplified | 2.662 | 2.676 |
| V004b | Baker (2003) ex. 2 slope, Mohr-Coulomb iteration parameters | 2.64; 2.66 | Janbu simplified | 2.610 | 2.609 |
| | | | Spencer | 2.696 | 2.666 |
| V005 | Baker (1993) dam, end of construction, infinite-slope face | 2.50; 2.41 | Bishop simplified | – | 2.524 |
| | | | Spencer | – | 2.524 |
| | | | Spencer, non-circular | – | 2.503 |
| V006 | Baker & Leshchinsky (2001) planar surface | 1.35 | Spencer | 1.351 | 1.352 |
| | | | Janbu simplified | 1.351 | 1.352 |
| V007a | Baker (2003) ex. 1, Mohr-Coulomb, Janbu simplified on the reference circle | 1.50 | Janbu simplified | 1.469 | 1.468 |
| V007b | Baker (2003) ex. 1 slope, Mohr-Coulomb iteration parameters | 0.97; 1.50 | Janbu simplified | 0.957 | 0.955 |
| | | | Spencer | 0.981 | 0.978 |
| V007c | Baker (2003) ex. 3, Janbu simplified | 1.35 | Janbu simplified | 1.291 | 1.290 |
| V008a | Baker (2003) example 1, Mohr-Coulomb fit | 1.50 | Spencer | 1.536 | 1.519 |
| V008b | Baker (2003) London clay slope, Mohr-Coulomb fit | 1.35 | Spencer | 1.366 | 1.365 |
| V009a | Baker & Leshchinsky clay-core dam, reservoir and crest crack | – | Spencer | 1.925 | 1.926 |
| V009b | Baker & Leshchinsky dam, Baker's non-circular surface | 1.91 | Spencer | – | 1.882 |
| V009c | Baker & Leshchinsky clay-core dam, critical surface search | 1.91 | Spencer, non-circular | 1.868 | 1.867 |
| V010 | Borges & Cardoso embankment 2 (geosynthetic) | – | Morgenstern-Price (half-sine) | 1.171 | 1.163 |
| | | | Bishop simplified | 1.170 | 1.164 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| V010b | Borges & Cardoso embankment 2 (geosynthetic), the paper's circle B | 1.15; 1.19; 1.22 | Bishop simplified | – | 1.159 |
| V011a | Borges & Cardoso embankment 3, H = 7 m, circle A | 1.25 | Bishop simplified | 1.230 | 1.220 |
| V011b | Borges & Cardoso embankment 3, H = 7 m, circle B | 1.19 | Bishop simplified | 1.220 | 1.217 |
| V012 | Borges & Cardoso embankment 3, H = 8.75 m, circle C | 0.99 | Bishop simplified | 0.980 | 0.982 |
| V013a | Cai & Ugai (2000) slope, no pile (see V013b, V013c, V013d and V013e for the pile rows) | 1.13 | Bishop simplified | 1.140 | 1.143 |
| V013b | Cai & Ugai slope, pile row D1/D = 2 | 1.54 | Bishop simplified | 1.540 | 1.541 |
| V013c | Cai & Ugai slope, pile row D1/D = 3 | 1.37 | Bishop simplified | 1.430 | 1.437 |
| V013d | Cai & Ugai slope, pile row D1/D = 4 | 1.31 | Bishop simplified | 1.330 | 1.331 |
| V013e | Cai & Ugai slope, pile row D1/D = 6 | 1.25 | Bishop simplified | 1.250 | 1.253 |
| V014 | Nailed wall, 14 rows, deep wedge | – | Janbu corrected | 1.448–1.460 | 1.448 |
| | | | Spencer | 1.576 | 1.575 |
| V015 | Soldier-pile tieback wall | – | Janbu corrected | 1.479 | 1.476 |
| | | | Janbu simplified | 1.446 | 1.442 |
| V016a | Stepped gabion wall, equivalent cohesion | – | Bishop simplified | 1.787–1.790 | 1.790 |
| | | | Spencer | 1.791 | 1.797 |
| V016b | Stepped gabion wall, mesh as geosynthetic supports | – | Bishop simplified | 1.830–1.835 | 1.830 |
| | | | Spencer | 1.839 | 1.835 |
| V016c | Gabion wall with weak joint layers | – | Bishop simplified | 1.790 | 1.790 |
| V017a | Ontario gabion wall (Cao et al. 2016), equivalent cohesion | – | Bishop simplified | 1.373 | 1.382 |
| | | | Spencer | 1.386 | 1.398 |
| V017b | Ontario gabion wall (Cao et al. 2016), mesh as geosynthetic supports | – | Bishop simplified | 1.378 | 1.382 |
| V018 | Chen & Shao (1988) frictionless 60° slope, Prandtl mechanism | 1.05 | Spencer | 1.051 | 1.052 |
| | | | Morgenstern-Price (specified f(x)) | 1.009–1.017 | 1.008 |
| V019 | Cheng et al. 11 m homogeneous slope | 1.383 | Spencer (40 slices, vertical pore-pressure head) | 1.380–1.398 | 1.399 |
| | | | Spencer (vertical pore-pressure head) | – | 1.398 |
| | | | Bishop simplified | 1.401 | 1.401 |
| V020 | Cheng et al. benched slope, global minimum | 1.33 | Spencer | 1.336–1.338 | 1.333 |
| | | | Bishop simplified | 1.342 | 1.337 |
| V021a | Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 2 | 1.1178 | Morgenstern-Price (half-sine) | 1.132 | 1.128 |
| | | | Bishop simplified | – | 1.128 |
| | | | Bishop simplified (60 slices) | 1.128 | 1.128 |
| V021b | Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 3 | 1.1088 | Morgenstern-Price (half-sine) | 1.116 | 1.110 |
| | | | Bishop simplified | – | 1.110 |
| | | | Bishop simplified (60 slices) | 1.109 | 1.110 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| V021c | Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 2 | 1.1096 | Morgenstern-Price (half-sine) | 1.110 | 1.108 |
| | | | Bishop simplified | 1.108 | 1.108 |
| V021d | Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 3 | 1.0639 | Morgenstern-Price (half-sine) | 1.063 | 1.058 |
| | | | Bishop simplified | 1.058 | 1.058 |
| V021e | Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 2 | 2.2343 | Morgenstern-Price (half-sine) | 2.248 | 2.245 |
| | | | Bishop simplified | 2.245 | 2.245 |
| V021f | Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 3 | 2.1396 | Morgenstern-Price (half-sine) | 2.138 | 2.128 |
| | | | Bishop simplified | 2.128 | 2.128 |
| V021g | Chowdhury & Xu (1995) example 4, Congress Street cut, circle tangent to the base of clay 2 | 1.4239 | Morgenstern-Price (half-sine) | 1.422 | 1.428 |
| | | | Bishop simplified | – | 1.428 |
| V022a | Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 15 | 1.1625 | Morgenstern-Price (half-sine) | 1.158 | 1.157 |
| | | | Bishop simplified | – | 1.159 |
| | | | Bishop simplified (60 slices) | 1.160 | 1.159 |
| V022b | Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 5 | 1.1479 | Morgenstern-Price (half-sine) | 1.178 | 1.175 |
| | | | Bishop simplified | 1.185 | 1.177 |
| V023 | LASH terminal trench, Duncan's Taylor series example | 1.17 | Spencer | 1.145 | 1.138 |
| V024 | Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39) | 1.57 | Bishop simplified | 1.563–1.572 | 1.572 |
| | | | Spencer | 1.557 | 1.562 |
| V025 | Duncan & Wright fig. 6.38, piezometric line | 1.141 | Bishop simplified | 1.142 | 1.132 |
| | | | Spencer | 1.142 | 1.133 |
| V026 | Duncan & Wright fig. 7.12, embankment on clay | 1.22; 1.19 | Bishop simplified | 1.228 | 1.220 |
| | | | Spencer | – | 1.195 |
| | | | Spencer, non-circular | 1.190 | 1.170 |
| V027 | Duncan & Wright fig. 14.3, undrained clay | 1.124 | Bishop simplified | 1.126 | 1.123 |
| | | | Spencer | – | 1.134 |
| V028 | Duncan & Wright fig. 14.3, undrained clay on a 46.5 ft foundation | 1.124 | Bishop simplified | 1.126 | 1.124 |
| | | | Spencer | – | 1.123 |
| V029 | Duncan & Wright fig. 14.3, undrained clay on a 60 ft foundation | 1.119 | Bishop simplified | 1.125 | 1.118 |
| | | | Spencer | – | 1.118 |
| V030 | Duncan & Wright fig. 14.4 | 1.40 | Bishop simplified | 1.412 | 1.408 |
| | | | Spencer | – | 1.399 |
| V031a | Duncan & Wright fig. 14.5, R = 87 ft | 2.56 | Bishop simplified | 2.549 | 2.534 |
| | | | Spencer | 2.545 | 2.531 |
| V031b | Duncan & Wright fig. 14.5, R = 102 ft | 1.35 | Bishop simplified | 1.398 | 1.389 |
| | | | Spencer | 1.359 | 1.352 |
| V032a | Duncan & Wright fig. 14.7, deep circle | 1.21 | Bishop simplified | 1.230 | 1.224 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| | | | Spencer | 1.209 | 1.205 |
| V032b | Duncan & Wright fig. 14.7, infinite-slope mechanism | 1.15 | Bishop simplified | 1.155 | 1.156 |
| | | | Spencer | 1.155 | 1.156 |
| V033 | Duncan & Wright fig. 14.20a, water table | 1.529 | Bishop simplified | 1.533 | 1.521 |
| | | | Spencer | – | 1.533 |
| | | | Spencer, non-circular | 1.534 | 1.522 |
| V034a | Duncan & Wright fig. 14.20b, Su = 200 + 15·depth | 1.276 | Bishop simplified | 1.313 | 1.306 |
| | | | Spencer | – | 1.276 |
| | | | Spencer, non-circular | 1.262 | 1.260 |
| V034b | Duncan & Wright fig. 14.20b, Su = 300 psf | 1.328 | Bishop simplified | 1.335 | 1.330 |
| | | | Spencer | – | 1.328 |
| | | | Spencer, non-circular | 1.182 | 1.169 |
| V035a | Duncan & Wright tieback, active | 1.51 | Morgenstern-Price (half-sine) | 1.575 | 1.568 |
| V035b | Duncan & Wright tieback, passive | 1.32 | Bishop simplified | 1.324 | 1.321 |
| V036a | Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9) | 0.75 | Bishop simplified | 0.761 | 0.757 |
| | | | Spencer | 0.756 | 0.752 |
| | | | Spencer, non-circular | 0.740 | 0.736 |
| V036b | Embankment on an undrained foundation, cz = 5 psf/ft | 0.90 | Bishop simplified | 0.909 | 0.906 |
| | | | Spencer | 0.898 | 0.898 |
| V036c | Embankment on an undrained foundation, cz = 10 psf/ft | 1.03 | Bishop simplified | 1.045 | 1.042 |
| | | | Spencer | 1.032 | 1.029 |
| | | | Spencer, non-circular | 1.018 | 1.017 |
| V036d | Embankment on an undrained foundation, cz = 15 psf/ft | 1.13 | Bishop simplified | 1.154 | 1.152 |
| | | | Spencer | 1.134 | 1.132 |
| | | | Spencer, non-circular | 1.116 | 1.119 |
| V037 | Geogrid-reinforced fill on rock (D&W fig. 7.28) | 1.62 | Bishop simplified | 1.629 | 1.613 |
| | | | Spencer | 1.620 | 1.607 |
| V038 | Homogeneous dam with a piezometric line (D&W fig. 7.19) | 1.16 | Bishop simplified, circles only | 1.070–1.090 | 1.069 |
| | | | Spencer, circles only | 1.100 | 1.077 |
| V038b | Homogeneous dam with a piezometric line, Duncan & Wright's input | 1.16; 1.19 | Spencer | – | 1.157 |
| V039 | James Bay dyke (Duncan & Wright fig. 7.16) | 1.45; 1.17; 1.16 | Spencer, circles only | – | 1.447 |
| | | | Spencer | – | 1.166 |
| | | | Spencer, non-circular | 1.167 | 1.162 |
| V040 | James Bay dyke, reference model (deterministic) | 1.46 | Bishop simplified, circles only | 1.461 | 1.459 |
| V041a | Submerged slope, pool 30 ft above the crest (D&W fig. 6.27) | 1.60 | Bishop simplified | 1.603 | 1.597 |
| | | | Spencer | 1.599 | 1.594 |
| | | | Spencer, non-circular | 1.592 | 1.591 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| V041b | Submerged slope, pool 60 ft above the crest (D&W fig. 6.27) | 1.60 | Bishop simplified | 1.603 | 1.596 |
| | | | Spencer | 1.599 | 1.593 |
| | | | Spencer, non-circular | 1.590 | 1.589 |
| V042 | Thick-core dam with a piezometric line (D&W fig. 7.24) | 1.67 | Bishop simplified | 1.584 | 1.582 |
| | | | Spencer | 1.648 | 1.649 |
| V043 | Syncrude tailings dyke (El-Ramly et al.), composite circle | 1.31 | Bishop simplified | 1.320 | 1.320 |
| V044 | Fredlund & Krahn (1977), dry | 2.08; 2.076; 2.073 | Bishop simplified | – | 2.076 |
| | | | Morgenstern-Price (half-sine) | – | 2.072 |
| | | | Spencer | – | 2.072 |
| V045 | Fredlund & Krahn (1977), with water table | 1.834; 1.832; 1.83 | Bishop simplified | – | 1.829 |
| | | | Morgenstern-Price (half-sine) | – | 1.827 |
| | | | Spencer | – | 1.827 |
| V046a | Fredlund & Krahn slope on a weak seam, composite surface | 1.37; 1.377; 1.373 | Bishop simplified | 1.382 | 1.380 |
| | | | Spencer | 1.382 | 1.379 |
| | | | Morgenstern-Price (half-sine) | – | 1.370 |
| V046b | Fredlund & Krahn slope on a weak seam, with a piezometric line | 1.248; 1.245 | Bishop simplified | – | 1.241 |
| | | | Spencer | – | 1.242 |
| | | | Morgenstern-Price (half-sine) | – | 1.235 |
| V047a | ACADS 1(a), homogeneous slope, dry | 1.00 | Bishop simplified | 0.987–0.992 | 0.985 |
| | | | Spencer | 0.986 | 0.984 |
| | | | Janbu corrected | 0.990 | 0.986 |
| | | | Morgenstern-Price (half-sine) | 0.991 | 0.984 |
| V047b | ACADS tension crack, reference crack depth 3.6 m (results correspond to a dry crack) | 1.65–1.70 | Bishop simplified | 1.664 | 1.659 |
| | | | Morgenstern-Price (half-sine) | 1.660 | 1.656 |
| V048 | ACADS 1(b), water-filled tension crack | 1.65 | Bishop simplified | 1.596 | 1.587 |
| | | | Spencer | 1.592 | 1.582 |
| | | | Janbu corrected | 1.489 | 1.475 |
| V049 | ACADS 1(c), three-layer slope | 1.39 | Bishop simplified | 1.403–1.405 | 1.403 |
| | | | Spencer | 1.375 | 1.372 |
| | | | Janbu corrected | 1.357 | 1.355 |
| | | | Morgenstern-Price (half-sine) | 1.382 | 1.372 |
| V050a | ACADS 3(b), weak layer, specified non-circular surface | 1.34 | Janbu corrected | 1.294 | 1.294 |
| | | | Spencer | 1.277 | 1.276 |
| | | | Morgenstern-Price (half-sine) | 1.261 | 1.260 |
| V050b | ACADS 3(a), weak layer, non-circular search | 1.24–1.27 | Spencer, non-circular | 1.246 | 1.225 |
| V051 | ACADS 4, weak layer, water table and surcharges | 0.78 | Spencer, non-circular | 0.707 | 0.714 |
| V052a | Talbingo Dam, end of construction | 1.95 | Bishop simplified | 1.948–1.951 | 1.964 |
| | | | Spencer | 1.948 | 1.964 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| | | | Janbu corrected | 1.949 | 1.974 |
| V052b | Talbingo Dam, specified circle | 2.29 | Bishop simplified | 2.208 | 2.206 |
| | | | Spencer | 2.292 | 2.290 |
| | | | Janbu corrected | 2.073 | 2.073 |
| | | | Morgenstern-Price (half-sine) | 2.299 | 2.299 |
| V053 | Greco (1996) ex. 5 / Chen & Shao weak-layer slope, reference geometry | 1.009–1.025; 0.97; 1.08 | Spencer, non-circular | 1.010 | 1.006 |
| | | | Spencer | – | 1.091 |
| V054 | Greco (1996) example 5 / Chen & Shao (1988) | 1.08 | Bishop simplified | 1.087 | 1.086 |
| | | | Spencer | 1.093 | 1.091 |
| V055 | Griffiths & Lane example 2, slope with a foundation layer | 1.40 | Spencer | 1.366 | 1.366 |
| V056 | Griffiths & Lane example 3, thin weak layer, cu2/cu1 = 0.2 | 0.485 | Janbu corrected | – | 0.499 |
| V057a | Griffiths & Lane example 4, cu2/cu1 = 1 (deep base circle) | 1.46; 1.47; 1.45 | Spencer | 1.468 | 1.472 |
| V057b | Griffiths & Lane example 4, cu2/cu1 = 2 (toe circle) | 2.04; 2.10; 2.03 | Spencer | 2.022 | 2.023 |
| V058 | Griffiths & Lane example 6, earth dam with full reservoir | 1.90 | Spencer | 1.915 | 1.915 |
| V059a | Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.4 | – | Spencer, non-circular | 1.215 | 1.214 |
| V059b | Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.5 | – | Spencer, non-circular | 1.290 | 1.289 |
| V059c | Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.6 | – | Spencer, non-circular | 1.315 | 1.312 |
| V060a | Cannon Dam end of construction (Hassan & Wolff), critical circle | – | Bishop simplified | 2.529 | 2.530 |
| V060b | Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β circle | – | Bishop simplified | 2.973 | 2.974 |
| V061 | Clarence Cannon Dam (Wolff & Harr), specified surface | 2.36 | Morgenstern-Price (half-sine) | 2.343 | 2.348 |
| | | | Spencer | 2.383 | 2.388 |
| V062a | Clarence Cannon Dam, surface 7A | 2.753 | Bishop simplified | 2.551–2.560 | 2.548 |
| V062b | Clarence Cannon Dam, surface 7B | 2.352 | Bishop simplified | 2.806–2.820 | 2.815 |
| V062c | Clarence Cannon Dam, surface 7C | 2.523 | Bishop simplified | 2.771–2.777 | 2.779 |
| V062d | Clarence Cannon Dam, surface 7D | 2.457 | Bishop simplified | 2.583–2.589 | 2.577 |
| V062e | Clarence Cannon Dam, surface 7E | 2.602 | Bishop simplified | 2.692–2.703 | 2.690 |
| V062f | Clarence Cannon Dam, surface 8B | 2.995 | Bishop simplified | 2.672–2.673 | 2.673 |
| V062g | Clarence Cannon Dam, surface 8F | 3.916 | Bishop simplified | 3.586–3.598 | 3.592 |
| V062h | Clarence Cannon Dam, surface 8G | 10.576 | Bishop simplified | 6.069–6.074 | 6.063 |
| V062i | Clarence Cannon Dam, surface 8H | 6.293 | Bishop simplified | 11.583 | 11.567 |
| V063 | Huang & Jia earth dam, dry | 2.43; 2.50 | Spencer | 2.455 | 2.451 |
| | | | Bishop simplified | – | 2.453 |
| V064 | Leshchinsky & Han MSE wall, five tiers | 1.00 | Bishop simplified | 1.020 | 1.017 |
| V065 | Leshchinsky & Han MSE wall, weak foundation | 1.00; 0.86; 0.98; 0.92 | Spencer | 0.964 | 0.959 |
| | | | Bishop simplified | – | 0.977 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| V066a | Leshchinsky & Han three-tier MSE wall, baseline | 1.00; 0.99 | Bishop simplified | 1.040 | 1.031 |
| V066b | Leshchinsky & Han MSE wall, L = 4.2 m | 1.00; 0.98 | Spencer | 1.011 | 1.011 |
| | | | Bishop simplified | – | 1.013 |
| V066c | Leshchinsky & Han MSE wall, two reinforcement types | 1.00; 1.01 | Bishop simplified | – | 1.012 |
| V066d | Leshchinsky & Han MSE wall, water hw = 3 m | 1.00; 1.01 | Bishop simplified | 1.010 | 1.010 |
| V066e | Leshchinsky & Han MSE wall, surcharge | 1.00; 1.02 | Bishop simplified | 0.958 | 0.961 |
| V067 | Low (1989), layered undrained clay | 1.44 | Bishop simplified | 1.439 | 1.436 |
| V068 | Low (1989), stiff crust over soft clays, published circle | 1.14; 1.36; 1.31; 1.38; 1.17; 1.13–1.27 | Bishop simplified | – | 1.143 |
| | | | Spencer, non-circular | – | – |
| V069a | Morgenstern (1963) complete drawdown, 3:1 | 1.20 | Bishop simplified | – | 1.200 |
| | | | Spencer | – | 1.205 |
| V069b | Morgenstern complete drawdown, the reference circle | 1.20 | Bishop simplified | 1.212 | 1.209 |
| V070 | Partial drawdown, $\bar{B} = 1$ (Morgenstern chart) | 1.41; 1.52; 1.48 | Bishop simplified | 1.417 | 1.415 |
| | | | Spencer | 1.422 | 1.421 |
| V071a | Pockoski & Duncan (2000) slope 1 | – | Bishop simplified | 1.293 | 1.290 |
| | | | Spencer | 1.300 | 1.296 |
| V071b | Pockoski & Duncan slope 1, circular search | – | Bishop simplified | 1.293 | 1.288 |
| | | | Spencer | 1.300 | 1.294 |
| | | | Janbu simplified | 1.151 | 1.148 |
| V072a | Pockoski & Duncan test slope 2, dry tension crack | – | Bishop simplified | 1.285 | 1.282 |
| | | | Spencer | 1.290 | 1.288 |
| V072b | Pockoski & Duncan slope 2, circular search | – | Bishop simplified | 1.285 | 1.282 |
| | | | Spencer | 1.290 | 1.287 |
| | | | Janbu simplified | 1.141 | 1.135 |
| V073 | Pockoski & Duncan test slope 3, clay seam and dry crack | – | Bishop simplified, circles only | 1.417 | 1.410 |
| | | | Spencer, circles only | 1.422 | 1.415 |
| | | | Bishop simplified | 1.392 | 1.387 |
| | | | Spencer | 1.400 | 1.395 |
| V074 | Soil-nailed wall with tension crack and surcharges | – | Spencer | 1.009 | 1.010 |
| | | | Janbu corrected | 1.073 | 1.075 |
| V075 | Tieback wall in sand, drawdown water table | – | Spencer | 0.570 | 0.578 |
| V076 | Tied-back wall in layered soil | – | Bishop simplified | 1.147 | 1.143 |
| | | | Spencer | 1.145 | 1.141 |
| V077 | Prandtl bearing capacity on level ground | 1.00 | Spencer | 0.940 | 0.943 |
| V078 | Prandtl bearing capacity, fully specified Prandtl surface (reference values) | 1.00 | Morgenstern-Price (half-sine) | 0.960 | 0.958 |
| V079 | Priest (1993) rigid block on a plane, water-filled crack | 1.049 | Janbu simplified | 1.049 | 1.048 |
| | | | Spencer | 1.049 | 1.048 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| V080 | Priest (1993) rigid block, reference geometry (60° face) | 1.049; 1.049 | Morgenstern-Price (half-sine) | 1.049 | 1.049 |
| | | | Spencer | – | 1.049 |
| V081a | Sharma (1996) reinforcement example, unreinforced | – | Bishop simplified | 0.764 | 0.764 |
| V081b | Sharma (1996) slope with a reinforced zone (φ = 54.93°) | – | Bishop simplified | 1.504 | 1.501 |
| V082a | Sharma (1996) slope, 11 ft dry tension crack instead of the zero-strength cap | – | Bishop simplified | 1.532 | 1.534 |
| | | | Spencer | 1.532 | 1.533 |
| | | | Morgenstern-Price (half-sine) | 1.532 | 1.533 |
| | | | Janbu corrected | 1.544 | 1.546 |
| V082b | Sharma (1996) slope, 11 ft tension crack with 6 ft of water | – | Bishop simplified | 1.511 | 1.513 |
| | | | Spencer | 1.510 | 1.512 |
| | | | Morgenstern-Price (half-sine) | 1.510 | 1.512 |
| | | | Janbu corrected | 1.520 | 1.521 |
| V082c | Sharma (1996) slope over undulating bedrock, zero-strength cap | – | Bishop simplified | 1.396 | 1.396 |
| | | | Spencer | 1.402 | 1.402 |
| | | | Janbu corrected | 1.391 | 1.392 |
| | | | Morgenstern-Price (half-sine) | 1.398 | 1.398 |
| V082d | Sharma (1996) slope over undulating bedrock, circular search with exit limits | – | Bishop simplified | 1.376 | 1.369 |
| | | | Spencer | 1.382 | 1.376 |
| | | | Morgenstern-Price (half-sine) | 1.378 | 1.371 |
| | | | Janbu corrected | 1.345 | 1.343 |
| V083 | Soil-nailed wall in clay (Amherst test wall) | 0.887 | Janbu corrected | 0.890 | 0.885 |
| V084a | Soil-nailed wall in sand (Clouterre test wall) | 0.989 | Janbu corrected | 0.989 | 0.988 |
| V084b | Soil-nailed wall in sand (Clouterre test wall), 45° plane | 1.176 | Janbu corrected | 1.123 | 1.124 |
| V084c | Soil-nailed wall in sand (Clouterre test wall), 50° plane | 1.07 | Janbu corrected | 1.043 | 1.043 |
| V084d | Soil-nailed wall in sand (Clouterre test wall), 60° plane | 0.929 | Janbu corrected | 0.945 | 0.944 |
| V084e | Soil-nailed wall in sand (Clouterre test wall), 65° plane | 0.893 | Janbu corrected | 0.922 | 0.920 |
| V084f | Soil-nailed wall in sand (Clouterre test wall), 70° plane | 0.887 | Janbu corrected | 0.923 | 0.920 |
| V085 | Bradwell excavated slope, stiff-fissured London clay | 1.76; 1.63 | Bishop simplified | 1.762 | 1.759 |
| | | | Spencer | 1.758 | 1.759 |
| | | | Janbu corrected | 1.736 | 1.735 |
| | | | Janbu simplified | 1.628 | 1.627 |
| V086a | Tandjiria embankment, clay fill, geosynthetic T = 169 kN/m | 1.35 | Spencer | 1.332–1.350 | 1.333 |
| V086b | Tandjiria embankment, clay fill, water-filled crack | 0.981 | Spencer (2 m tension crack) | 0.975 | 0.968 |
| | | | Spencer (tension crack of automatic depth) | 0.975 | 0.966 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| | | | Morgenstern-Price (half-sine) | 0.975 | 0.968 |
| V086c | Tandjiria embankment, sand fill, dry crack | 1.219 | Spencer | 1.209 | 1.206 |
| | | | Morgenstern-Price (half-sine) | 1.218 | 1.215 |
| V086d | Tandjiria embankment, clay fill, the paper's geometry | 0.981 | Spencer | – | 0.973 |
| V086e | Tandjiria embankment, sand fill, the paper's geometry | 1.219 | Spencer | – | 1.204 |
| V087 | USACE dam, chart-check properties (EM 1110-2-1902 fig. 4-3) | 2.30 | Janbu corrected | 2.290 | 2.248 |
| V088 | USACE dam, end of construction (EM 1110-2-1902 fig. 4-1) | 2.44 | Spencer | 2.445 | 2.442 |
| | | | Bishop simplified | 2.447 | 2.442 |
| | | | Janbu corrected | 2.430 | 2.444 |
| | | | Morgenstern-Price (half-sine) | 2.447 | 2.442 |
| V089 | USACE dam, upstream low pool (EM 1110-2-1902 fig. 4-2) | 2.71 | Bishop simplified | 2.716 | 2.698 |
| | | | Spencer | 2.736 | 2.721 |
| | | | Janbu corrected | 2.650 | 2.647 |
| | | | Morgenstern-Price (half-sine) | 2.744 | 2.727 |
| V090 | USACE EM 1110-2-1902, example F-5 | 1.33 | Bishop simplified | 1.332 | 1.320 |
| | | | Spencer | 1.328 | 1.317 |
| | | | Janbu corrected | 1.345 | 1.340 |
| | | | Morgenstern-Price (half-sine) | 1.327 | 1.316 |
| V091 | USACE example E-2 (fig. E-10), ponded water | 1.24 | Bishop simplified | 1.241 | 1.236 |
| | | | Spencer | 1.241 | 1.236 |
| | | | Morgenstern-Price (half-sine) | 1.244 | 1.236 |
| | | | Janbu corrected | 1.385 | 1.390 |
| V092 | USACE example F-6, steady-seepage dam | 2.01; 2.07 | Bishop simplified | 2.011 | 1.990 |
| | | | Spencer | 2.026 | 2.005 |
| | | | Janbu corrected | 1.830 | 1.823 |
| | | | Morgenstern-Price (half-sine) | 2.027 | 2.005 |
| V093 | Yamagami & Ueta (1988) | 1.348; 1.282; 1.185; 1.339; 1.33 | Bishop simplified | 1.344 | 1.343 |
| | | | Spencer, non-circular | 1.325 | 1.322 |
| | | | Janbu simplified, non-circular | 1.178 | 1.176 |
| V094 | Yamagami & Ueta (1988) four-layer slope, non-circular | 1.40–1.42 | Spencer, non-circular | 1.398 | 1.400 |
| | | | Morgenstern-Price (half-sine), non-circular | 1.389 | 1.389 |
| V095a | Yamagami micro-pile slope, no pile | 1.10 | Bishop simplified | 1.102 | 1.101 |
| V095c | Homogeneous slope with a micro-pile row | 1.20 | Bishop simplified | 1.193 | 1.196 |
| V096a | Zhu & Lee heterogeneous slope, dry, deep circle | – | Bishop simplified | 1.804 | 1.794 |
| | | | Spencer | 1.804 | 1.794 |
| | | | Morgenstern-Price (half-sine) | 1.790 | 1.780 |
| V096b | Zhu & Lee heterogeneous slope with a water table, deep circle | – | Bishop simplified | 1.176 | 1.173 |
| | | | Spencer | 1.189 | 1.187 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| | | | Morgenstern-Price (half-sine) | 1.174 | 1.172 |
| V096c | Zhu & Lee slope, surface 3 (the reference circle), dry, 1 m crack | – | Spencer | 1.804 | 1.800 |
| | | | Bishop simplified | 1.804 | 1.799 |
| V096d | Zhu & Lee slope, the paper's surface 3, dry | 1.836; 1.823; 1.744; 1.429; 1.229 | Spencer | – | 1.824 |
| | | | Morgenstern-Price (half-sine) | – | 1.808 |
| | | | Bishop simplified | – | 1.823 |
| V096e | Zhu & Lee slope, the paper's surface 3, with the water table | 1.211; 1.197; 1.166; 1.079; 0.922 | Spencer | – | 1.212 |
| | | | Morgenstern-Price (half-sine) | – | 1.197 |
| | | | Bishop simplified | – | 1.199 |
| V097a | Eurocode 7 cutting with surcharge | – | Bishop simplified | 1.173 | 1.172 |
| V097b | Eurocode 7 cutting, partial factors applied by cSlope | – | Bishop simplified | 1.173 | 1.172 |
| | | | Spencer | 1.174 | 1.174 |
| V098 | Example slope without seismic load | – | Spencer, non-circular | 1.360 | 1.357 |
| | | | Spencer | 1.372 | 1.372 |
| V021a | Congress Street cut, probabilistic (tangent to clay 2) | – | Pf, share FOS < 1 (Monte Carlo) | 0.246 | 0.239 |
| V021a | Congress Street cut, probabilistic (tangent to clay 2) | – | β lognormal (Monte Carlo) | 0.650 | 0.672 |
| V021a | Chowdhury & Xu (1995) example 1, probability of failure | – | Pf, share FOS < 1 (Monte Carlo) | 0.253 | 0.245 |
| V021b | Congress Street cut, probabilistic (tangent to clay 3) | – | Pf, share FOS < 1 (Monte Carlo) | 0.279 | 0.282 |
| V021b | Congress Street cut, probabilistic (tangent to clay 3) | – | β lognormal (Monte Carlo) | 0.537 | 0.553 |
| V022a | Embankment on soft clay, probabilistic (tangent to the interface) | – | Pf, share FOS < 1 (Monte Carlo) | 0.206–0.212 | 0.206 |
| V022a | Embankment on soft clay, probabilistic (tangent to the interface) | – | β lognormal (Monte Carlo) | 0.799–0.800 | 0.812 |
| V022a | Embankment on soft clay, probabilistic (tangent to the interface) | – | β lognormal (Taylor series) | 0.787 | 0.788 |
| V022a | Chowdhury & Xu (1995) example 5, probability of failure | – | Pf, share FOS < 1 (Monte Carlo) | 0.210 | 0.208 |
| V022b | Embankment on soft clay, probabilistic (tangent to the clay base) | – | Pf, share FOS < 1 (Monte Carlo) | 0.199–0.209 | 0.212 |
| V022b | Embankment on soft clay, probabilistic (tangent to the clay base) | – | β lognormal (Monte Carlo) | 0.789–0.820 | 0.790 |
| V022b | Embankment on soft clay, probabilistic (tangent to the clay base) | – | β lognormal (Taylor series) | 0.798 | 0.803 |
| V022b | Chowdhury & Xu (1995) example 5, probability of failure | – | Pf, share FOS < 1 (Monte Carlo) | 0.214 | 0.216 |
| V060b | Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β circle | – | β lognormal (Taylor series) | 3.353 | 3.354 |
| V023 | LASH terminal trench, Duncan's Taylor series example | 0.900 | β lognormal (Taylor series) | – | 0.892 |
| V023 | LASH terminal trench, Duncan's Taylor series example | 0.180 | Pf lognormal (Taylor series) | – | 0.186 |
| V062a | Clarence Cannon Dam, surface 7A | – | β lognormal (Monte Carlo) | 10.953 | 11.034 |

| Problem | Description | Published | cSlope analysis | Other programs | cSlope |
|---|---|---|---|---|---|
| V062b | Clarence Cannon Dam, surface 7B | – | β lognormal (Monte Carlo) | 4.351 | 4.402 |
| V062c | Clarence Cannon Dam, surface 7C | – | β lognormal (Monte Carlo) | 4.263 | 4.325 |
| V062d | Clarence Cannon Dam, surface 7D | – | β lognormal (Monte Carlo) | 11.092 | 11.076 |
| V062e | Clarence Cannon Dam, surface 7E | – | β lognormal (Monte Carlo) | 10.281 | 10.329 |
| V062f | Clarence Cannon Dam, surface 8B | – | β lognormal (Monte Carlo) | 4.858 | 4.933 |
| V062g | Clarence Cannon Dam, surface 8F | – | β lognormal (Monte Carlo) | 5.485 | 5.528 |
| V062h | Clarence Cannon Dam, surface 8G | – | β lognormal (Monte Carlo) | 5.563 | 5.490 |
| V062i | Clarence Cannon Dam, surface 8H | – | β lognormal (Monte Carlo) | 6.394 | 6.431 |

# 3 Verification problems

## 3.1 V001 — Arai & Tagyo (1985) example 1

*Arai & Tagyo (1985), example 1.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Homogeneous 1.5:1 slope, 20 m high; critical circle search.

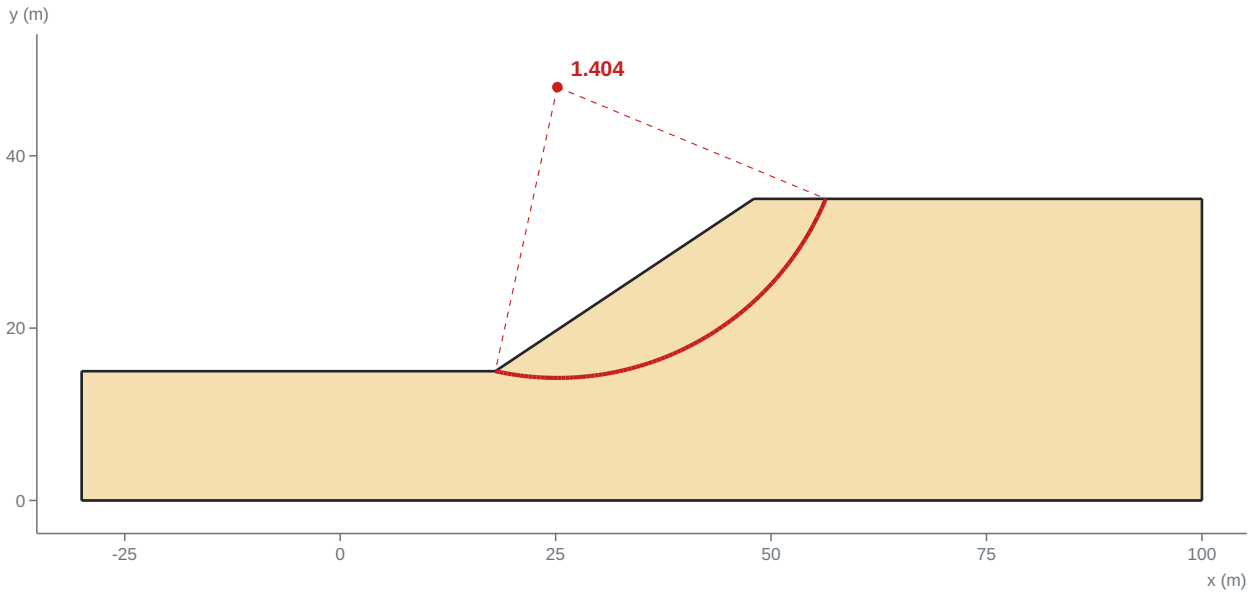



*Figure 1. V001: critical circle (Bishop simplified, FOS = 1.404).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 18.82 | c' = 41.65 | 15.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.451 | Bishop simplified | Arai & Tagyo (1985) | 1.404 |
| 1.265 | Janbu simplified, non-circular | Arai & Tagyo (1985) | 1.252 |
| 1.357 | Janbu corrected, non-circular | Arai & Tagyo (1985) | 1.345 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.409–1.417 | 1.404 |
| Morgenstern-Price (half-sine) | 1.414 | 1.401 |
| Spencer, non-circular | 1.386 | 1.384 |
| Janbu simplified, non-circular | – | 1.252 |
| Janbu corrected, non-circular | – | 1.345 |

*Arai & Tagyo's Bishop value 1.451 is 2–3 % above the other programs (1.409–1.417) and cSlope on the same slope; their non-circular Janbu values are within 1 % of cSlope's.*

## 3.2 V002 — Arai & Tagyo (1985) example 2, three layers

*Arai & Tagyo (1985), example 2.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Slope with a weak middle layer between a stronger upper layer and a very strong base; critical circle search.

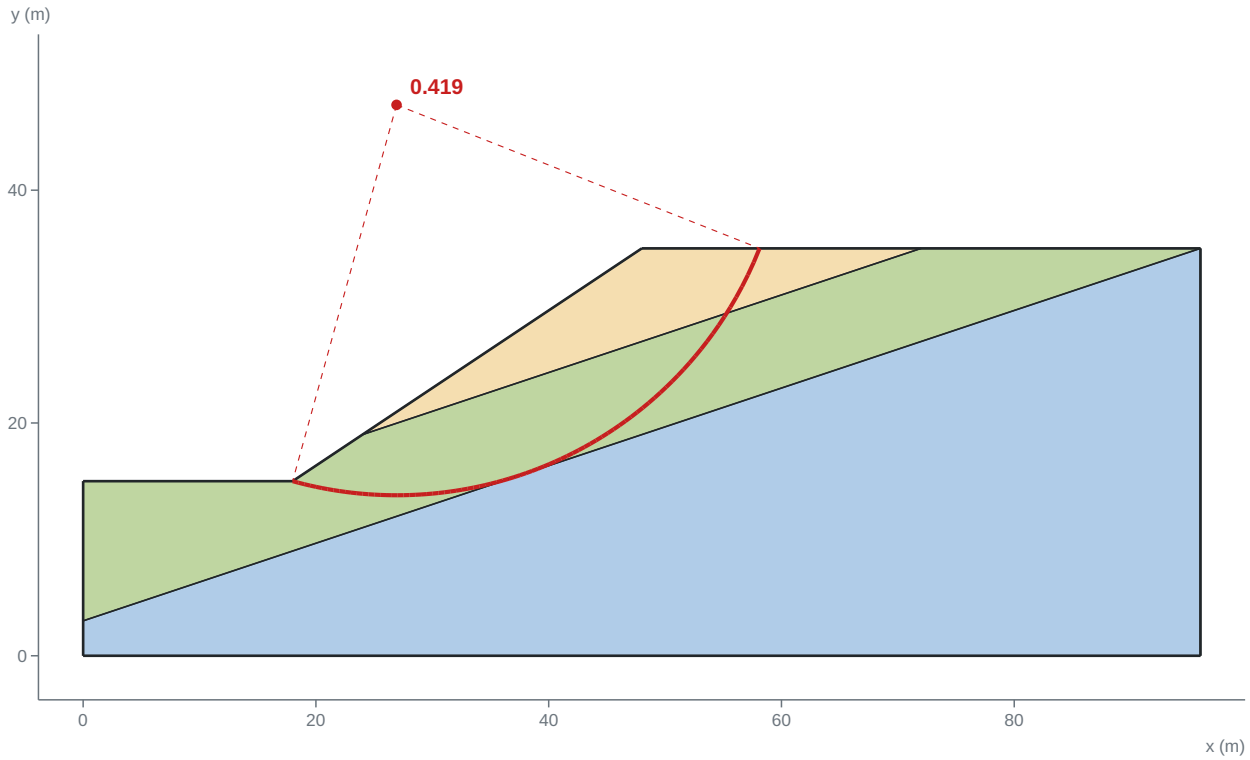



*Figure 2. V002: critical circle (Bishop simplified, FOS = 0.419).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Upper | Mohr-Coulomb | 18.82 | c' = 29.40 | 12.00° |
| Middle | Mohr-Coulomb | 18.82 | c' = 9.80 | 5.00° |
| Lower | Mohr-Coulomb | 18.82 | c' = 294.0 | 40.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.417 | Bishop simplified | Arai & Tagyo (1985) | 0.419 |
| 0.405 | Janbu simplified, non-circular | Arai & Tagyo (1985) | 0.393 |
| 0.43 | Janbu corrected, non-circular | Arai & Tagyo (1985) | 0.413 |
| 0.43 | Bishop simplified | Kim et al. (2002) | 0.419 |
| 0.44 | Spencer, non-circular | Kim et al. (2002) | 0.411 |
| 0.40–0.45 | limit analysis, lower and upper bound | Kim et al. (2002) | – |
| 0.39 | Spencer, non-circular | Greco (1996) | 0.411 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 0.420 | 0.419 |
| Spencer, non-circular | 0.414 | 0.411 |
| Janbu simplified, non-circular | – | 0.393 |
| Janbu corrected, non-circular | – | 0.413 |

## 3.3 V003 — Arai & Tagyo (1985) example 3, water table

*Arai & Tagyo (1985), Soils and Foundations 25(1).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Homogeneous slope with a water table; critical-circle search.

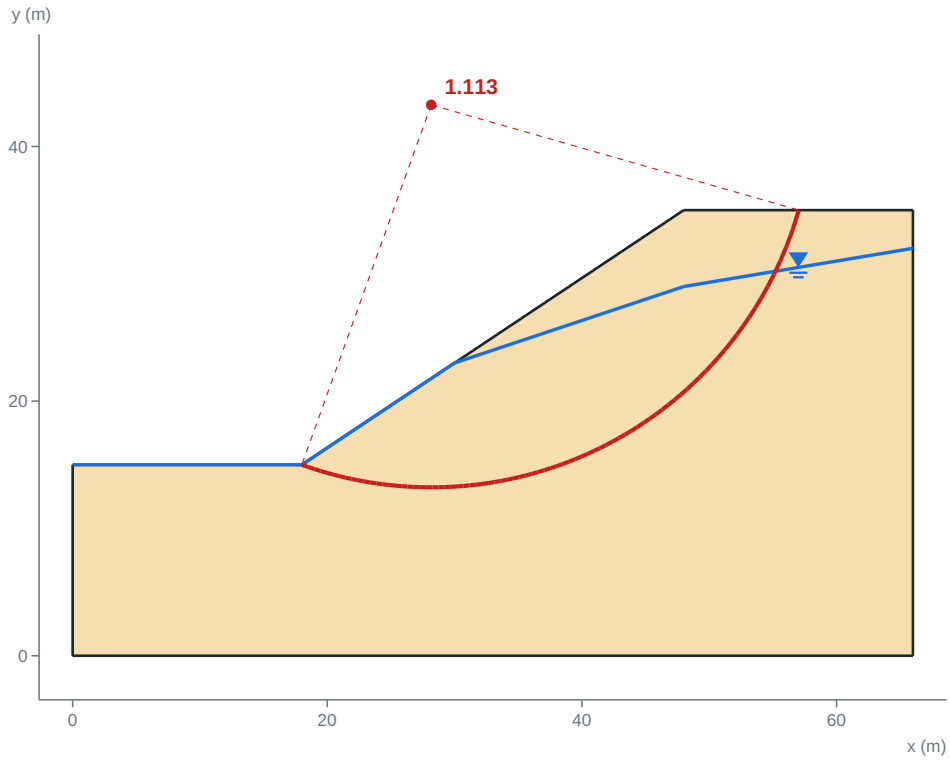



*Figure 3. V003: critical circle (Bishop simplified, FOS = 1.113).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| A&T ex3 soil | Mohr-Coulomb | 18.82 | c' = 41.65 | 15.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.138 | Bishop simplified | Arai & Tagyo (1985) | 1.113 |
| 0.995 | Janbu simplified, non-circular | Arai & Tagyo (1985) | 0.966 |
| 1.071 | Janbu corrected, non-circular | Arai & Tagyo (1985) | 1.047 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.112–1.118 | 1.113 |
| Spencer | 1.118 | 1.114 |
| Janbu corrected | 1.131 | 1.129 |
| Janbu simplified, non-circular | – | 0.966 |
| Janbu corrected, non-circular | – | 1.047 |

*Arai & Tagyo's Bishop value 1.138 is about 2 % above the other programs (1.112–1.118 with Bishop).*

## 3.4 V004a — 4:1 slope, Mohr-Coulomb fit

*Verification manual of a commercial slope stability program.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

12 m high 4:1 slope with the same soil as V008a; critical circle search.

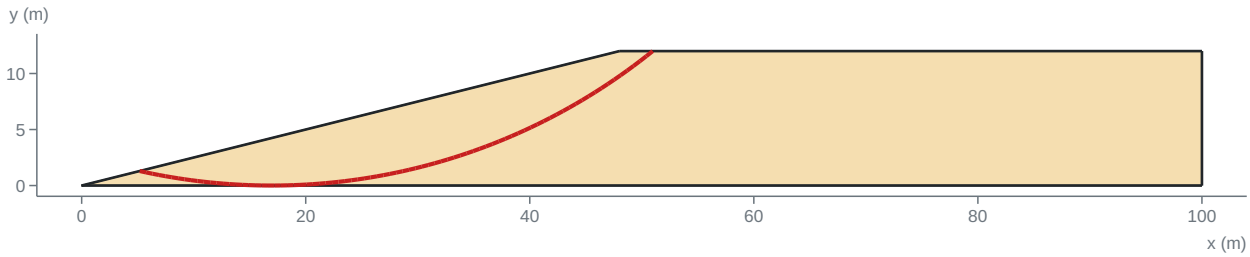



*Figure 4. V004a: critical circle (Spencer, FOS = 2.802).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 18.00 | c' = 11.64 | 24.70° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.66 | Mohr-Coulomb strength | Baker (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 2.794 | 2.802 |
| Janbu simplified | 2.662 | 2.676 |

*Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).*

## 3.5 V004b — Baker (2003) ex. 2 slope, Mohr-Coulomb iteration parameters

*Verification manual of a commercial slope stability program.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V004a 4:1 slope with Baker's iterated Mohr-Coulomb parameters (c′ = 2.439 kPa, φ′ = 30.392°); critical-circle search.

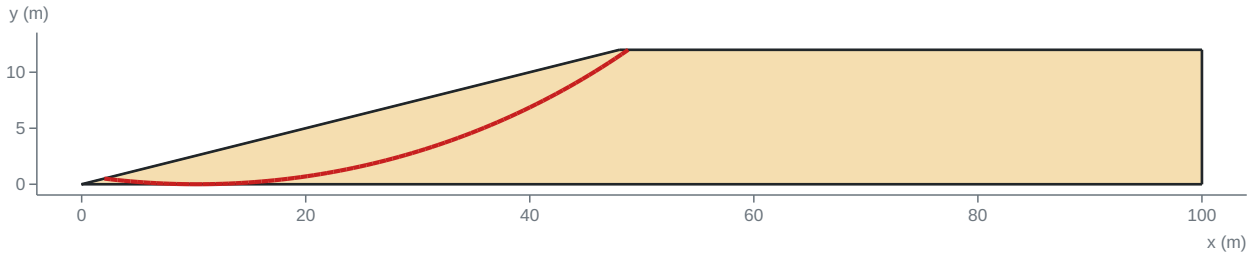



*Figure 5. V004b: critical circle (Janbu simplified, FOS = 2.609).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 18.00 | c' = 2.44 | 30.39° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.64 | non-linear strength envelope | Baker (2003) | – |
| 2.66 | Mohr-Coulomb strength | Baker (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu simplified | 2.610 | 2.609 |
| Spencer | 2.696 | 2.666 |

*Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).*

## 3.6 V005 — Baker (1993) dam, end of construction, infinite-slope face

*Baker, R., Frydman, S. & Talesnick, M. (1993). Slope stability analysis for undrained loading conditions. Int. J. Numer. Anal. Meth. Geomech. 17, 15–43.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Compacted-clay embankment (c′ = 6.5 kPa, φ′ = 40°) on a deep natural-clay foundation (c′ = 0, φ′ = 32°) with a 6 m crest and a straight 4H:1V downstream face, whose natural-clay part governs; dry; circular search. Theoretical (infinite slope) FOS tan 32°/tan 14.04° = 2.50.

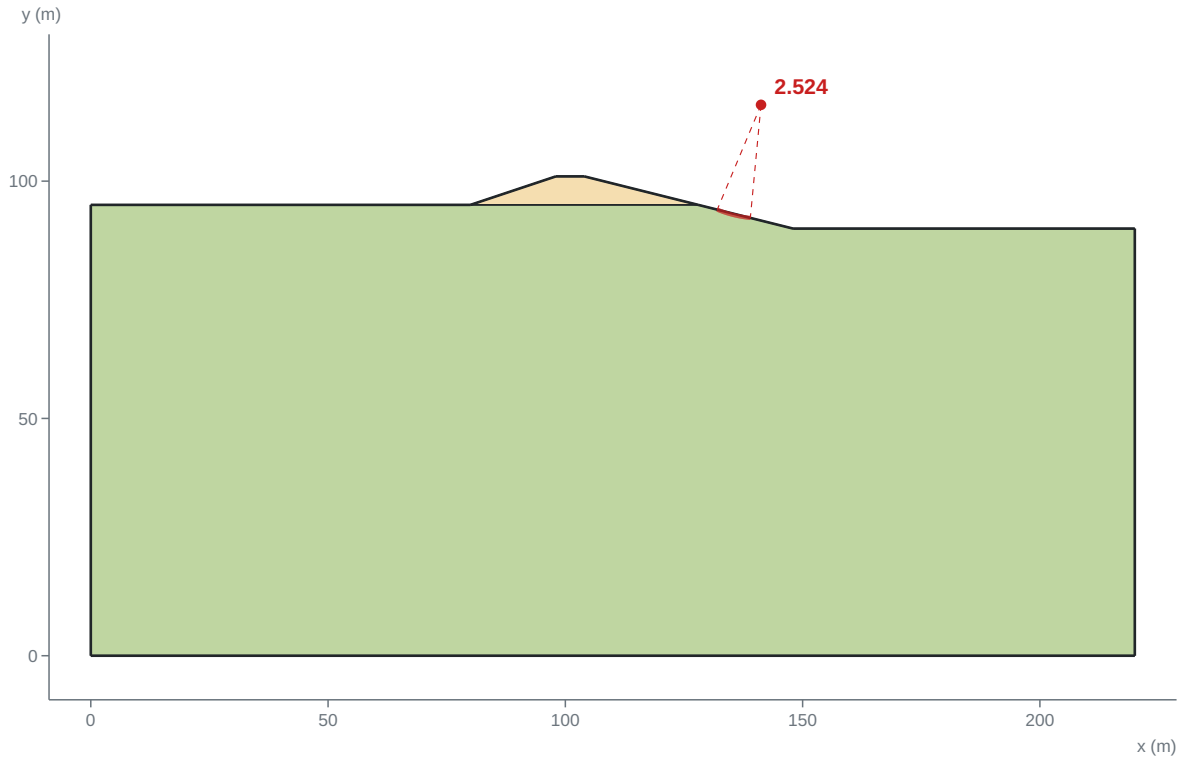



*Figure 6. V005: critical circle (Bishop simplified, FOS = 2.524).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Compacted clay | Mohr-Coulomb | 18.00 | c' = 6.50 | 40.00° |
| Natural clay | Mohr-Coulomb | 18.00 | c' = 0.00 | 32.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.50 | infinite slope | Theory | 2.524 |
| 2.41 | Spencer, non-circular (dynamic programming) | Baker, Frydman & Talesnick (1993) | 2.503 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | – | 2.524 |
| Spencer | – | 2.524 |
| Spencer, non-circular | – | 2.503 |

## 3.7 V006 — Baker & Leshchinsky (2001) planar surface

*Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Homogeneous 10 m slope on the critical plane from the toe to the crest at 49.6°.

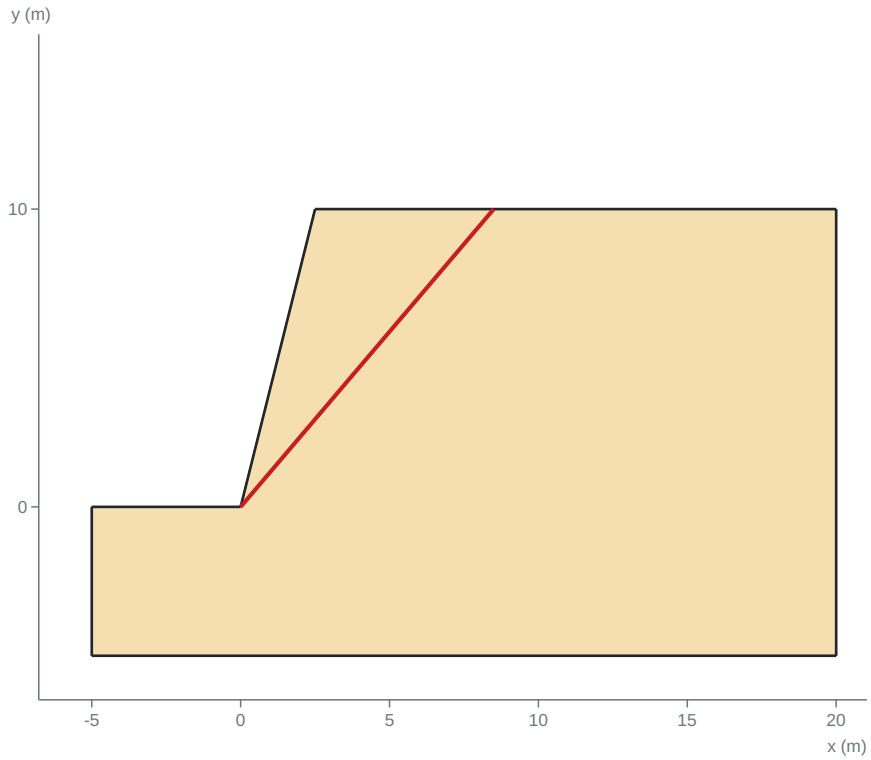



*Figure 7. V006: analysed surface (Spencer, FOS = 1.352).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 30.00 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.35 | Culmann (planar wedge) | Baker & Leshchinsky (2001) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.351 | 1.352 |
| Janbu simplified | 1.351 | 1.352 |

## 3.8 V007a — Baker (2003) ex. 1, Mohr-Coulomb, Janbu simplified on the reference circle

*Verification manual of a commercial slope stability program.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V008a slope (Mohr-Coulomb fit c′ = 11.64 kPa, φ′ = 24.7°) with Janbu simplified (uncorrected) on the reference critical circle (Fig. 44.3).

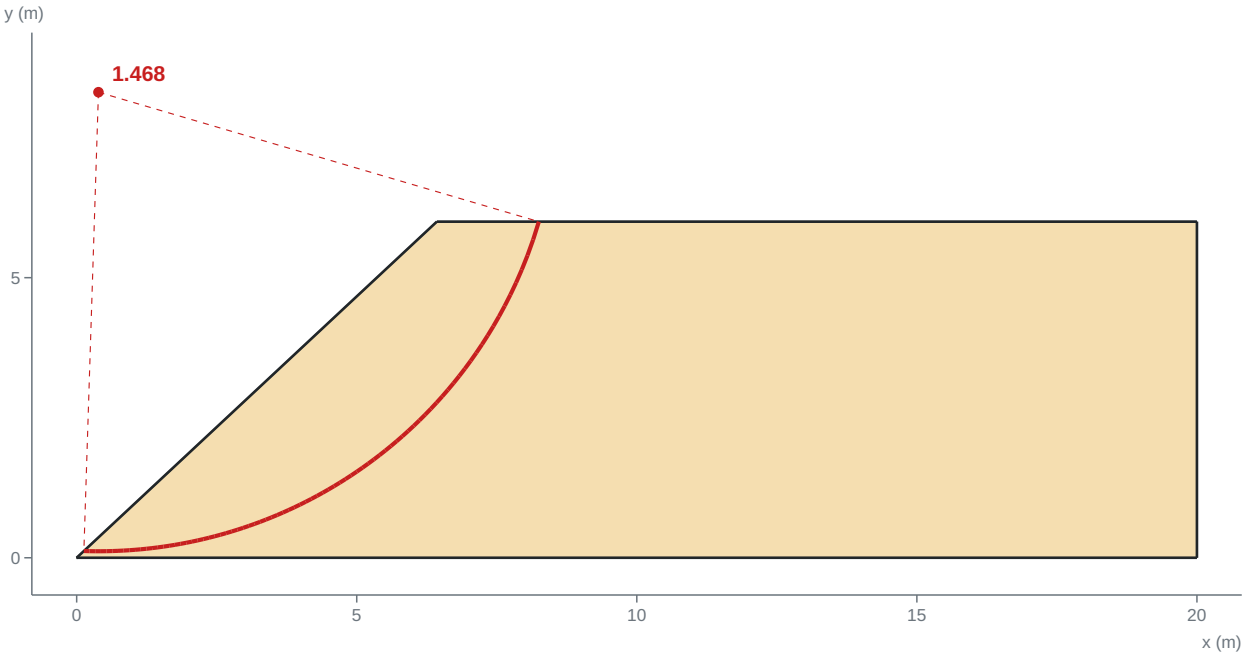



*Figure 8. V007a: analysed circle (Janbu simplified, FOS = 1.468).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 18.00 | c' = 11.64 | 24.70° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.50 | Mohr-Coulomb strength | Baker (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu simplified | 1.469 | 1.468 |

*Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).*

## 3.9 V007b — Baker (2003) ex. 1 slope, Mohr-Coulomb iteration parameters

*Verification manual of a commercial slope stability program.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V008a slope with Baker's Mohr-Coulomb parameters from his iterative linearisation (c′ = 0.39 kPa, φ′ = 38.6°); critical-circle search.

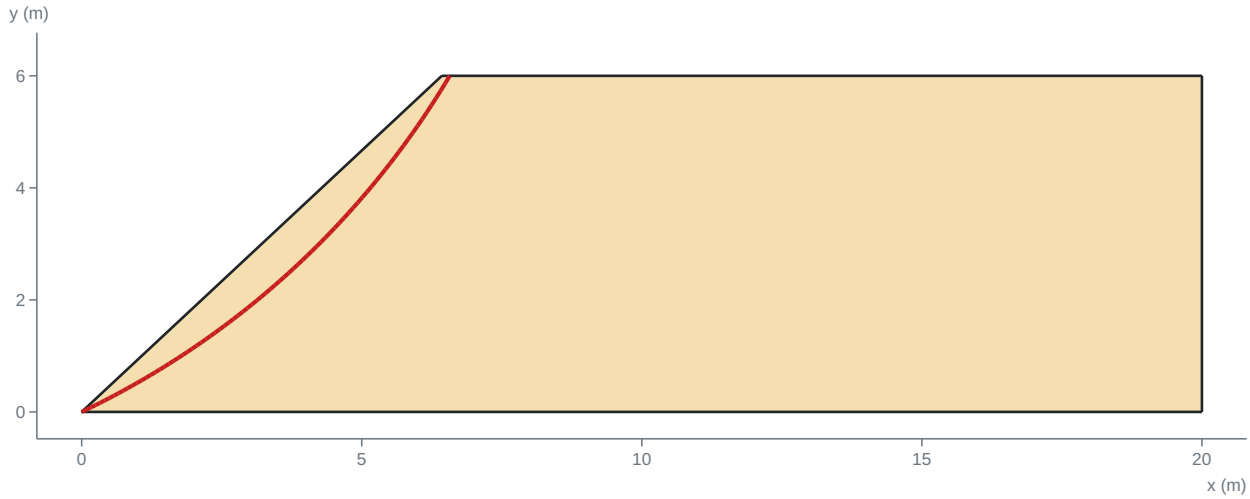



*Figure 9. V007b: critical circle (Janbu simplified, FOS = 0.955).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 18.00 | c' = 0.39 | 38.60° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.97 | non-linear strength envelope | Baker (2003) | – |
| 1.50 | Mohr-Coulomb strength | Baker (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu simplified | 0.957 | 0.955 |
| Spencer | 0.981 | 0.978 |

*Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387); 1.50 is his Mohr-Coulomb value for the same slope.*

## 3.10 V007c — Baker (2003) ex. 3, Janbu simplified

*Verification manual of a commercial slope stability program.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V008b London clay slope (c′ = 6 kPa, φ′ = 32°) with Janbu simplified (uncorrected); critical-circle search.

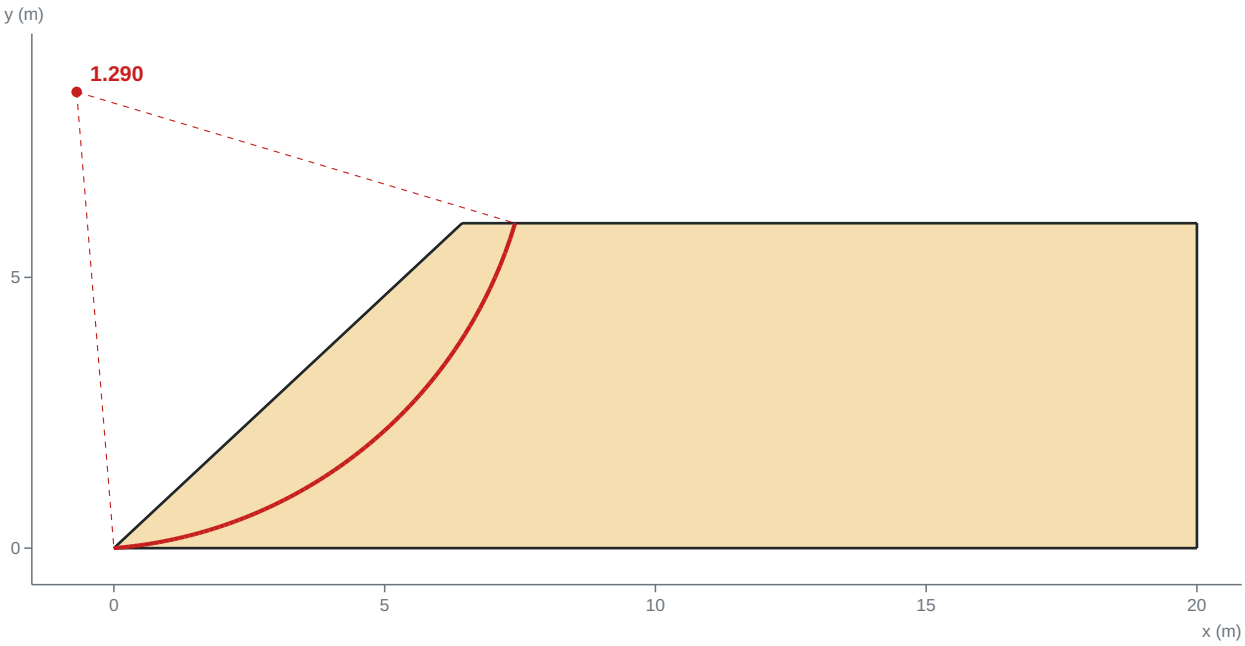



*Figure 10. V007c: critical circle (Janbu simplified, FOS = 1.290).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 18.00 | c' = 6.00 | 32.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.35 | Mohr-Coulomb strength | Baker (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu simplified | 1.291 | 1.290 |

*Baker (2003) gives the factor of safety for the Mohr-Coulomb fit with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).*

## 3.11 V008a — Baker (2003) example 1, Mohr-Coulomb fit

*Baker (2003), example 1.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

6 m high slope at about 43° in soil with a linear Mohr-Coulomb fit to a curved envelope; critical circle search.

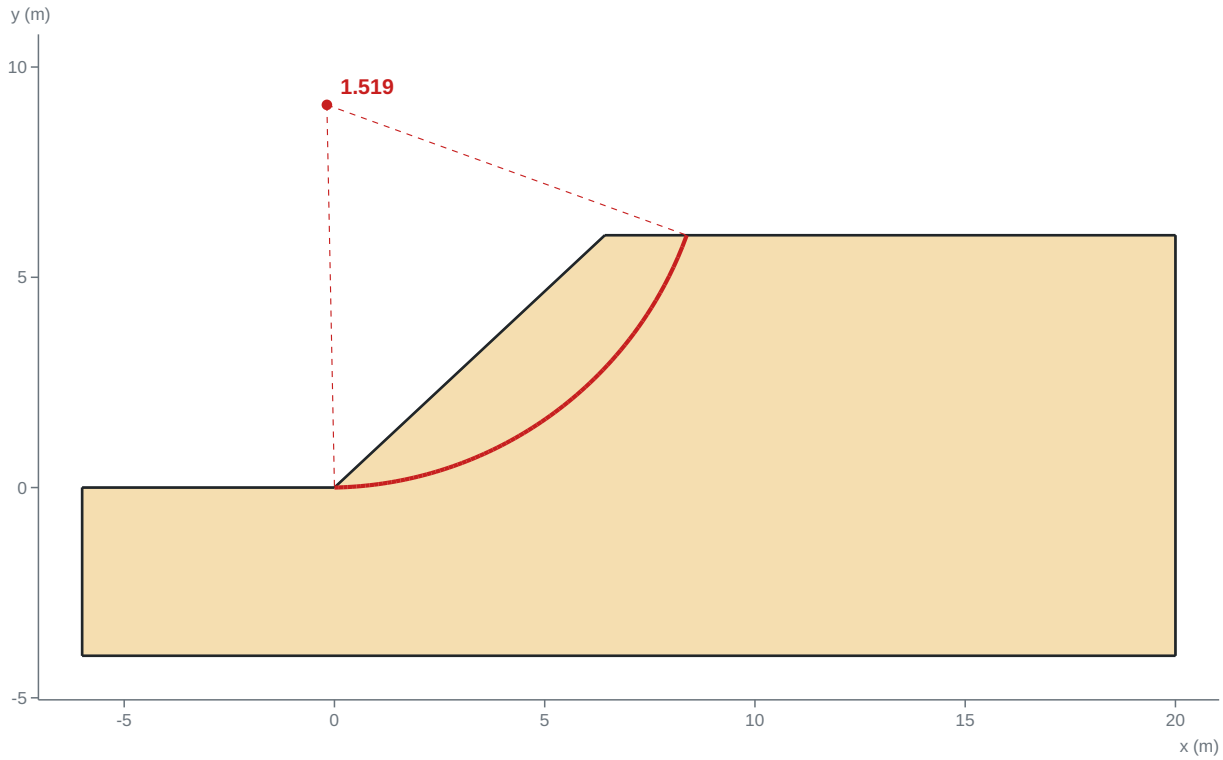



*Figure 11. V008a: critical circle (Spencer, FOS = 1.519).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 18.00 | c' = 11.64 | 24.70° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.50 | Mohr-Coulomb strength | Baker (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.536 | 1.519 |

*Baker (2003) gives the factor of safety for the Mohr-Coulomb fit with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).*

## 3.12 V008b — Baker (2003) London clay slope, Mohr-Coulomb fit

*Baker (2003) example 3.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

43° slope, 6 m high, with the Mohr-Coulomb envelope fitted to London clay data (c' = 6 kPa, φ' = 32°); critical circle search.

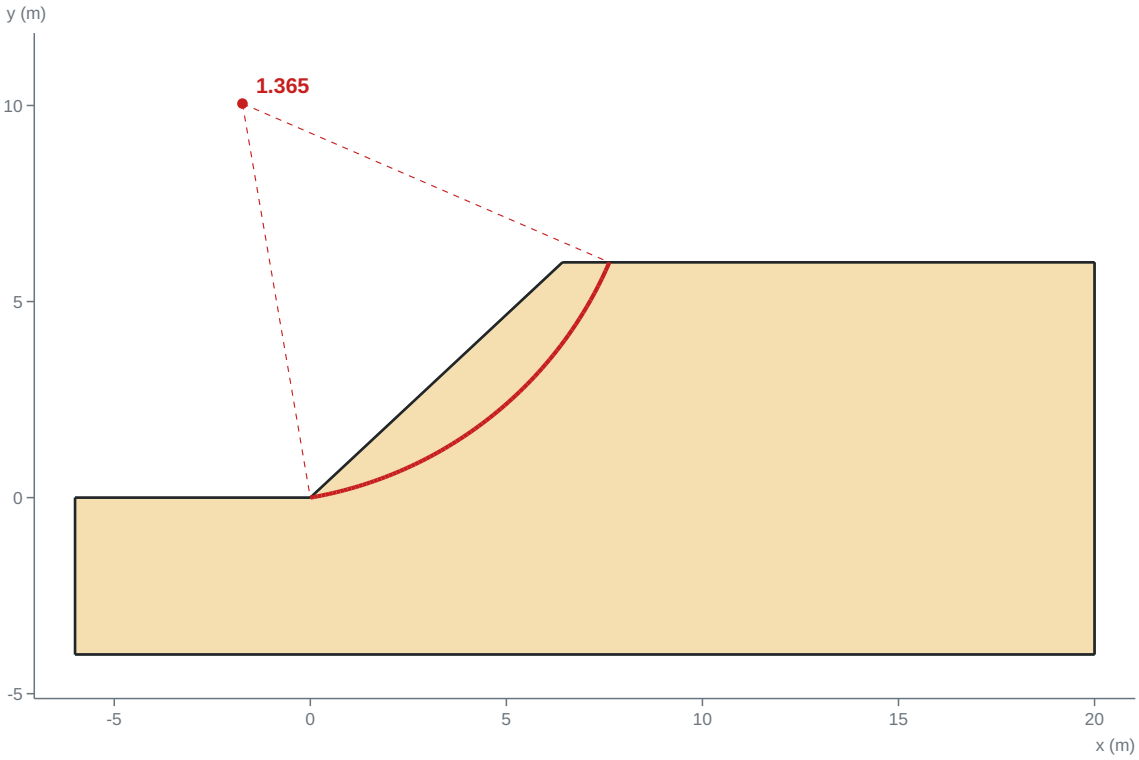



*Figure 12. V008b: critical circle (Spencer, FOS = 1.365).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| London clay | Mohr-Coulomb | 18.00 | c' = 6.00 | 32.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.35 | Mohr-Coulomb strength | Baker (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.366 | 1.365 |

*Baker (2003) gives the factor of safety for the Mohr-Coulomb fit with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).*

## 3.13 V009a — Baker & Leshchinsky clay-core dam, reservoir and crest crack

*Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Granular dam with a clay core (a hexagon reaching the base) on a hard base, half-full reservoir standing on the face (free water), phreatic surface through the core and a 5 m dry tension crack at the crest; a commercial program's critical circle.

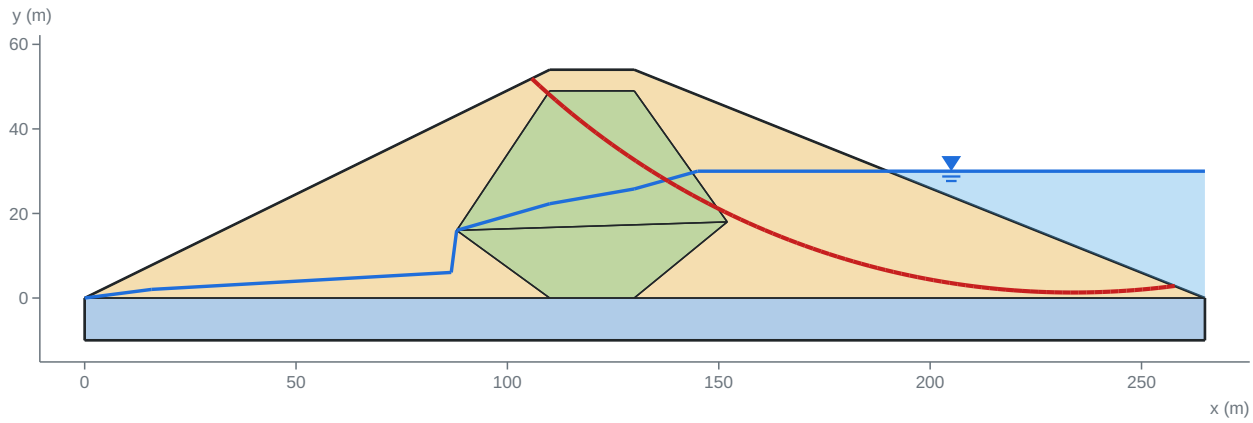



*Figure 13. V009a: analysed circle (Spencer, FOS = 1.926).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Granular fill | Mohr-Coulomb | 21.50 | c' = 0.00 | 40.00° |
| Clay core | Mohr-Coulomb | 20.00 | c' = 20.00 | 20.00° |
| Hard base | Mohr-Coulomb | 24.00 | c' = 200.0 | 45.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.925 | 1.926 |

*The published 1.91 belongs to Baker & Leshchinsky's own non-circular surface (V009b), not to this circle.*

## 3.14 V009b — Baker & Leshchinsky dam, Baker's non-circular surface

*Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

As V009a, on the non-circular surface published by Baker & Leshchinsky.

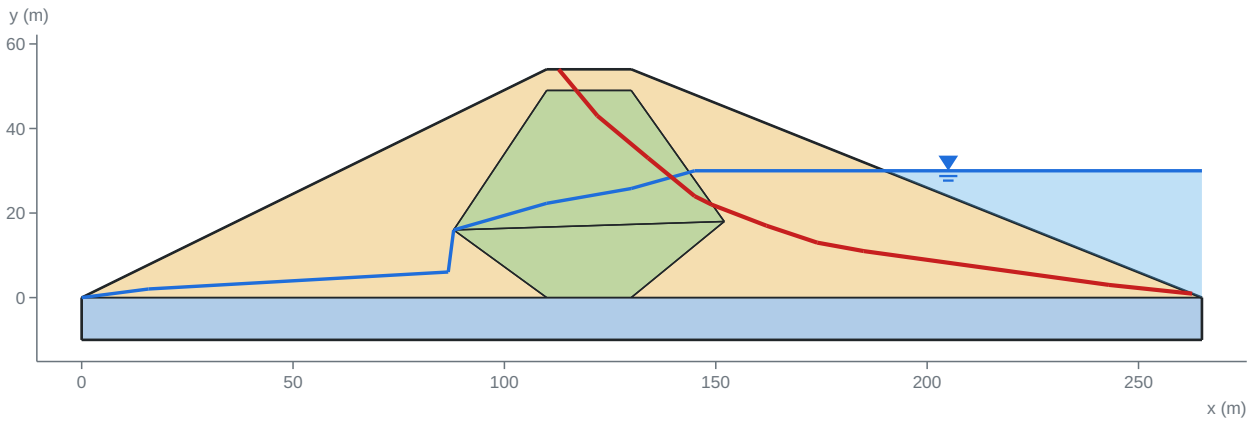



*Figure 14. V009b: analysed surface (Spencer, FOS = 1.882).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Granular fill | Mohr-Coulomb | 21.50 | c' = 0.00 | 40.00° |
| Clay core | Mohr-Coulomb | 20.00 | c' = 20.00 | 20.00° |
| Hard base | Mohr-Coulomb | 24.00 | c' = 200.0 | 45.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.91 | Spencer | Baker & Leshchinsky | 1.882 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | – | 1.882 |

## 3.15 V009c — Baker & Leshchinsky clay-core dam, critical surface search

*Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.*

**Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)**

The V009a clay-core dam (reservoir on the face, 5 m dry crest crack) with a circular search and non-circular optimisation.

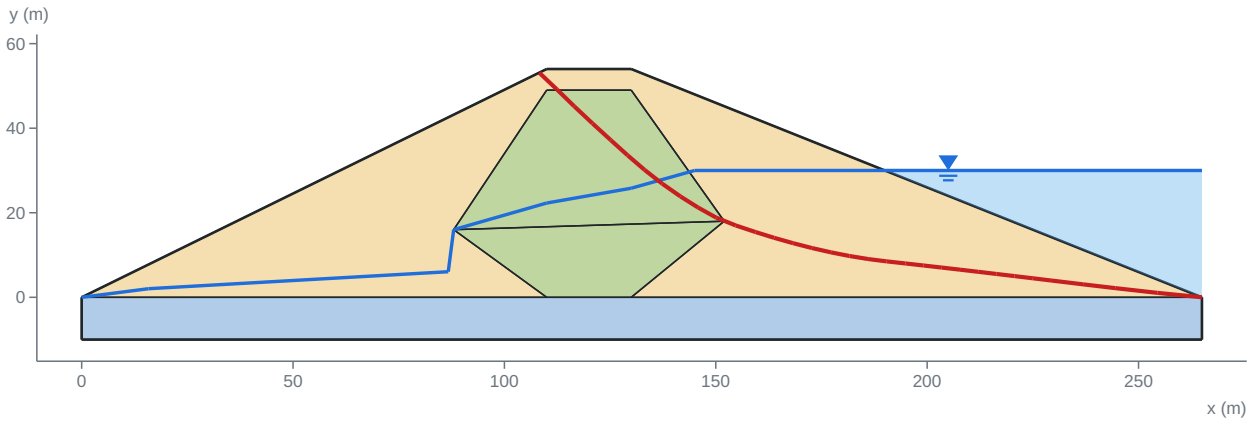



*Figure 15. V009c: critical surface (Spencer, non-circular, FOS = 1.867).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Granular fill | Mohr-Coulomb | 21.50 | c' = 0.00 | 40.00° |
| Clay core | Mohr-Coulomb | 20.00 | c' = 20.00 | 20.00° |
| Hard base | Mohr-Coulomb | 24.00 | c' = 200.0 | 45.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.91 | Spencer, non-circular | Baker & Leshchinsky (2001) | 1.867 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, non-circular | 1.868 | 1.867 |

*Baker & Leshchinsky's 1.91 is for their own surface (V009b); searches find lower surfaces.*

## 3.16 V010 — Borges & Cardoso embankment 2 (geosynthetic)

*Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

5 m embankment (φ′ = 35°) on four soft clays with Su increasing with depth (Borges & Cardoso Table 6, p. 408: clay 3 from 16 to 18.4 kPa, clay 4 from 18.4 to 55.1 kPa), a 200 kN/m geosynthetic at y = 1 m (axial, active, not anchored, interface friction 33.7° on both faces); a commercial program's critical circle.

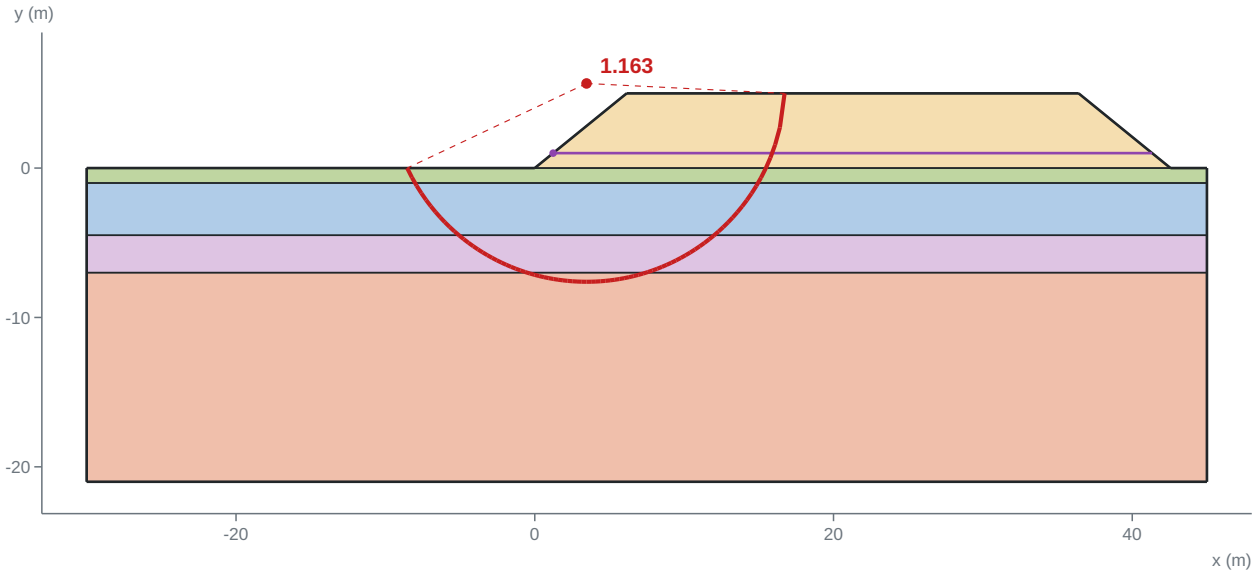



*Figure 16. V010: analysed circle (Morgenstern-Price (half-sine), FOS = 1.163).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 20.00 | c' = 0.00 | 35.00° |
| Clay 1 | Undrained (φ = 0) | 17.00 | Su = 33.00 | 0° |
| Clay 2 | Undrained (φ = 0) | 17.00 | Su = 16.00 | 0° |
| Clay 3 | Undrained (φ = 0) | 17.00 | Su = 16.00 at y = -4.50, +0.95 kPa/m below | 0° |
| Clay 4 | Undrained (φ = 0) | 17.00 | Su = 18.38 at y = -7.00, +2.63 kPa/m below | 0° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.171 | 1.163 |
| Bishop simplified | 1.170 | 1.164 |

## 3.17 V010b — Borges & Cardoso embankment 2 (geosynthetic), the paper's circle B

*Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421; case 2, Table 11, Fig. 14.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V010 embankment as Borges & Cardoso analyse it: the 200 kN/m geosynthetic at 0.9 m height (p. 407) with its full horizontal force in the resisting moment (passive), on their circle B (3.0, 8.0, R = 15.04).

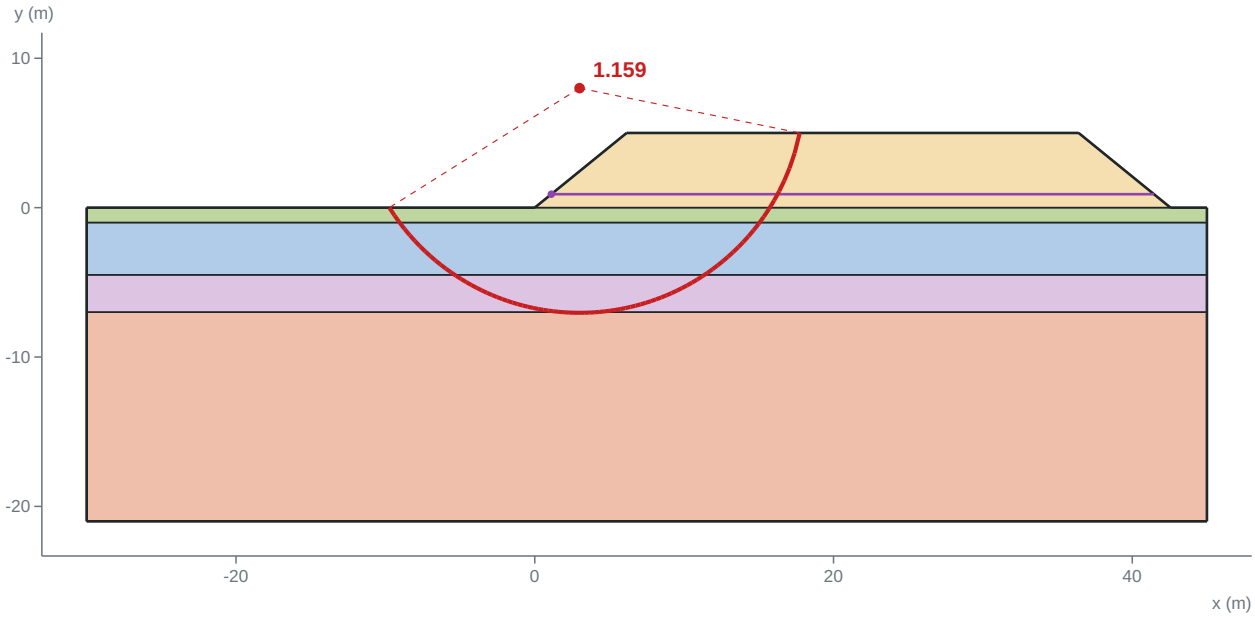



*Figure 17. V010b: analysed circle (Bishop simplified, FOS = 1.159).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 20.00 | c' = 0.00 | 35.00° |
| Clay 1 | Undrained (φ = 0) | 17.00 | Su = 33.00 | 0° |
| Clay 2 | Undrained (φ = 0) | 17.00 | Su = 16.00 | 0° |
| Clay 3 | Undrained (φ = 0) | 17.00 | Su = 16.00 at y = -4.50, +0.95 kPa/m below | 0° |
| Clay 4 | Undrained (φ = 0) | 17.00 | Su = 18.38 at y = -7.00, +2.63 kPa/m below | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.15 | Bishop simplified | Borges & Cardoso | 1.159 |
| 1.19 | authors' moment method, circle A | Borges & Cardoso (2002) | – |
| 1.22 | finite elements, circle A | Borges & Cardoso (2002) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | – | 1.159 |

*Borges & Cardoso use their own moment-equilibrium method of slices on circles (Kaniraj & Abdullah 1993 with their improvements); simplified Bishop is the nearest of cSlope's methods. Circle B's coordinates and the geosynthetic level are printed in the paper (Table 11, p. 407).*

## 3.18 V011a — Borges & Cardoso embankment 3, H = 7 m, circle A

*Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Granular fill (two stages) on five soft clay layers with a T = 200 kN/m geosynthetic at y = 0.9 m (axial, passive as in a commercial program, pullout over 1.3 m from each end); Bishop on circle A of the paper.

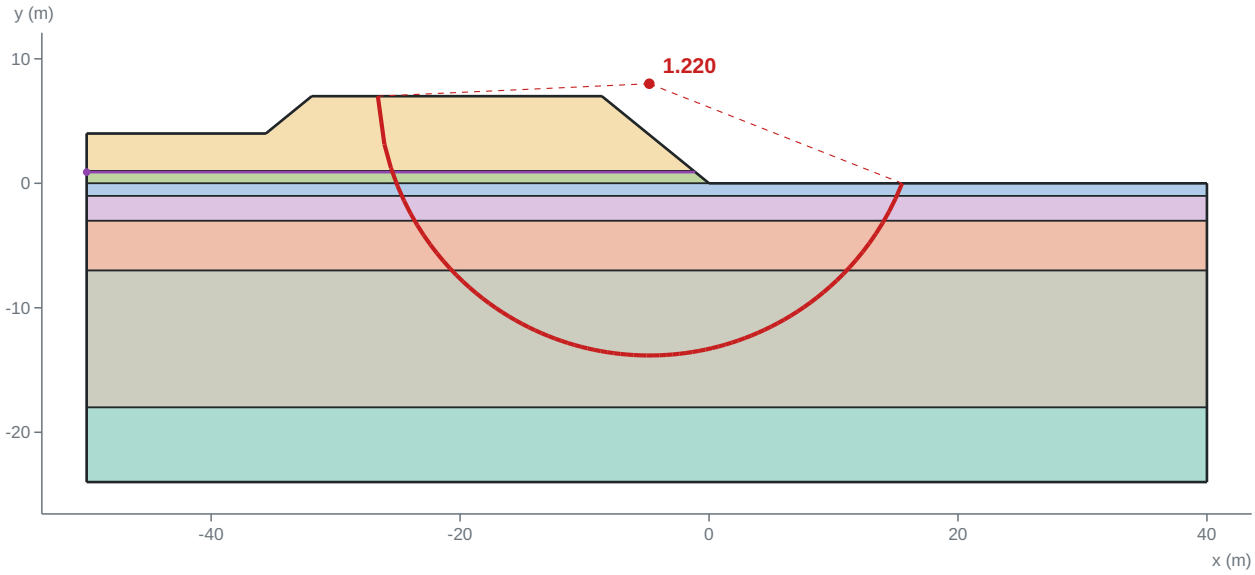



*Figure 18. V011a: analysed circle (Bishop simplified, FOS = 1.220).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Upper embankment | Mohr-Coulomb | 21.90 | c' = 0.00 | 35.00° |
| Lower embankment | Mohr-Coulomb | 17.20 | c' = 0.00 | 33.00° |
| Clay 1 | Undrained (φ = 0) | 18.00 | Su = 43.00 | 0° |
| Clay 2 | Undrained (φ = 0) | 16.60 | Su = 31.00 | 0° |
| Clay 3 | Undrained (φ = 0) | 13.50 | Su = 30.00 | 0° |
| Clay 4 | Undrained (φ = 0) | 17.00 | Su = 32.00 | 0° |
| Clay 5 | Undrained (φ = 0) | 17.50 | Su = 32.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.25 | not stated | Borges & Cardoso (2002) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.230 | 1.220 |

*Borges & Cardoso print two decimals and do not name the method; the other programs differ from them by up to 2.5 %.*

## 3.19 V011b — Borges & Cardoso embankment 3, H = 7 m, circle B

*Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V011a on circle B.

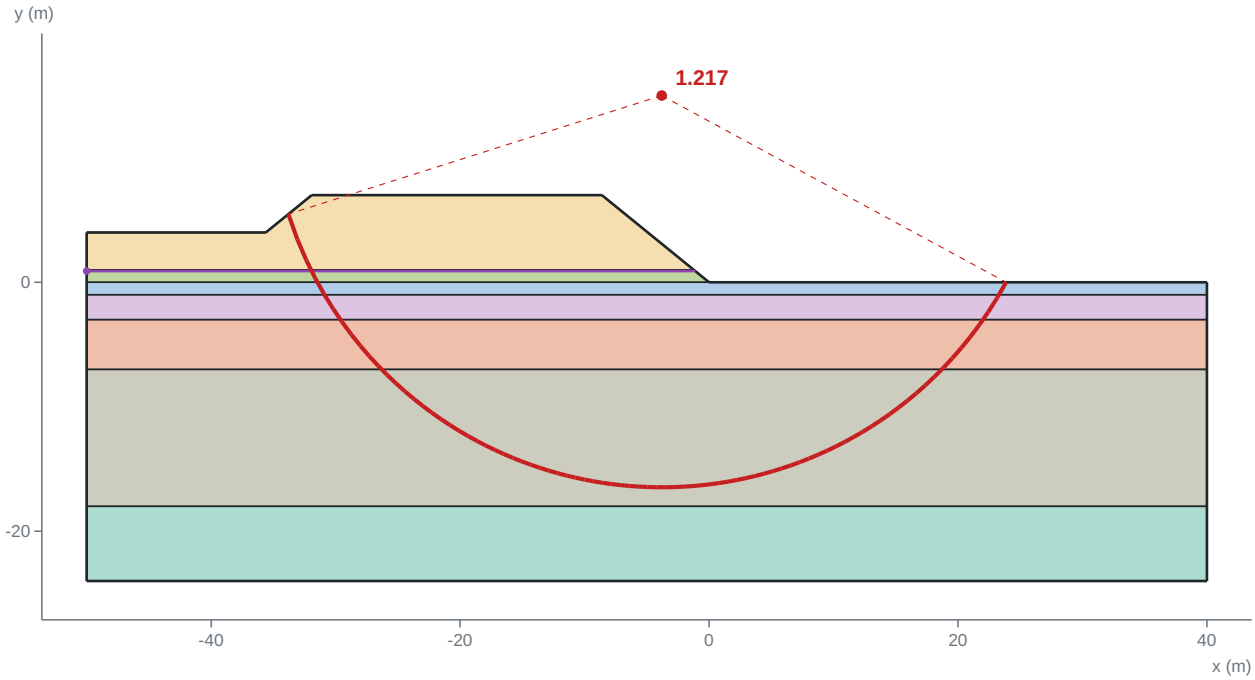



*Figure 19. V011b: analysed circle (Bishop simplified, FOS = 1.217).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Upper embankment | Mohr-Coulomb | 21.90 | c' = 0.00 | 35.00° |
| Lower embankment | Mohr-Coulomb | 17.20 | c' = 0.00 | 33.00° |
| Clay 1 | Undrained (φ = 0) | 18.00 | Su = 43.00 | 0° |
| Clay 2 | Undrained (φ = 0) | 16.60 | Su = 31.00 | 0° |
| Clay 3 | Undrained (φ = 0) | 13.50 | Su = 30.00 | 0° |
| Clay 4 | Undrained (φ = 0) | 17.00 | Su = 32.00 | 0° |
| Clay 5 | Undrained (φ = 0) | 17.50 | Su = 32.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.19 | not stated | Borges & Cardoso (2002) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.220 | 1.217 |

*Borges & Cardoso print two decimals and do not name the method; the other programs differ from them by up to 2.5 %.*

## 3.20 V012 — Borges & Cardoso embankment 3, H = 8.75 m, circle C

*Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The second fill stage (8.75 m crest) on circle C.

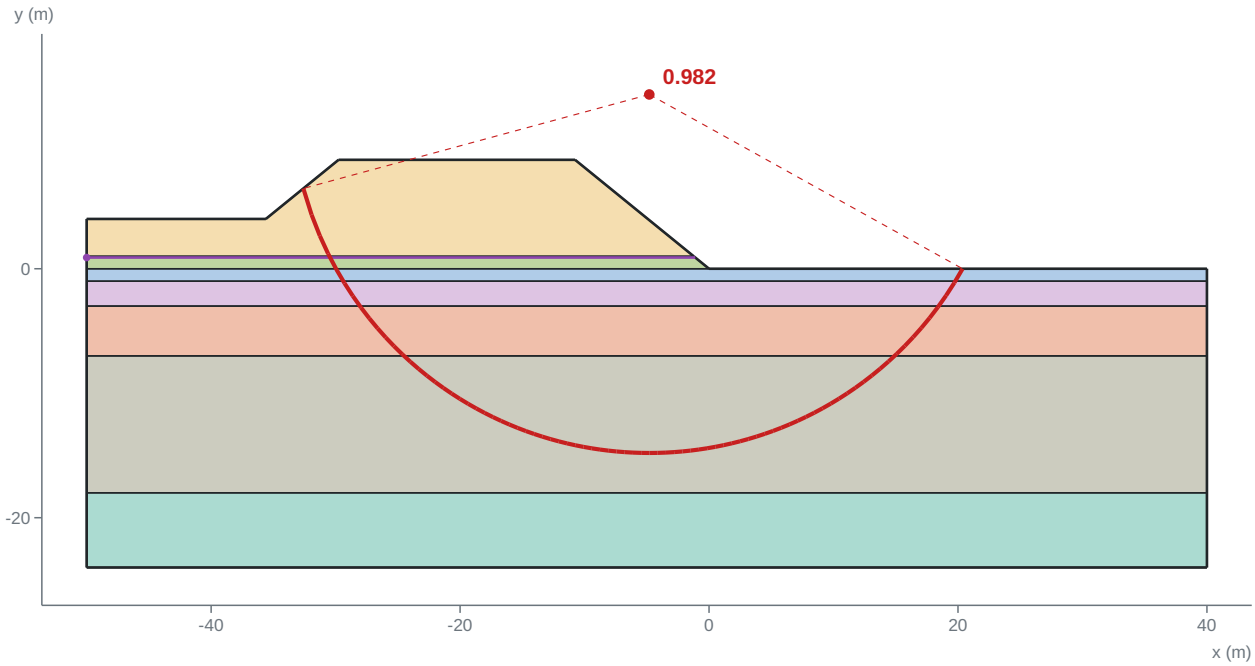



*Figure 20. V012: analysed circle (Bishop simplified, FOS = 0.982).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Upper embankment | Mohr-Coulomb | 21.90 | c' = 0.00 | 35.00° |
| Lower embankment | Mohr-Coulomb | 17.20 | c' = 0.00 | 33.00° |
| Clay 1 | Undrained (φ = 0) | 18.00 | Su = 43.00 | 0° |
| Clay 2 | Undrained (φ = 0) | 16.60 | Su = 31.00 | 0° |
| Clay 3 | Undrained (φ = 0) | 13.50 | Su = 30.00 | 0° |
| Clay 4 | Undrained (φ = 0) | 17.00 | Su = 32.00 | 0° |
| Clay 5 | Undrained (φ = 0) | 17.50 | Su = 32.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.99 | not stated | Borges & Cardoso (2002) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 0.980 | 0.982 |

*Borges & Cardoso print two decimals and do not name the method.*

## 3.21 V013a — Cai & Ugai (2000) slope, no pile (see V013b, V013c, V013d and V013e for the pile r

*Cai & Ugai (2000), Soils and Foundations 40(1).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Homogeneous 1:1.5 slope, 10 m high, without piles; critical circle search.

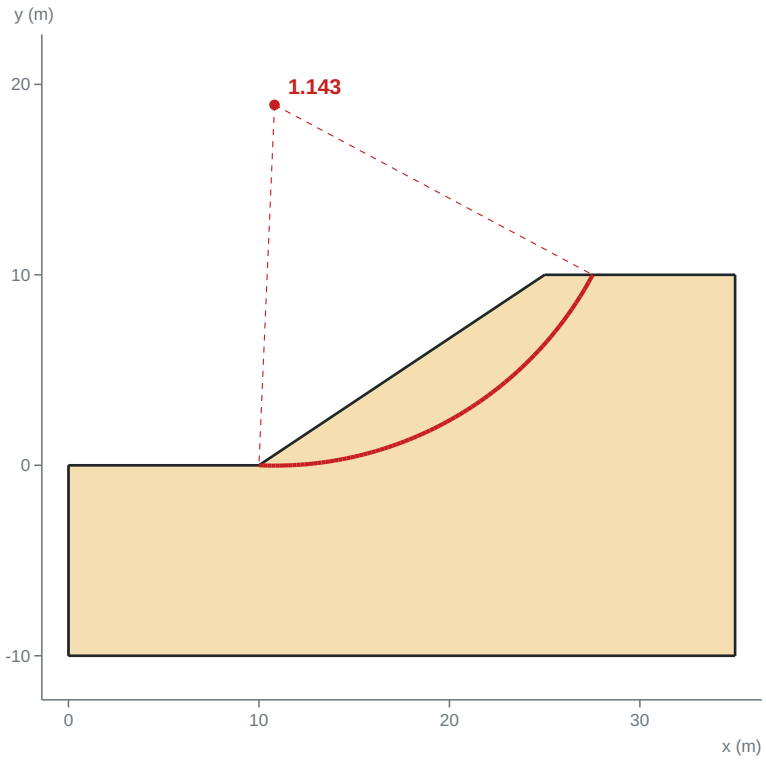



*Figure 21. V013a: critical circle (Bishop simplified, FOS = 1.143).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 10.00 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.13 | Bishop simplified | Cai & Ugai | 1.143 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.140 | 1.143 |

## 3.22 V013b — Cai & Ugai slope, pile row D1/D = 2

*Cai & Ugai (2000), Soils and Foundations 40(1).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

A row of 0.8 m piles at mid-slope, centre-to-centre 1.6 m, with the Ito & Matsui force (passive); Bishop search.

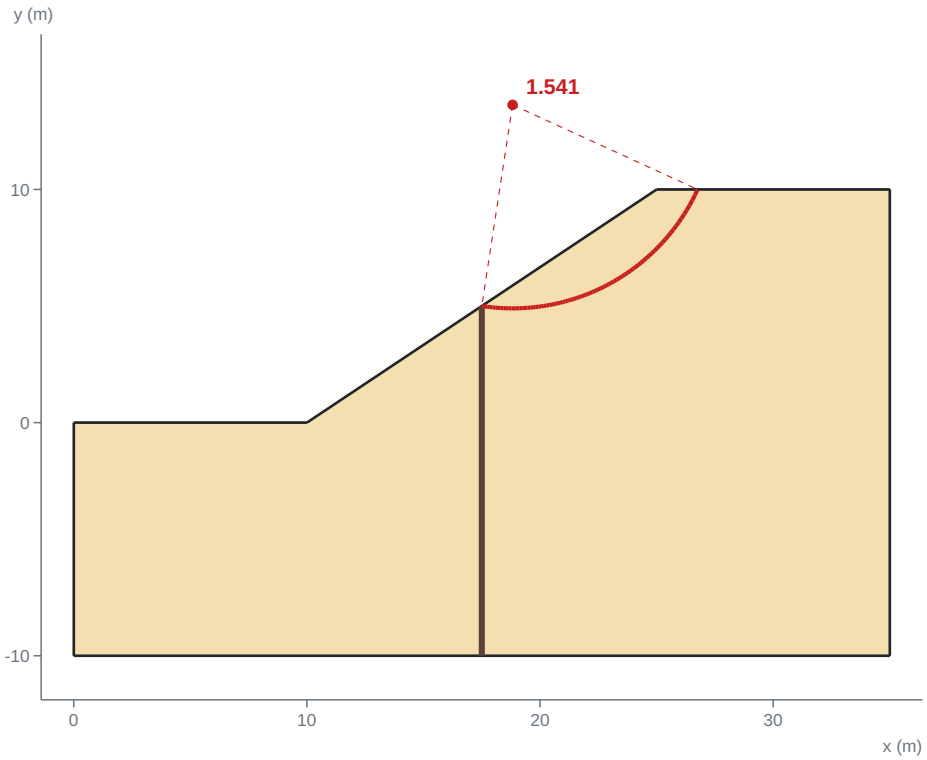



*Figure 22. V013b: critical circle (Bishop simplified, FOS = 1.541).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 10.00 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.54 | not stated | Cai & Ugai (2000) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.540 | 1.541 |

*Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.*

## 3.23 V013c — Cai & Ugai slope, pile row D1/D = 3

*Cai & Ugai (2000), Soils and Foundations 40(1).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

As V013b at 2.4 m spacing.

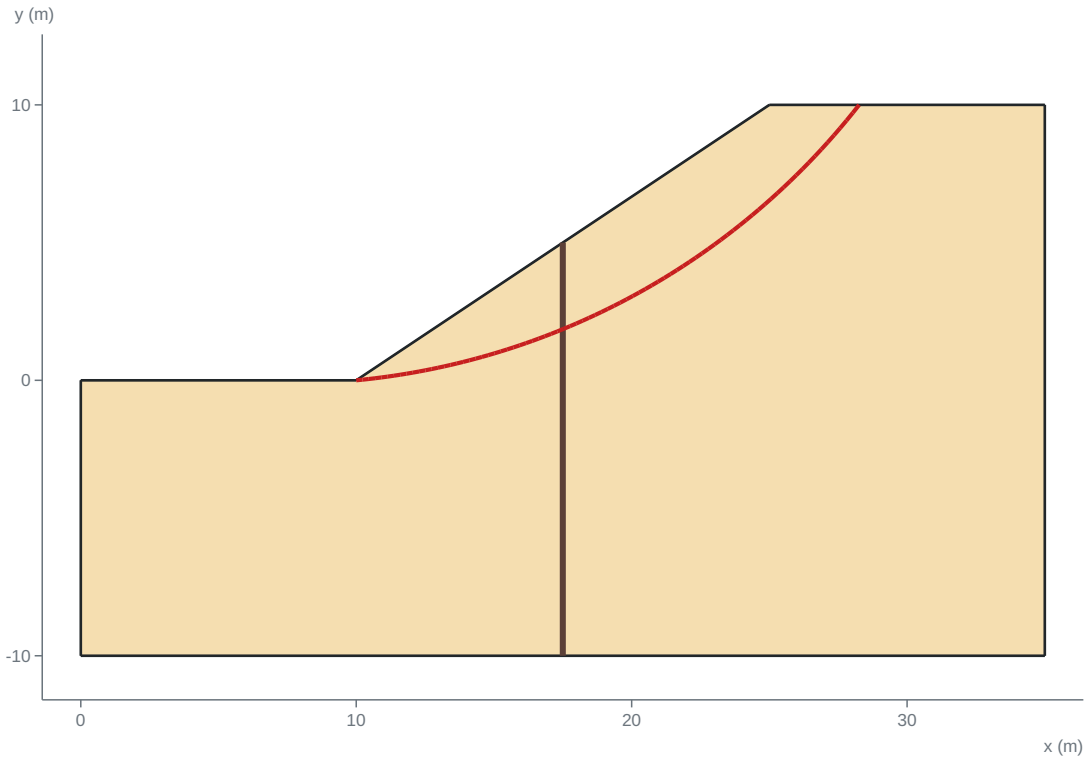



*Figure 23. V013c: critical circle (Bishop simplified, FOS = 1.437).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 10.00 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.37 | not stated | Cai & Ugai (2000) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.430 | 1.437 |

*Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.*

## 3.24 V013d — Cai & Ugai slope, pile row D1/D = 4

*Cai & Ugai (2000), Soils and Foundations 40(1).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

As V013b at 3.2 m spacing.

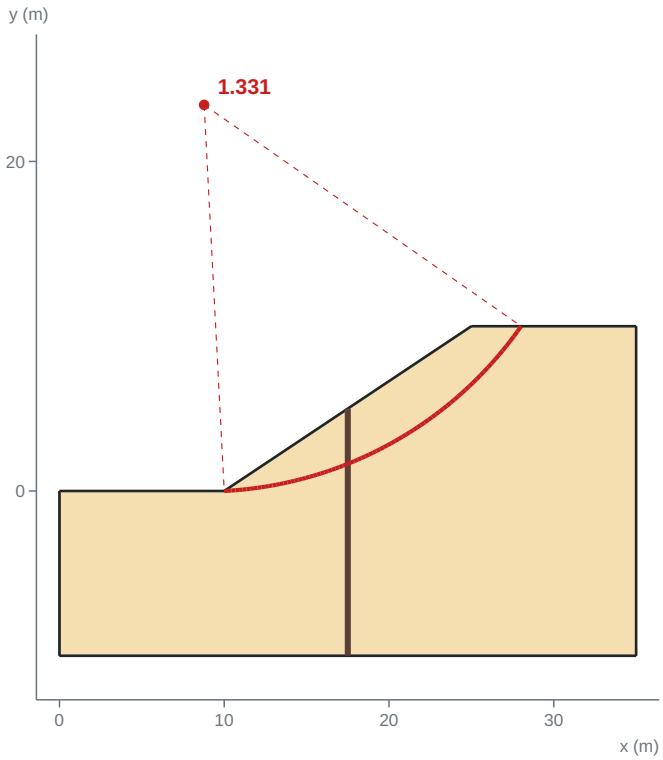



*Figure 24. V013d: critical circle (Bishop simplified, FOS = 1.331).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 10.00 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.31 | not stated | Cai & Ugai (2000) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.330 | 1.331 |

*Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.*

## 3.25 V013e — Cai & Ugai slope, pile row D1/D = 6

*Cai & Ugai (2000), Soils and Foundations 40(1).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

As V013b at 4.8 m spacing.

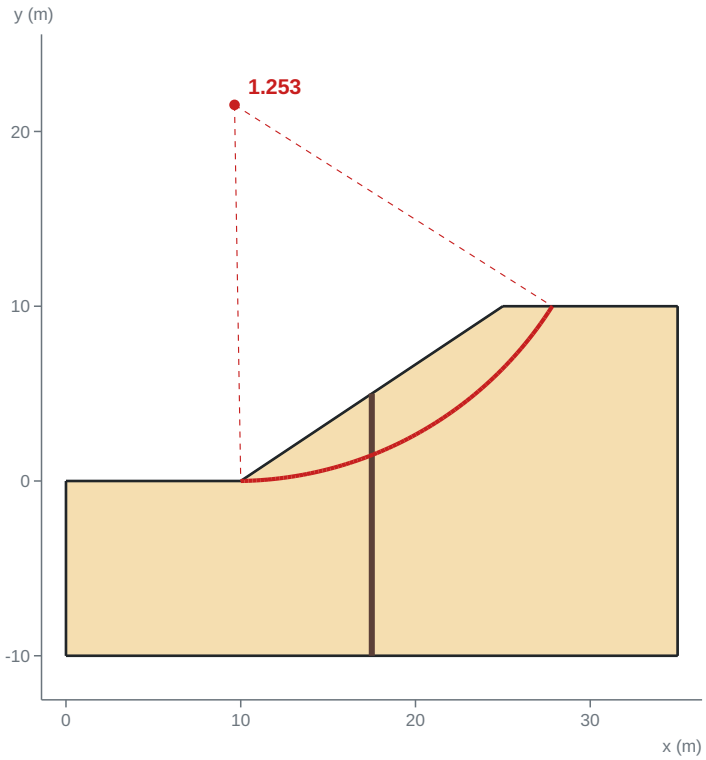



*Figure 25. V013e: critical circle (Bishop simplified, FOS = 1.253).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 10.00 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.25 | not stated | Cai & Ugai (2000) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.250 | 1.253 |

*Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.*

## 3.26 V014 — Nailed wall, 14 rows, deep wedge

*Caltrans soil nail reference manual.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Nail-reinforced wall with 14 horizontal rows (per-row length, capacity and bond; fully anchored at the wall, tangent, passive) on the printed bilinear wedge. Converted to SI.

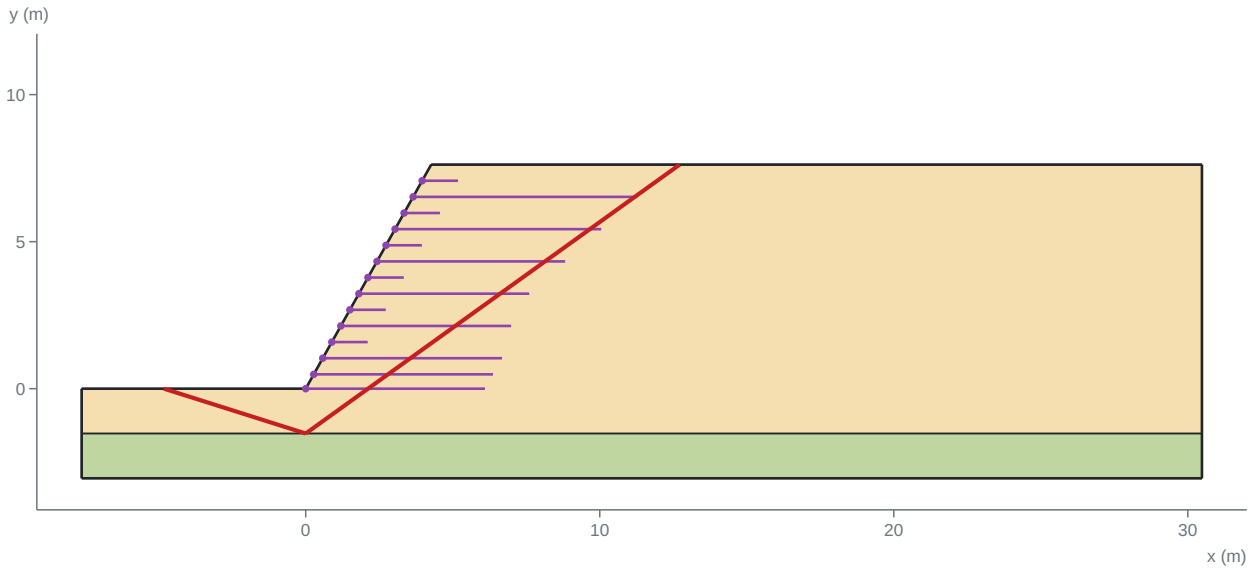



*Figure 26. V014: analysed surface (Janbu corrected, FOS = 1.448).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 | Mohr-Coulomb | 19.64 | c' = 0.00 | 32.00° |
| Layer 2 | Mohr-Coulomb | 20.11 | c' = 23.94 | 35.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 1.448–1.460 | 1.448 |
| Spencer | 1.576 | 1.575 |

*The reference results come from design programs quoted in the soil nail manual; no independent value is published.*

## 3.27 V015 — Soldier-pile tieback wall

*Caltrans soil nail reference manual.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Two-layer slope cut by a soldier-pile wall with two grouted tieback rows (8 ft spacing); the soldier pile is a shear force of 5900 lb/ft where the surface passes, as a commercial program models it; the manual's bilinear wedge from the wall toe. Converted to SI.

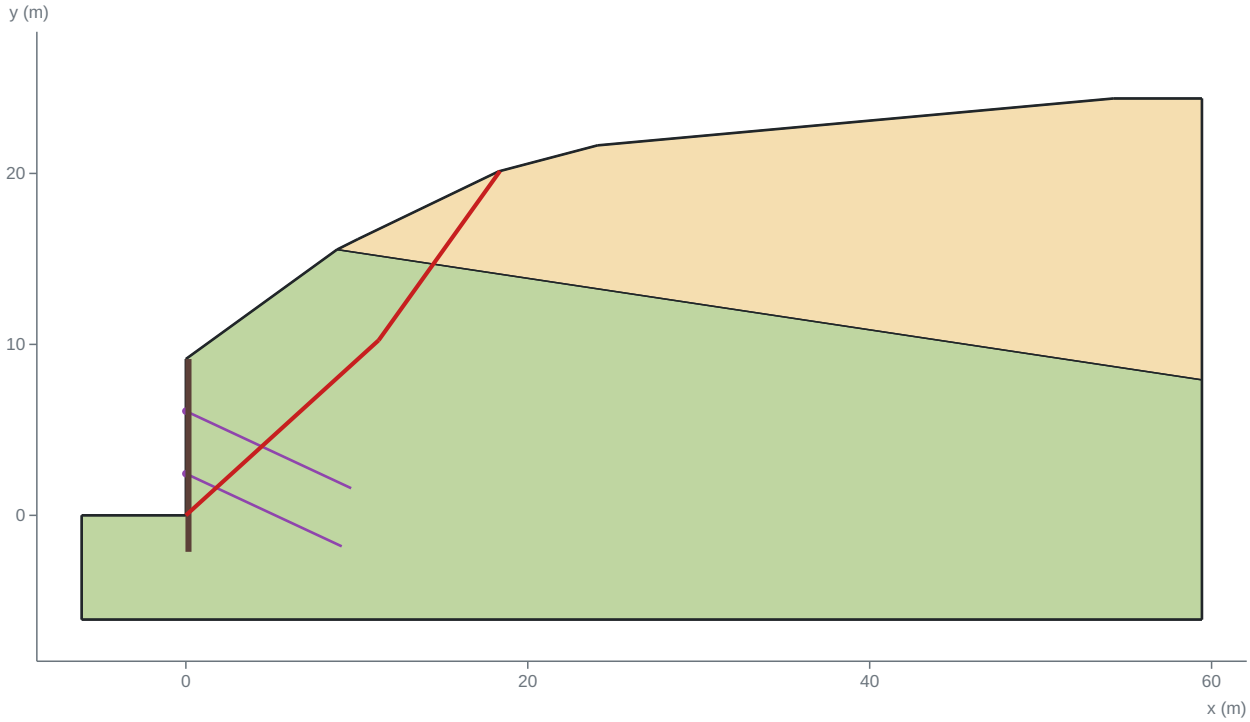



*Figure 27. V015: analysed surface (Janbu corrected, FOS = 1.476).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 | Mohr-Coulomb | 18.85 | c' = 28.73 | 24.00° |
| Layer 2 | Mohr-Coulomb | 20.42 | c' = 14.36 | 34.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 1.479 | 1.476 |
| Janbu simplified | 1.446 | 1.442 |

*The reference results come from design programs quoted in the soil nail manual; no independent value is published.*

## 3.28 V016a — Stepped gabion wall, equivalent cohesion

*Cao et al. (2016); Grodecki (2017).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

4 m gabion wall of 1 m cubes with the steps facing outwards on a sloping two-layer profile, dry; mesh as an equivalent cohesion (59.7 kPa); the reference circle.

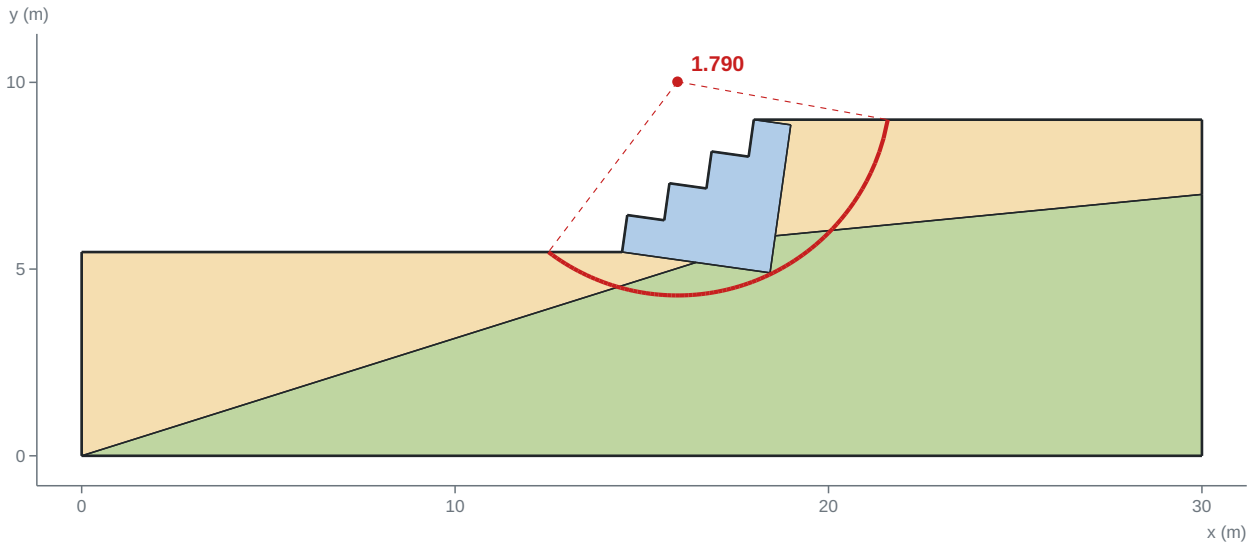



*Figure 28. V016a: analysed circle (Bishop simplified, FOS = 1.790).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil #1 | Mohr-Coulomb | 21.00 | c' = 5.00 | 30.00° |
| Soil #2 | Mohr-Coulomb | 20.00 | c' = 0.00 | 25.00° |
| Gabion wall | Mohr-Coulomb | 23.00 | c' = 59.70 | 42.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.787–1.790 | 1.790 |
| Spencer | 1.791 | 1.797 |

*The paper gives its results in figures only; the comparison is with the other programs.*

## 3.29 V016b — Stepped gabion wall, mesh as geosynthetic supports

*Cao et al. (2016); Grodecki (2017).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V016a with the mesh as geosynthetic supports (T = 100 kN/m) at the course interfaces.

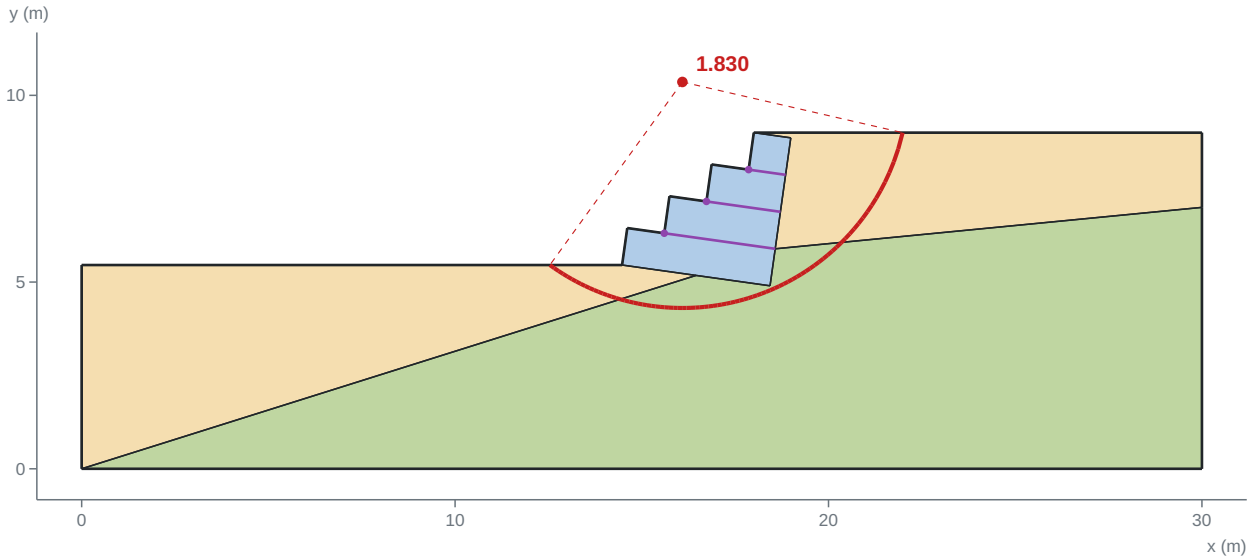



*Figure 29. V016b: analysed circle (Bishop simplified, FOS = 1.830).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil #1 | Mohr-Coulomb | 21.00 | c' = 5.00 | 30.00° |
| Soil #2 | Mohr-Coulomb | 20.00 | c' = 0.00 | 25.00° |
| Gabion wall | Mohr-Coulomb | 23.00 | c' = 0.00 | 42.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.830–1.835 | 1.830 |
| Spencer | 1.839 | 1.835 |

*The paper gives its results in figures only; the comparison is with the other programs.*

## 3.30 V016c — Gabion wall with weak joint layers

*Cao et al. (2016); Grodecki (2017).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V016 wall with 6 cm weak bands at the course interfaces (φ = 37.8°, c = 20.4 kPa from the joint tensile strength); the V016a reference circle, which passes below the wall and crosses no joint.

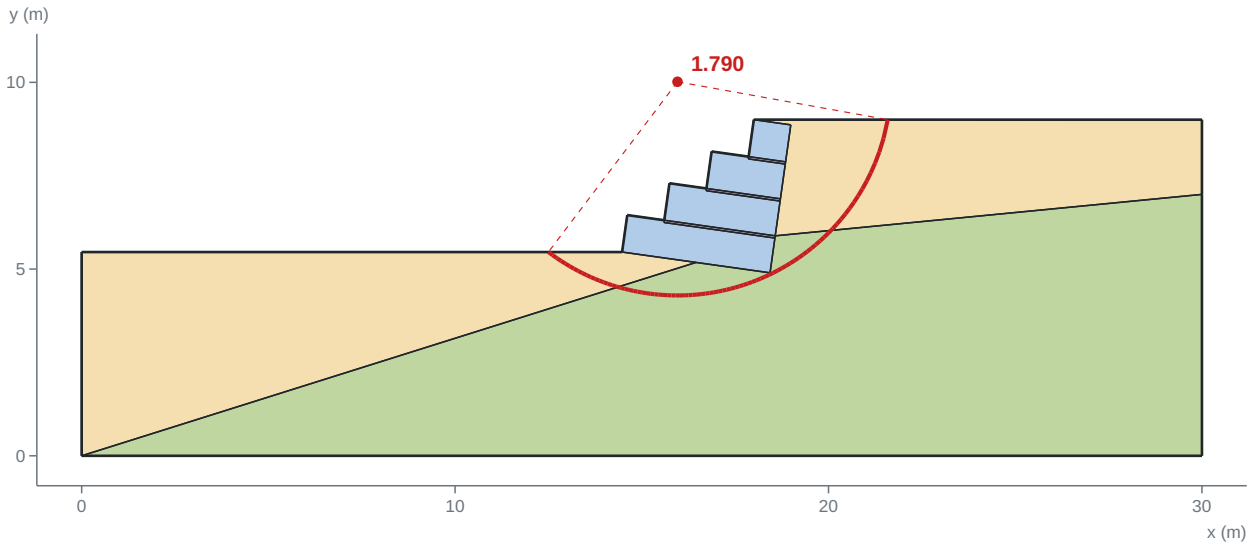



*Figure 30. V016c: analysed circle (Bishop simplified, FOS = 1.790).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil #1 | Mohr-Coulomb | 21.00 | c' = 5.00 | 30.00° |
| Soil #2 | Mohr-Coulomb | 20.00 | c' = 0.00 | 25.00° |
| Gabion wall | Mohr-Coulomb | 23.00 | c' = 59.70 | 42.00° |
| Weak joint | Mohr-Coulomb | 23.00 | c' = 20.40 | 37.80° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.790 | 1.790 |

*The paper gives its results in figures only; the comparison is with the other programs.*

## 3.31 V017a — Ontario gabion wall (Cao et al. 2016), equivalent cohesion

*Cao et al. (2016); Grodecki (2017).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

6 m battered gabion wall retaining a 30° backfill with a 12 kPa crest surcharge and a water table in the retained slope; the gabion mesh as an equivalent cohesion (100 kPa); the reference circle. Courses from the base 4, 3, 2.5, 2, 1.5 and 1 m wide.

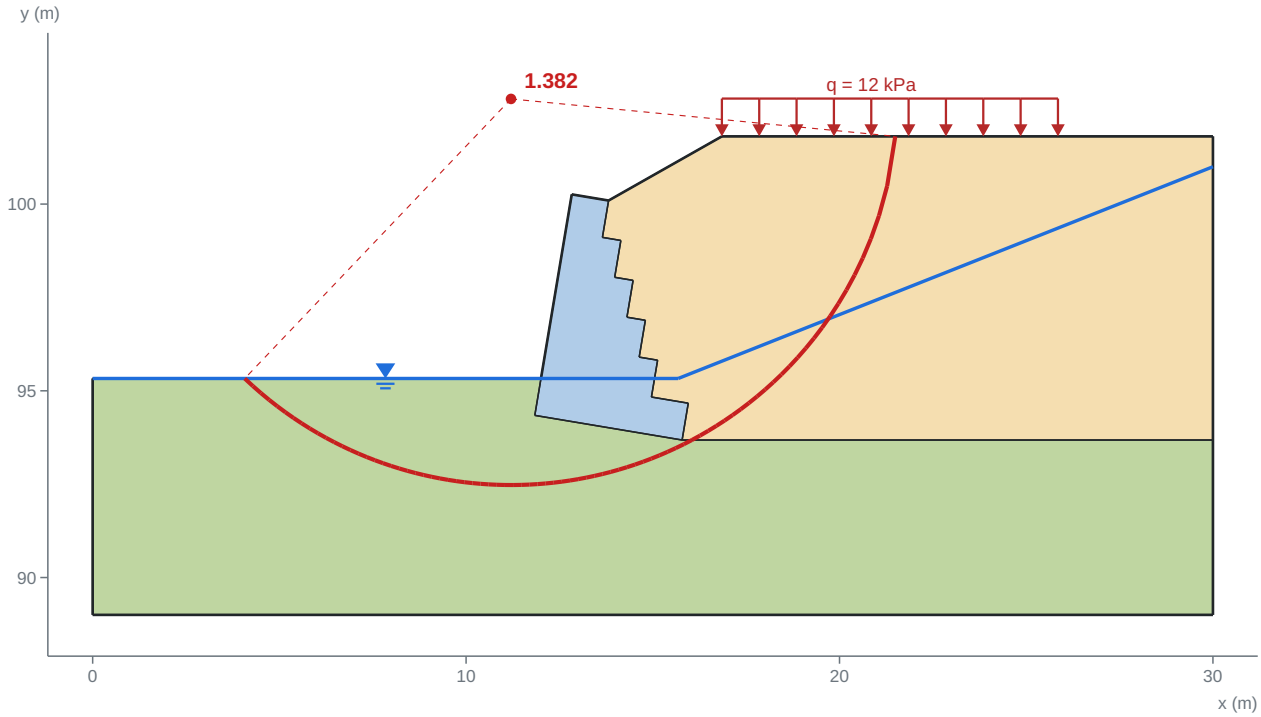



*Figure 31. V017a: analysed circle (Bishop simplified, FOS = 1.382).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil #1 | Mohr-Coulomb | 21.00 | c' = 0.00 | 32.00° |
| Soil #2 | Mohr-Coulomb | 20.00 | c' = 0.00 | 30.00° |
| Gabion wall | Mohr-Coulomb | 20.00 | c' = 100.0 | 45.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.373 | 1.382 |
| Spencer | 1.386 | 1.398 |

*The paper gives its results in figures only; the comparison is with the other programs.*

## 3.32 V017b — Ontario gabion wall (Cao et al. 2016), mesh as geosynthetic supports

*Cao et al. (2016); Grodecki (2017).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V017a with the steel mesh as geosynthetic supports at every course interface instead of the equivalent cohesion.

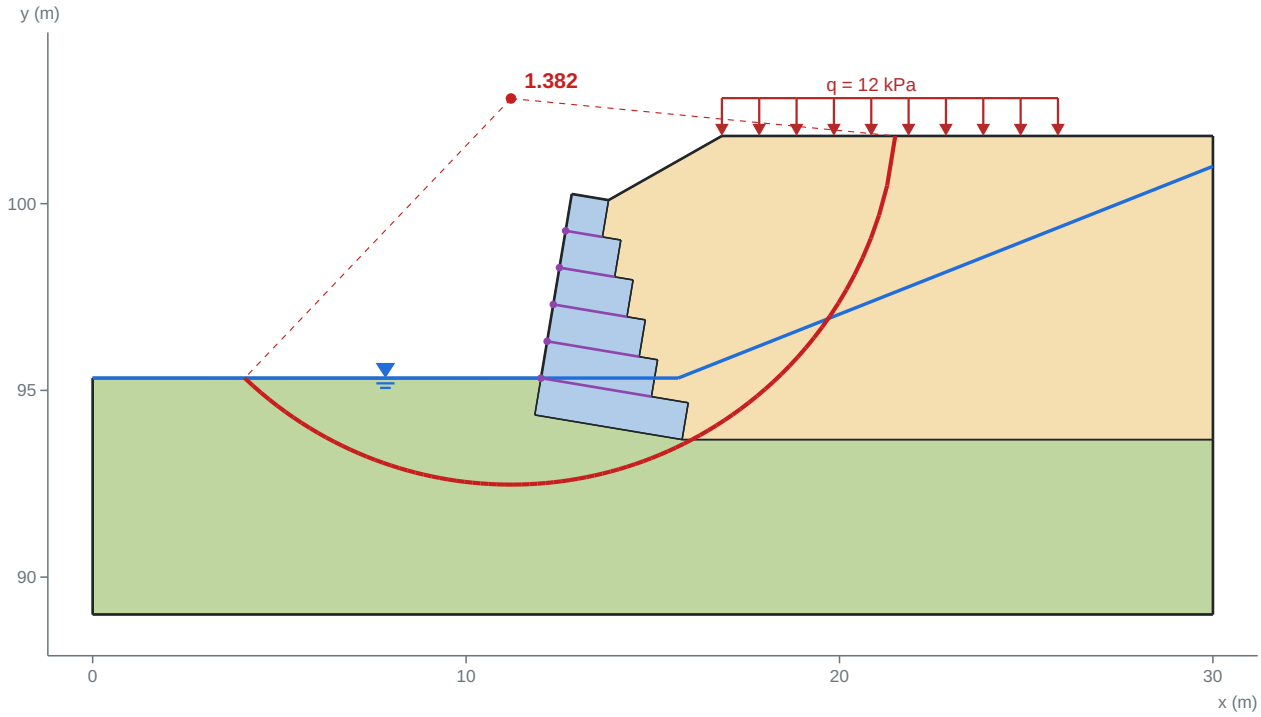



*Figure 32. V017b: analysed circle (Bishop simplified, FOS = 1.382).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil #1 | Mohr-Coulomb | 21.00 | c' = 0.00 | 32.00° |
| Soil #2 | Mohr-Coulomb | 20.00 | c' = 0.00 | 30.00° |
| Gabion wall | Mohr-Coulomb | 20.00 | c' = 0.00 | 45.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.378 | 1.382 |

*The paper gives its results in figures only; the comparison is with the other programs.*

## 3.33 V018 — Chen & Shao (1988) frictionless 60° slope, Prandtl mechanism

*Chen, Z. & Shao, C. (1988). Evaluation of minimum factor of safety in slope stability analysis. Can. Geotech. J. 25(4), 735–748.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Weightless, frictionless 10 m slope at 60° under the critical strip load (149.31 kPa) on the Prandtl surface (wedge, fan and passive wedge); theoretical FOS 1.0.

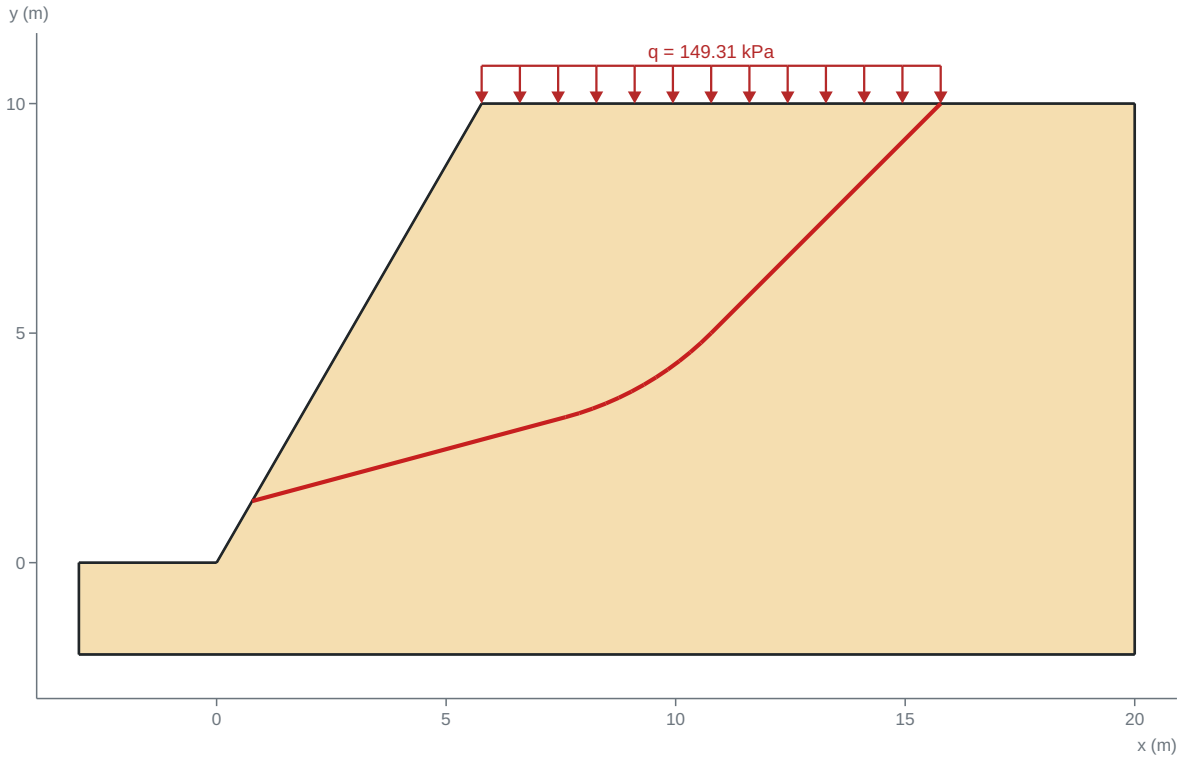



*Figure 33. V018: analysed surface (Spencer, FOS = 1.052).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Clay | Mohr-Coulomb | 0.00 | c' = 49.00 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.05 | Spencer | Chen & Shao (1988) | 1.052 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.051 | 1.052 |
| Morgenstern-Price (specified f(x)) | 1.009–1.017 | 1.008 |

## 3.34 V019 — Cheng et al. 11 m homogeneous slope

*Cheng, Y.M., Lansivaara, T. & Wei, W.B. (2007). Two-dimensional slope stability analysis by limit equilibrium and strength reduction methods. Computers and Geotechnics 34(3), 137–150.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

11 m homogeneous slope (c = 10 kPa, φ = 30°, γ = 20 kN/m³); circular search.

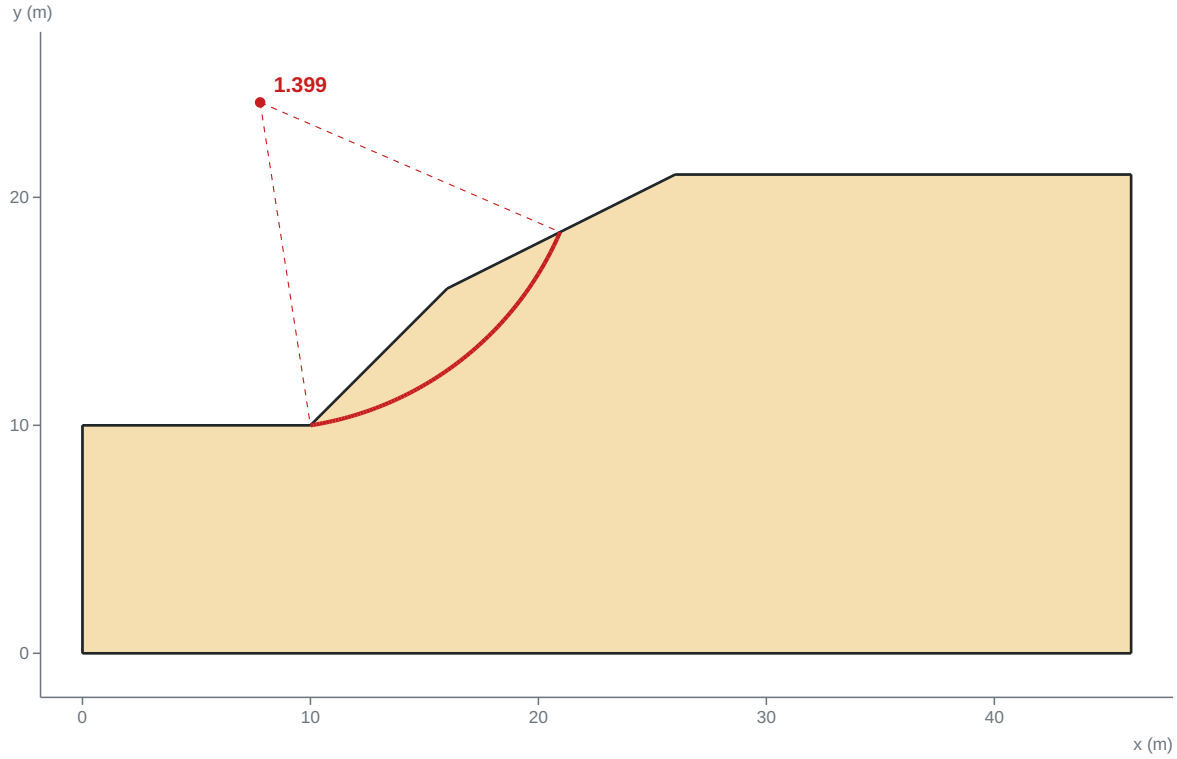



*Figure 34. V019: critical circle (Spencer (40 slices, vertical pore-pressure head), FOS = 1.399).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| soil | Mohr-Coulomb | 20.00 | c' = 10.00 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.383 | Spencer | Cheng et al. | 1.398 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer (40 slices, vertical pore-pressure head) | 1.380–1.398 | 1.399 |
| Spencer (vertical pore-pressure head) | – | 1.398 |
| Bishop simplified | 1.401 | 1.401 |

## 3.35 V020 — Cheng et al. benched slope, global minimum

*Cheng, Y.M., Lansivaara, T. & Wei, W.B. (2007). Two-dimensional slope stability analysis by limit equilibrium and strength reduction methods. Computers and Geotechnics 34(3), 137–150.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Homogeneous benched slope; the global minimum on the mid-lower face by circular search.

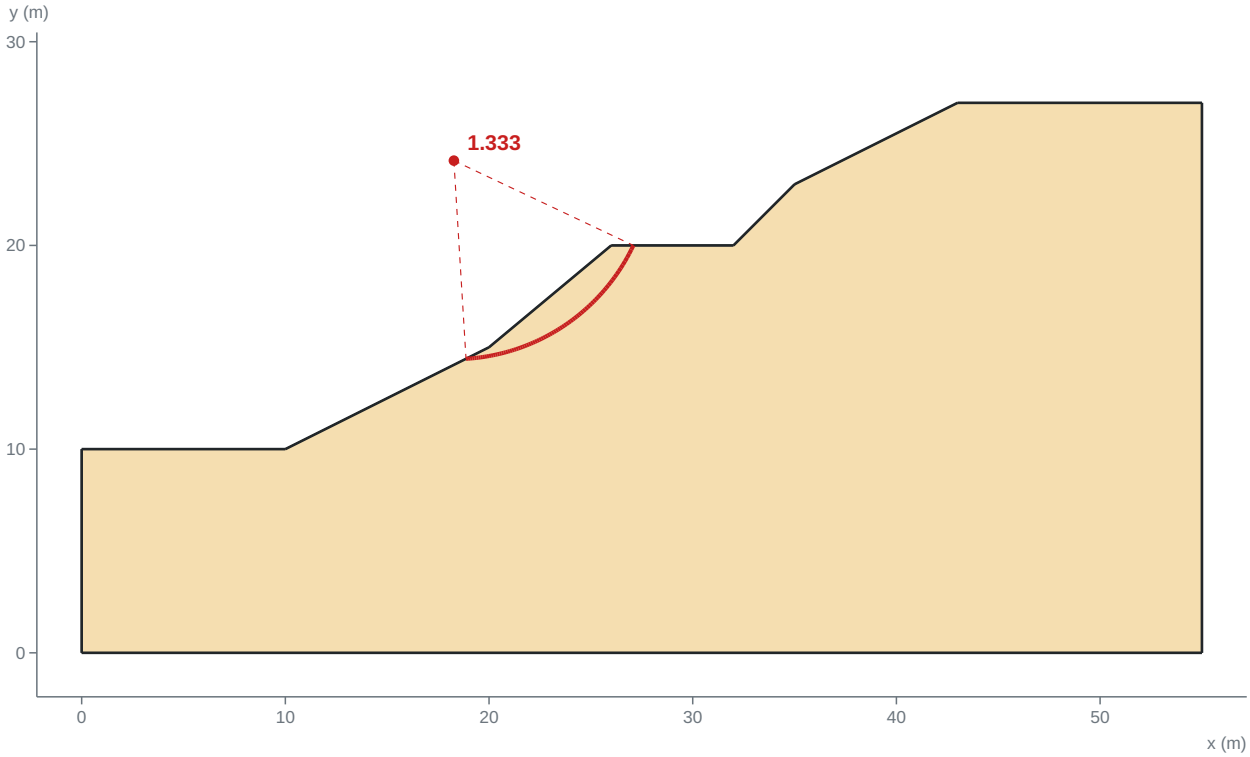



*Figure 35. V020: critical circle (Spencer, FOS = 1.333).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| soil | Mohr-Coulomb | 20.00 | c' = 5.00 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.33 | Spencer | Cheng et al. | 1.333 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.336–1.338 | 1.333 |
| Bishop simplified | 1.342 | 1.337 |

## 3.36 V021a — Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the bas

*Chowdhury & Xu (1995); Ireland (1954).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Chowdhury & Xu example 1, the Congress Street cut as a commercial program models it (Su = 55, 43, 56 kPa under a strengthless sand cap, digitised piezometric line), on the critical circle tangent to the base of clay 2.

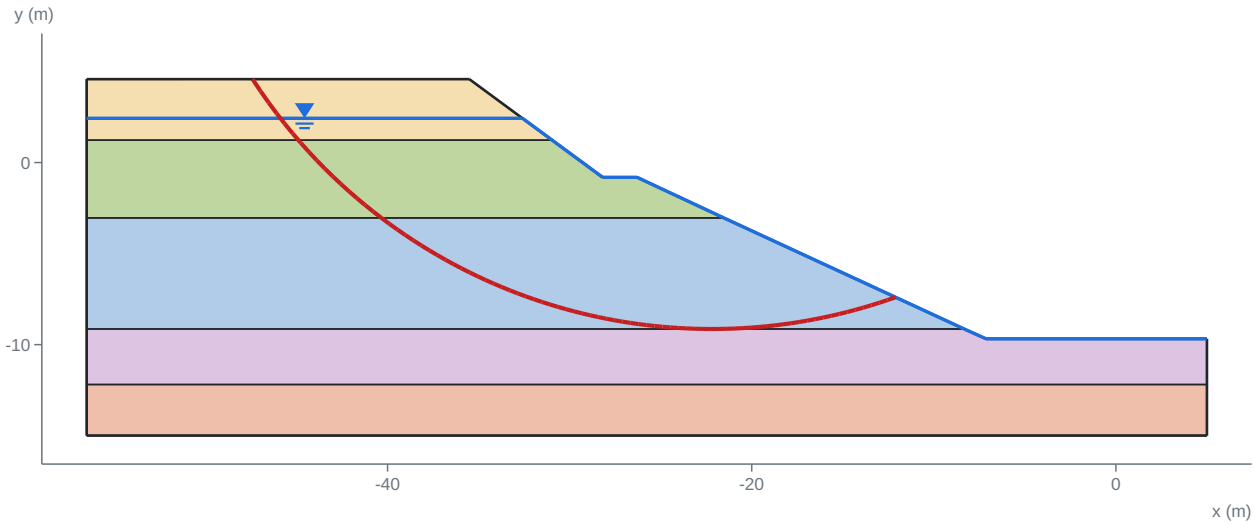



*Figure 36. V021a: analysed circle (Morgenstern-Price (half-sine), FOS = 1.128).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sand | Undrained (φ = 0) | 22.00 | Su = 0.00 | 0° |
| Clay 1 | Undrained (φ = 0) | 21.00 | Su = 55.00 | 0° |
| Clay 2 | Undrained (φ = 0) | 22.00 | Su = 43.00 | 0° |
| Clay 3 | Undrained (φ = 0) | 22.00 | Su = 56.00 | 0° |
| Bed | Mohr-Coulomb | 22.00 | c' = 200.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.1178 | Bishop simplified | Chowdhury & Xu | 1.128 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.132 | 1.128 |
| Bishop simplified | – | 1.128 |
| Bishop simplified (60 slices) | 1.128 | 1.128 |

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| Pf, share FOS < 1 (Monte Carlo) | – | 0.246 | 0.239 |
| β lognormal (Monte Carlo) | – | 0.650 | 0.672 |

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| Pf, share FOS < 1 (Monte Carlo) | – | 0.253 | 0.245 |

## 3.37 V021b — Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the bas

*Chowdhury & Xu (1995); Ireland (1954).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V021a on the critical circle tangent to the base of clay 3.

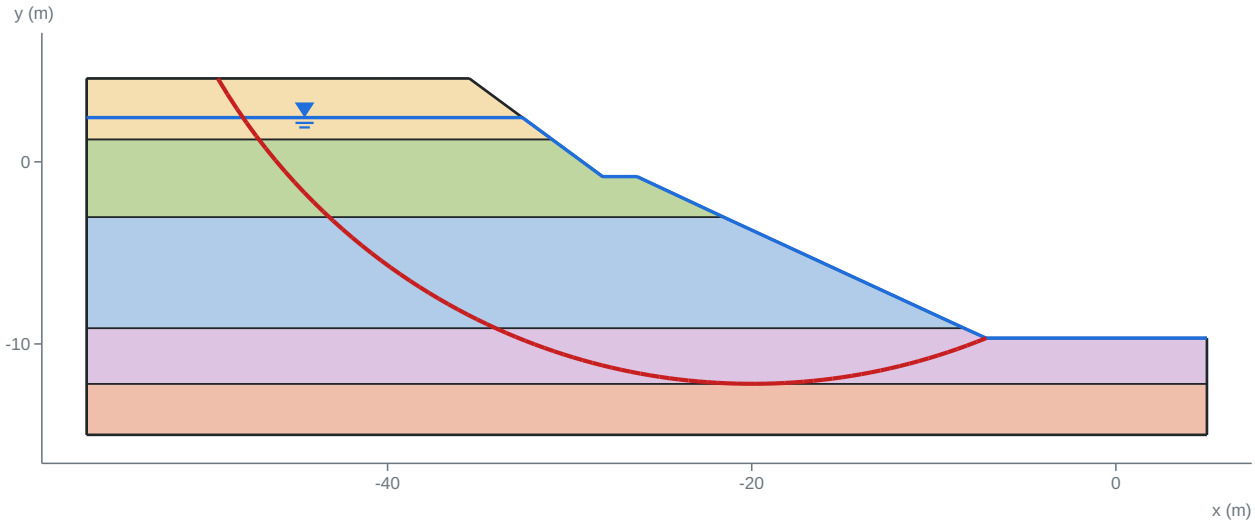



*Figure 37. V021b: analysed circle (Morgenstern-Price (half-sine), FOS = 1.110).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sand | Undrained (φ = 0) | 22.00 | Su = 0.00 | 0° |
| Clay 1 | Undrained (φ = 0) | 21.00 | Su = 55.00 | 0° |
| Clay 2 | Undrained (φ = 0) | 22.00 | Su = 43.00 | 0° |
| Clay 3 | Undrained (φ = 0) | 22.00 | Su = 56.00 | 0° |
| Bed | Mohr-Coulomb | 22.00 | c' = 200.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.1088 | Bishop simplified | Chowdhury & Xu | 1.110 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.116 | 1.110 |
| Bishop simplified | – | 1.110 |
| Bishop simplified (60 slices) | 1.109 | 1.110 |

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| Pf, share FOS < 1 (Monte Carlo) | – | 0.279 | 0.282 |
| β lognormal (Monte Carlo) | – | 0.537 | 0.553 |

## 3.38 V021c — Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the bas

*Chowdhury & Xu (1995); Ireland (1954).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Chowdhury & Xu example 2 (Su = 68.1, 39.3, 50.8 kPa), circle tangent to the base of clay 2.

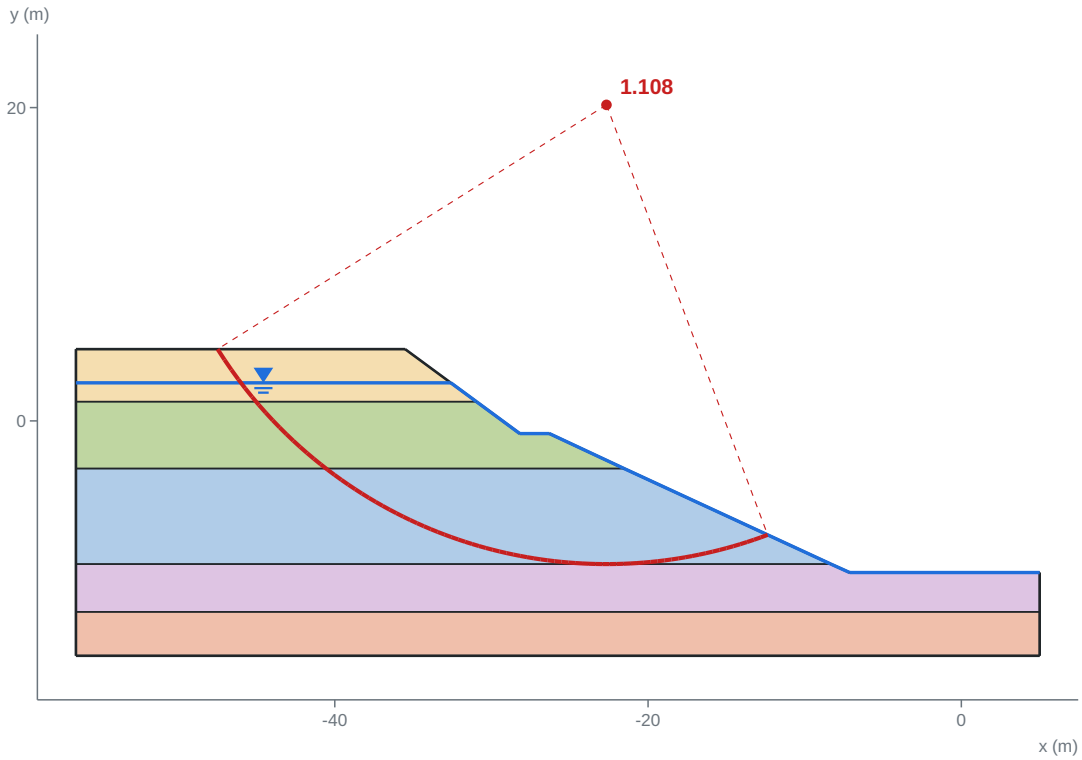



*Figure 38. V021c: analysed circle (Morgenstern-Price (half-sine), FOS = 1.108).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sand | Undrained (φ = 0) | 22.00 | Su = 0.00 | 0° |
| Clay 1 | Undrained (φ = 0) | 21.00 | Su = 68.10 | 0° |
| Clay 2 | Undrained (φ = 0) | 22.00 | Su = 39.30 | 0° |
| Clay 3 | Undrained (φ = 0) | 22.00 | Su = 50.80 | 0° |
| Bed | Mohr-Coulomb | 22.00 | c' = 200.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.1096 | Bishop simplified | Chowdhury & Xu | 1.108 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.110 | 1.108 |
| Bishop simplified | 1.108 | 1.108 |

## 3.39 V021d — Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the bas

*Chowdhury & Xu (1995); Ireland (1954).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Chowdhury & Xu example 2, circle tangent to the base of clay 3.

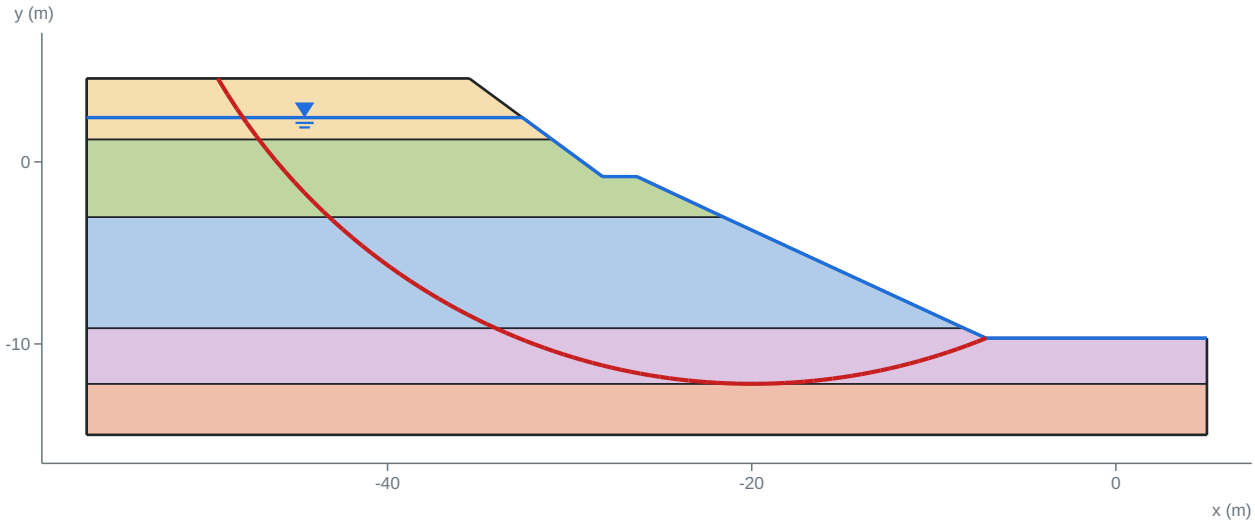



*Figure 39. V021d: analysed circle (Morgenstern-Price (half-sine), FOS = 1.058).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sand | Undrained (φ = 0) | 22.00 | Su = 0.00 | 0° |
| Clay 1 | Undrained (φ = 0) | 21.00 | Su = 68.10 | 0° |
| Clay 2 | Undrained (φ = 0) | 22.00 | Su = 39.30 | 0° |
| Clay 3 | Undrained (φ = 0) | 22.00 | Su = 50.80 | 0° |
| Bed | Mohr-Coulomb | 22.00 | c' = 200.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.0639 | Bishop simplified | Chowdhury & Xu | 1.058 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.063 | 1.058 |
| Bishop simplified | 1.058 | 1.058 |

## 3.40 V021e — Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the bas

*Chowdhury & Xu (1995); Ireland (1954).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Chowdhury & Xu example 3 (Su = 136, 80, 102 kPa), circle tangent to the base of clay 2.

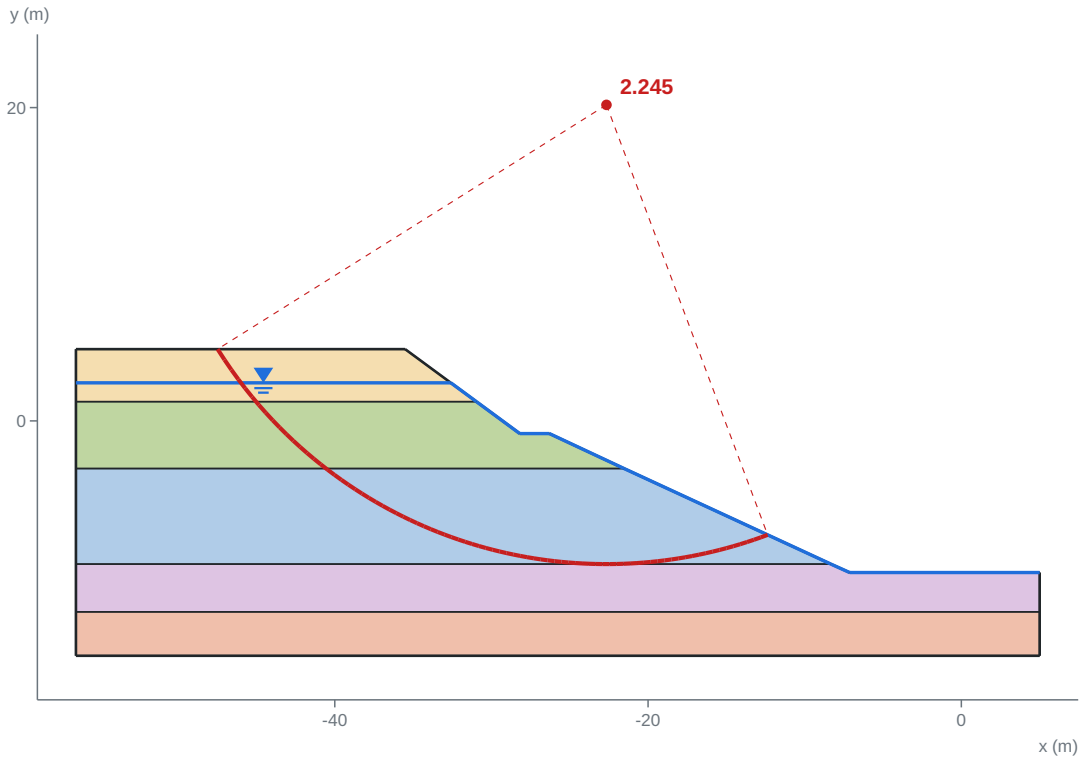



*Figure 40. V021e: analysed circle (Morgenstern-Price (half-sine), FOS = 2.245).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sand | Undrained (φ = 0) | 22.00 | Su = 0.00 | 0° |
| Clay 1 | Undrained (φ = 0) | 21.00 | Su = 136.0 | 0° |
| Clay 2 | Undrained (φ = 0) | 22.00 | Su = 80.00 | 0° |
| Clay 3 | Undrained (φ = 0) | 22.00 | Su = 102.0 | 0° |
| Bed | Mohr-Coulomb | 22.00 | c' = 200.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.2343 | Bishop simplified | Chowdhury & Xu | 2.245 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 2.248 | 2.245 |
| Bishop simplified | 2.245 | 2.245 |

## 3.41 V021f — Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base

*Chowdhury & Xu (1995); Ireland (1954).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Chowdhury & Xu example 3, circle tangent to the base of clay 3.

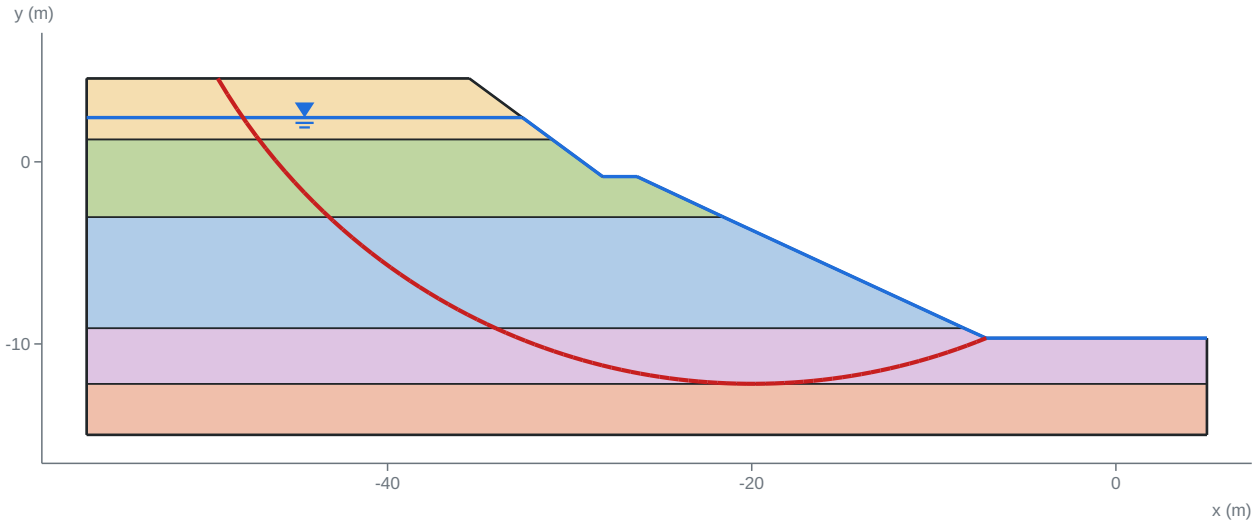



*Figure 41. V021f: analysed circle (Morgenstern-Price (half-sine), FOS = 2.128).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sand | Undrained (φ = 0) | 22.00 | Su = 0.00 | 0° |
| Clay 1 | Undrained (φ = 0) | 21.00 | Su = 136.0 | 0° |
| Clay 2 | Undrained (φ = 0) | 22.00 | Su = 80.00 | 0° |
| Clay 3 | Undrained (φ = 0) | 22.00 | Su = 102.0 | 0° |
| Bed | Mohr-Coulomb | 22.00 | c' = 200.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.1396 | Bishop simplified | Chowdhury & Xu | 2.128 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 2.138 | 2.128 |
| Bishop simplified | 2.128 | 2.128 |

## 3.42 V021g — Chowdhury & Xu (1995) example 4, Congress Street cut, circle tangent to the bas

*Chowdhury & Xu (1995); Ireland (1954).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Chowdhury & Xu example 4 (c-φ clays with the digitised piezometric line), circle tangent to the base of clay 2.

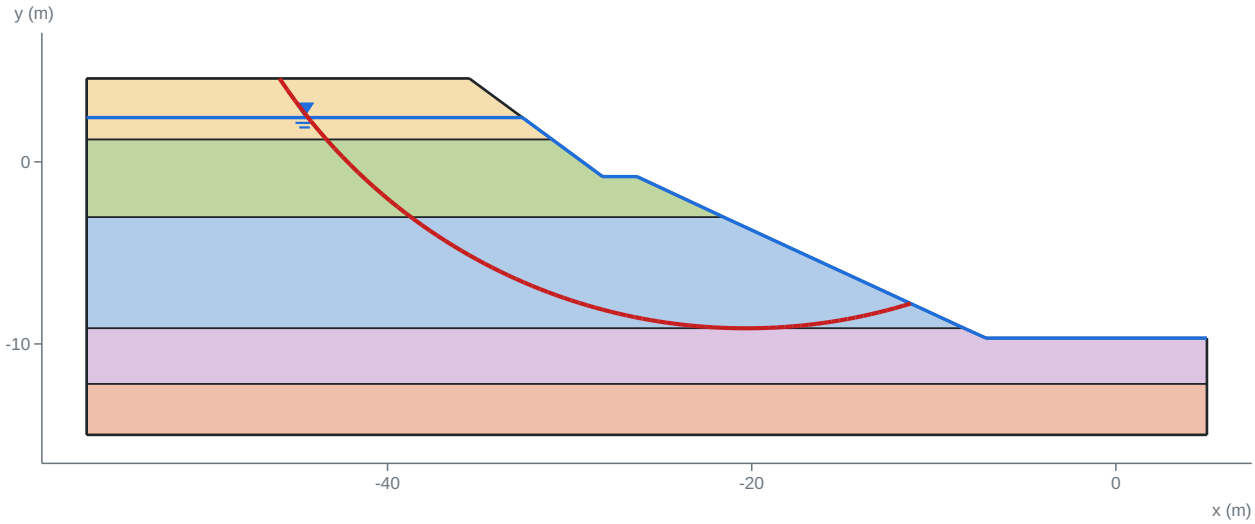



*Figure 42. V021g: analysed circle (Morgenstern-Price (half-sine), FOS = 1.428).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sand | Undrained (φ = 0) | 22.00 | Su = 0.00 | 0° |
| Clay 1 | Mohr-Coulomb | 17.00 | c' = 55.00 | 5.00° |
| Clay 2 | Mohr-Coulomb | 22.00 | c' = 43.00 | 7.00° |
| Clay 3 | Mohr-Coulomb | 22.00 | c' = 56.00 | 8.00° |
| Bed | Mohr-Coulomb | 22.00 | c' = 200.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.4239 | Bishop simplified | Chowdhury & Xu | 1.428 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.422 | 1.428 |
| Bishop simplified | – | 1.428 |

## 3.43 V022a — Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y =

*Chowdhury & Xu (1995).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Chowdhury & Xu example 5: embankment fill (c′ = 10 kPa, φ′ = 12°) on soft clay (Su = 40 kPa), the geometry as a commercial program's verification manual draws it; circle tangent to the fill–clay interface.

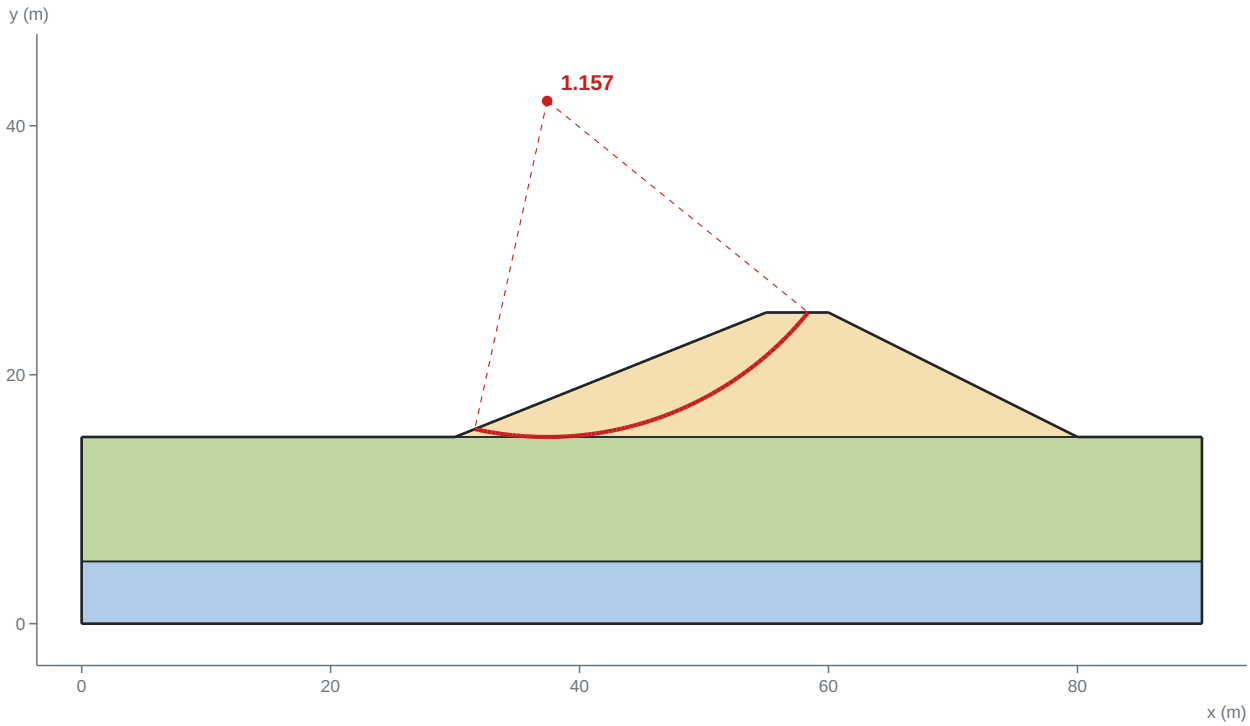



*Figure 43. V022a: analysed circle (Morgenstern-Price (half-sine), FOS = 1.157).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 | Mohr-Coulomb | 20.00 | c' = 10.00 | 12.00° |
| Layer 2 | Undrained (φ = 0) | 18.00 | Su = 40.00 | 0° |
| Hard layer | Undrained (φ = 0) | 20.00 | Su = 1000.0 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.1625 | Bishop simplified | Chowdhury & Xu | 1.159 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.158 | 1.157 |
| Bishop simplified | – | 1.159 |
| Bishop simplified (60 slices) | 1.160 | 1.159 |

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| Pf, share FOS < 1 (Monte Carlo) | – | 0.206–0.212 | 0.206 |
| β lognormal (Monte Carlo) | – | 0.799–0.800 | 0.812 |
| β lognormal (Taylor series) | – | 0.787 | 0.788 |

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| Pf, share FOS < 1 (Monte Carlo) | – | 0.210 | 0.208 |

## 3.44 V022b — Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y =

*Chowdhury & Xu (1995).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V022a with the circle tangent to the base of the soft clay.

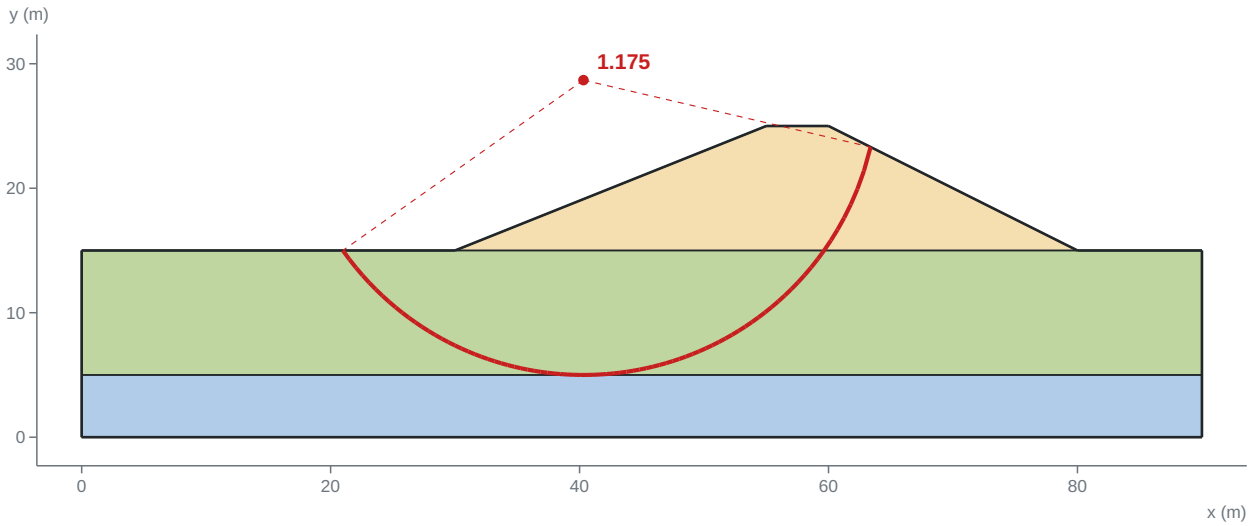



*Figure 44. V022b: analysed circle (Morgenstern-Price (half-sine), FOS = 1.175).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 | Mohr-Coulomb | 20.00 | c' = 10.00 | 12.00° |
| Layer 2 | Undrained (φ = 0) | 18.00 | Su = 40.00 | 0° |
| Hard layer | Undrained (φ = 0) | 20.00 | Su = 1000.0 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.1479 | Bishop simplified | Chowdhury & Xu (1995) | 1.177 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.178 | 1.175 |
| Bishop simplified | 1.185 | 1.177 |

*Chowdhury & Xu's 1.1479 is about 2.5 % below the other programs on this family of circles.*

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| Pf, share FOS < 1 (Monte Carlo) | – | 0.199–0.209 | 0.212 |
| β lognormal (Monte Carlo) | – | 0.789–0.820 | 0.790 |
| β lognormal (Taylor series) | – | 0.798 | 0.803 |

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| Pf, share FOS < 1 (Monte Carlo) | – | 0.214 | 0.216 |

## 3.45 V023 — LASH terminal trench, Duncan's Taylor series example

*Duncan (2000), J. Geotech. Geoenviron. Eng. 126(4).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Underwater trench in San Francisco Bay Mud: Su = 100 psf at el. −20 ft increasing 9.8 psf/ft, γ = 100 pcf, water at el. 0. Duncan's surface as a circle, Spencer; σγ = 3.3 pcf and σ of the Su gradient 1.2 psf/ft (a commercial program's Table 29.2). Converted to SI.

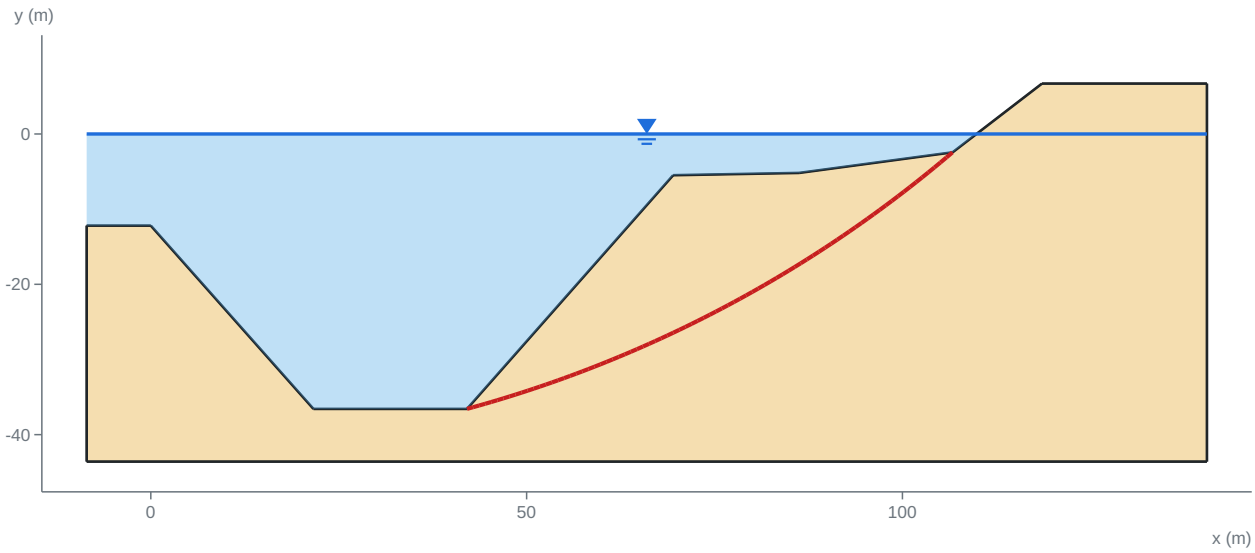



*Figure 45. V023: analysed circle (Spencer, FOS = 1.138).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| San Francisco Bay Mud | Undrained (φ = 0) | 15.71 | Su = 4.79 at y = -6.10, +1.54 kPa/m below | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.17 | not stated | Duncan (2000) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.145 | 1.138 |

*Duncan (2000) prints two decimals; the other programs give 1.145 on the same input.*

Probabilistic analysis on the same surface: 2 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Taylor series) | 0.900 Duncan (2000) | – | 0.892 |
| Pf lognormal (Taylor series) | 0.180 Duncan (2000) | – | 0.186 |

## 3.46 V024 — Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39)

*Duncan & Wright, Soil Strength and Slope Stability, figs. 6.39–6.40.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Symmetric dam with a narrow clay core on 30 ft of clay over permeable sand; a commercial program's piezometric line from the underseepage; the circle tangent to el. 197 ft. Converted to SI.

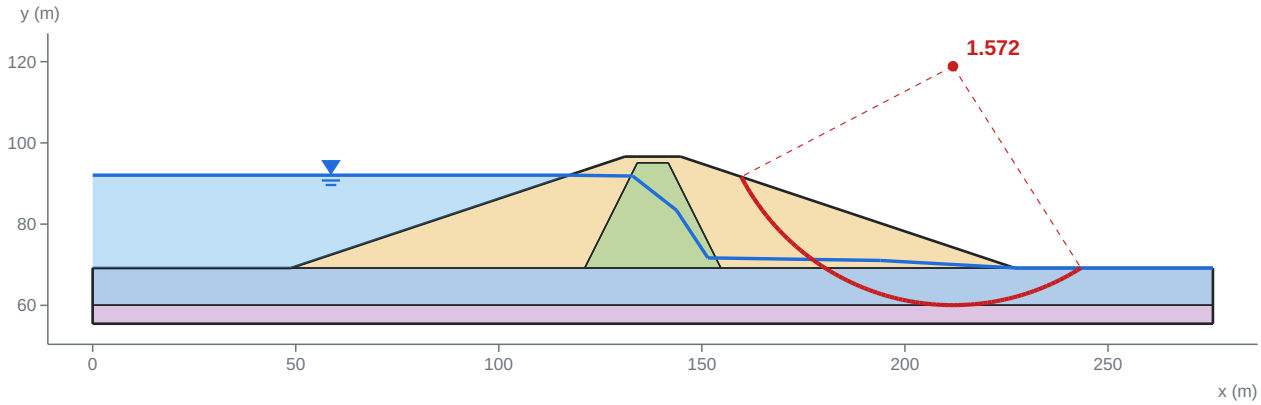



*Figure 46. V024: analysed circle (Bishop simplified, FOS = 1.572).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Outer shell | Mohr-Coulomb | 19.64 | c' = 0.00 | 34.00° |
| Clay core | Mohr-Coulomb | 19.16 | c' = 4.79 | 26.00° |
| Foundation clay | Mohr-Coulomb | 19.32 | c' = 0.00 | 24.00° |
| Foundation sand | Mohr-Coulomb | 19.95 | c' = 0.00 | 32.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.57 | Bishop simplified | Duncan & Wright | 1.572 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.563–1.572 | 1.572 |
| Spencer | 1.557 | 1.562 |

## 3.47 V025 — Duncan & Wright fig. 6.38, piezometric line

*Duncan & Wright, Soil Strength and Slope Stability, fig. 6.38.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Homogeneous 2:1 slope with a nearly flat piezometric line; critical circle search. Converted to SI; the water line was traced from a figure.

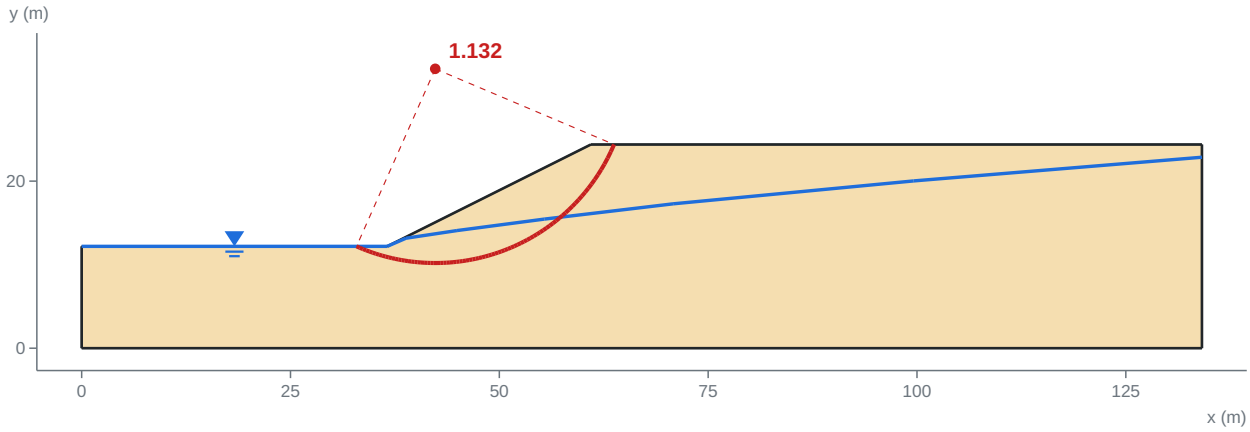



*Figure 47. V025: critical circle (Bishop simplified, FOS = 1.132).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 19.64 | c' = 9.58 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.141 | not stated | Duncan & Wright | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.142 | 1.132 |
| Spencer | 1.142 | 1.133 |

## 3.48 V026 — Duncan & Wright fig. 7.12, embankment on clay

*Duncan & Wright, Soil Strength and Slope Stability, fig. 7.12.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

Sand embankment on a thick undrained clay foundation; critical circle search.

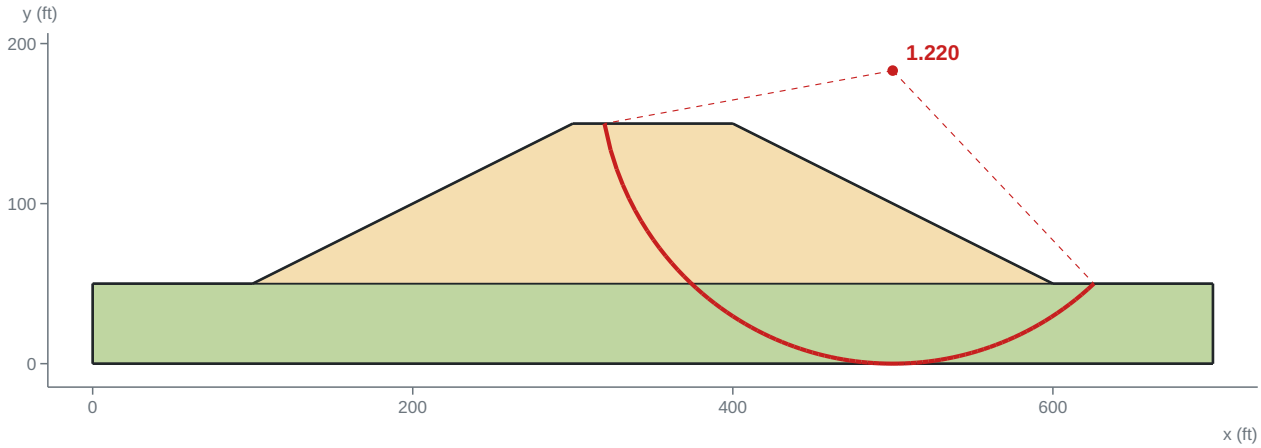



*Figure 48. V026: critical circle (Bishop simplified, FOS = 1.220).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Sand | Mohr-Coulomb | 140.0 | c' = 0.00 | 40.00° |
| Clay | Mohr-Coulomb | 140.0 | c' = 2500.0 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.22 | Bishop simplified | Duncan & Wright | 1.220 |
| 1.19 | Spencer | Duncan & Wright | 1.195 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.228 | 1.220 |
| Spencer | – | 1.195 |
| Spencer, non-circular | 1.190 | 1.170 |

## 3.49 V027 — Duncan & Wright fig. 14.3, undrained clay

*Duncan & Wright, Soil Strength and Slope Stability, fig. 14.3.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

Homogeneous undrained clay slope (φ = 0); critical circle search.

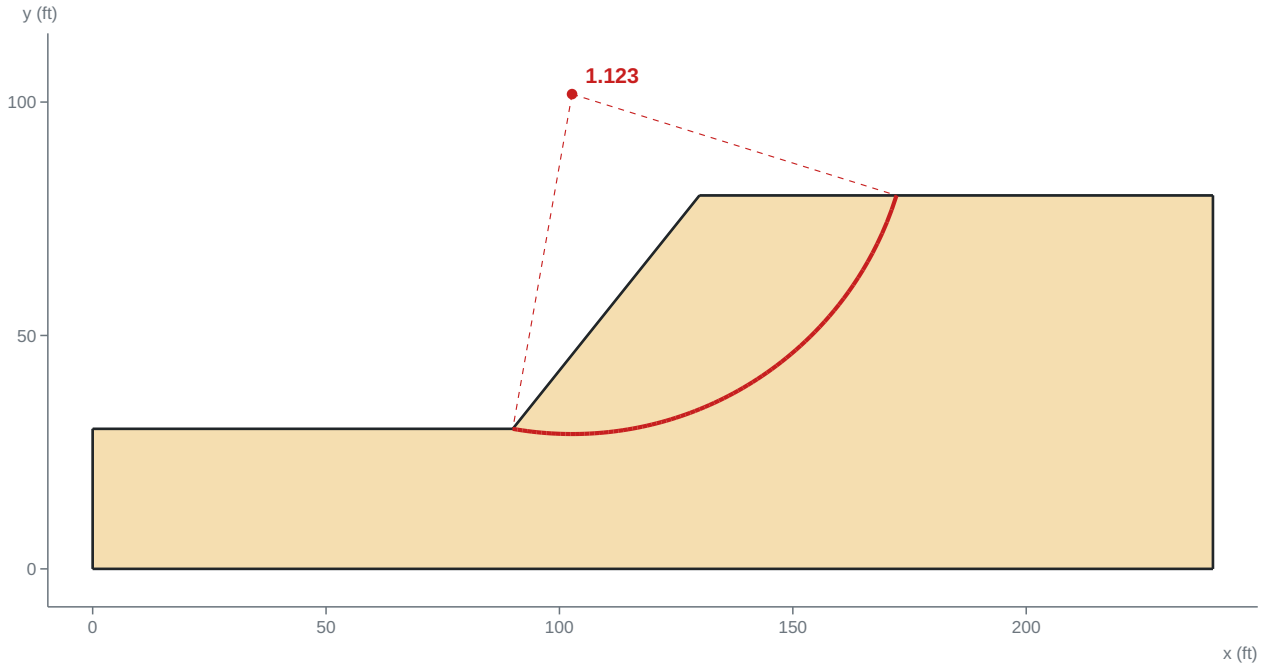



*Figure 49. V027: critical circle (Bishop simplified, FOS = 1.123).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Clay | Mohr-Coulomb | 100.0 | c' = 1000.0 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.124 | Spencer | Duncan & Wright | 1.134 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.126 | 1.123 |
| Spencer | – | 1.134 |

## 3.50 V028 — Duncan & Wright fig. 14.3, undrained clay on a 46.5 ft foundation

*Verification manual of a commercial slope stability program.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

The V027 φ = 0 clay slope with a 46.5 ft thick foundation; critical-circle search (the minimum of the toe and base-tangent circles of Tables 78.4–78.5).

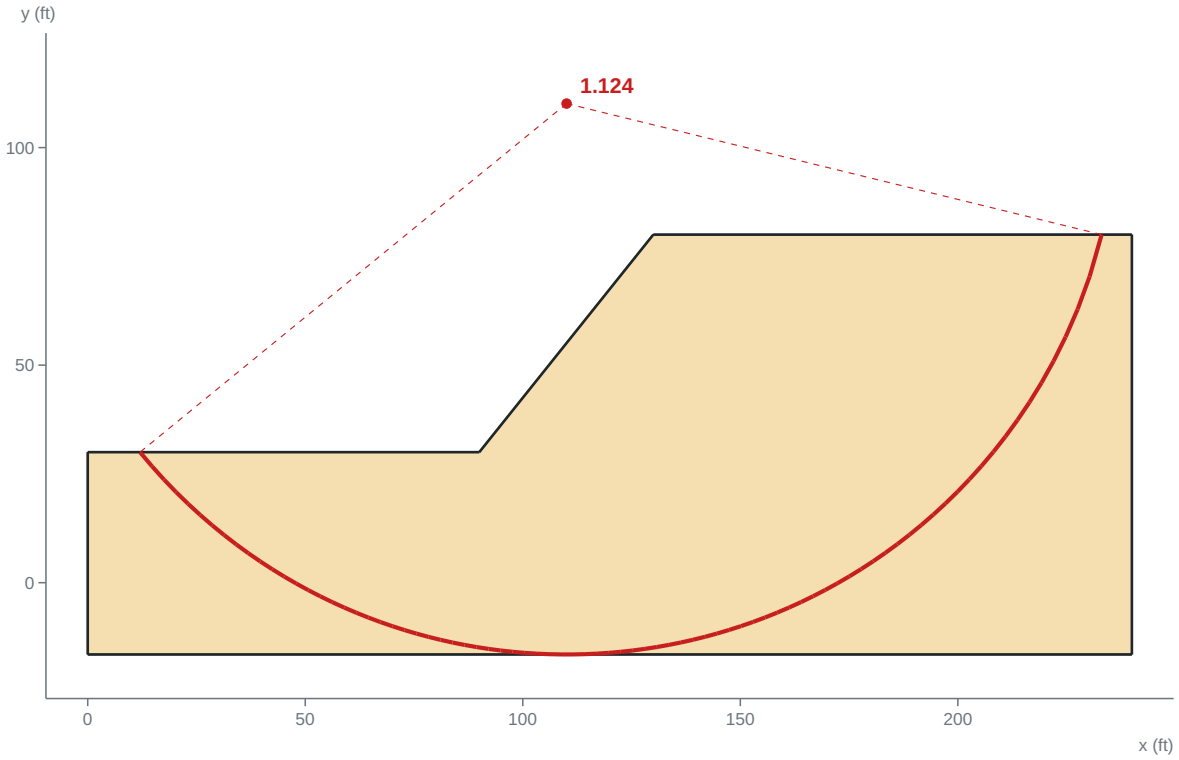



*Figure 50. V028: critical circle (Bishop simplified, FOS = 1.124).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Clay | Mohr-Coulomb | 100.0 | c' = 1000.0 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.124 | Spencer | Duncan & Wright | 1.123 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.126 | 1.124 |
| Spencer | – | 1.123 |

## 3.51 V029 — Duncan & Wright fig. 14.3, undrained clay on a 60 ft foundation

*Verification manual of a commercial slope stability program.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

As V028 with a 60 ft thick foundation (Tables 78.6–78.7).

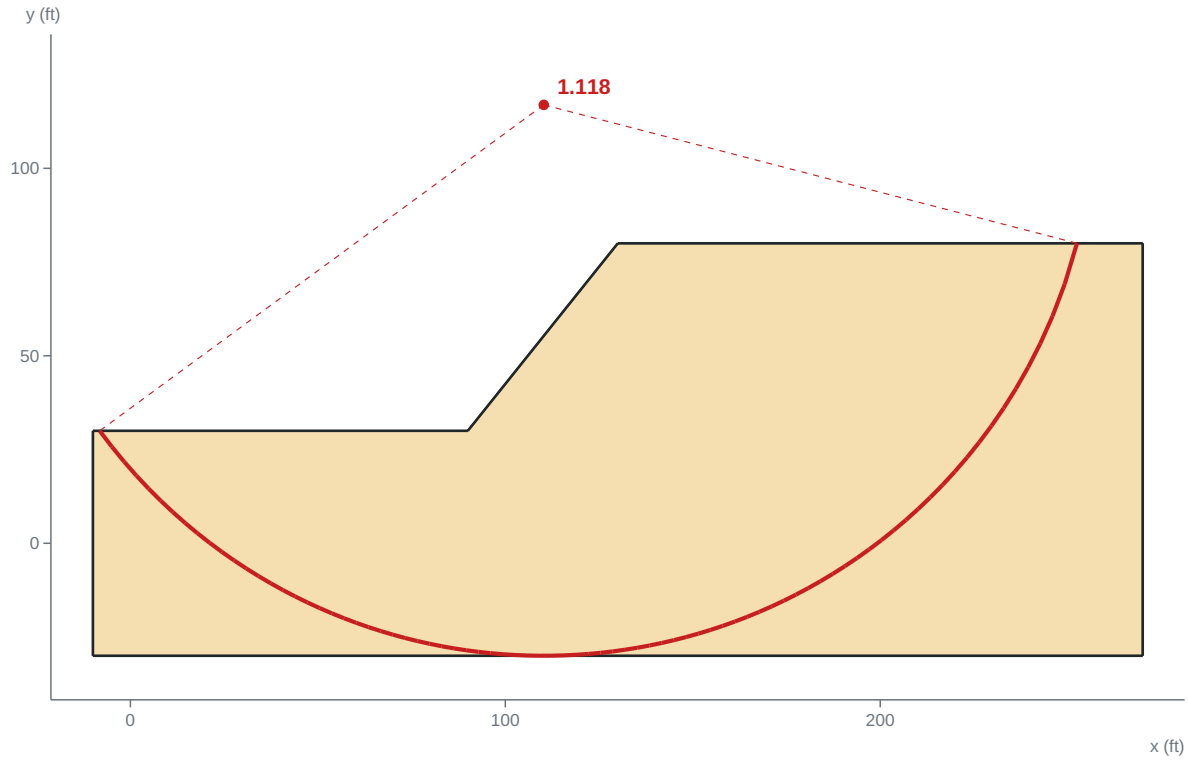



*Figure 51. V029: critical circle (Bishop simplified, FOS = 1.118).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Clay | Mohr-Coulomb | 100.0 | c' = 1000.0 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.119 | Spencer | Duncan & Wright | 1.118 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.125 | 1.118 |
| Spencer | – | 1.118 |

## 3.52 V030 — Duncan & Wright fig. 14.4

*Duncan & Wright, Soil Strength and Slope Stability, fig. 14.4.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

Cohesionless fill on undrained clay; critical circle search.

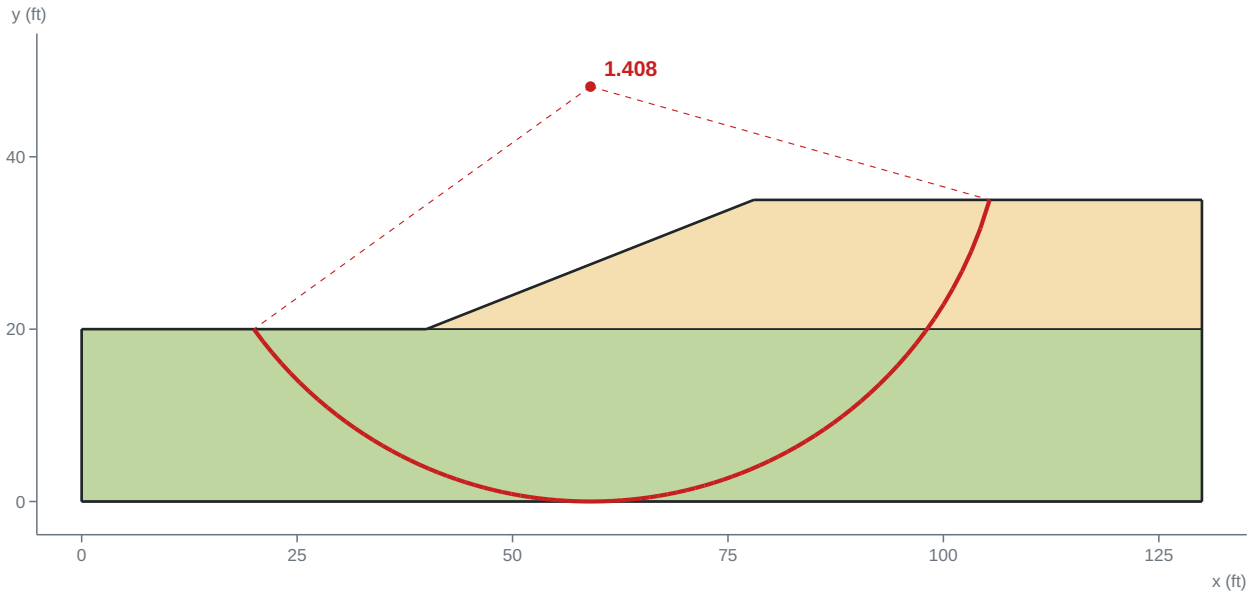



*Figure 52. V030: critical circle (Bishop simplified, FOS = 1.408).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Fill | Mohr-Coulomb | 120.0 | c' = 0.00 | 30.00° |
| Clay | Mohr-Coulomb | 120.0 | c' = 450.0 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.40 | Spencer | Duncan & Wright | 1.399 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.412 | 1.408 |
| Spencer | – | 1.399 |

## 3.53 V031a — Duncan & Wright fig. 14.5, R = 87 ft

*Duncan & Wright, Soil Strength and Slope Stability, fig. 14.5.*

**Fixed circle · Units: imperial (ft, psf, pcf)**

Embankment over alternating cohesive and frictional layers; circle of radius 87 ft.

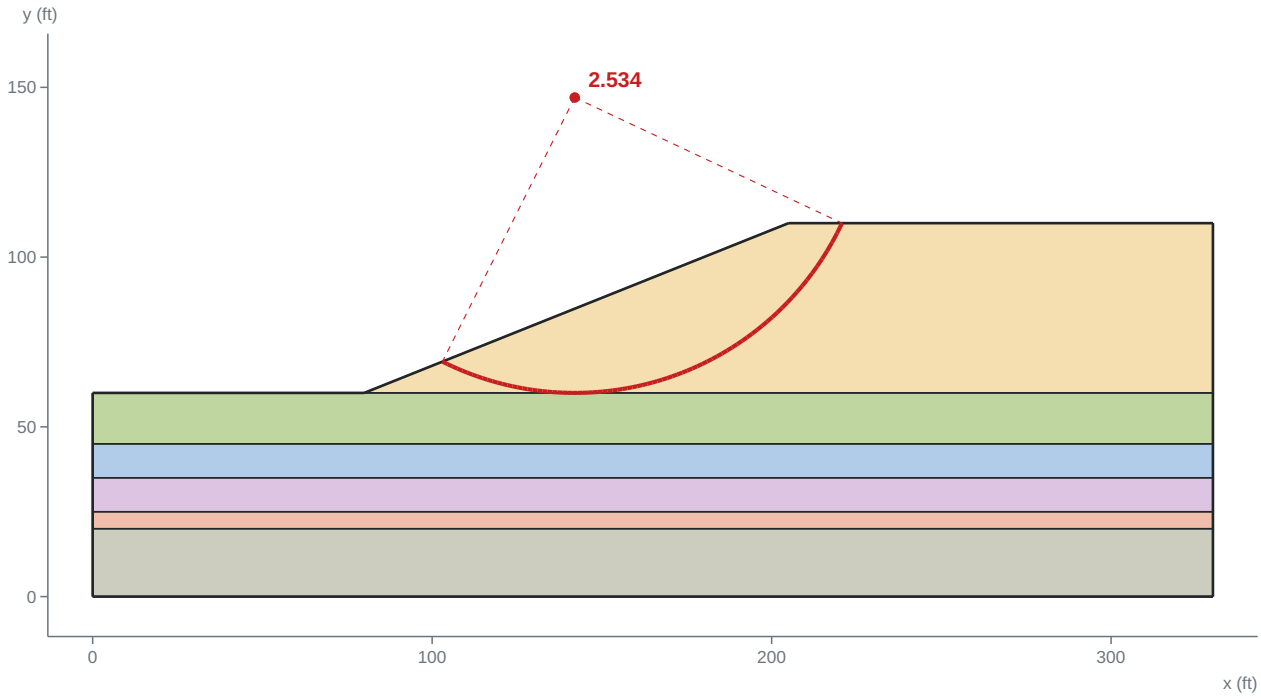



*Figure 53. V031a: analysed circle (Bishop simplified, FOS = 2.534).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 120.0 | c' = 1.00 | 35.00° |
| L60 | Mohr-Coulomb | 110.0 | c' = 950.0 | 0.00° |
| L45 | Mohr-Coulomb | 122.0 | c' = 1.00 | 32.00° |
| L35 | Mohr-Coulomb | 98.00 | c' = 500.0 | 0.00° |
| L25 | Mohr-Coulomb | 131.0 | c' = 1.00 | 37.00° |
| L20 | Mohr-Coulomb | 103.0 | c' = 600.0 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.56 | Spencer | Duncan & Wright | 2.531 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.549 | 2.534 |
| Spencer | 2.545 | 2.531 |

## 3.54 V031b — Duncan & Wright fig. 14.5, R = 102 ft

*Duncan & Wright, Soil Strength and Slope Stability, fig. 14.5.*

**Fixed circle · Units: imperial (ft, psf, pcf)**

As V031a with a deeper circle of radius 102 ft, reaching the weak clay layers.

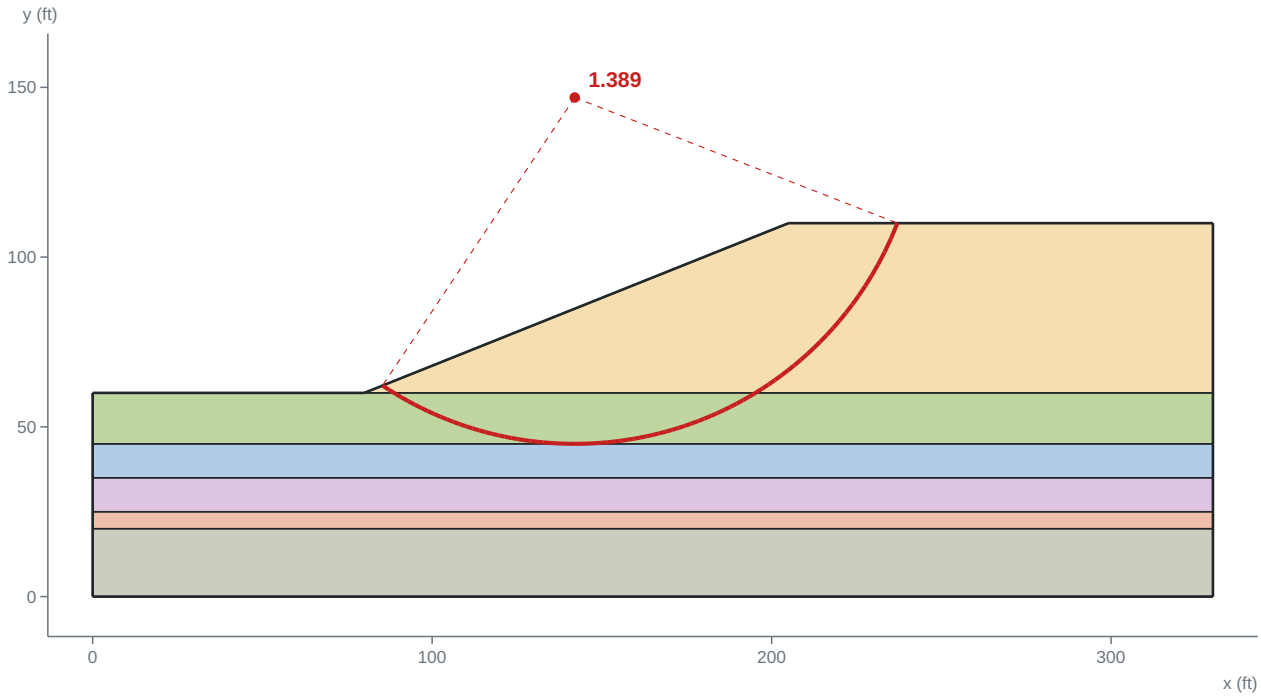



*Figure 54. V031b: analysed circle (Bishop simplified, FOS = 1.389).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 120.0 | c' = 1.00 | 35.00° |
| L60 | Mohr-Coulomb | 110.0 | c' = 950.0 | 0.00° |
| L45 | Mohr-Coulomb | 122.0 | c' = 1.00 | 32.00° |
| L35 | Mohr-Coulomb | 98.00 | c' = 500.0 | 0.00° |
| L25 | Mohr-Coulomb | 131.0 | c' = 1.00 | 37.00° |
| L20 | Mohr-Coulomb | 103.0 | c' = 600.0 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.35 | Spencer | Duncan & Wright | 1.352 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.398 | 1.389 |
| Spencer | 1.359 | 1.352 |

## 3.55 V032a — Duncan & Wright fig. 14.7, deep circle

*Duncan & Wright, Soil Strength and Slope Stability, fig. 14.7.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

Cohesionless embankment on an undrained clay foundation. Entry is limited to the crest and exit to the toe flat to find the published deep circle.

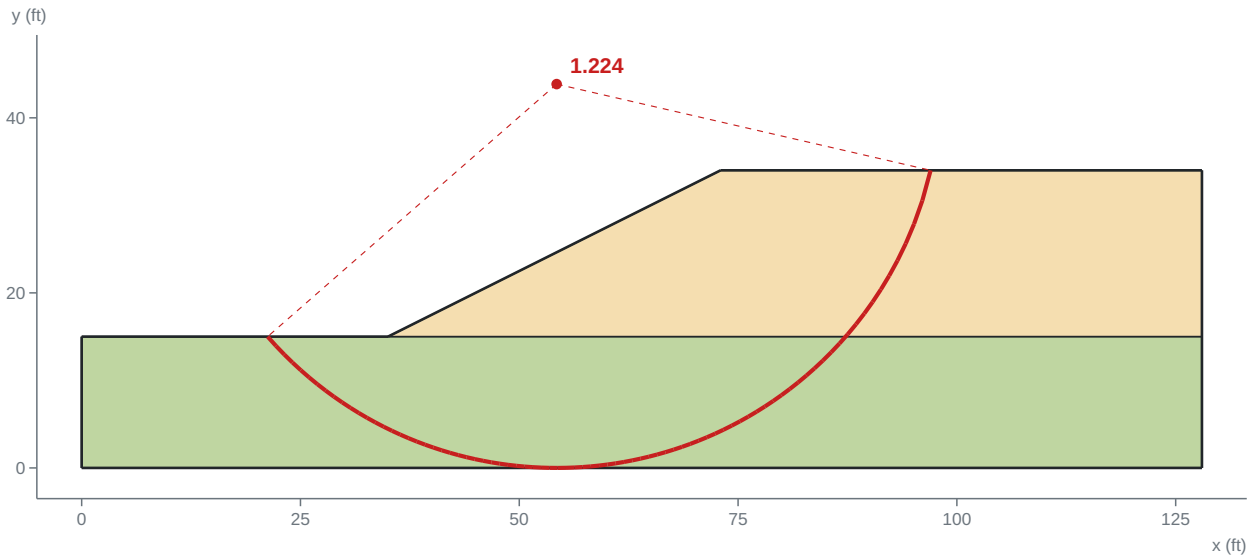



*Figure 55. V032a: critical circle (Bishop simplified, FOS = 1.224).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 124.0 | c' = 0.00 | 30.00° |
| Foundation | Mohr-Coulomb | 98.00 | c' = 500.0 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.21 | Spencer | Duncan & Wright | 1.205 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.230 | 1.224 |
| Spencer | 1.209 | 1.205 |

## 3.56 V032b — Duncan & Wright fig. 14.7, infinite-slope mechanism

*Duncan & Wright, Soil Strength and Slope Stability, fig. 14.7. Verification manual of a commercial slope stability program.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

The V032a embankment without search limits: the critical surface is the shallow infinite-slope sliver in the cohesionless fill (tan 30° / tan 26.6° = 1.155).

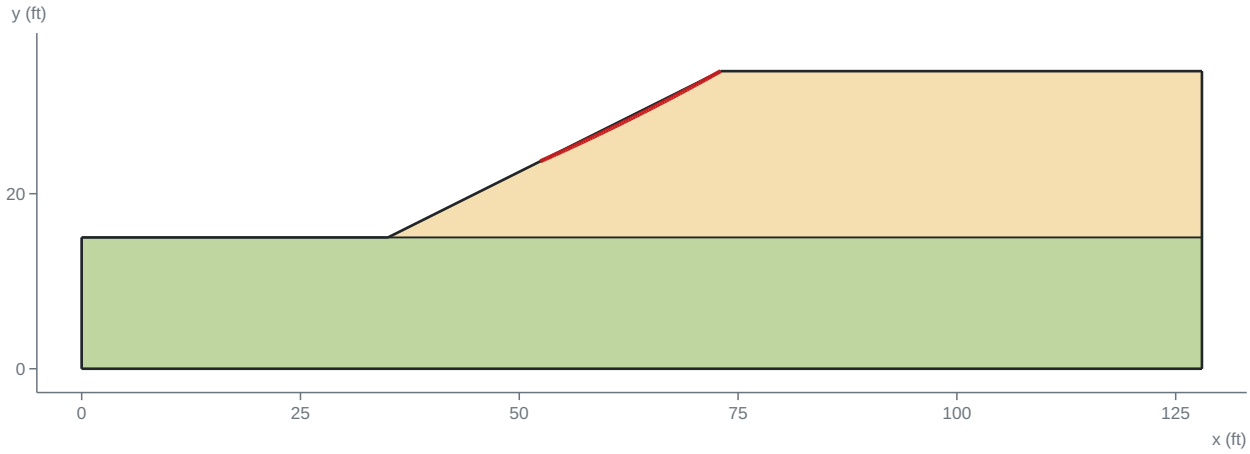



*Figure 56. V032b: critical circle (Bishop simplified, FOS = 1.156).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 124.0 | c' = 0.00 | 30.00° |
| Foundation | Mohr-Coulomb | 98.00 | c' = 500.0 | 0.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.15 | Spencer | Duncan & Wright | 1.156 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.155 | 1.156 |
| Spencer | 1.155 | 1.156 |

## 3.57 V033 — Duncan & Wright fig. 14.20a, water table

*Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20a.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Cohesive-frictional fill over sand with a piezometric line; critical circle search. Converted to SI.

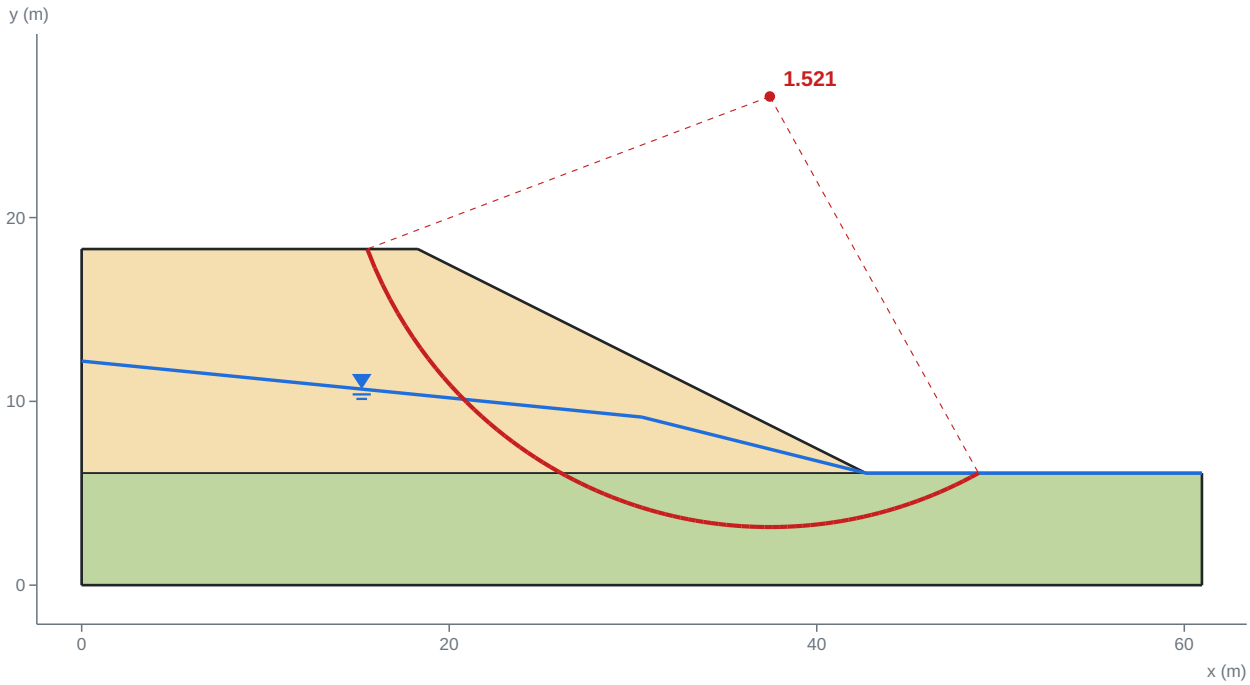



*Figure 57. V033: critical circle (Bishop simplified, FOS = 1.521).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fill | Mohr-Coulomb | 19.64 | c' = 28.73 | 25.00° |
| Sand | Mohr-Coulomb | 20.74 | c' = 0.00 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.529 | Spencer | Duncan & Wright | 1.533 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.533 | 1.521 |
| Spencer | – | 1.533 |
| Spencer, non-circular | 1.534 | 1.522 |

## 3.58 V034a — Duncan & Wright fig. 14.20b, Su = 200 + 15·depth

*Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20b.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

Granular fill over clay with undrained strength increasing linearly with depth; critical circle search.

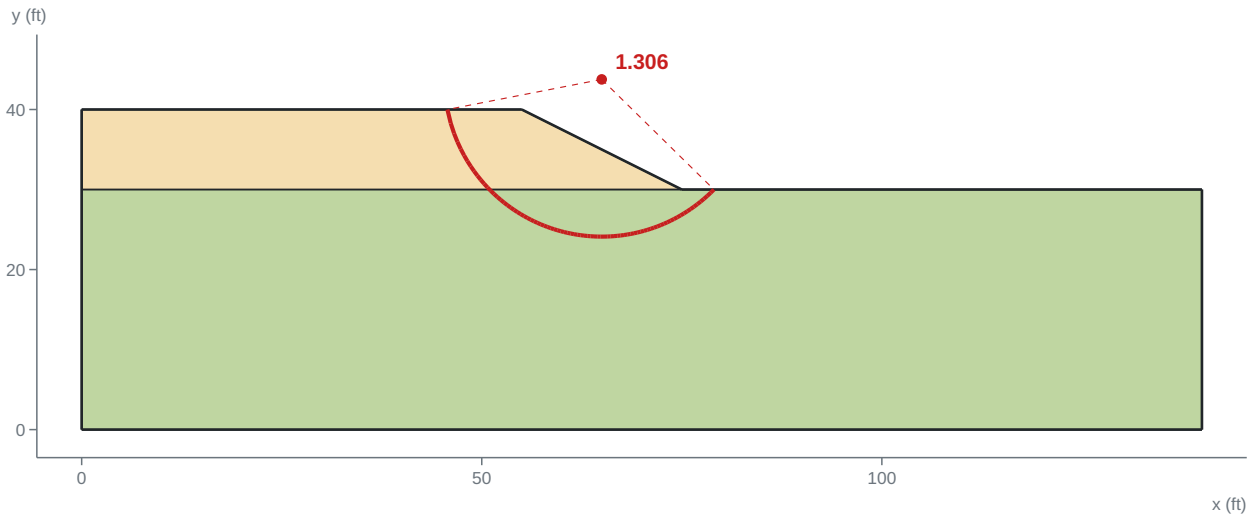



*Figure 58. V034a: critical circle (Bishop simplified, FOS = 1.306).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Fill | Mohr-Coulomb | 123.0 | c' = 0.00 | 36.00° |
| Clay | Undrained (φ = 0) | 97.00 | Su = 200.0 at y = 30.00, +15.00 psf/ft below | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.276 | Spencer | Duncan & Wright | 1.276 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.313 | 1.306 |
| Spencer | – | 1.276 |
| Spencer, non-circular | 1.262 | 1.260 |

## 3.59 V034b — Duncan & Wright fig. 14.20b, Su = 300 psf

*Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20b.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

As V034a with a constant undrained strength of 300 psf.

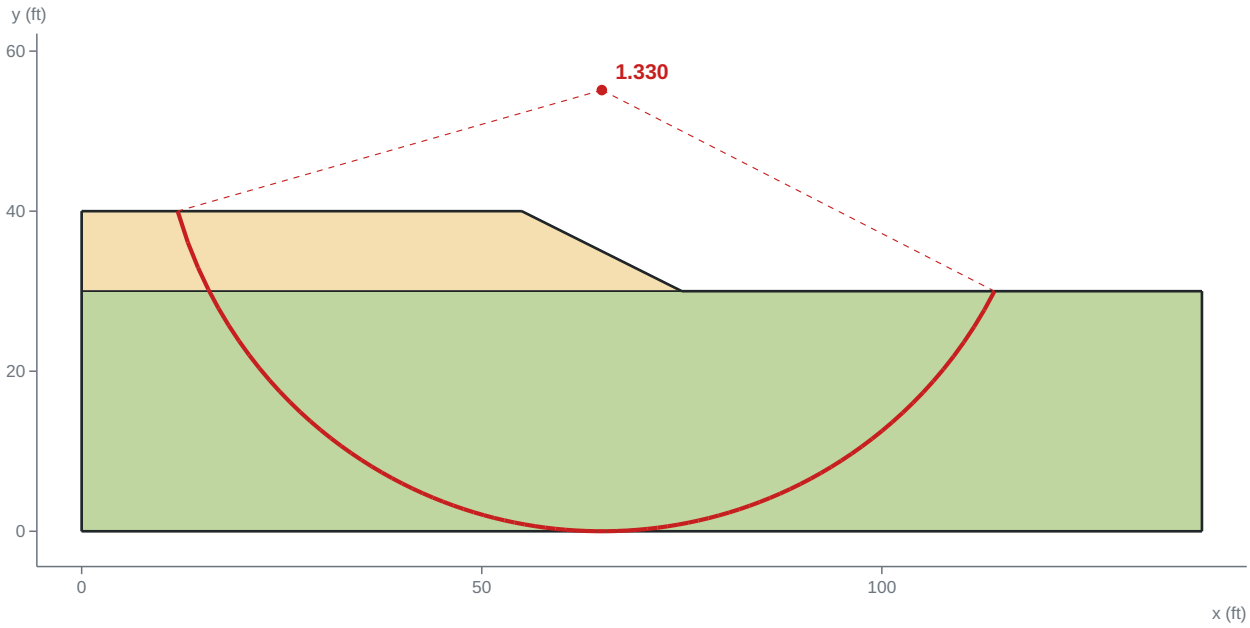



*Figure 59. V034b: critical circle (Bishop simplified, FOS = 1.330).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Fill | Mohr-Coulomb | 123.0 | c' = 0.00 | 36.00° |
| Clay | Undrained (φ = 0) | 97.00 | Su = 300.0 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.328 | Spencer | Duncan & Wright | 1.328 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.335 | 1.330 |
| Spencer | – | 1.328 |
| Spencer, non-circular | 1.182 | 1.169 |

## 3.60 V035a — Duncan & Wright tieback, active

*Duncan & Wright, Soil Strength and Slope Stability, fig. 6.34.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Homogeneous slope with a 9000 lb/ft tieback applied horizontally at mid-height as an active force, on the reference printed GLE circle. Converted to SI.

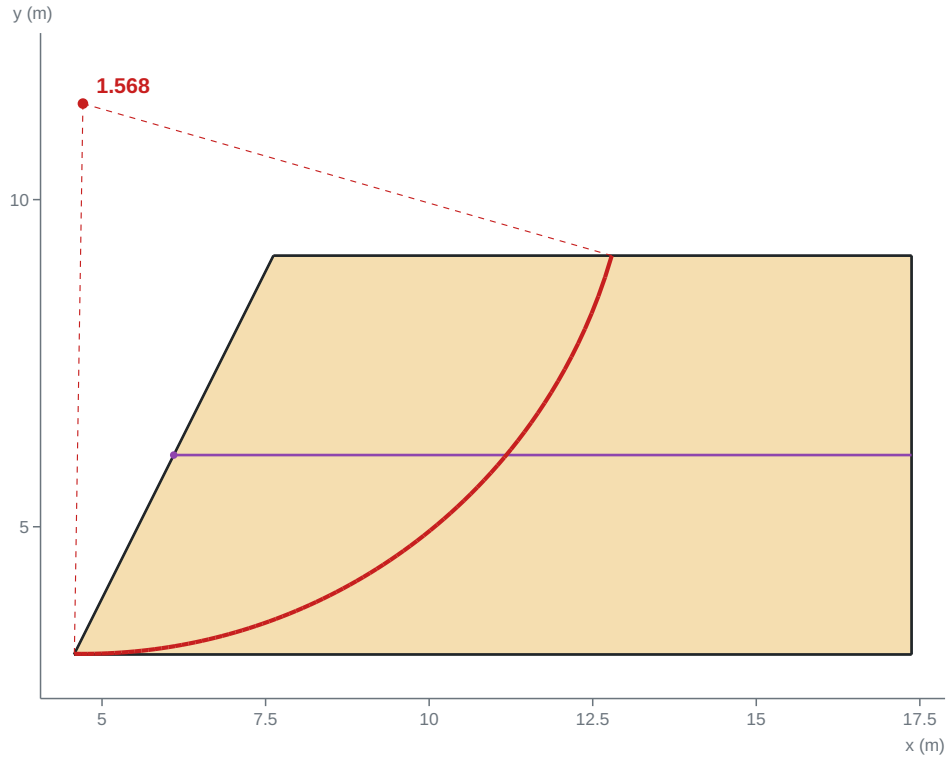



*Figure 60. V035a: analysed circle (Morgenstern-Price (half-sine), FOS = 1.568).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| saturated clay | Undrained (φ = 0) | 15.39 | Su = 16.76 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.51 | not stated | Duncan & Wright | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.575 | 1.568 |

*Duncan & Wright's 1.51 is about 4 % below the other programs on this surface.*

## 3.61 V035b — Duncan & Wright tieback, passive

*Duncan & Wright, Soil Strength and Slope Stability, fig. 6.34.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V035a with the tieback applied as a passive force (divided by F), on the reference printed Bishop circle.

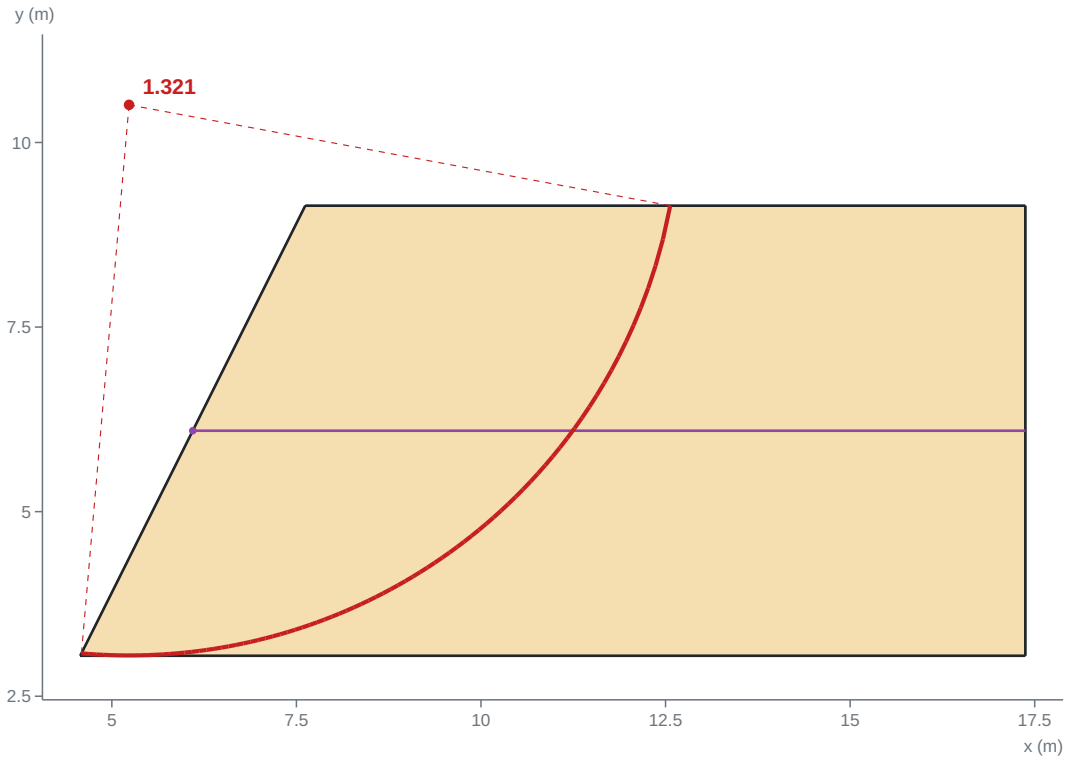



*Figure 61. V035b: analysed circle (Bishop simplified, FOS = 1.321).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| saturated clay | Undrained (φ = 0) | 15.39 | Su = 16.76 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.32 | Bishop simplified | Duncan & Wright | 1.321 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.324 | 1.321 |

## 3.62 V036a — Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9)

*Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Embankment (c′ = 0, φ′ = 35°, γ = 125 pcf) on a 20 ft undrained foundation, cu = 300 psf, constant; circular search. Converted to SI.

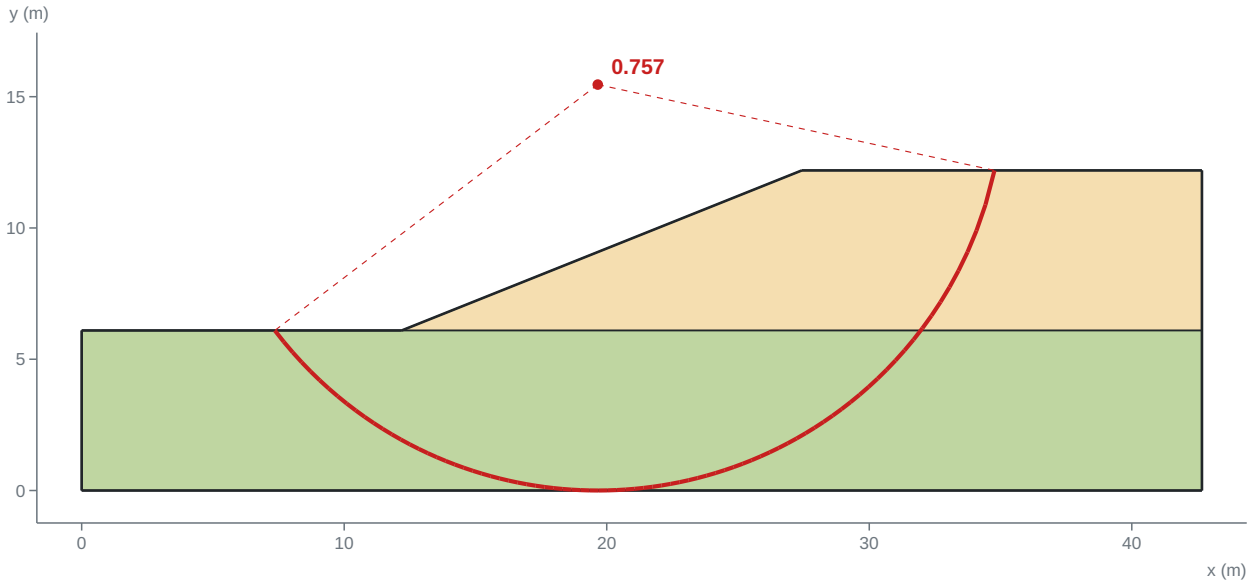



*Figure 62. V036a: critical circle (Bishop simplified, FOS = 0.757).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 19.64 | c' = 0.00 | 35.00° |
| Foundation | Undrained (φ = 0) | 15.71 | Su = 14.36 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.75 | Spencer | Duncan & Wright | 0.752 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 0.761 | 0.757 |
| Spencer | 0.756 | 0.752 |
| Spencer, non-circular | 0.740 | 0.736 |

## 3.63 V036b — Embankment on an undrained foundation, cz = 5 psf/ft

*Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

As V036a with cu increasing 5 psf/ft with depth.

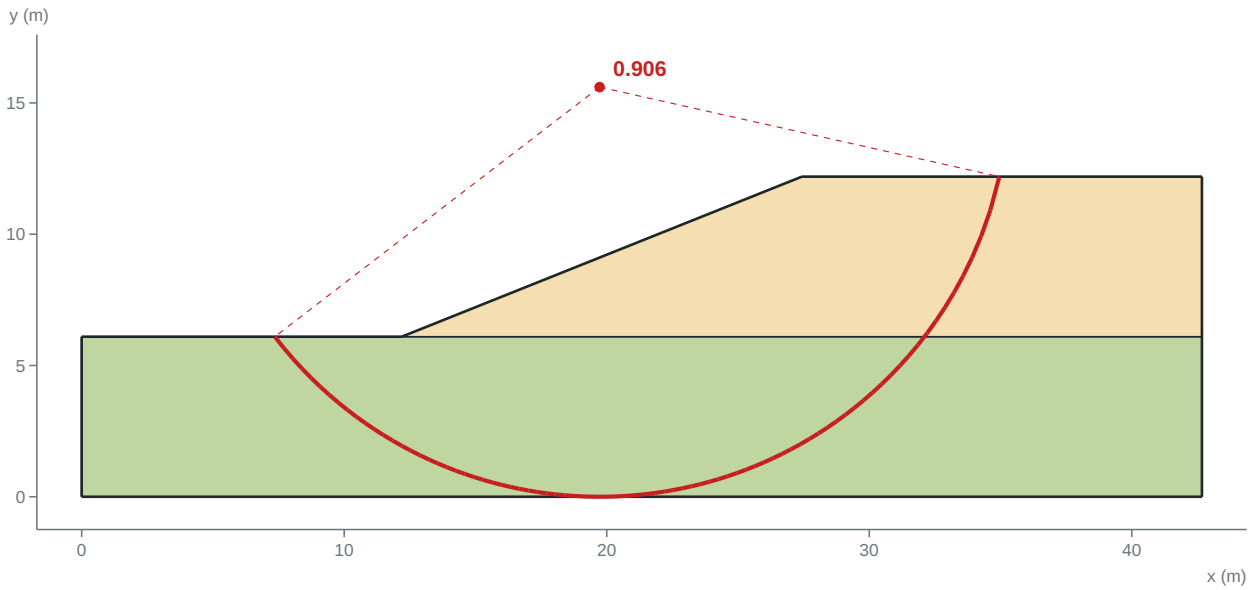



*Figure 63. V036b: critical circle (Bishop simplified, FOS = 0.906).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 19.64 | c' = 0.00 | 35.00° |
| Foundation | Undrained (φ = 0) | 15.71 | Su = 14.36 at y = 6.10, +0.79 kPa/m below | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.90 | Spencer | Duncan & Wright | 0.898 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 0.909 | 0.906 |
| Spencer | 0.898 | 0.898 |

## 3.64 V036c — Embankment on an undrained foundation, cz = 10 psf/ft

*Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

As V036a with cu increasing 10 psf/ft with depth.

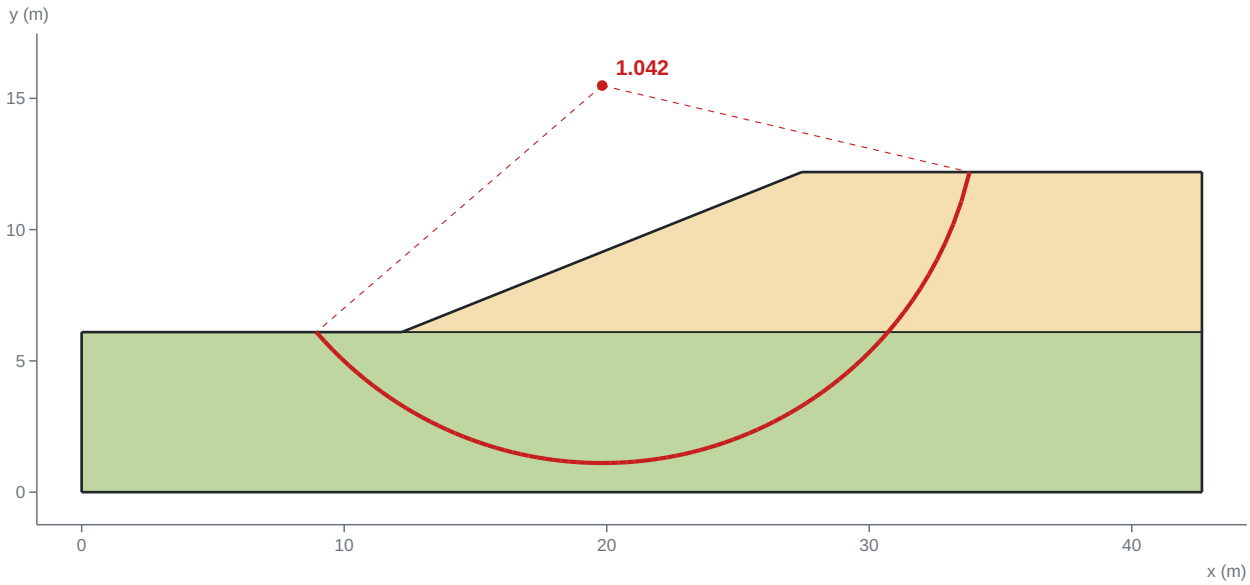



*Figure 64. V036c: critical circle (Bishop simplified, FOS = 1.042).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 19.64 | c' = 0.00 | 35.00° |
| Foundation | Undrained (φ = 0) | 15.71 | Su = 14.36 at y = 6.10, +1.57 kPa/m below | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.03 | Spencer | Duncan & Wright | 1.029 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.045 | 1.042 |
| Spencer | 1.032 | 1.029 |
| Spencer, non-circular | 1.018 | 1.017 |

## 3.65 V036d — Embankment on an undrained foundation, cz = 15 psf/ft

*Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

As V036a with cu increasing 15 psf/ft with depth.

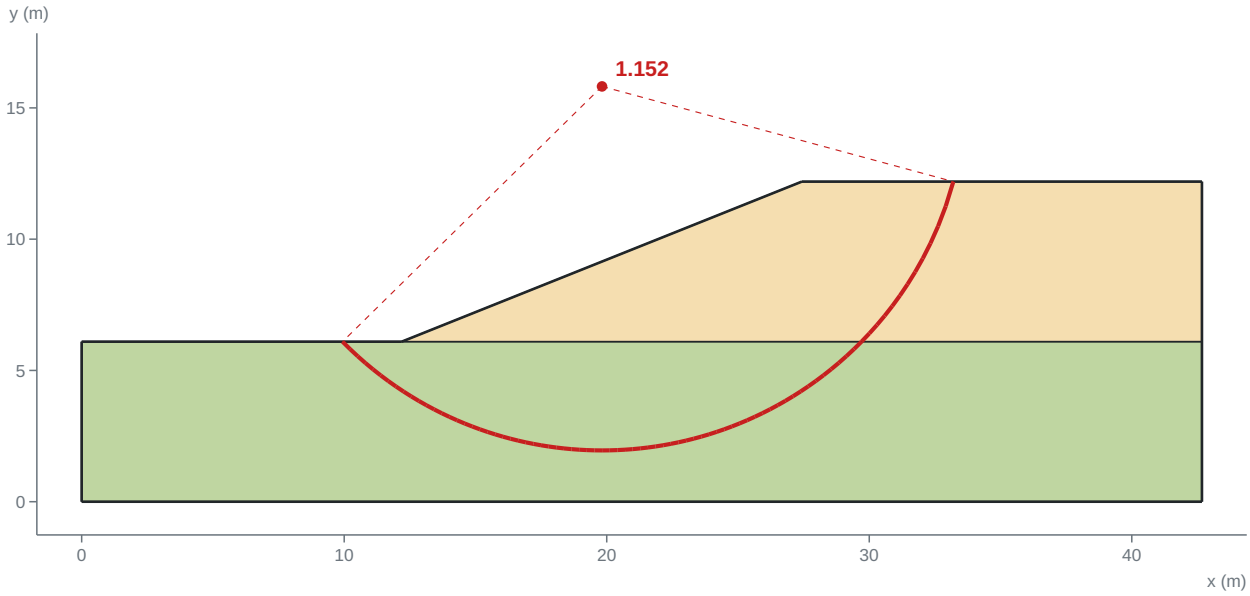



*Figure 65. V036d: critical circle (Bishop simplified, FOS = 1.152).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 19.64 | c' = 0.00 | 35.00° |
| Foundation | Undrained (φ = 0) | 15.71 | Su = 14.36 at y = 6.10, +2.36 kPa/m below | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.13 | Spencer | Duncan & Wright | 1.132 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.154 | 1.152 |
| Spencer | 1.134 | 1.132 |
| Spencer, non-circular | 1.116 | 1.119 |

## 3.66 V037 — Geogrid-reinforced fill on rock (D&W fig. 7.28)

*Duncan & Wright, Soil Strength and Slope Stability, fig. 7.28.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

24 ft fill slope (φ′ = 37°) on a strong rock foundation with five 800 lb/ft geogrids, 20 ft long, every 4 ft; circular search for surfaces through the toe. Converted to SI.

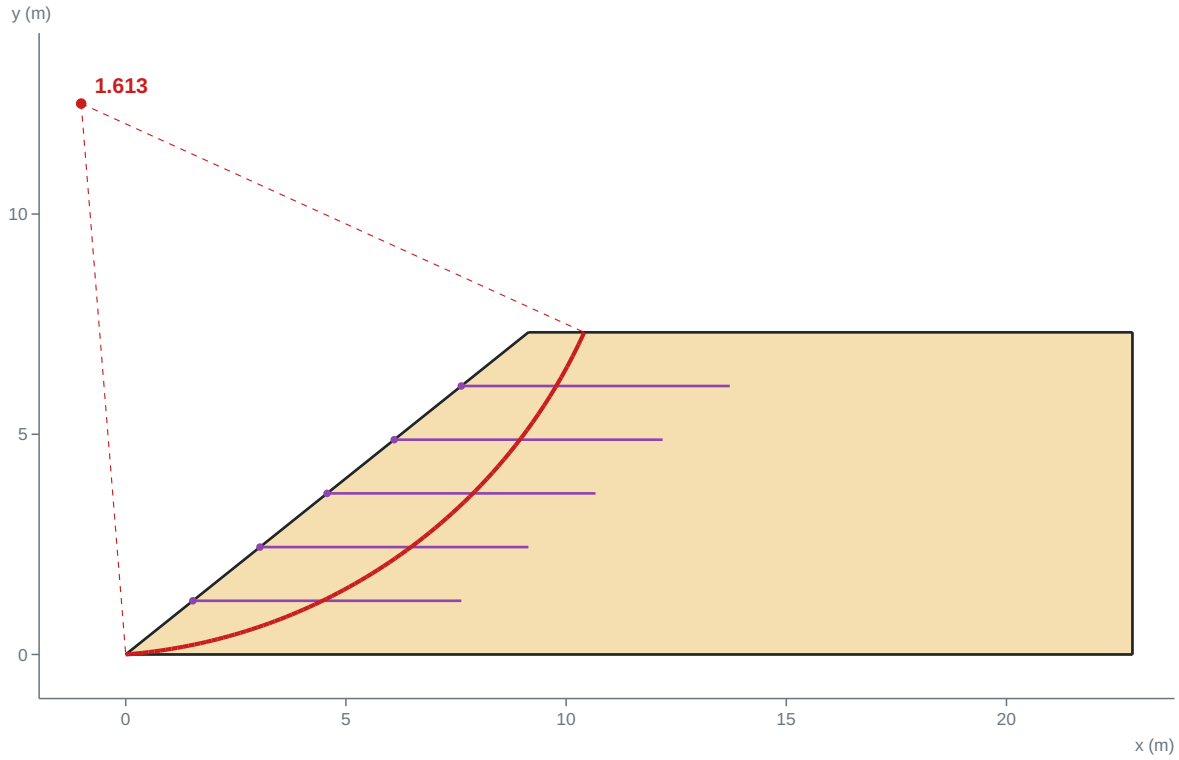



*Figure 66. V037: critical circle (Bishop simplified, FOS = 1.613).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| fill | Mohr-Coulomb | 20.42 | c' = 0.00 | 37.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.62 | Spencer | Duncan, Wright & Brandon (2014) | 1.607 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.629 | 1.613 |
| Spencer | 1.620 | 1.607 |

## 3.67 V038 — Homogeneous dam with a piezometric line (D&W fig. 7.19)

*Duncan & Wright, Soil Strength and Slope Stability, fig. 7.19.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Homogeneous embankment (c′ = 100 psf, φ′ = 30°, γ = 100 pcf) with the pool at el. 40 ft and a commercial program's piezometric line; critical-circle search. Converted to SI.

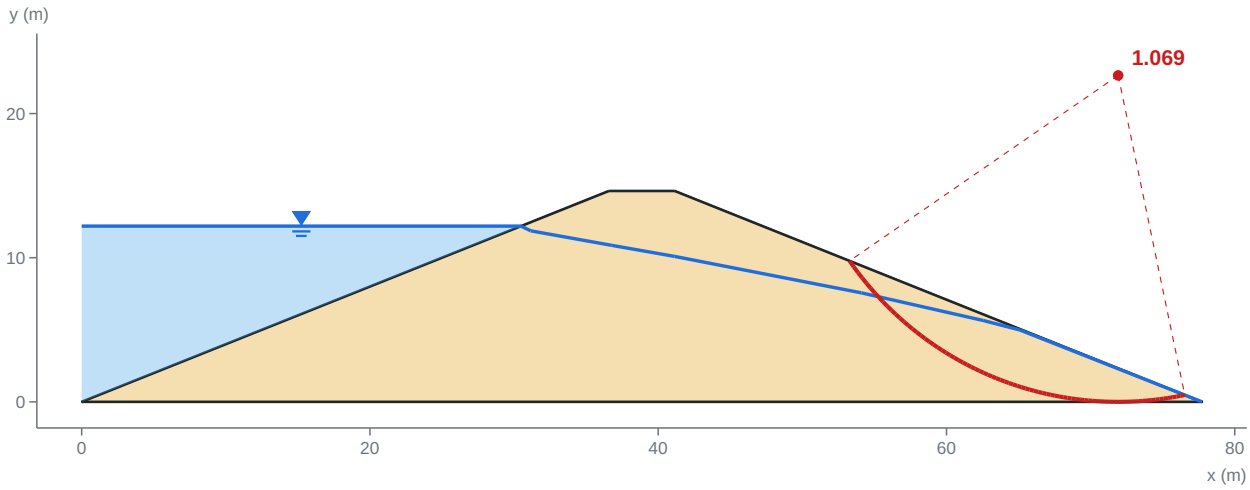



*Figure 67. V038: critical circle (Bishop simplified, circles only, FOS = 1.069).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Material 1 | Mohr-Coulomb | 15.71 | c' = 4.79 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.16 | not stated | Duncan & Wright | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified, circles only | 1.070–1.090 | 1.069 |
| Spencer, circles only | 1.100 | 1.077 |

*Duncan & Wright's 1.16 is for γ = 125 pcf and their printed phreatic line, not this program input (100 pcf, a 9-point line); on their input cSlope gives 1.157 (V038b).*

## 3.68 V038b — Homogeneous dam with a piezometric line, Duncan & Wright's input

*Duncan, J.M., Wright, S.G. & Brandon, T.L. (2014). Soil Strength and Slope Stability, 2nd ed., Wiley; §7.7, Figs. 7.30–7.36, Table 7.8.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V038 dam with the unit weight Duncan, Wright & Brandon's slice calculations use (125 pcf) and their printed phreatic line (30 points, Fig. 7.31); critical-circle search. Converted to SI.

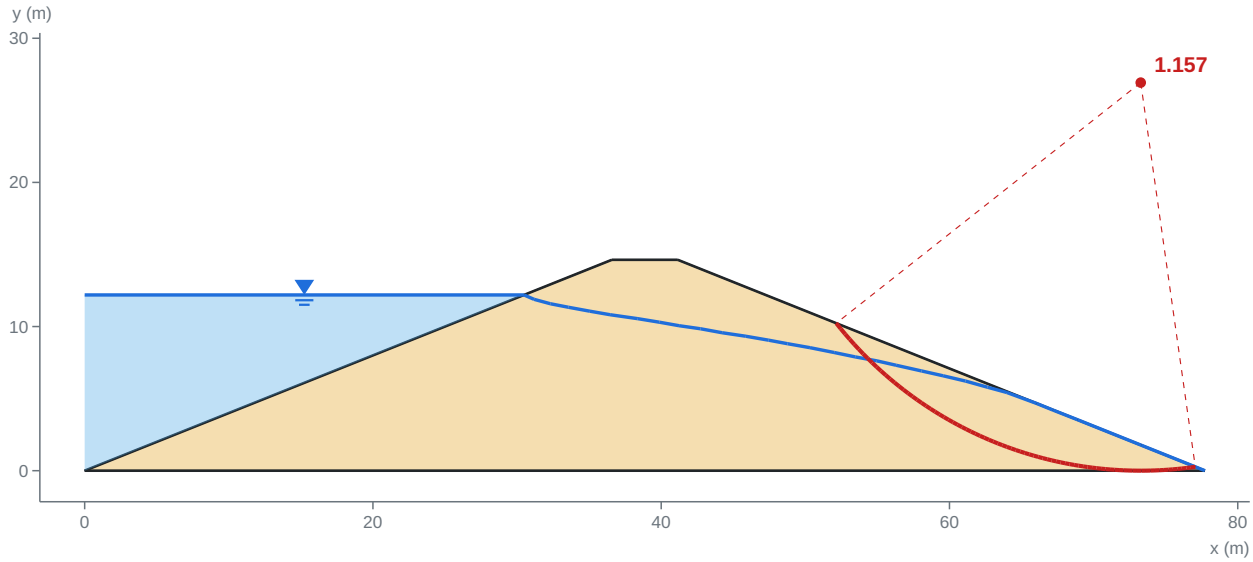



*Figure 68. V038b: critical circle (Spencer, FOS = 1.157).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Material 1 | Mohr-Coulomb | 19.64 | c' = 4.79 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.16 | Spencer | Duncan & Wright | 1.157 |
| 1.19 | Spencer, finite-element pore pressures | Duncan, Wright & Brandon (2014) | 1.157 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | – | 1.157 |

*The book contradicts itself on the unit weight: its figures say 100 pcf, its hand calculations and slice weights use 125 pcf. With 125 pcf and the book's line cSlope reproduces the 1.16 that all three of the book's programs give (Table 7.8); with 100 pcf it gives 1.03.*

## 3.69 V039 — James Bay dyke (Duncan & Wright fig. 7.16)

*Duncan & Wright, Soil Strength and Slope Stability, fig. 7.16.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Granular fill embankment with a wide berm (c′ = 0, φ′ = 30°) on a 4 m clay crust (Su = 41), 8 m marine clay (34.5) and 6.5 m lacustrine clay (31.2); circular search. The critical circle is deep and tangent to the base of the lacustrine clay.

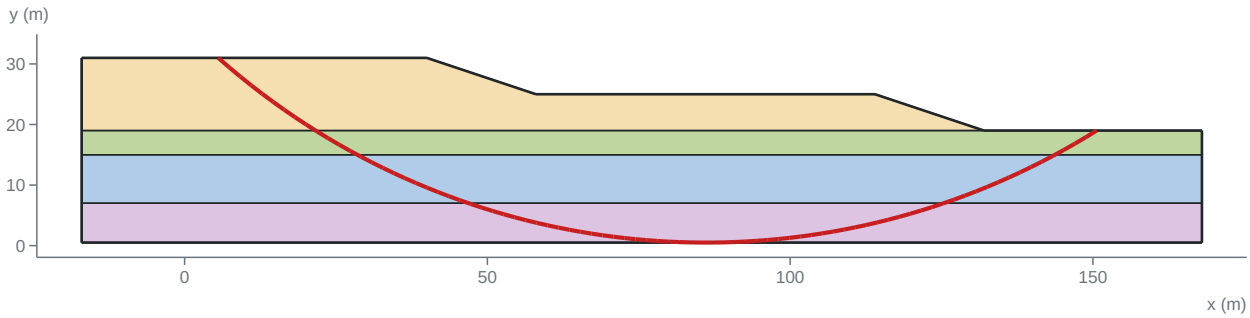



*Figure 69. V039: critical circle (Spencer, circles only, FOS = 1.447).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fill | Mohr-Coulomb | 20.00 | c' = 0.00 | 30.00° |
| Clay crust | Undrained (φ = 0) | 20.00 | Su = 41.00 | 0° |
| Marine clay | Undrained (φ = 0) | 18.80 | Su = 34.50 | 0° |
| Lacustrine clay | Undrained (φ = 0) | 20.30 | Su = 31.20 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.45 | Spencer | Duncan & Wright | 1.447 |
| 1.17 | Spencer, composite circle, composite surface | Duncan & Wright | 1.166 |
| 1.16 | Spencer, non-circular | Duncan & Wright | 1.162 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, circles only | – | 1.447 |
| Spencer | – | 1.166 |
| Spencer, non-circular | 1.167 | 1.162 |

## 3.70 V040 — James Bay dyke, reference model (deterministic)

*El-Ramly et al. (2002); Duncan & Wright fig. 7.16.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The James Bay dyke as a commercial program models it (crest 36 m, berm at 30 m, clay crust Su = 43 kPa, till below y = 5.5); critical-circle search, deterministic.

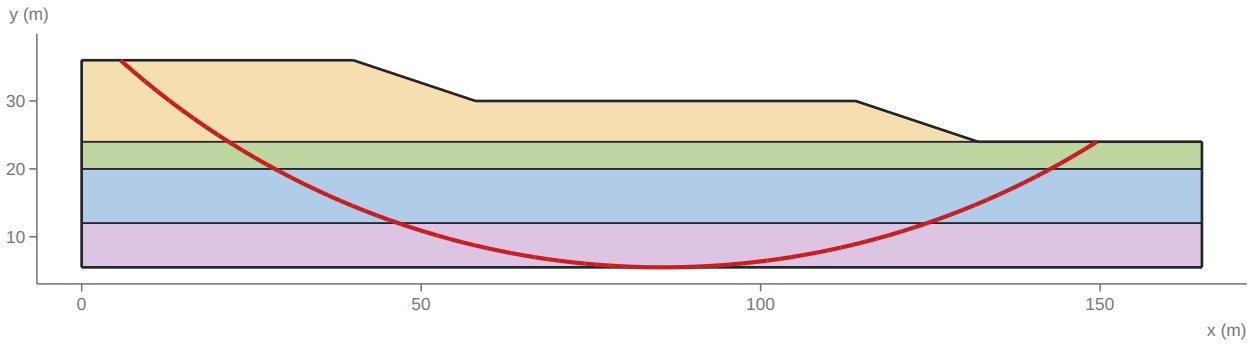



*Figure 70. V040: critical circle (Bishop simplified, circles only, FOS = 1.459).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 20.00 | c' = 0.00 | 30.00° |
| Clay crust | Undrained (φ = 0) | 18.80 | Su = 43.00 | 0° |
| Marine clay | Undrained (φ = 0) | 18.80 | Su = 34.50 | 0° |
| Lacustrine clay | Undrained (φ = 0) | 20.30 | Su = 31.20 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.46 | Bishop simplified | El-Ramly et al. (2002) | 1.459 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified, circles only | 1.461 | 1.459 |

## 3.71 V041a — Submerged slope, pool 30 ft above the crest (D&W fig. 6.27)

*Duncan & Wright, Soil Strength and Slope Stability, fig. 6.27.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Fully submerged homogeneous slope (c = 100 psf, φ = 20°, γ = 128 pcf) with the pool 30 ft above the crest: the extra water weight and pore pressure cancel. Converted to SI.

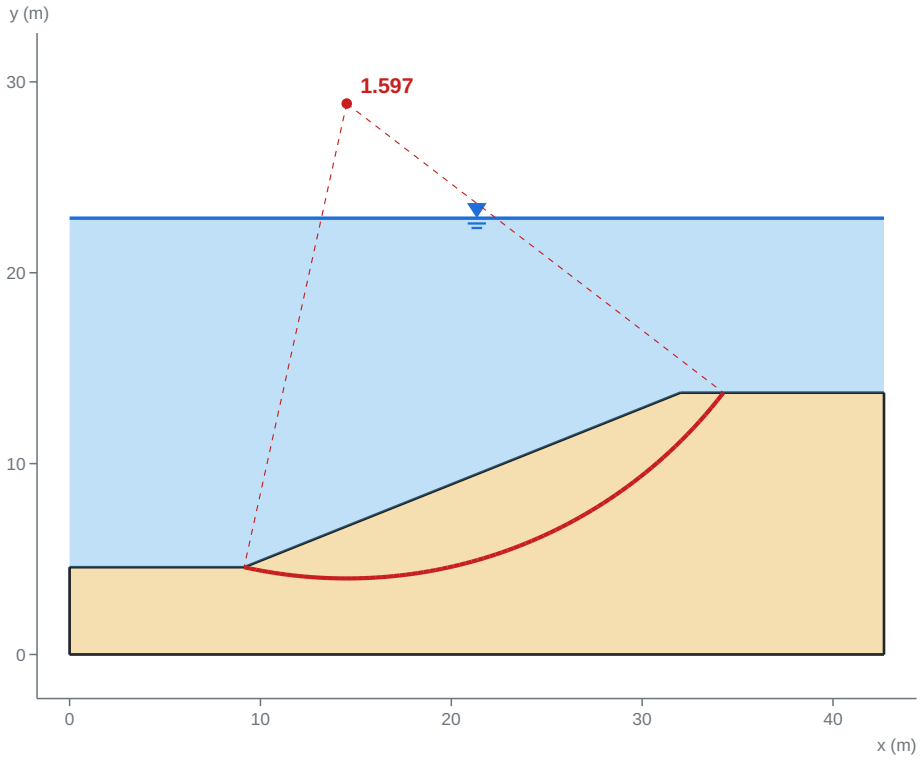



*Figure 71. V041a: critical circle (Bishop simplified, FOS = 1.597).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Material 1 | Mohr-Coulomb | 20.11 | c' = 4.79 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.60 | Spencer | Duncan & Wright | 1.594 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.603 | 1.597 |
| Spencer | 1.599 | 1.594 |
| Spencer, non-circular | 1.592 | 1.591 |

## 3.72 V041b — Submerged slope, pool 60 ft above the crest (D&W fig. 6.27)

*Duncan & Wright, Soil Strength and Slope Stability, fig. 6.27.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

As V041a with the pool 60 ft above the crest: the same factor of safety.

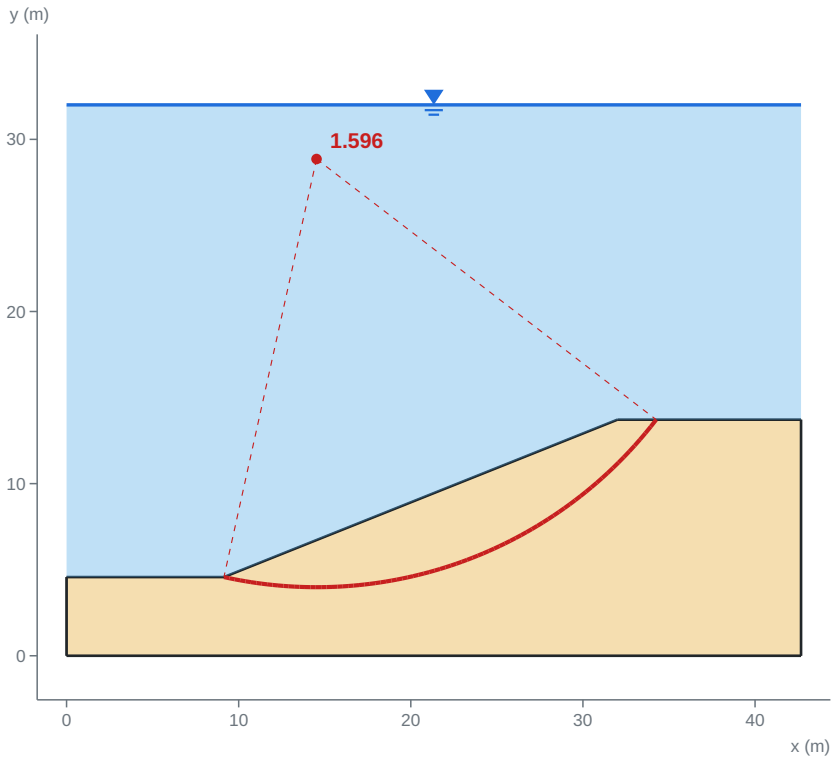



*Figure 72. V041b: critical circle (Bishop simplified, FOS = 1.596).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Material 1 | Mohr-Coulomb | 20.11 | c' = 4.79 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.60 | Spencer | Duncan & Wright | 1.593 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.603 | 1.596 |
| Spencer | 1.599 | 1.593 |
| Spencer, non-circular | 1.590 | 1.589 |

## 3.73 V042 — Thick-core dam with a piezometric line (D&W fig. 7.24)

*Duncan & Wright, Soil Strength and Slope Stability, fig. 7.24.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Symmetric dam with a thick clay core (φ′ = 20°) in a shell (φ′ = 38°), pond at el. 315 ft, a commercial program's piezometric line; the deep circle. Converted to SI.

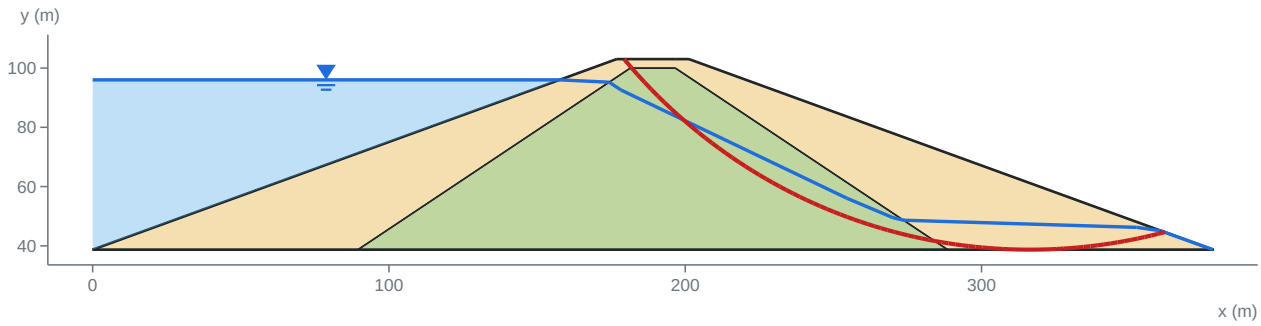



*Figure 73. V042: analysed circle (Bishop simplified, FOS = 1.582).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Shell | Mohr-Coulomb | 21.99 | c' = 0.00 | 38.00° |
| Core | Mohr-Coulomb | 18.85 | c' = 0.00 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.67 | not stated | Duncan & Wright | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.584 | 1.582 |
| Spencer | 1.648 | 1.649 |

## 3.74 V043 — Syncrude tailings dyke (El-Ramly et al.), composite circle

*El-Ramly, H., Morgenstern, N.R. & Cruden, D.M. (2003). Probabilistic stability analysis of a tailings dyke on presheared clay-shale. Can. Geotech. J. 40(1), 192–208.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Cohesionless tailings sand, glacio-fluvial sands and tills (φ = 34° here; El-Ramly et al. give the sandy till 35.7°) on a presheared clay-shale (φ = 7.5°) with a water table; the reference circle (327.5, 394, R = 124) reaches below the model base, so the surface follows the base there (composite).

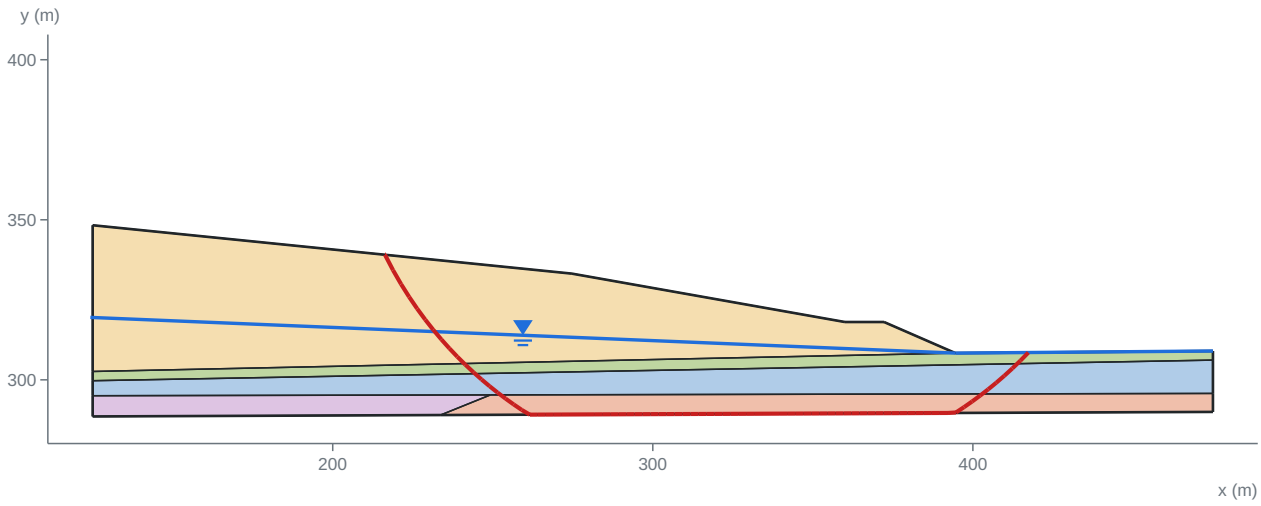



*Figure 74. V043: analysed surface (Bishop simplified, FOS = 1.320).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Tailing sand (TS) | Mohr-Coulomb | 20.00 | c' = 0.00 | 34.00° |
| Glacio-fluvial sand (Pf4) | Mohr-Coulomb | 17.00 | c' = 0.00 | 34.00° |
| Sandy till (Pgs) | Mohr-Coulomb | 17.00 | c' = 0.00 | 34.00° |
| Clayey till (Pgc) | Mohr-Coulomb | 17.00 | c' = 0.00 | 7.50° |
| Disturbed clay-shale (Kca) | Mohr-Coulomb | 17.00 | c' = 0.00 | 7.50° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.31 | Bishop simplified | El-Ramly et al. | 1.320 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.320 | 1.320 |

## 3.75 V044 — Fredlund & Krahn (1977), dry

*Fredlund, D.G. & Krahn, J. (1977). Comparison of slope stability methods of analysis. Can. Geotech. J. 14(3).*

**Fixed circle · Units: imperial (ft, psf, pcf)**

Homogeneous 2:1 slope, 40 ft high, on a firm base; the classic comparison problem of Fredlund & Krahn analysed with the published circle and no pore pressure.

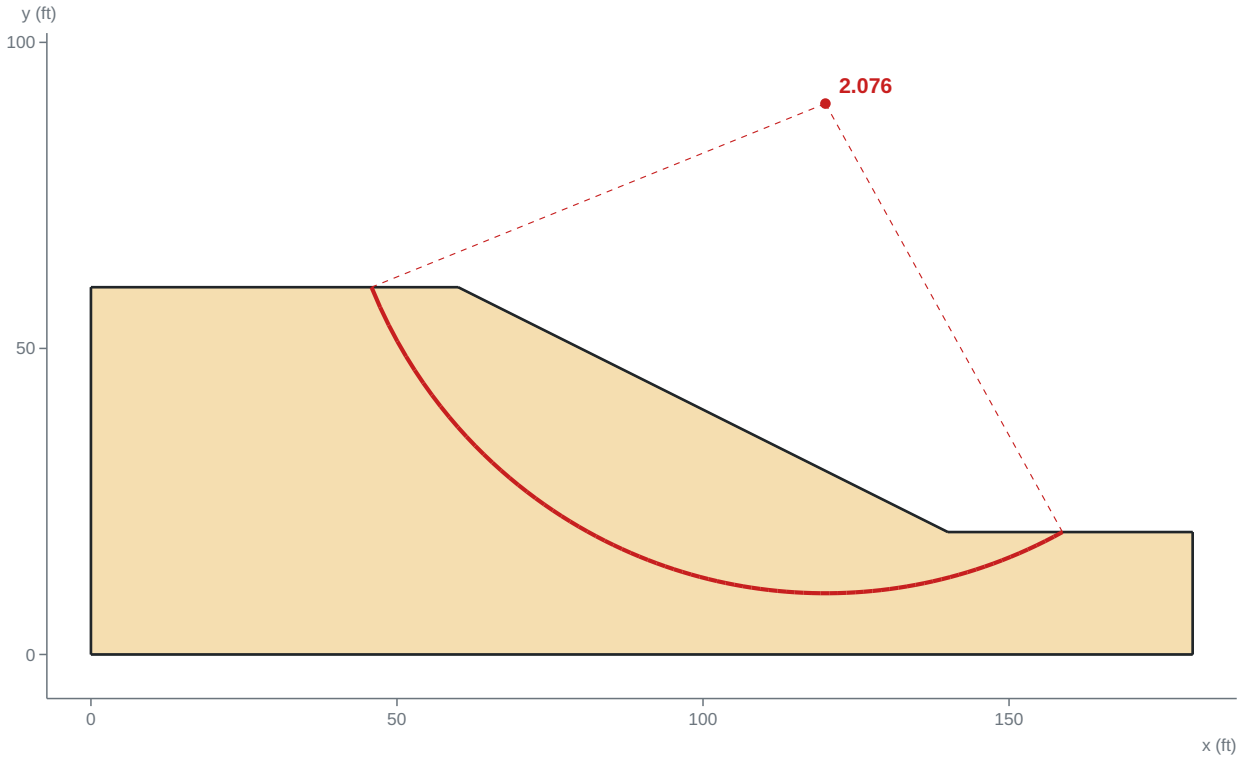



*Figure 75. V044: analysed circle (Bishop simplified, FOS = 2.076).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 120.0 | c' = 600.0 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.08 | Bishop simplified | Fredlund & Krahn | 2.076 |
| 2.076 | Morgenstern-Price (half-sine) | Fredlund & Krahn | 2.072 |
| 2.073 | Spencer | Fredlund & Krahn | 2.072 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | – | 2.076 |
| Morgenstern-Price (half-sine) | – | 2.072 |
| Spencer | – | 2.072 |

## 3.76 V045 — Fredlund & Krahn (1977), with water table

*Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The Fredlund & Krahn slope with a piezometric line, analysed on the published circle. Converted to SI.

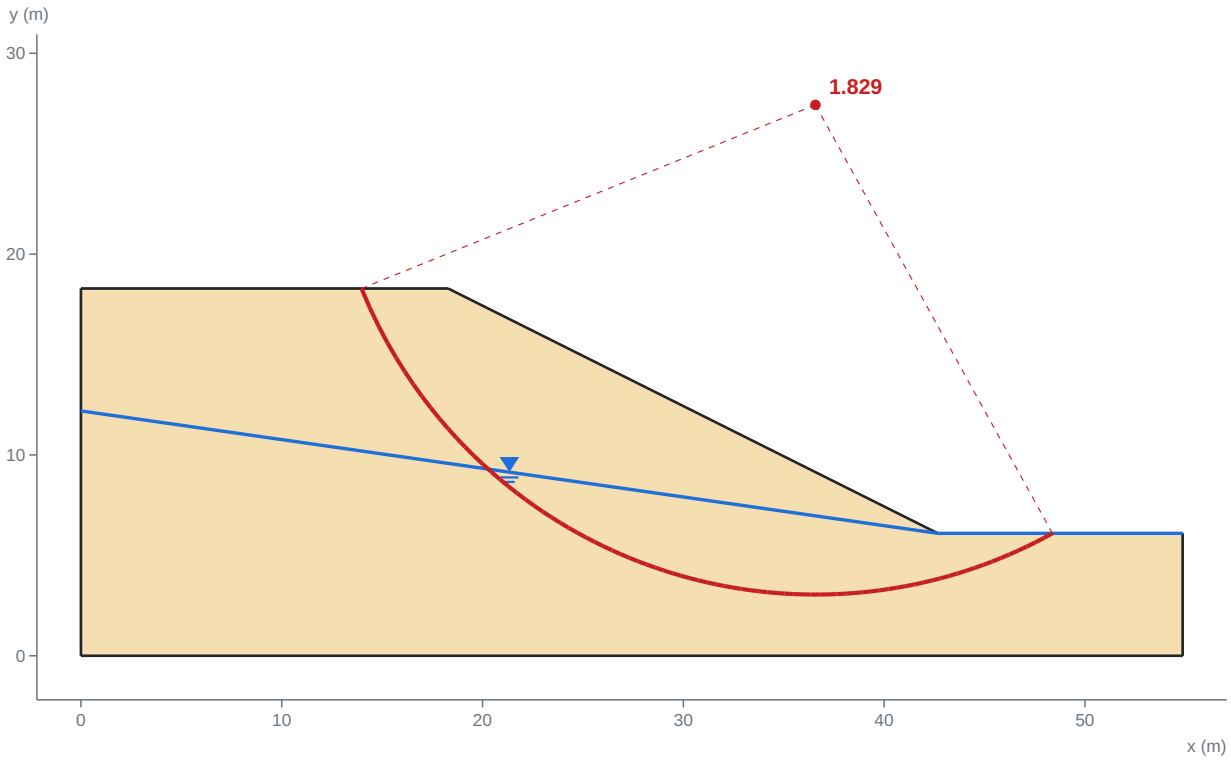



*Figure 76. V045: analysed circle (Bishop simplified, FOS = 1.829).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 18.85 | c' = 28.73 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.834 | Bishop simplified | Fredlund & Krahn | 1.829 |
| 1.832 | Morgenstern-Price (half-sine) | Fredlund & Krahn | 1.827 |
| 1.83 | Spencer | Fredlund & Krahn | 1.827 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | – | 1.829 |
| Morgenstern-Price (half-sine) | – | 1.827 |
| Spencer | – | 1.827 |

## 3.77 V046a — Fredlund & Krahn slope on a weak seam, composite surface

*Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).*

**Specified non-circular surface · Units: imperial (ft, psf, pcf)**

The Fredlund & Krahn slope with a 1 ft weak seam (c′ = 0, φ′ = 10°) on bedrock at el. 15 (their case 2); the published circle (120, 90, R = 80) reaches below the seam, so the surface follows the arc down to the bedrock, runs along the seam and climbs back on the arc. Dry.

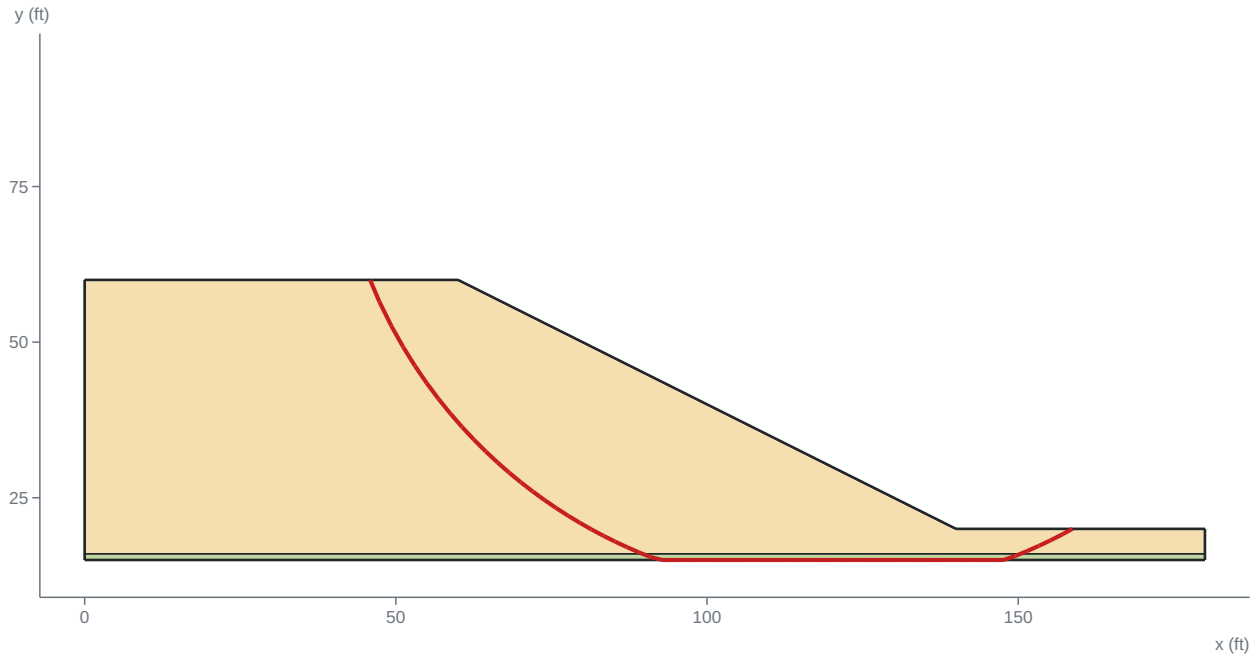



*Figure 77. V046a: analysed surface (Bishop simplified, FOS = 1.380).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Upper soil | Mohr-Coulomb | 120.0 | c' = 600.0 | 20.00° |
| Weak seam | Mohr-Coulomb | 120.0 | c' = 0.00 | 10.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.37 | Morgenstern-Price (half-sine) | Fredlund & Krahn | 1.370 |
| 1.377 | Bishop simplified | Fredlund & Krahn | 1.380 |
| 1.373 | Spencer | Fredlund & Krahn | 1.379 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.382 | 1.380 |
| Spencer | 1.382 | 1.379 |
| Morgenstern-Price (half-sine) | – | 1.370 |

## 3.78 V046b — Fredlund & Krahn slope on a weak seam, with a piezometric line

*Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

As V046a with the piezometric line of V045 (from (0, 40) ft to the toe) for both materials: Fredlund & Krahn's case 6. Converted to SI.

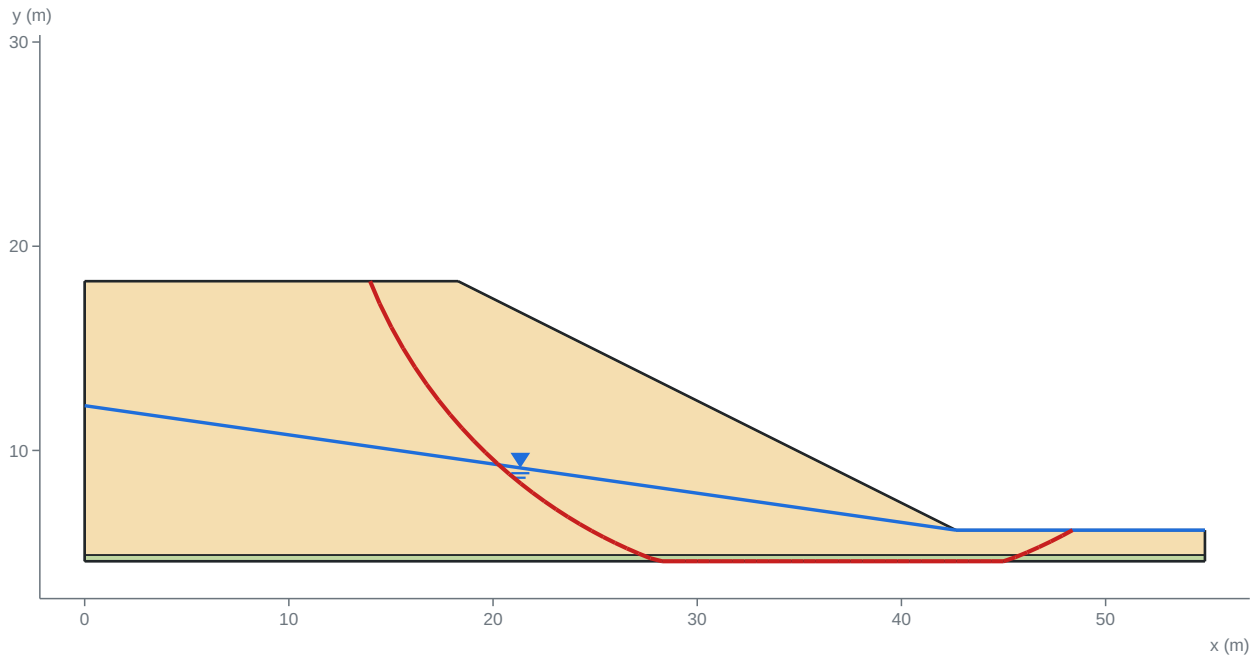



*Figure 78. V046b: analysed surface (Bishop simplified, FOS = 1.241).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Upper soil | Mohr-Coulomb | 18.85 | c' = 28.73 | 20.00° |
| Weak seam | Mohr-Coulomb | 18.85 | c' = 0.00 | 10.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.248 | Bishop simplified | Fredlund & Krahn | 1.241 |
| 1.245 | Spencer | Fredlund & Krahn | 1.242 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | – | 1.241 |
| Spencer | – | 1.242 |
| Morgenstern-Price (half-sine) | – | 1.235 |

## 3.79 V047a — ACADS 1(a), homogeneous slope, dry

*Giam & Donald (1989).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

ACADS 1(a): homogeneous 2:1 slope, 10 m high (c′ = 3 kPa, φ′ = 19.6°, γ = 20 kN/m³), dry; critical-circle search.

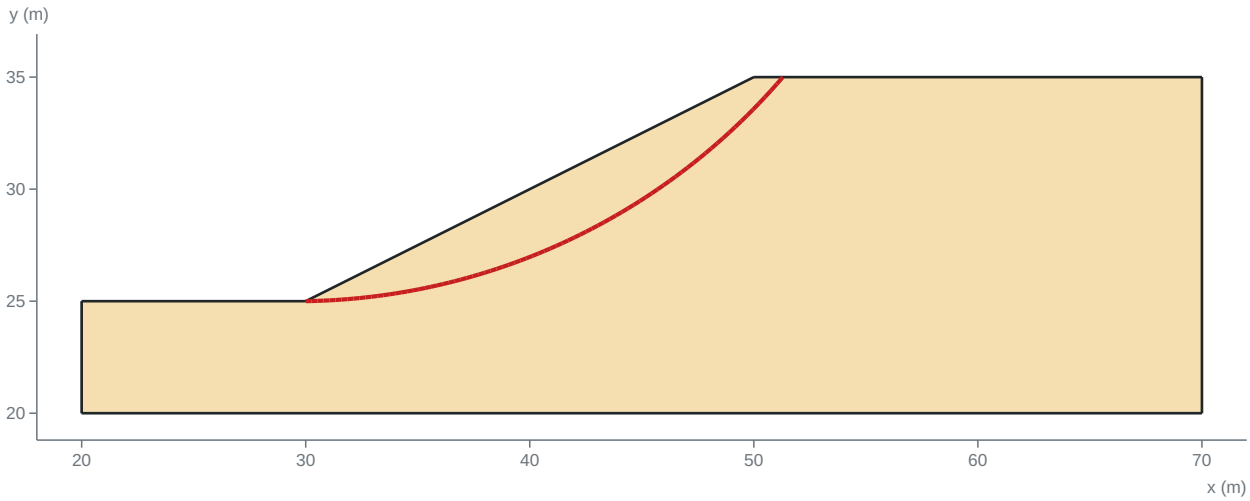



*Figure 79. V047a: critical circle (Bishop simplified, FOS = 0.985).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 3.00 | 19.60° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | Spencer | ACADS | 0.984 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 0.987–0.992 | 0.985 |
| Spencer | 0.986 | 0.984 |
| Janbu corrected | 0.990 | 0.986 |
| Morgenstern-Price (half-sine) | 0.991 | 0.984 |

## 3.80 V047b — ACADS tension crack, reference crack depth 3.6 m (results correspond to a dry c

*Giam & Donald (1989), ACADS problem 1(b).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

ACADS 1(b) as a commercial program models it: c′ = 32 kPa, φ′ = 10°, a 3.6 m tension crack (dry); critical-circle search.

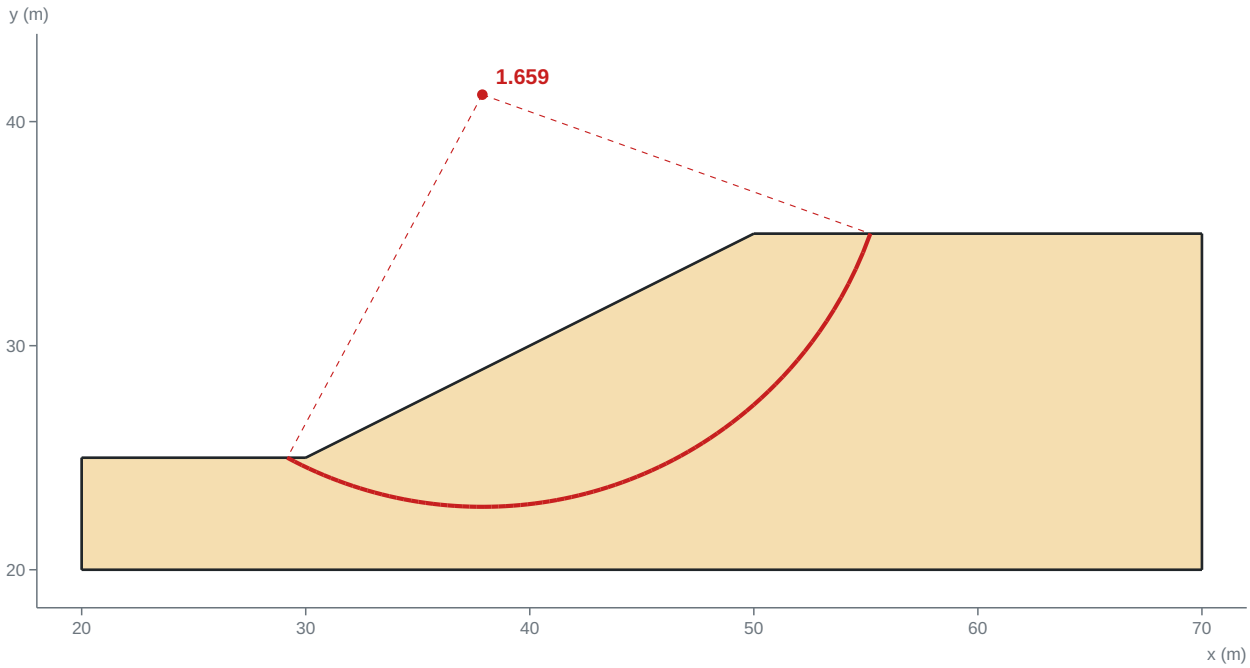



*Figure 80. V047b: critical circle (Bishop simplified, FOS = 1.659).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 32.00 | 10.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.65–1.70 | not stated | Giam & Donald (1989), ACADS | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.664 | 1.659 |
| Morgenstern-Price (half-sine) | 1.660 | 1.656 |

## 3.81 V048 — ACADS 1(b), water-filled tension crack

*Giam & Donald (1989), ACADS problem 1(b).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The ACADS 1(a) slope (10 m high, 2:1) with c' = 32 kPa, φ' = 10° and a water-filled tension crack 4 m deep, the crack zone of a commercial program's Fig. 2.1 (35 → 31 m; the formula 2c/(γ√Ka) gives 3.814 m); critical circle search.

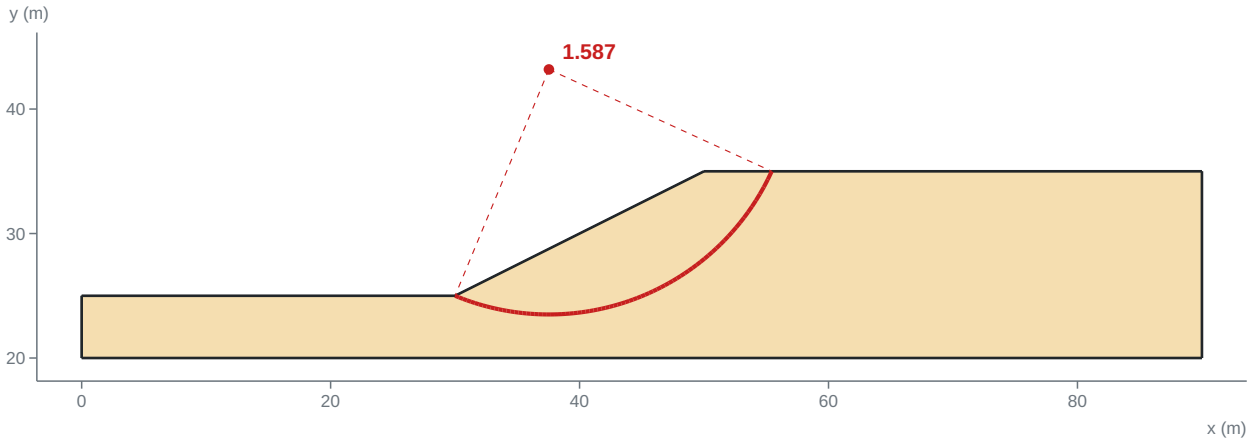



*Figure 81. V048: critical circle (Bishop simplified, FOS = 1.587).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 32.00 | 10.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.65 | not stated | Giam & Donald (1989), ACADS | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.596 | 1.587 |
| Spencer | 1.592 | 1.582 |
| Janbu corrected | 1.489 | 1.475 |

*The ACADS reference 1.65 corresponds to a dry tension crack; with the crack filled with water the programs give about 1.59–1.60.*

## 3.82 V049 — ACADS 1(c), three-layer slope

*Giam & Donald (1989), ACADS slope stability programs review.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Non-homogeneous three-layer slope (ACADS 1c); critical-circle search.

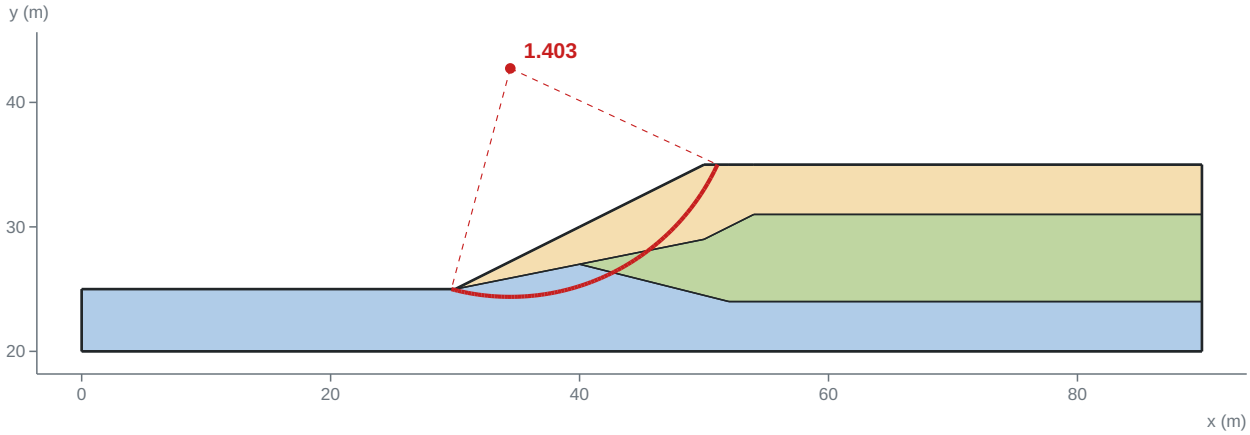



*Figure 82. V049: critical circle (Bishop simplified, FOS = 1.403).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil #1 | Mohr-Coulomb | 19.50 | c' = 0.00 | 38.00° |
| Soil #2 | Mohr-Coulomb | 19.50 | c' = 5.30 | 23.00° |
| Soil #3 | Mohr-Coulomb | 19.50 | c' = 7.20 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.39 | not stated | Giam & Donald (1989), ACADS | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.403–1.405 | 1.403 |
| Spencer | 1.375 | 1.372 |
| Janbu corrected | 1.357 | 1.355 |
| Morgenstern-Price (half-sine) | 1.382 | 1.372 |

## 3.83 V050a — ACADS 3(b), weak layer, specified non-circular surface

*Giam & Donald (1989), ACADS problem 3(b).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Slope with a 0.5 m weak layer (c' = 0, φ' = 10°) and a water table at its base, analysed on the published four-point non-circular surface that runs along the weak layer.

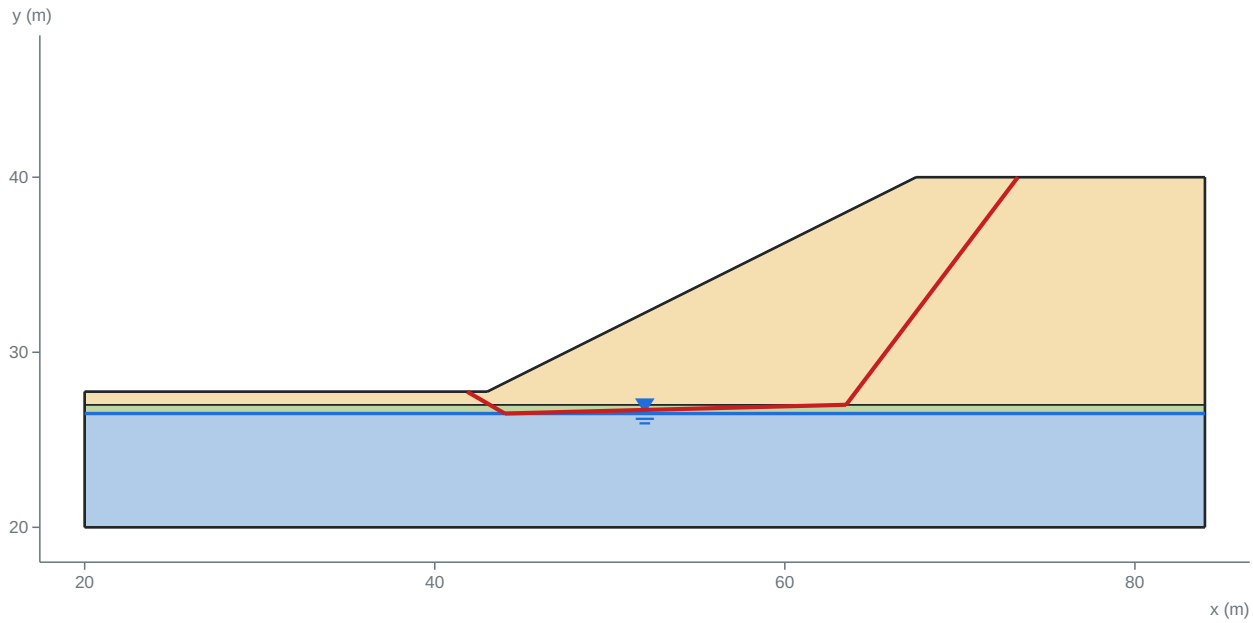



*Figure 83. V050a: analysed surface (Janbu corrected, FOS = 1.294).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil 1 | Mohr-Coulomb | 18.84 | c' = 28.50 | 20.00° |
| Weak layer | Mohr-Coulomb | 18.84 | c' = 0.00 | 10.00° |
| Soil 1 below | Mohr-Coulomb | 18.84 | c' = 28.50 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.34 | not stated | Giam & Donald (1989), ACADS | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 1.294 | 1.294 |
| Spencer | 1.277 | 1.276 |
| Morgenstern-Price (half-sine) | 1.261 | 1.260 |

*The ACADS referee value comes from the 1989 survey; programs today give 1.26–1.29 on the specified surface.*

## 3.84 V050b — ACADS 3(a), weak layer, non-circular search

*Giam & Donald (1989), ACADS problem 3(a).*

**Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)**

As V050a, with a circular search followed by non-circular optimisation; the critical surface slides along the weak layer with a back scarp to the crest.

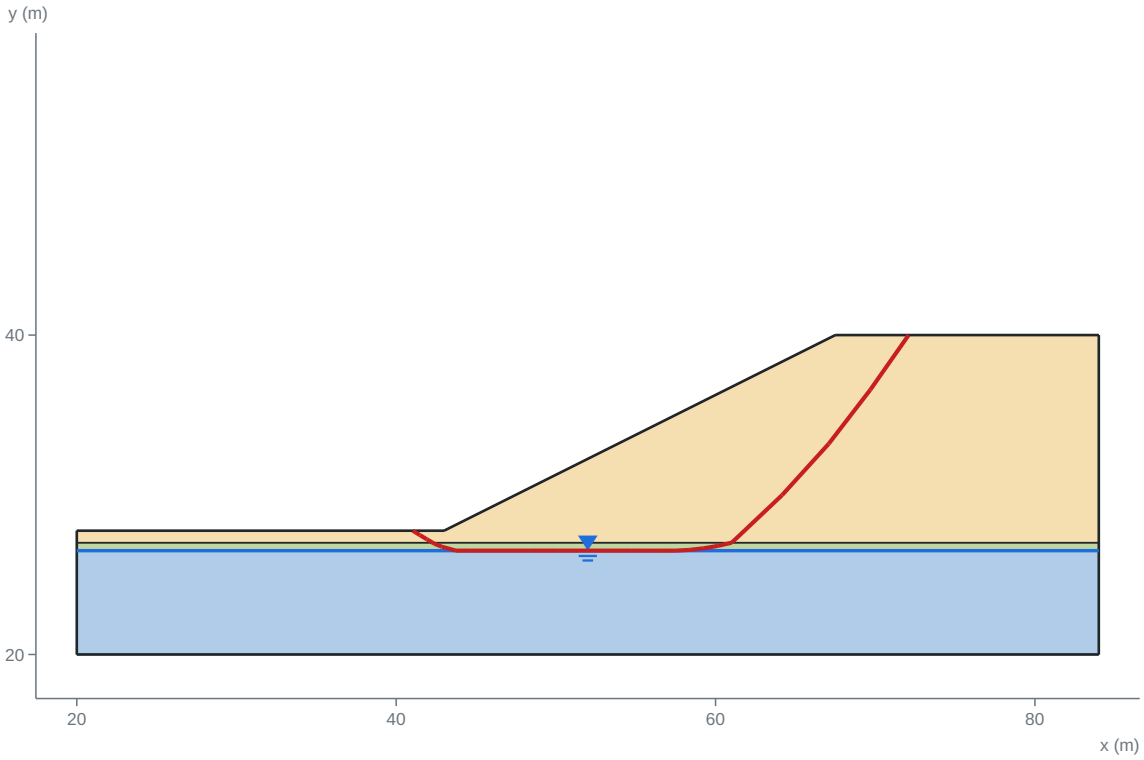



*Figure 84. V050b: critical surface (Spencer, non-circular, FOS = 1.225).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil 1 | Mohr-Coulomb | 18.84 | c' = 28.50 | 20.00° |
| Weak layer | Mohr-Coulomb | 18.84 | c' = 0.00 | 10.00° |
| Soil 1 below | Mohr-Coulomb | 18.84 | c' = 28.50 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.24–1.27 | not stated | Giam & Donald (1989), ACADS | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, non-circular | 1.246 | 1.225 |

*ACADS accepted 1.24–1.27; cSlope's search finds a surface along the weak layer at 1.228 (Spencer), unchanged at 50, 100 and 200 slices.*

## 3.85 V051 — ACADS 4, weak layer, water table and surcharges

*Giam & Donald (1989), ACADS slope stability programs review.*

**Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)**

Weak-layer slope with a piezometric line and two surcharge strips (20 kPa on the bench, 20–40 kPa ramp on the crest); non-circular search. The published spread is wide: a search-difficulty benchmark.

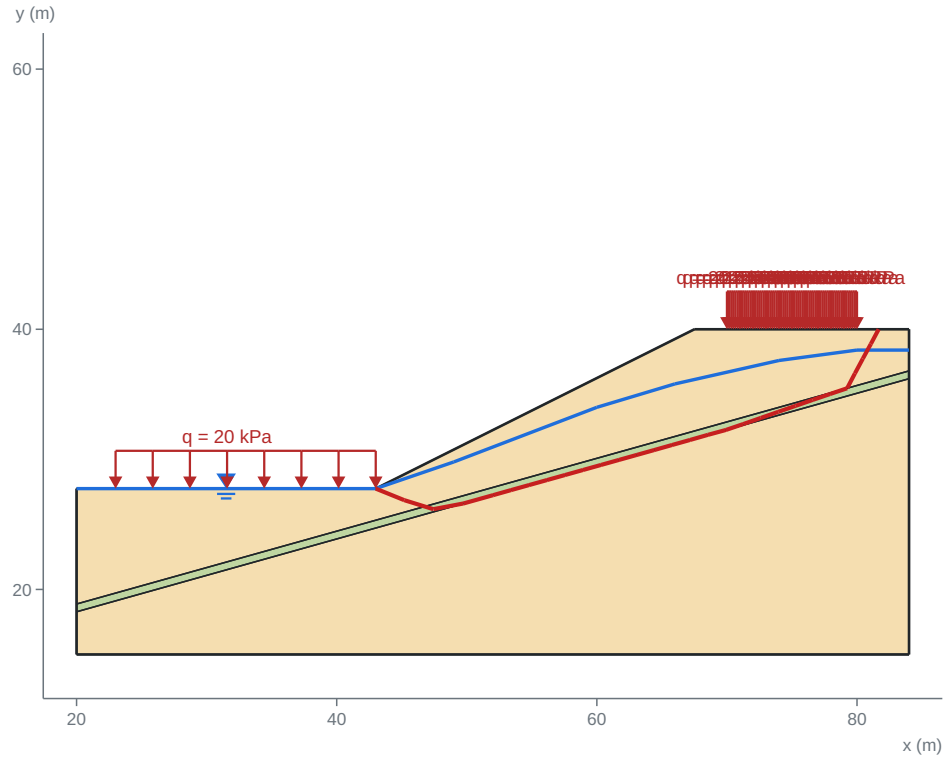



*Figure 85. V051: critical surface (Spencer, non-circular, FOS = 0.714).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil 1 | Mohr-Coulomb | 18.84 | c' = 28.50 | 20.00° |
| Weak Layer | Mohr-Coulomb | 18.84 | c' = 0.00 | 10.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.78 | not stated | Giam & Donald (1989), ACADS | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, non-circular | 0.707 | 0.714 |

*The ACADS referee value 0.78 comes from the 1989 survey, where the participating programs averaged 0.81; optimised non-circular searches today give about 0.68–0.71.*

## 3.86 V052a — Talbingo Dam, end of construction

*Giam & Donald (1989), ACADS slope stability programs review.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

ACADS 2(a): Talbingo Dam, four zones, critical-circle search. The minimum is a shallow slide parallel to the steeper upstream face.

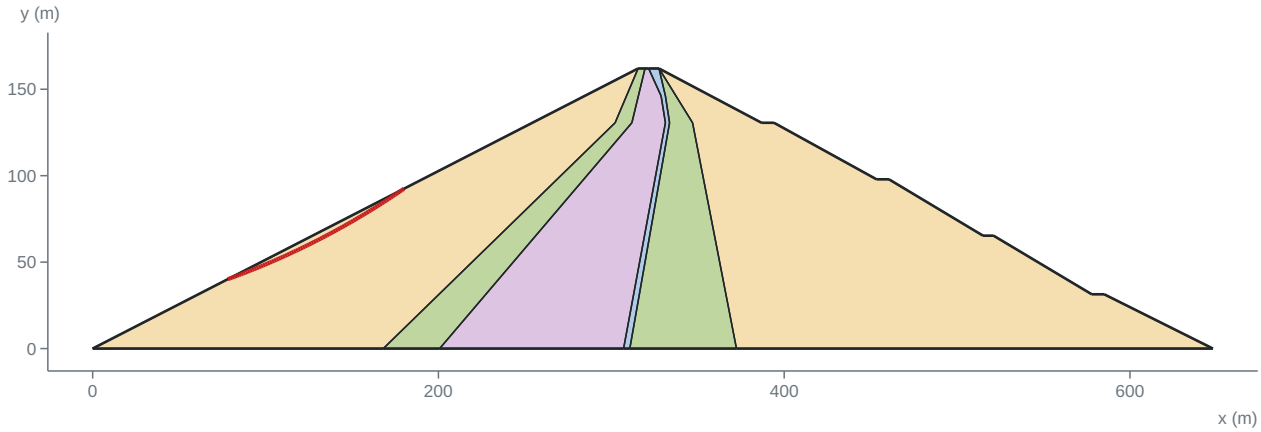



*Figure 86. V052a: critical circle (Bishop simplified, FOS = 1.964).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Rockfill | Mohr-Coulomb | 20.40 | c' = 0.00 | 45.00° |
| Transitions | Mohr-Coulomb | 20.40 | c' = 0.00 | 45.00° |
| Filter | Mohr-Coulomb | 20.40 | c' = 0.00 | 45.00° |
| Core | Mohr-Coulomb | 18.10 | c' = 85.00 | 23.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.95 | not stated | Giam & Donald (1989), ACADS | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.948–1.951 | 1.964 |
| Spencer | 1.948 | 1.964 |
| Janbu corrected | 1.949 | 1.974 |

## 3.87 V052b — Talbingo Dam, specified circle

*Giam & Donald (1989), ACADS slope stability programs review.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

ACADS 2(b): Talbingo Dam on the circle (100.3, 291.0, R = 278.8).

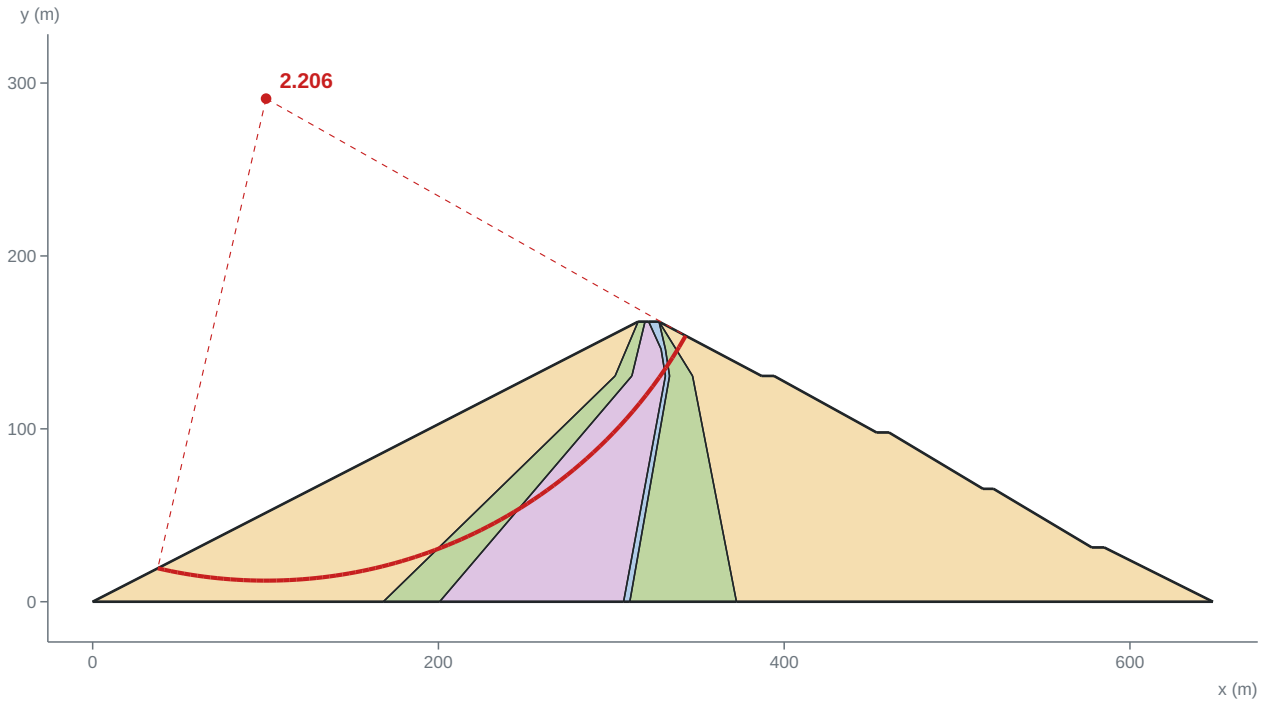



*Figure 87. V052b: analysed circle (Bishop simplified, FOS = 2.206).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Rockfill | Mohr-Coulomb | 20.40 | c' = 0.00 | 45.00° |
| Transitions | Mohr-Coulomb | 20.40 | c' = 0.00 | 45.00° |
| Filter | Mohr-Coulomb | 20.40 | c' = 0.00 | 45.00° |
| Core | Mohr-Coulomb | 18.10 | c' = 85.00 | 23.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.29 | not stated | Giam & Donald (1989), ACADS | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.208 | 2.206 |
| Spencer | 2.292 | 2.290 |
| Janbu corrected | 2.073 | 2.073 |
| Morgenstern-Price (half-sine) | 2.299 | 2.299 |

## 3.88 V099 — ACADS 5, excavated slope with seepage

*Giam & Donald (1989), ACADS problem 5.*

**Finite elements only · Units: SI (m, kPa, kN/m³)**

1:2 cut, 10 m deep, below an initially horizontal ground with the water table 2 m down; long-term steady seepage towards the excavation, whose floor is under 1 m of water. Checked by finite elements only: the limit equilibrium values come from a flow net that the input cannot reproduce (a straight water table gives a different pore pressure field).

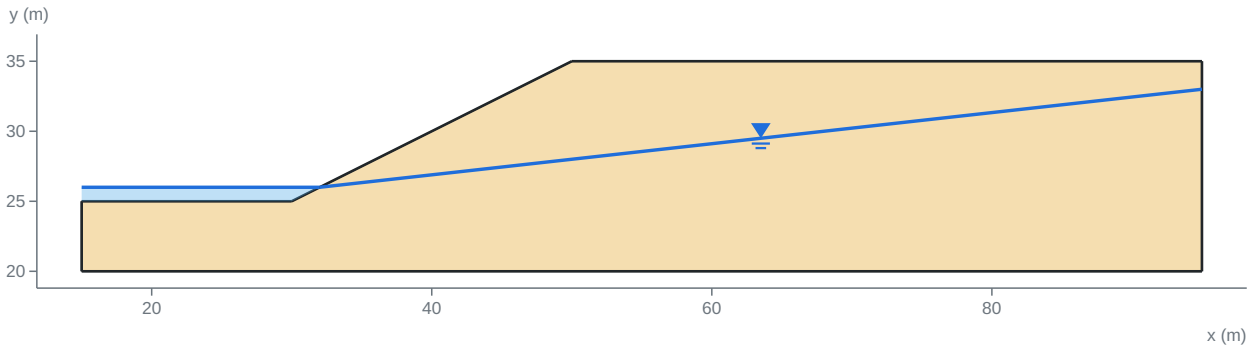



*Figure 88. V099: geometry and water*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 20.00 | c' = 11.00 | 28.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|

## 3.89 V053 — Greco (1996) ex. 5 / Chen & Shao weak-layer slope, reference geometry

*Greco (1996) example 5; Chen & Shao (1988).*

**Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)**

Greco's weak-seam slope (V054) with a commercial program's geometry and water table; circular search with non-circular optimisation, vertical pore-pressure head.

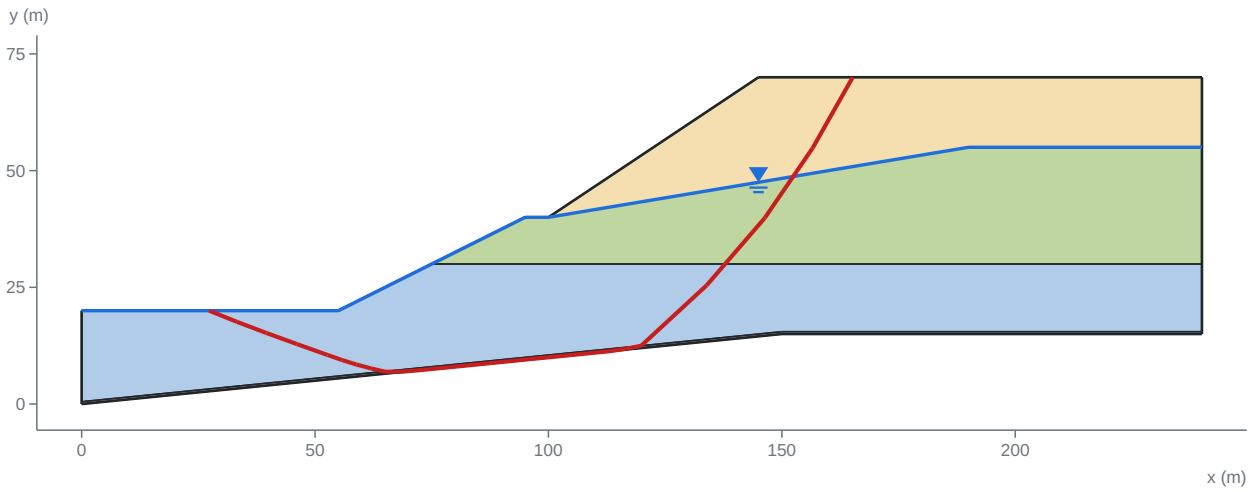



*Figure 89. V053: critical surface (Spencer, non-circular, FOS = 1.006).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 | Mohr-Coulomb | 20.00 | c' = 9.80 | 35.00° |
| Layer 2 | Mohr-Coulomb | 19.00 | c' = 58.80 | 25.00° |
| Layer 3 | Mohr-Coulomb | 21.50 | c' = 19.80 | 30.00° |
| Layer 4 | Mohr-Coulomb | 21.50 | c' = 9.80 | 16.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.009–1.025 | Spencer, non-circular (test problem 3) | Chen & Shao (1988) | 1.006 |
| 0.97 | Spencer, non-circular (global minimum along the weak layer) | Greco (1996) | 1.006 |
| 1.08 | Spencer, local minimum (nearly circular surface from the toe) | Greco (1996) | 1.091 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, non-circular | 1.010 | 1.006 |
| Spencer | – | 1.091 |

*Chen & Shao give 1.009–1.025 with Spencer's assumption for a surface sliding along the weak seam (their test problem 3, Tables 5–6). Greco's Monte Carlo search finds a global minimum of 0.97 along the weak layer, entering the level ground far from the toe, and a local minimum of 1.08 on a nearly circular surface from the toe; the programs' block searches and cSlope give 1.01 on a commercial program's version of the geometry, 4 % above Greco's global minimum.*

## 3.90 V054 — Greco (1996) example 5 / Chen & Shao (1988)

*Greco (1996), example 5; Chen & Shao (1988).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Four layers including a weak seam, with a water table; critical circle search, vertical pore-pressure head.

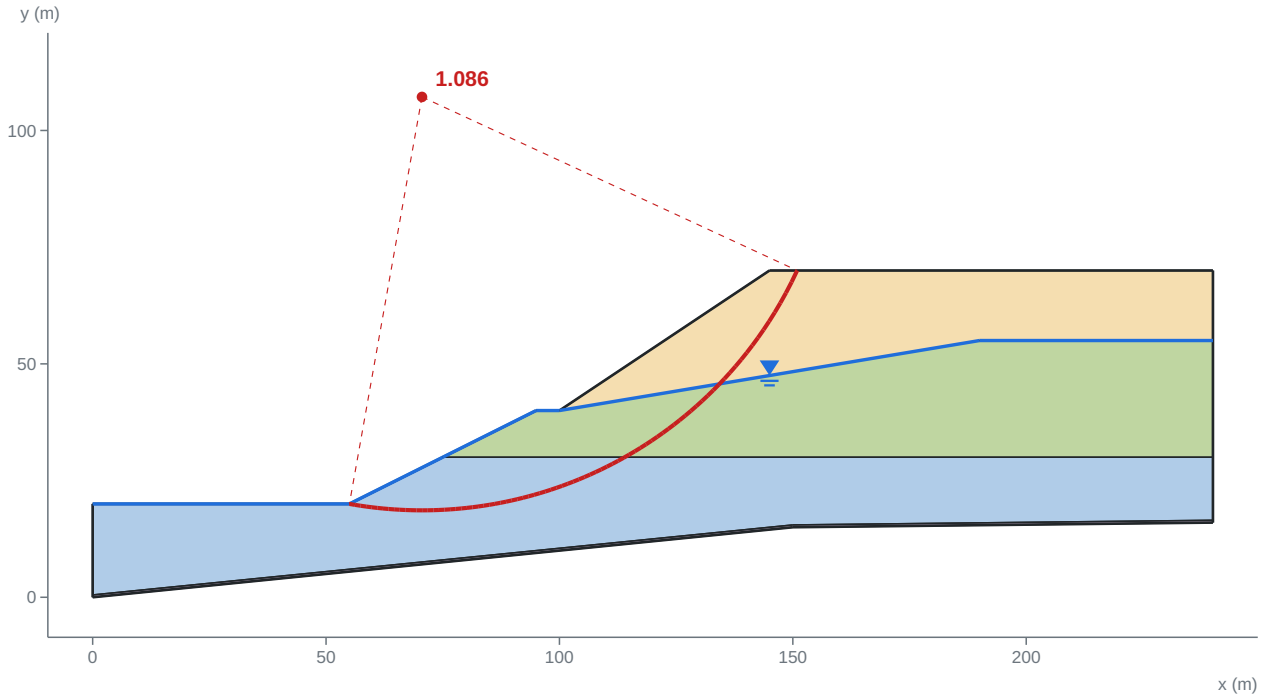



*Figure 90. V054: critical circle (Bishop simplified, FOS = 1.086).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 | Mohr-Coulomb | 20.00 | c' = 9.80 | 35.00° |
| Layer 2 | Mohr-Coulomb | 19.00 | c' = 58.80 | 25.00° |
| Layer 3 | Mohr-Coulomb | 21.50 | c' = 19.80 | 30.00° |
| Seam | Mohr-Coulomb | 21.50 | c' = 9.80 | 16.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.08 | Spencer, local minimum (nearly circular surface from the toe) | Greco (1996) | 1.091 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.087 | 1.086 |
| Spencer | 1.093 | 1.091 |

*Greco's 1.08 is the local minimum of his search, a nearly circular surface from the toe; his global minimum, 0.97 along the weak layer, is compared in V053.*

## 3.91 V055 — Griffiths & Lane example 2, slope with a foundation layer

*Griffiths & Lane (1999), Géotechnique 49(3).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The 2:1 slope (c/γH = 0.05, φ = 20°) on a foundation of the same soil; unconstrained circular search finds the toe circle. Converted to SI.

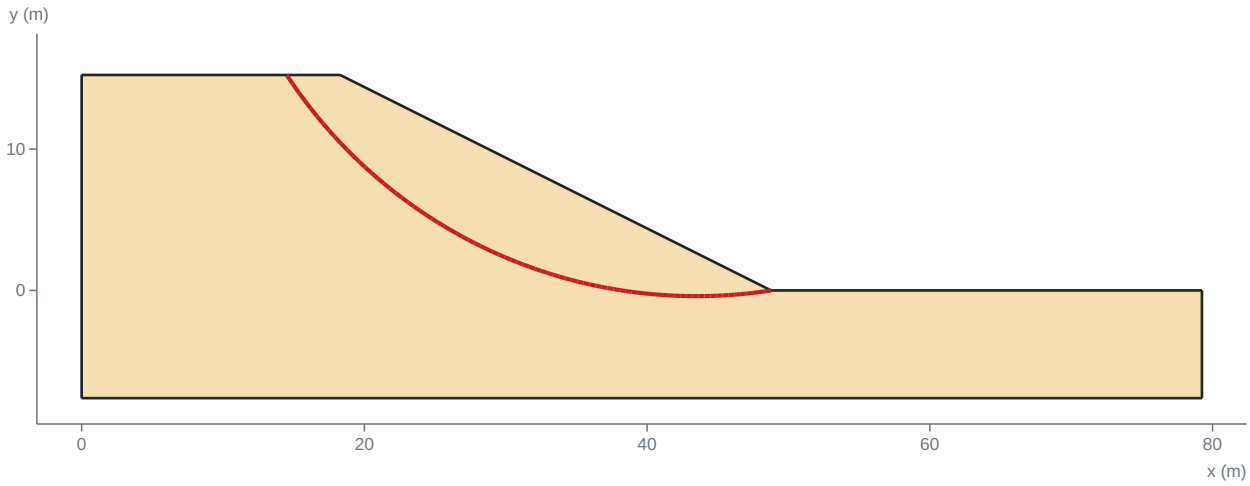



*Figure 91. V055: critical circle (Spencer, FOS = 1.366).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| soil | Mohr-Coulomb | 19.64 | c' = 14.96 | 20.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.40 | finite elements | Griffiths & Lane | 1.366 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.366 | 1.366 |

## 3.92 V056 — Griffiths & Lane example 3, thin weak layer, cu2/cu1 = 0.2

*Griffiths & Lane (1999), Géotechnique 49(3).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Undrained clay slope with a thin weak layer at a fifth of the strength, on the three-line wedge through the layer (the paper's Janbu mechanism). Converted to SI.

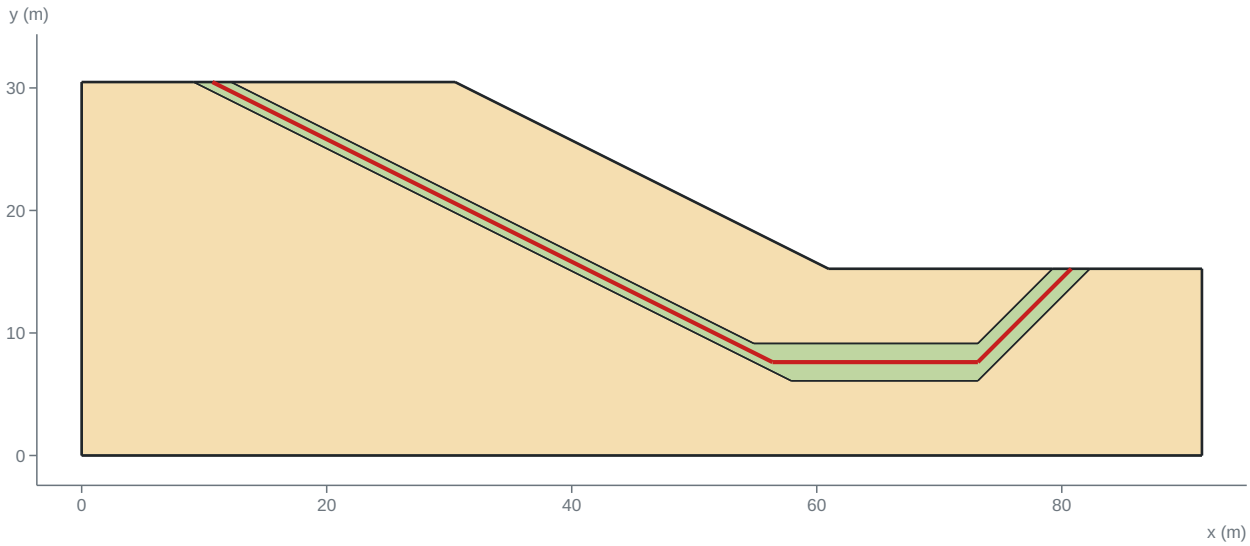



*Figure 92. V056: analysed surface (Janbu corrected, FOS = 0.499).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Surrounding clay | Undrained (φ = 0) | 19.64 | Su = 74.81 | 0° |
| Thin weak layer | Undrained (φ = 0) | 19.64 | Su = 14.96 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.485 | Janbu wedge | Griffiths & Lane | 0.499 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | – | 0.499 |

## 3.93 V057a — Griffiths & Lane example 4, cu2/cu1 = 1 (deep base circle)

*Griffiths & Lane (1999), Géotechnique 49(3).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Undrained clay slope over a foundation of the same strength; the deep base circle. Converted to SI.

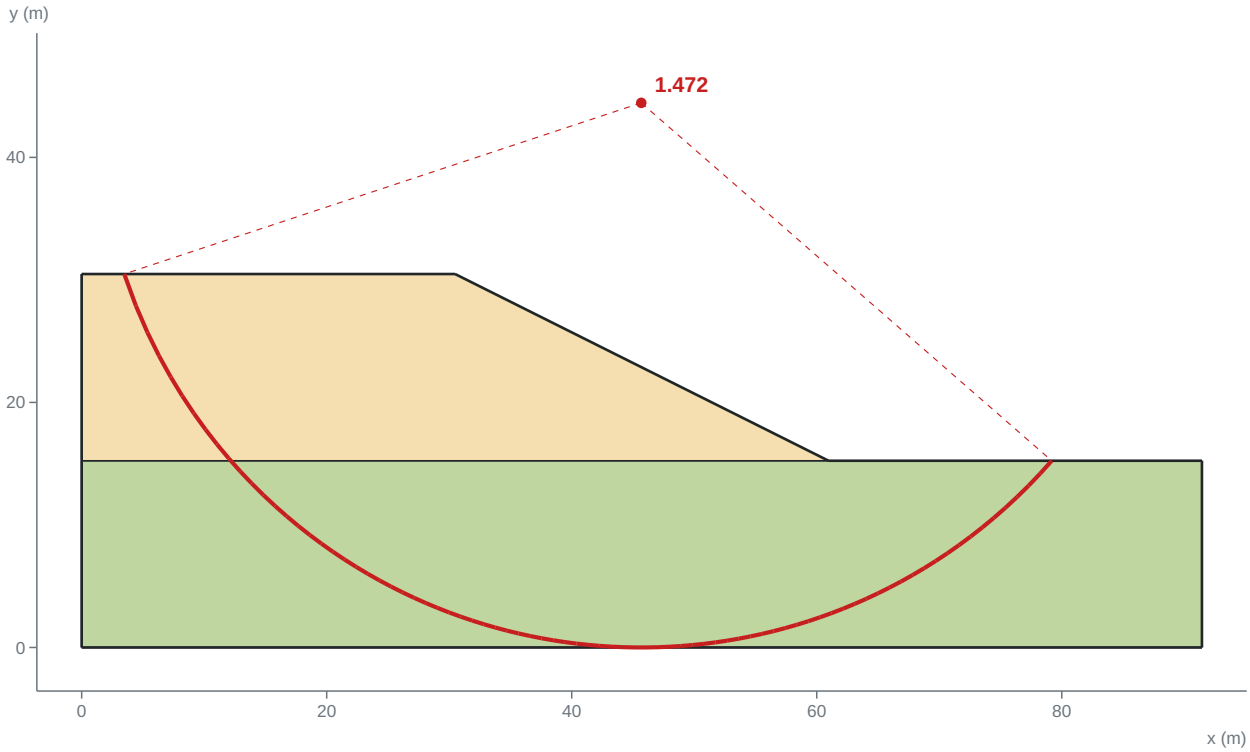



*Figure 93. V057a: critical circle (Spencer, FOS = 1.472).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Slope clay | Undrained (φ = 0) | 19.64 | Su = 74.81 | 0° |
| Foundation clay | Undrained (φ = 0) | 19.64 | Su = 74.81 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.46 | limit-equilibrium chart | Griffiths & Lane | 1.472 |
| 1.47 | stability chart (Taylor) | Griffiths & Lane (1999) | – |
| 1.45 | finite elements | Griffiths & Lane (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.468 | 1.472 |

## 3.94 V057b — Griffiths & Lane example 4, cu2/cu1 = 2 (toe circle)

*Griffiths & Lane (1999), Géotechnique 49(3).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The same slope over a foundation twice as strong; the shallow toe circle. Converted to SI.

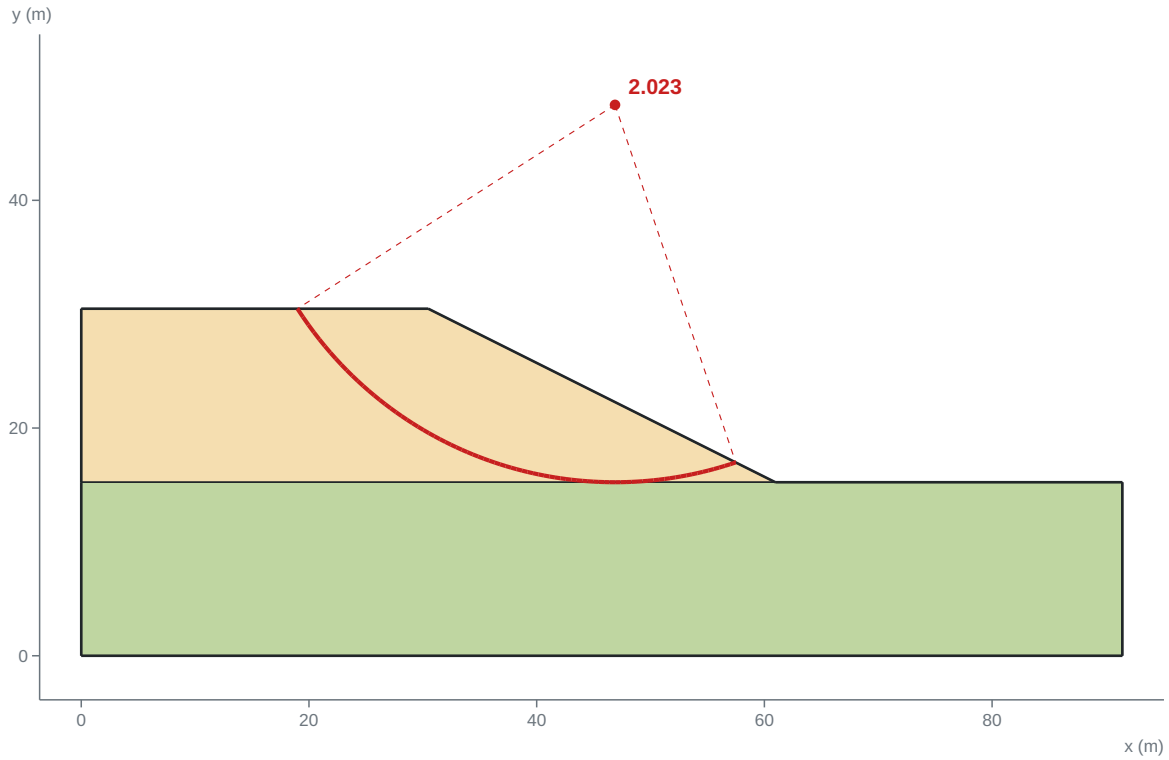



*Figure 94. V057b: critical circle (Spencer, FOS = 2.023).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Slope clay | Undrained (φ = 0) | 19.64 | Su = 74.81 | 0° |
| Foundation clay | Undrained (φ = 0) | 19.64 | Su = 149.6 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.04 | limit-equilibrium chart | Griffiths & Lane | 2.023 |
| 2.10 | stability chart (Taylor) | Griffiths & Lane (1999) | – |
| 2.03 | finite elements | Griffiths & Lane (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 2.022 | 2.023 |

## 3.95 V058 — Griffiths & Lane example 6, earth dam with full reservoir

*Griffiths & Lane (1999), Géotechnique 49(3).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Two-sided earth dam with the reservoir full and a piezometric line; circular search. Converted to SI.

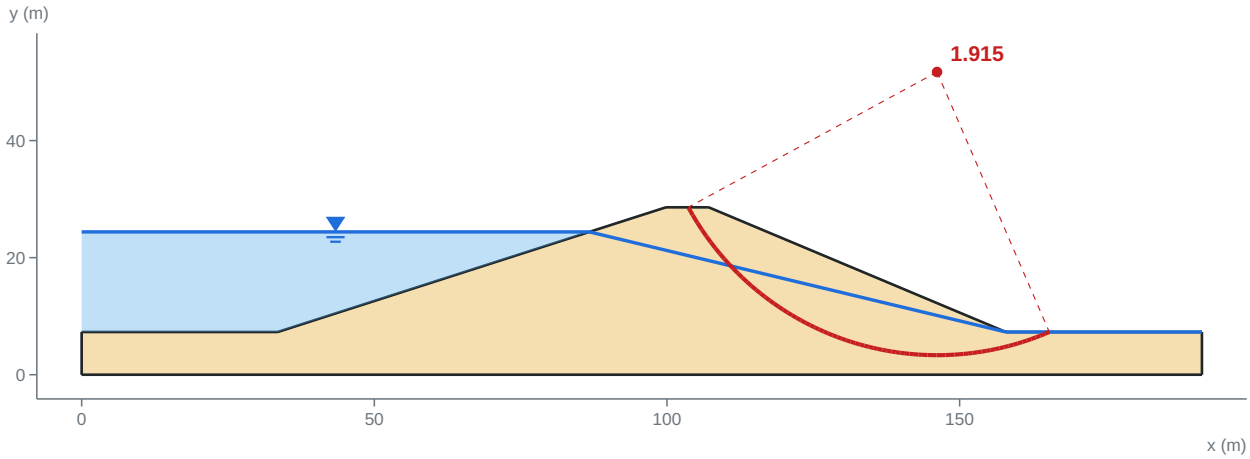



*Figure 95. V058: critical circle (Spencer, FOS = 1.915).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 18.20 | c' = 13.80 | 37.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.90 | finite elements | Griffiths & Lane | 1.915 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.915 | 1.915 |

## 3.96 V059a — Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.4

*Guo & Griffiths (2020), Can. Geotech. J. 57.*

**Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)**

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 84 kPa), γ = 20, firm base; optimised non-circular search. The deep mechanism governs.

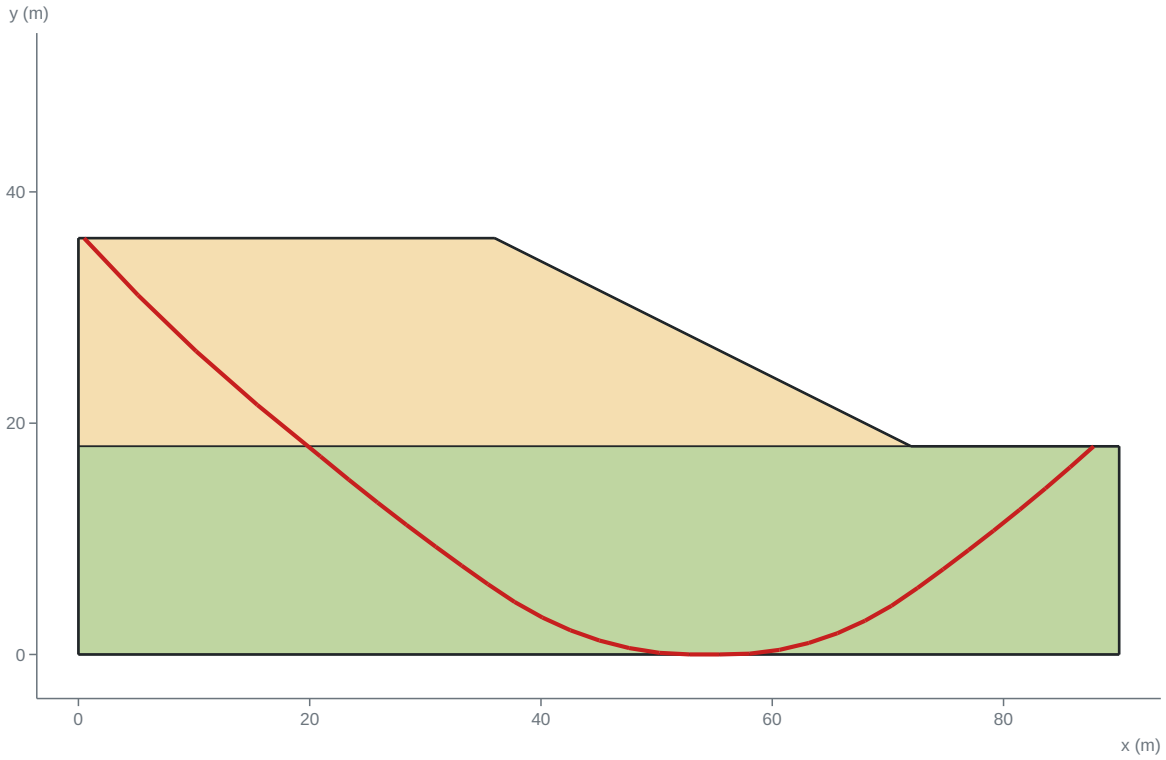



*Figure 96. V059a: critical surface (Spencer, non-circular, FOS = 1.214).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Undrained (φ = 0) | 20.00 | Su = 60.00 | 0° |
| Foundation | Undrained (φ = 0) | 20.00 | Su = 84.00 | 0° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, non-circular | 1.215 | 1.214 |

*The paper gives its results in figures only; the comparison is with the other programs.*

## 3.97 V059b — Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.5

*Guo & Griffiths (2020), Can. Geotech. J. 57.*

**Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)**

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 90 kPa), γ = 20, firm base; optimised non-circular search. The deep mechanism governs.

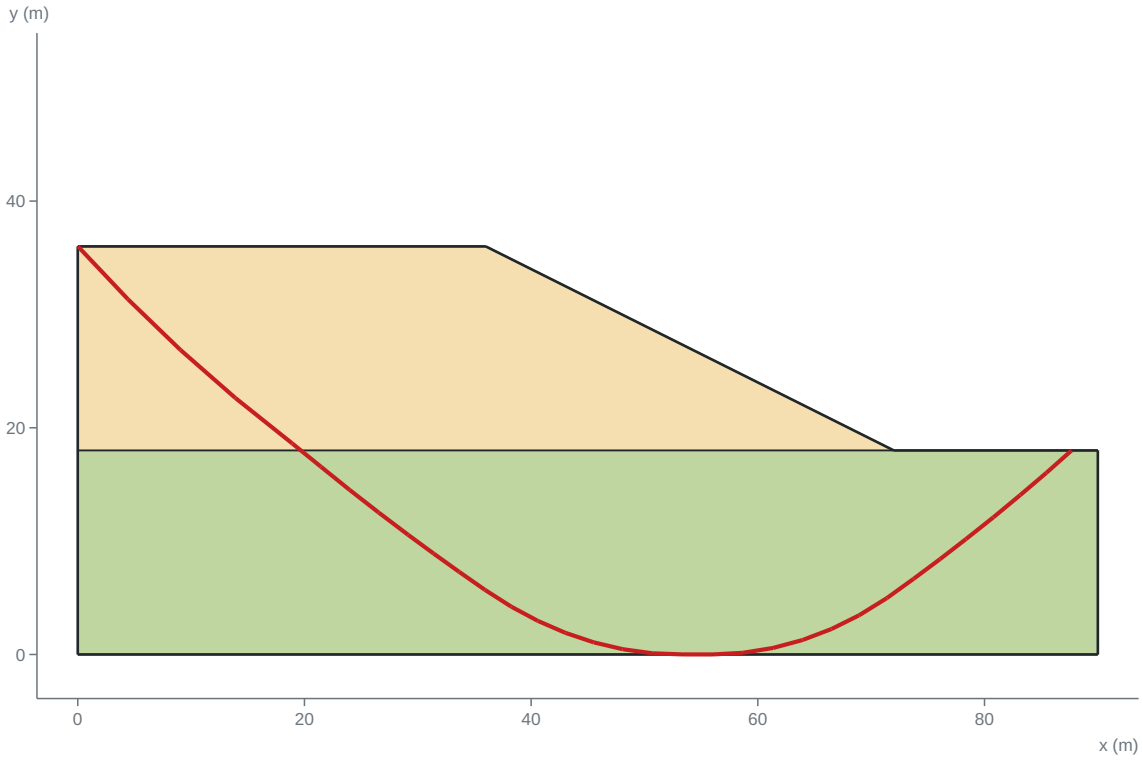



*Figure 97. V059b: critical surface (Spencer, non-circular, FOS = 1.289).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Undrained (φ = 0) | 20.00 | Su = 60.00 | 0° |
| Foundation | Undrained (φ = 0) | 20.00 | Su = 90.00 | 0° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, non-circular | 1.290 | 1.289 |

*The paper gives its results in figures only; the comparison is with the other programs.*

## 3.98 V059c — Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.6

*Guo & Griffiths (2020), Can. Geotech. J. 57.*

**Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)**

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 96 kPa), γ = 20, firm base; optimised non-circular search. The shallow mechanism governs.

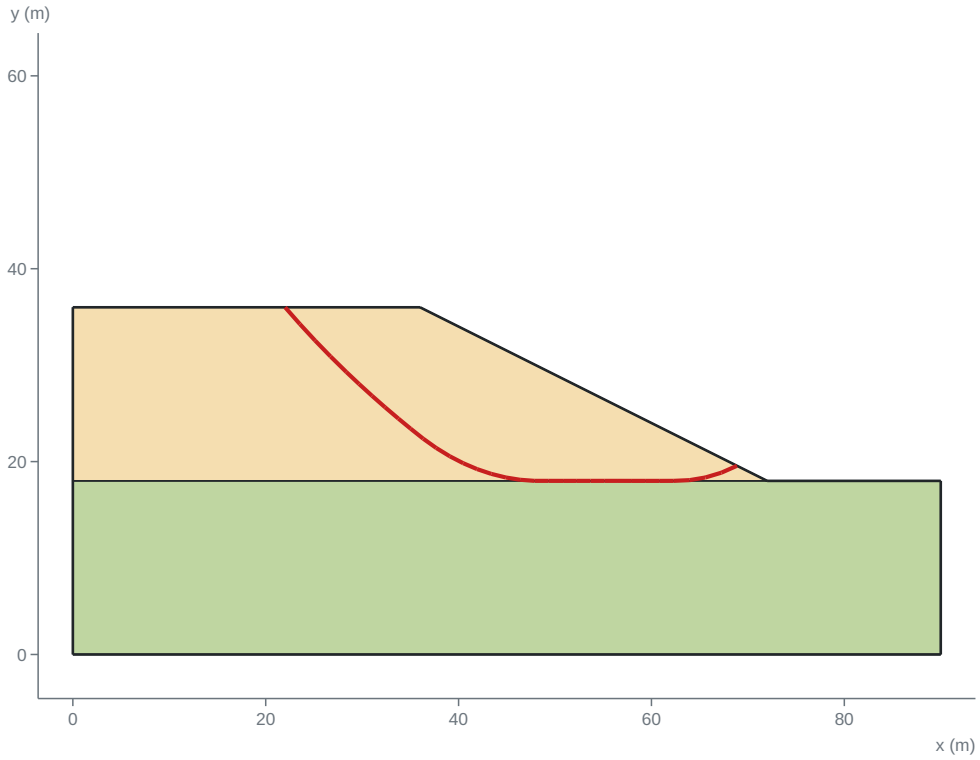



*Figure 98. V059c: critical surface (Spencer, non-circular, FOS = 1.312).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Undrained (φ = 0) | 20.00 | Su = 60.00 | 0° |
| Foundation | Undrained (φ = 0) | 20.00 | Su = 96.00 | 0° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, non-circular | 1.315 | 1.312 |

*The paper gives its results in figures only; the comparison is with the other programs.*

## 3.99 V060a — Cannon Dam end of construction (Hassan & Wolff), critical circle

*Hassan & Wolff (1999), J. Geotech. Geoenviron. Eng. 125(4).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Two clay fills with a sand-filter strip under the crest and a spoil-covered downstream toe; Bishop search for the minimum factor of safety at the mean strengths. The reliability index on the minimum-β surface is V060b.

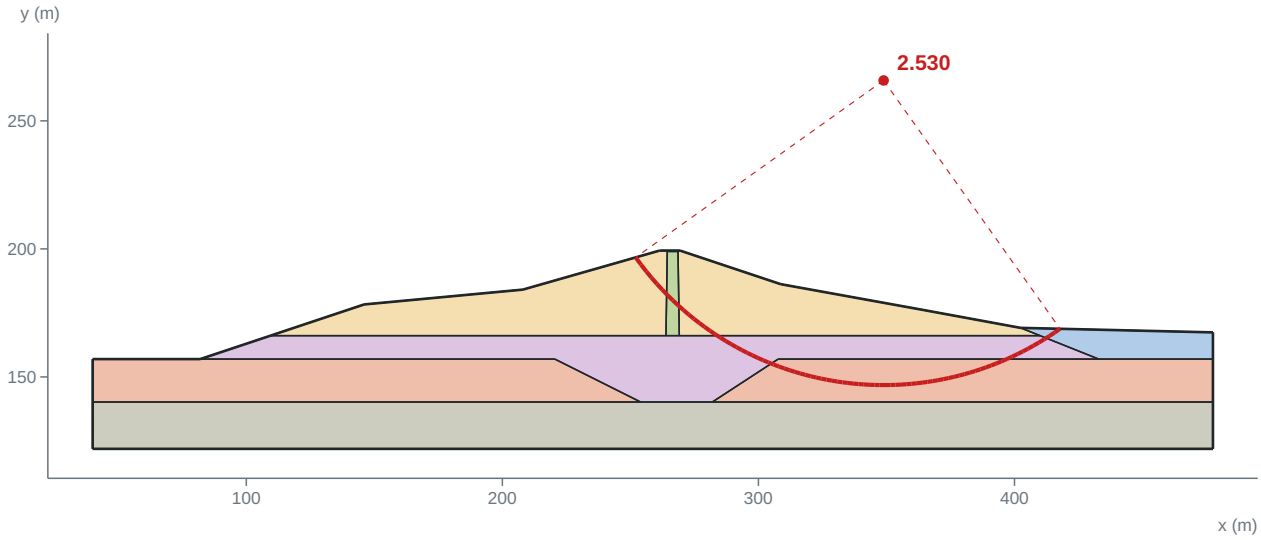



*Figure 99. V060a: critical circle (Bishop simplified, FOS = 2.530).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II clay fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Phase I clay fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 23.00 | c' = 1000.0 | 45.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.529 | 2.530 |

*Hassan & Wolff give 2.753 on a fixed circle (V062a), not a search result.*

## 3.100 V060b — Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β

*Hassan & Wolff (1999), J. Geotech. Geoenviron. Eng. 125(4).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V060a dam on the circle of minimum reliability index: Phase I fill φ = 8.5° ± 8.5°, c = 117.8 ± 58.9 kPa; Phase II fill c = 143.6 ± 79 kPa, φ = 15° ± 9°, treated as independent normal variables (another program's Taylor series has no correlation term); Bishop.

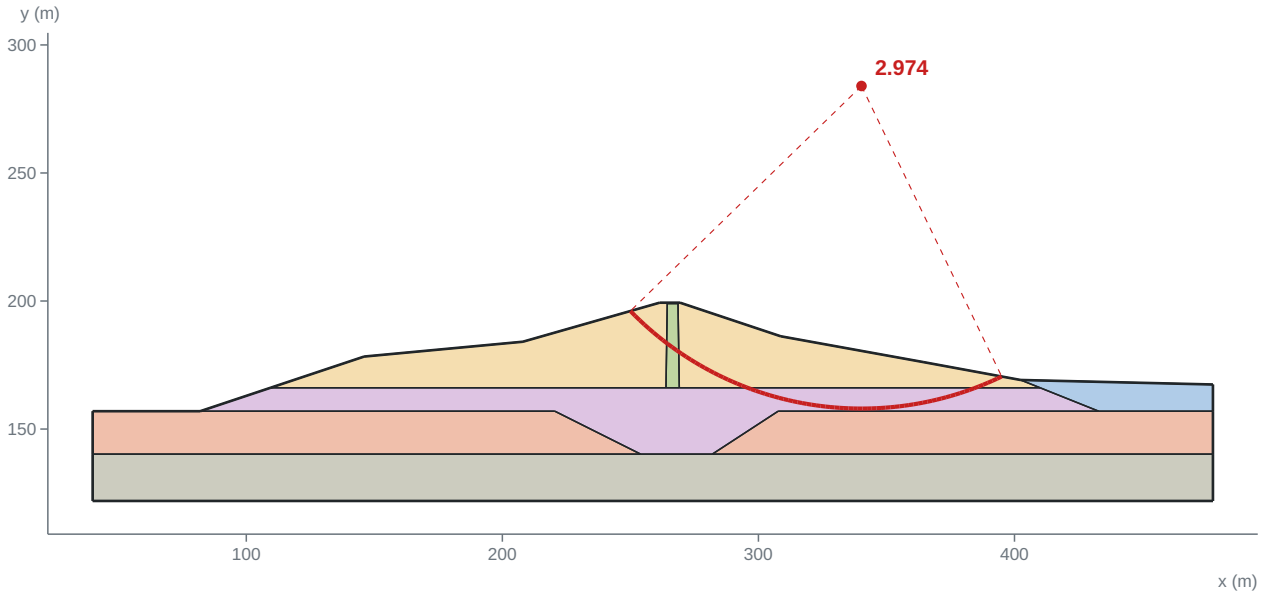



*Figure 100. V060b: analysed circle (Bishop simplified, FOS = 2.974).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II clay fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Phase I clay fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 23.00 | c' = 1000.0 | 45.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.973 | 2.974 |

*The source gives reliability indices on fixed circles; no factor of safety is published for the minimum-β circle.*

Probabilistic analysis on the same surface: 4 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Taylor series) | – | 3.353 | 3.354 |

## 3.101 V061 — Clarence Cannon Dam (Wolff & Harr), specified surface

*Wolff & Harr (1987).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Phase I and II fills with a chimney drain, a horizontal drain blanket (el. 548.2–552.8 ft) and a cutoff trench to rock, flat water table at el. 557 ft; the printed non-circular surface. Converted to SI.

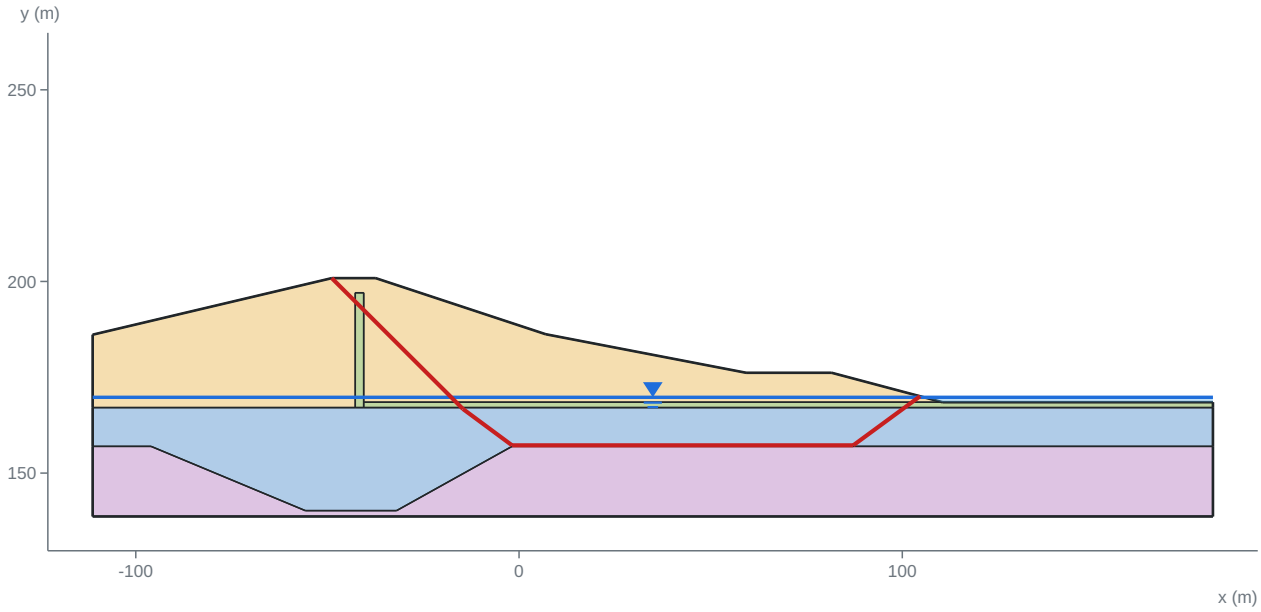



*Figure 101. V061: analysed surface (Morgenstern-Price (half-sine), FOS = 2.348).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 23.56 | c' = 138.9 | 14.80° |
| Sand drain | Mohr-Coulomb | 18.85 | c' = 0.00 | 30.00° |
| Phase I fill | Mohr-Coulomb | 23.56 | c' = 106.8 | 6.34° |
| Foundation sand | Mohr-Coulomb | 18.85 | c' = 0.00 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.36 | Morgenstern-Price (half-sine) | Wolff & Harr | 2.348 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 2.343 | 2.348 |
| Spencer | 2.383 | 2.388 |

## 3.102 V062a — Clarence Cannon Dam, surface 7A

*Hassan & Wolff (1999).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Six-zone embankment dam on sand over limestone, analysed on the published circle 7A (centre 320.175, 313.336; tangent elevation 145.347), Bishop.

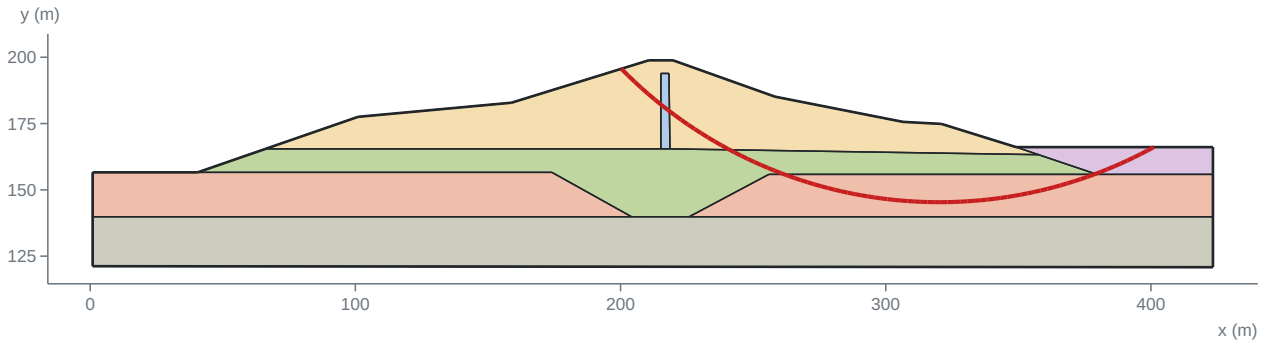



*Figure 102. V062a: analysed circle (Bishop simplified, FOS = 2.548).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Phase I fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 20.00 | c' = 100.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.753 | not stated | Hassan & Wolff (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.551–2.560 | 2.548 |

*Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.*

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Monte Carlo) | – | 10.953 | 11.034 |

## 3.103 V062b — Clarence Cannon Dam, surface 7B

*Hassan & Wolff (1999).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Six-zone embankment dam on sand over limestone, analysed on the published circle 7B (centre 295.235, 277.589; tangent elevation 156.298), Bishop.

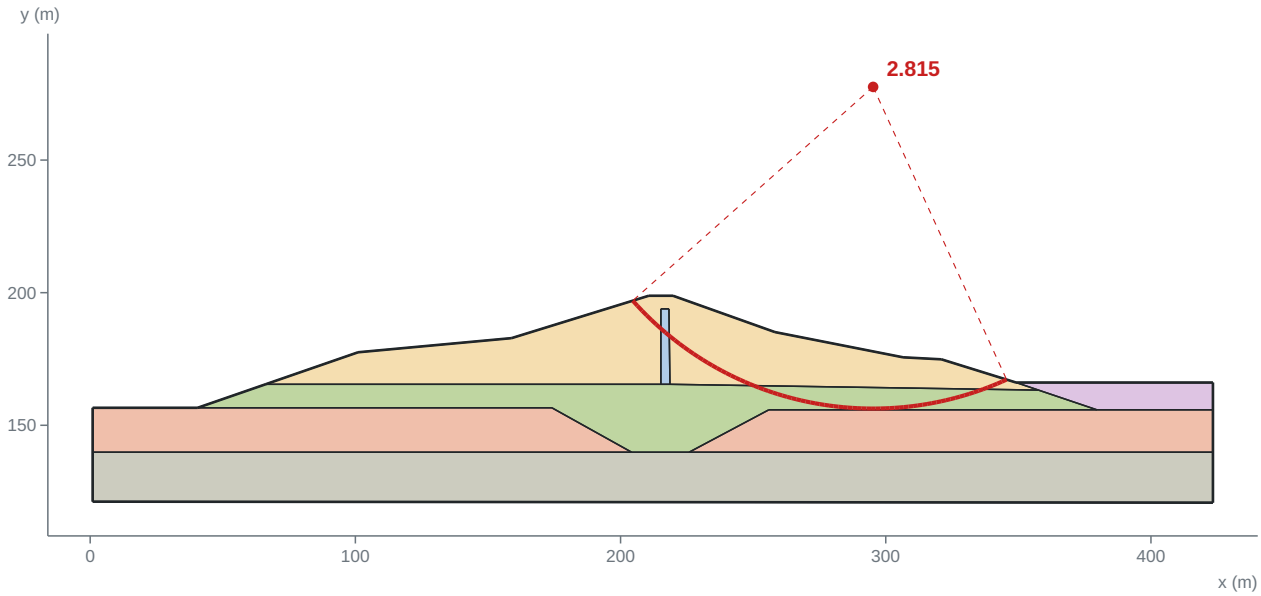



*Figure 103. V062b: analysed circle (Bishop simplified, FOS = 2.815).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Phase I fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 20.00 | c' = 100.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.352 | not stated | Hassan & Wolff (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.806–2.820 | 2.815 |

*Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.*

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Monte Carlo) | – | 4.351 | 4.402 |

## 3.104 V062c — Clarence Cannon Dam, surface 7C

*Hassan & Wolff (1999).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Six-zone embankment dam on sand over limestone, analysed on the published circle 7C (centre 303.683, 310.206; tangent elevation 155.928), Bishop.

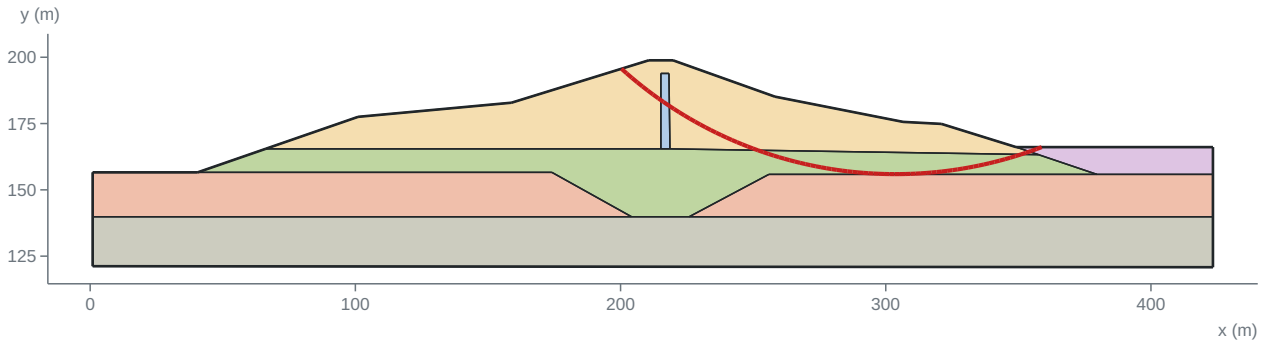



*Figure 104. V062c: analysed circle (Bishop simplified, FOS = 2.779).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Phase I fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 20.00 | c' = 100.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.523 | not stated | Hassan & Wolff (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.771–2.777 | 2.779 |

*Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.*

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Monte Carlo) | – | 4.263 | 4.325 |

## 3.105 V062d — Clarence Cannon Dam, surface 7D

*Hassan & Wolff (1999).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Six-zone embankment dam on sand over limestone, analysed on the published circle 7D (centre 322.152, 310.206; tangent elevation 145.36), Bishop.

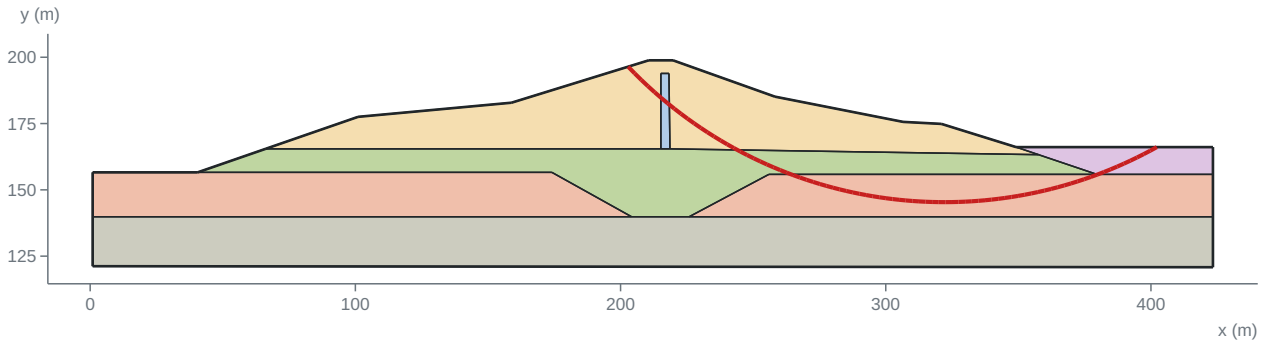



*Figure 105. V062d: analysed circle (Bishop simplified, FOS = 2.577).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Phase I fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 20.00 | c' = 100.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.457 | not stated | Hassan & Wolff (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.583–2.589 | 2.577 |

*Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.*

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Monte Carlo) | – | 11.092 | 11.076 |

## 3.106 V062e — Clarence Cannon Dam, surface 7E

*Hassan & Wolff (1999).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Six-zone embankment dam on sand over limestone, analysed on the published circle 7E (centre 327.693, 324.018; tangent elevation 147.163), Bishop.

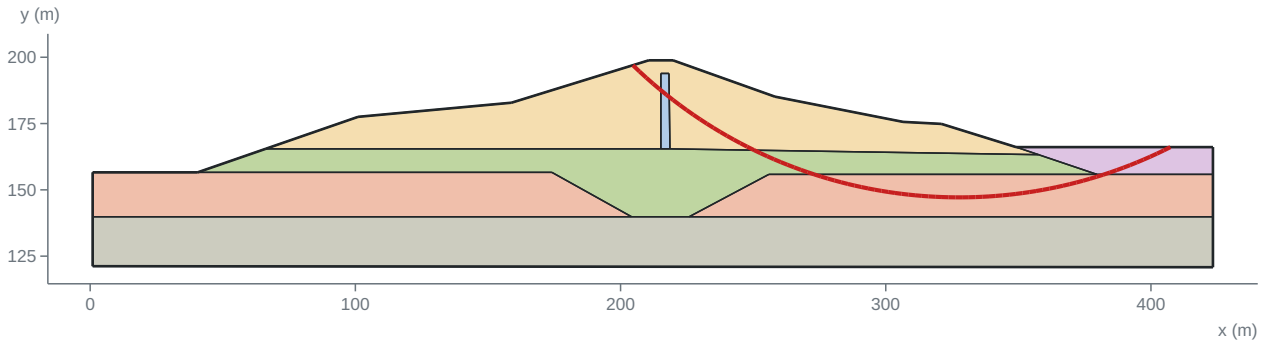



*Figure 106. V062e: analysed circle (Bishop simplified, FOS = 2.690).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Phase I fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 20.00 | c' = 100.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.602 | not stated | Hassan & Wolff (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.692–2.703 | 2.690 |

*Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.*

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Monte Carlo) | – | 10.281 | 10.329 |

## 3.107 V062f — Clarence Cannon Dam, surface 8B

*Hassan & Wolff (1999).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Six-zone embankment dam on sand over limestone, analysed on the published circle 8B (centre 304.239, 307.858; tangent elevation 155.311), Bishop.

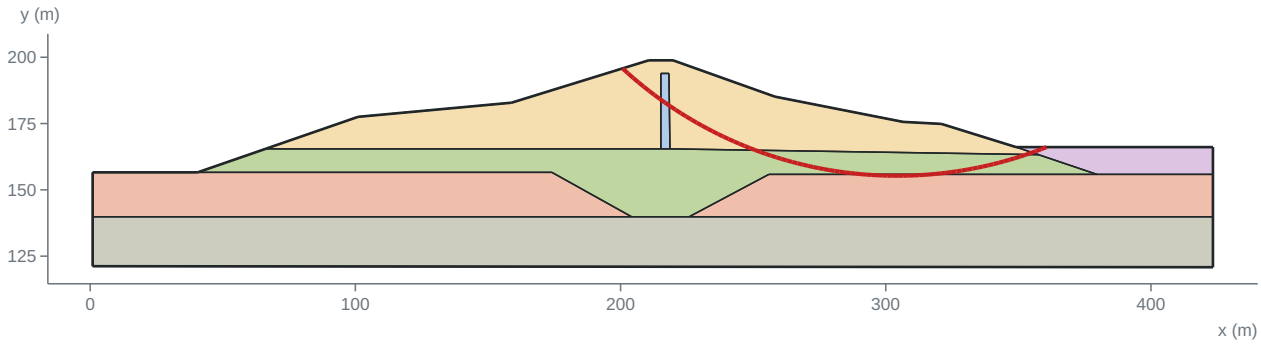



*Figure 107. V062f: analysed circle (Bishop simplified, FOS = 2.673).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Phase I fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 20.00 | c' = 100.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.995 | not stated | Hassan & Wolff (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.672–2.673 | 2.673 |

*Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.*

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Monte Carlo) | – | 4.858 | 4.933 |

## 3.108 V062g — Clarence Cannon Dam, surface 8F

*Hassan & Wolff (1999).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Six-zone embankment dam on sand over limestone, analysed on the published circle 8F (centre 313.114, 304.784; tangent elevation 160.368), Bishop.

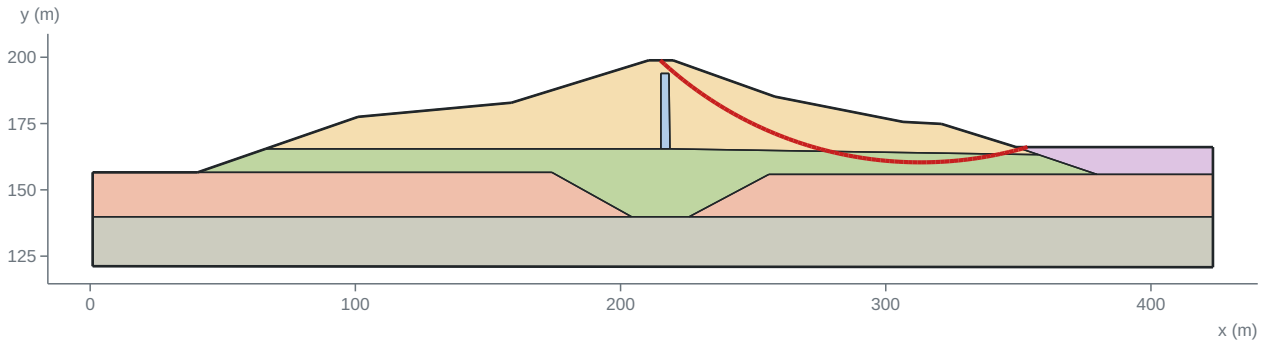



*Figure 108. V062g: analysed circle (Bishop simplified, FOS = 3.592).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Phase I fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 20.00 | c' = 100.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 3.916 | not stated | Hassan & Wolff (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 3.586–3.598 | 3.592 |

*Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.*

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Monte Carlo) | – | 5.485 | 5.528 |

## 3.109 V062h — Clarence Cannon Dam, surface 8G

*Hassan & Wolff (1999).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Six-zone embankment dam on sand over limestone, analysed on the published circle 8G (centre 314.856, 305.604; tangent elevation 165.756), Bishop.

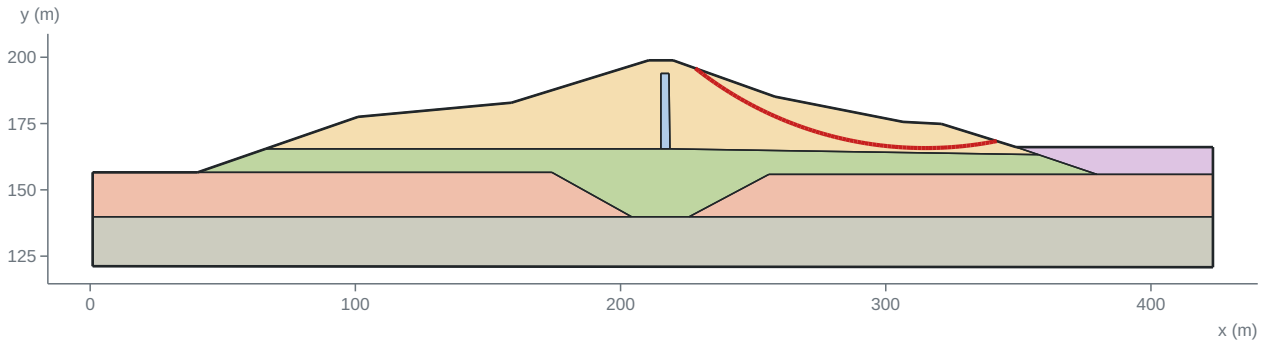



*Figure 109. V062h: analysed circle (Bishop simplified, FOS = 6.063).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Phase I fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 20.00 | c' = 100.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 10.576 | not stated | Hassan & Wolff (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 6.069–6.074 | 6.063 |

*Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.*

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Monte Carlo) | – | 5.563 | 5.490 |

## 3.110 V062i — Clarence Cannon Dam, surface 8H

*Hassan & Wolff (1999).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Six-zone embankment dam on sand over limestone, analysed on the published circle 8H (centre 316.198, 305.604; tangent elevation 170.537), Bishop.

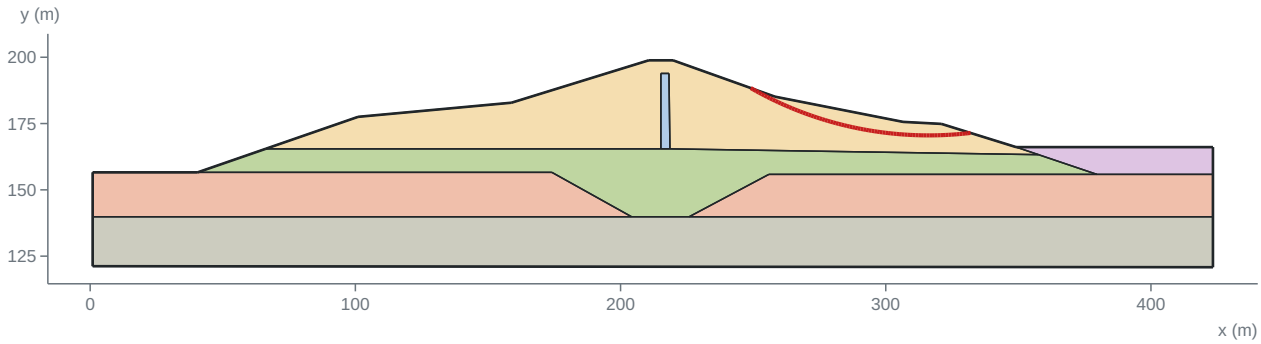



*Figure 110. V062i: analysed circle (Bishop simplified, FOS = 11.567).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Phase II fill | Mohr-Coulomb | 22.00 | c' = 143.6 | 15.00° |
| Phase I fill | Mohr-Coulomb | 22.00 | c' = 117.8 | 8.50° |
| Sand filter | Mohr-Coulomb | 22.00 | c' = 0.00 | 35.00° |
| Spoil fill | Mohr-Coulomb | 25.00 | c' = 5.00 | 35.00° |
| Foundation sand | Mohr-Coulomb | 20.00 | c' = 5.00 | 18.00° |
| Limestone | Mohr-Coulomb | 20.00 | c' = 100.0 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 6.293 | not stated | Hassan & Wolff (1999) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 11.583 | 11.567 |

*Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.*

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

| Quantity | Published | Other programs | cSlope |
|---|---|---|---|
| β lognormal (Monte Carlo) | – | 6.394 | 6.431 |

## 3.111 V063 — Huang & Jia earth dam, dry

*Huang, M. & Jia, C.-Q. (2009). Strength reduction FEM in stability analysis of soil slopes subjected to transient unsaturated seepage. Computers and Geotechnics 36(1–2), 93–101.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Homogeneous earth dam (c′ = 13.8 kPa, φ′ = 37°, γ = 18.2 kN/m³) before the reservoir is filled; circular search.

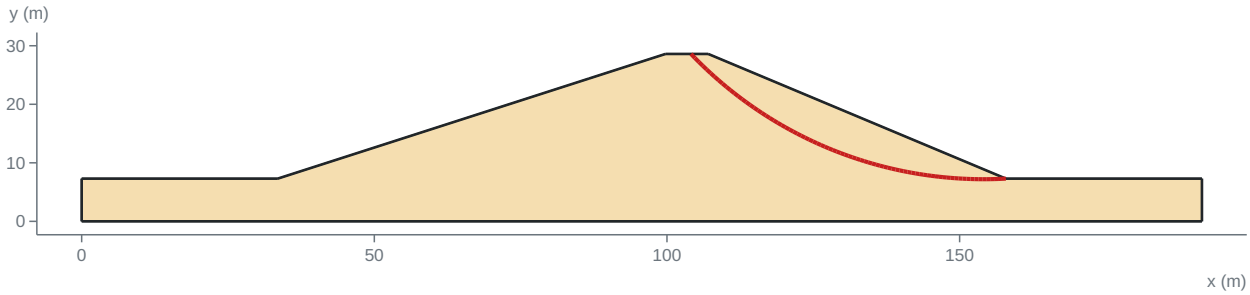



*Figure 111. V063: critical circle (Spencer, FOS = 2.451).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Material 1 | Mohr-Coulomb | 18.20 | c' = 13.80 | 37.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.43 | Bishop simplified | Huang & Jia | 2.453 |
| 2.50 | finite-element strength reduction | Huang & Jia (2009) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 2.455 | 2.451 |
| Bishop simplified | – | 2.453 |

*Huang & Jia's limit-equilibrium value 2.43 is by Bishop's method (their p. 95 and fig. 3a); their strength-reduction finite-element value is 2.50.*

## 3.112 V064 — Leshchinsky & Han MSE wall, five tiers

*Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Five 1.8 m tiers (Ta = 10.1 kN/m).

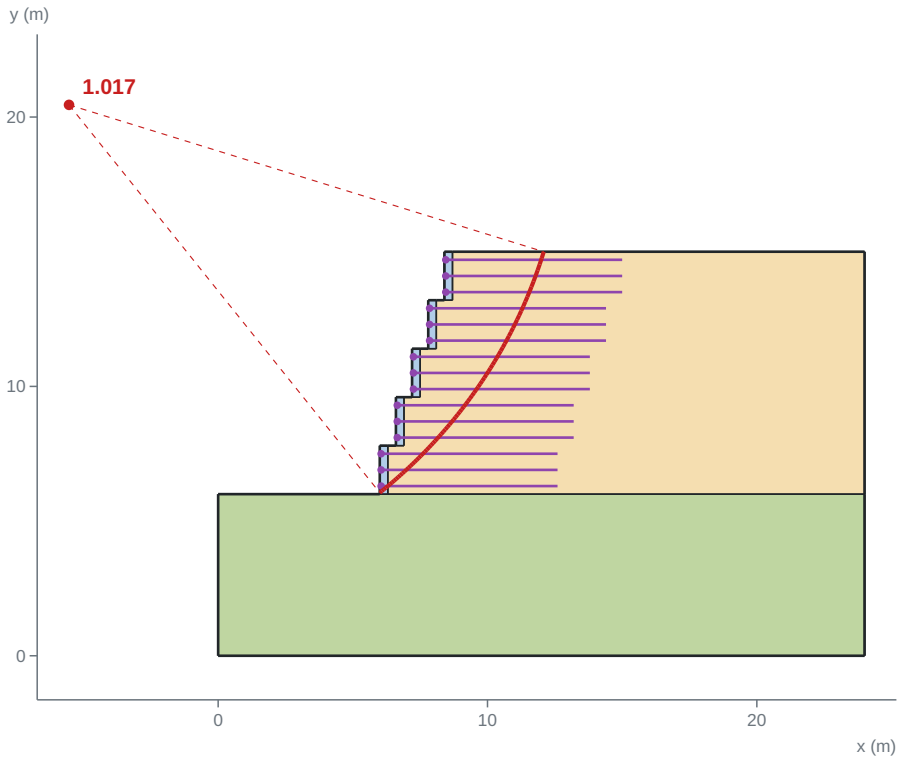



*Figure 112. V064: analysed circle (Bishop simplified, FOS = 1.017).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Reinforced and retained fill | Mohr-Coulomb | 18.00 | c' = 0.00 | 34.00° |
| Foundation soil | Mohr-Coulomb | 18.00 | c' = 10.00 | 34.00° |
| Blocks | Mohr-Coulomb | 18.00 | c' = 2.50 | 34.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | Bishop simplified, circular | Leshchinsky & Han (2004) | 1.017 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.020 | 1.017 |

*Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.*

## 3.113 V065 — Leshchinsky & Han MSE wall, weak foundation

*Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V066a on a c = 0, φ = 18° foundation.

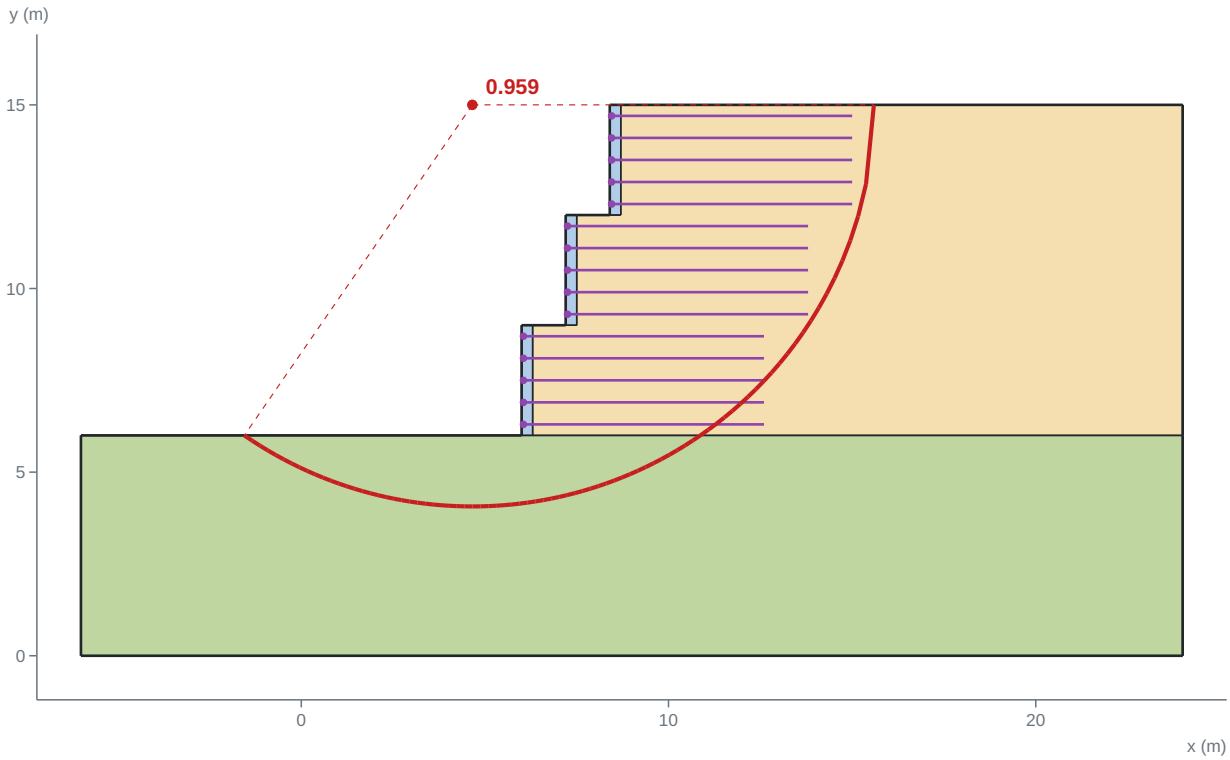



*Figure 113. V065: analysed circle (Spencer, FOS = 0.959).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Reinforced and retained fill | Mohr-Coulomb | 18.00 | c' = 0.00 | 34.00° |
| Foundation soil | Mohr-Coulomb | 18.00 | c' = 0.00 | 18.00° |
| Blocks | Mohr-Coulomb | 18.00 | c' = 2.50 | 34.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | Bishop simplified, circular | Leshchinsky & Han (2004) | 0.977 |
| 0.86 | finite differences | Leshchinsky & Han (2004) | – |
| 0.98 | Bishop simplified, deep circle (read from fig. 7a) | Leshchinsky & Han (2004) | 0.977 |
| 0.92 | Spencer, three-part wedge | Leshchinsky & Han (2004) | 0.959 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 0.964 | 0.959 |
| Bishop simplified | – | 0.977 |

*Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.*

## 3.114 V066a — Leshchinsky & Han three-tier MSE wall, baseline

*Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Three 3 m tiers offset 1.2 m, block facing, geotextiles every 0.6 m (L = 6.3 m, Ta = 10 kN/m, applied horizontally, passive), a commercial program's critical circle.

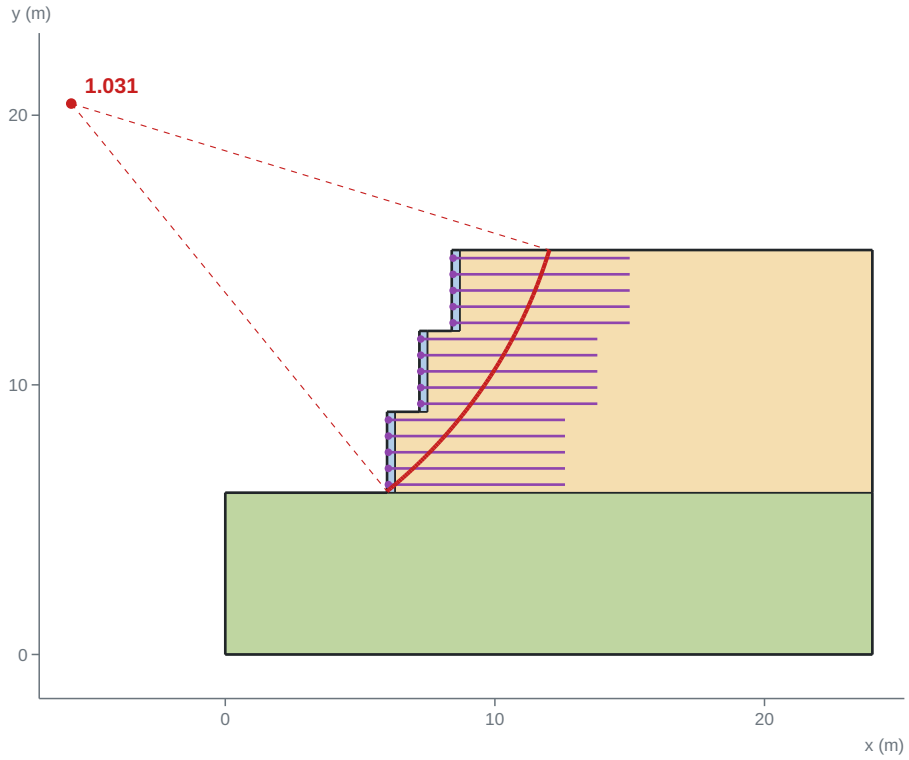



*Figure 114. V066a: analysed circle (Bishop simplified, FOS = 1.031).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Reinforced and retained fill | Mohr-Coulomb | 18.00 | c' = 0.00 | 34.00° |
| Foundation soil | Mohr-Coulomb | 18.00 | c' = 10.00 | 34.00° |
| Blocks | Mohr-Coulomb | 18.00 | c' = 2.50 | 34.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | Bishop simplified, circular | Leshchinsky & Han (2004) | 1.031 |
| 0.99 | finite differences | Leshchinsky & Han (2004) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.040 | 1.031 |

*Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.*

## 3.115 V066b — Leshchinsky & Han MSE wall, L = 4.2 m

*Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V066a with 4.2 m geotextiles (Ta = 11.4 kN/m).

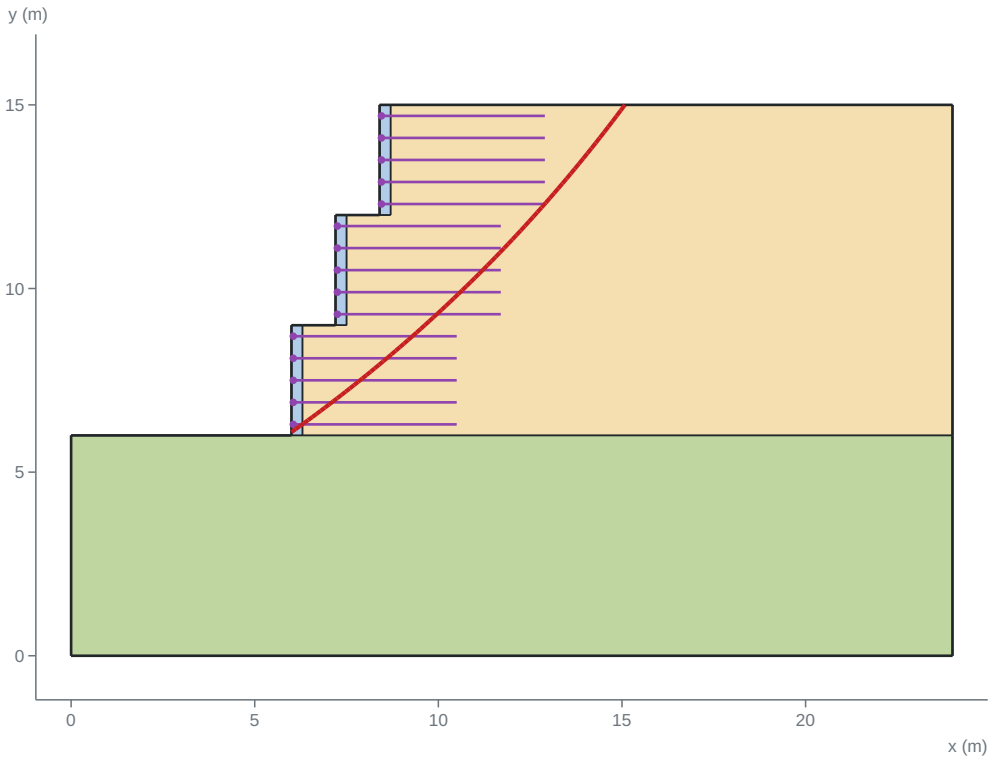



*Figure 115. V066b: analysed circle (Spencer, FOS = 1.011).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Reinforced and retained fill | Mohr-Coulomb | 18.00 | c' = 0.00 | 34.00° |
| Foundation soil | Mohr-Coulomb | 18.00 | c' = 10.00 | 34.00° |
| Blocks | Mohr-Coulomb | 18.00 | c' = 2.50 | 34.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | Bishop simplified, circular | Leshchinsky & Han (2004) | 1.013 |
| 0.98 | finite differences | Leshchinsky & Han (2004) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.011 | 1.011 |
| Bishop simplified | – | 1.013 |

*Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.*

## 3.116 V066c — Leshchinsky & Han MSE wall, two reinforcement types

*Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V066a with two reinforcement strengths (7.5 and 11.0 kN/m).

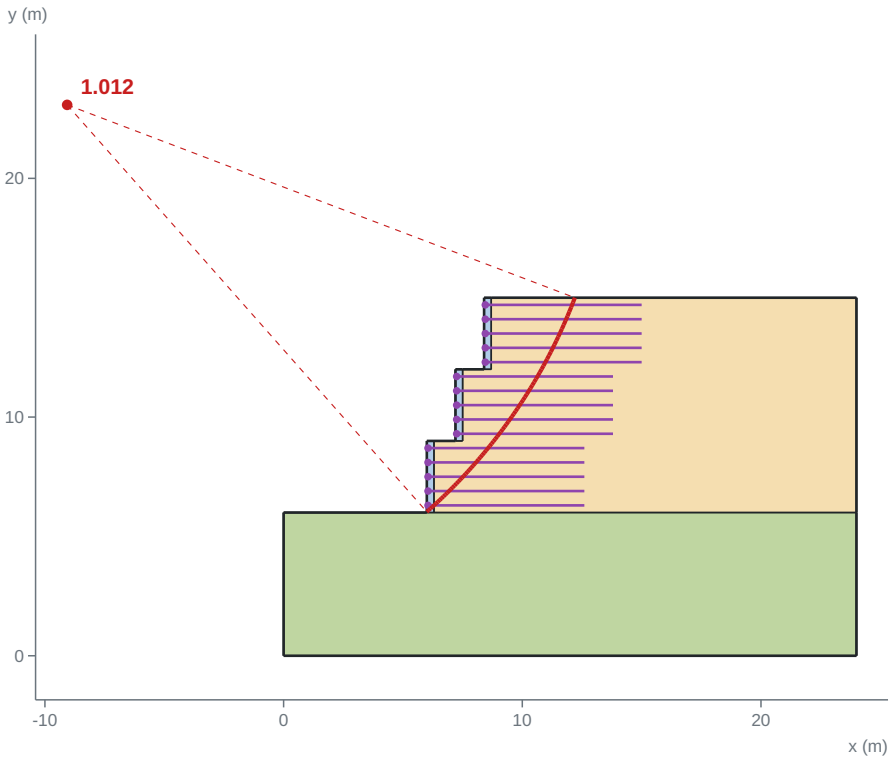



*Figure 116. V066c: analysed circle (Bishop simplified, FOS = 1.012).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Reinforced and retained fill | Mohr-Coulomb | 18.00 | c' = 0.00 | 34.00° |
| Foundation soil | Mohr-Coulomb | 18.00 | c' = 10.00 | 34.00° |
| Blocks | Mohr-Coulomb | 18.00 | c' = 2.50 | 34.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | Bishop simplified | Leshchinsky & Han | 1.012 |
| 1.01 | finite differences | Leshchinsky & Han (2004) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | – | 1.012 |

*Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.*

## 3.117 V066d — Leshchinsky & Han MSE wall, water hw = 3 m

*Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V066a with the water table 3 m above the toe: pore pressure in the foundation only (drained fill); Ta = 9.25 kN/m. The pond in front of the wall is left out of this comparison: another program's water loads apply no pressure to the vertical facing, whereas cSlope does.

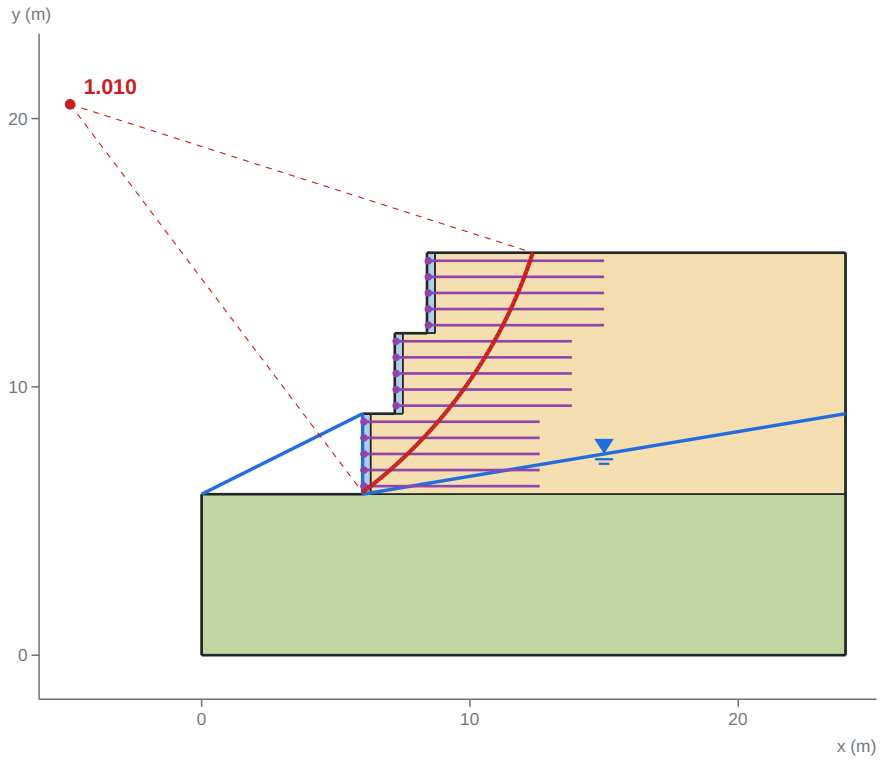



*Figure 117. V066d: analysed circle (Bishop simplified, FOS = 1.010).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Reinforced and retained fill | Mohr-Coulomb | 18.00 | c' = 0.00 | 34.00° |
| Foundation soil | Mohr-Coulomb | 18.00 | c' = 10.00 | 34.00° |
| Blocks | Mohr-Coulomb | 18.00 | c' = 2.50 | 34.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | Bishop simplified, circular | Leshchinsky & Han (2004) | 1.010 |
| 1.01 | finite differences | Leshchinsky & Han (2004) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.010 | 1.010 |

*Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.*

## 3.118 V066e — Leshchinsky & Han MSE wall, surcharge

*Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V066a (Ta = 10 kN/m) with a 20 kPa surcharge on the top tier (x = 8.7 to 24).

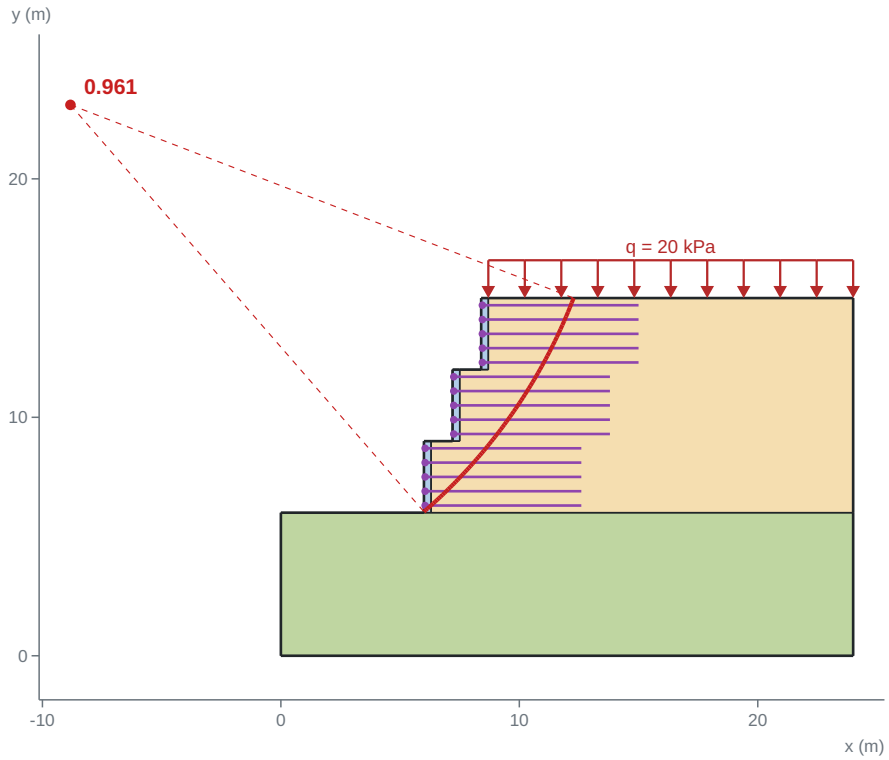



*Figure 118. V066e: analysed circle (Bishop simplified, FOS = 0.961).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Reinforced and retained fill | Mohr-Coulomb | 18.00 | c' = 0.00 | 34.00° |
| Foundation soil | Mohr-Coulomb | 18.00 | c' = 10.00 | 34.00° |
| Blocks | Mohr-Coulomb | 18.00 | c' = 2.50 | 34.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | Bishop simplified, circular | Leshchinsky & Han (2004) | 0.961 |
| 1.02 | finite differences | Leshchinsky & Han (2004) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 0.958 | 0.961 |

*Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value. Here the reinforcement keeps the baseline strength of 10 kN/m, as in the program compared with, not the 11.6 kN/m designed for the surcharge, so the published 1.00 is not directly comparable.*

## 3.119 V067 — Low (1989), layered undrained clay

*Low (1989).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Embankment on three undrained clay layers of different strength; critical circle search.

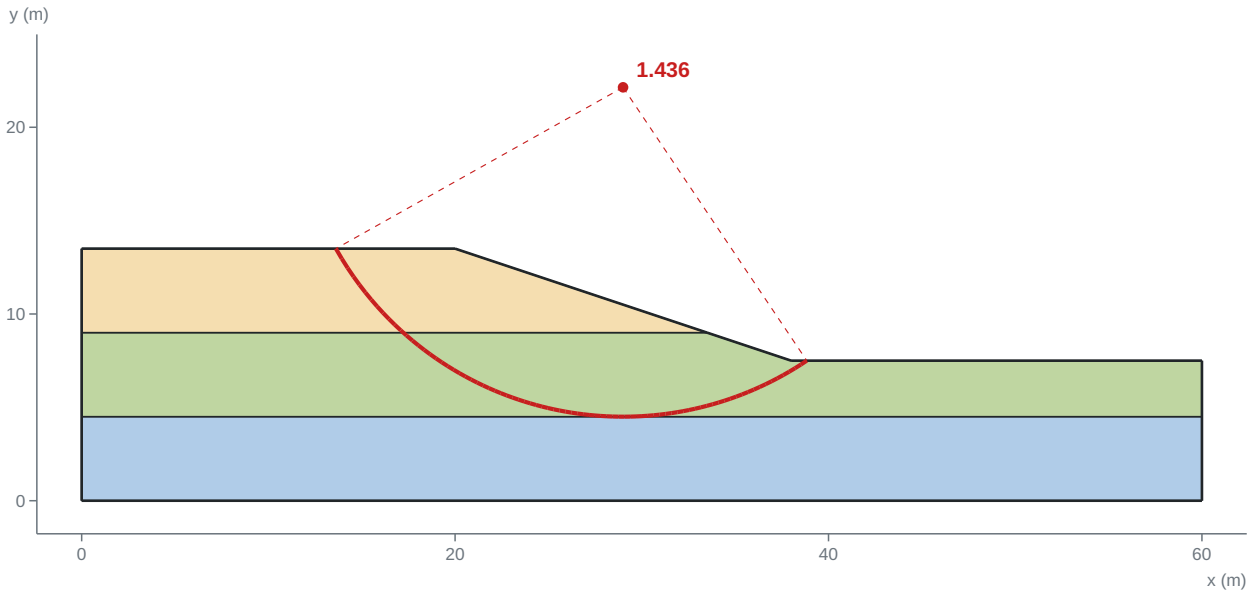



*Figure 119. V067: critical circle (Bishop simplified, FOS = 1.436).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| A | Undrained (φ = 0) | 18.00 | Su = 30.00 | 0° |
| B | Undrained (φ = 0) | 18.00 | Su = 20.00 | 0° |
| C | Undrained (φ = 0) | 18.00 | Su = 150.0 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.44 | Ordinary | Low (1989) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.439 | 1.436 |

## 3.120 V068 — Low (1989), stiff crust over soft clays, published circle

*Low, B.K. (1989). Stability analysis of embankments on soft ground. J. Geotech. Eng. 115(2), 211–227; example 2, fig. 11.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Stiff upper slope on two undrained clays, the lowest with Su increasing linearly with depth (S = f(datum)), on the critical circle a commercial program prints with its Bishop value (centre (18.001, 16.000), R 15.556).

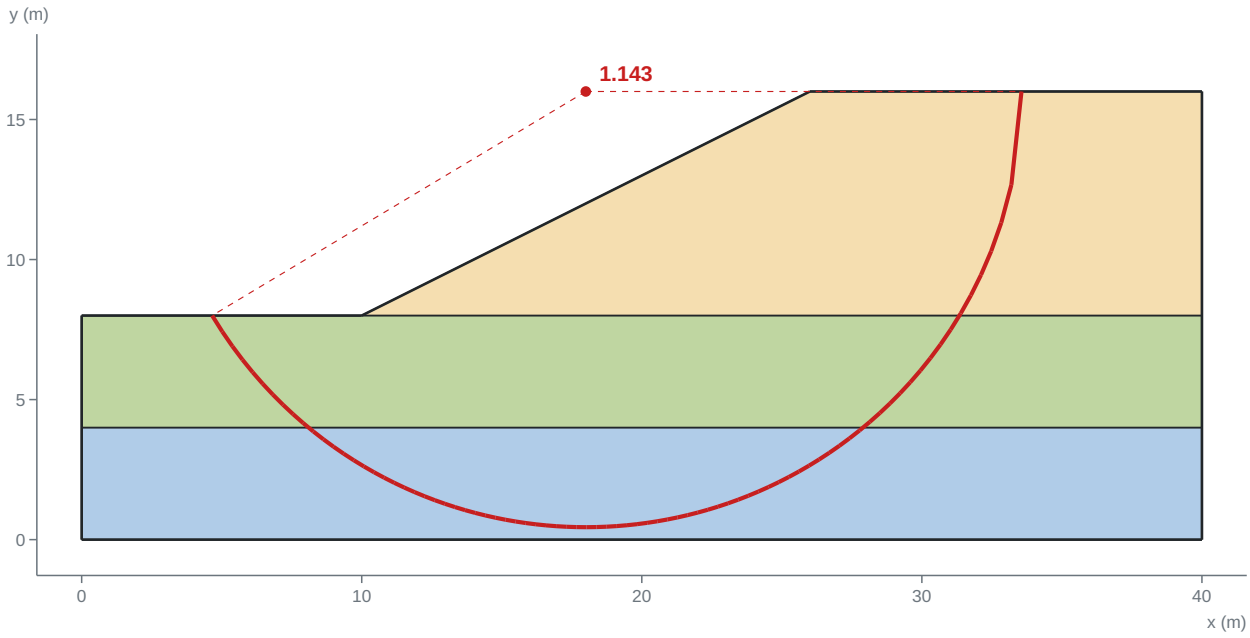



*Figure 120. V068: analysed circle (Bishop simplified, FOS = 1.143).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Upper | Mohr-Coulomb | 20.00 | c' = 95.00 | 15.00° |
| Middle | Mohr-Coulomb | 20.00 | c' = 15.00 | 0.00° |
| Lower | S = f(datum) | 20.00 | Su = 15.00 at y = 4.00, 3.75 kPa/m below | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.14 | Bishop simplified | Low | 1.143 |
| 1.36 | Ordinary | Low (1989) | – |
| 1.31 | Ordinary and Bishop with φ = 0 in the bank | Low (1989) | – |
| 1.38 | hand calculation (Low's method) | Low (1989) | – |
| 1.17 | Spencer, non-circular (random search) | Kim et al. (2002) | – |
| 1.13–1.27 | limit analysis, lower and upper bound | Kim et al. (2002) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | – | 1.143 |
| Spencer, non-circular | – | – |

*The comparison program's 1.19 is not reproduced on the circle it prints with that value: cSlope gives 1.13 there, close to Low's Bishop value 1.14, which is the check. Low himself calls that Bishop value misleading, an artefact of negative effective normal stresses in the stiff bank with φ = 15°: with φ = 0 in the bank both Ordinary and Bishop give 1.31, and his own hand method 1.38. Kim et al. (2002) give 1.17 with Spencer's method on a non-circular surface and limit-analysis bounds of 1.13–1.27.*

## 3.121 V069a — Morgenstern (1963) complete drawdown, 3:1

*Morgenstern, N. (1963). Stability charts for earth slopes during rapid drawdown. Géotechnique 13(2).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

3:1 slope after complete drawdown: piezometric line on the slope face, base at the toe (D = 1). Pore pressure from the vertical head. Converted to SI.

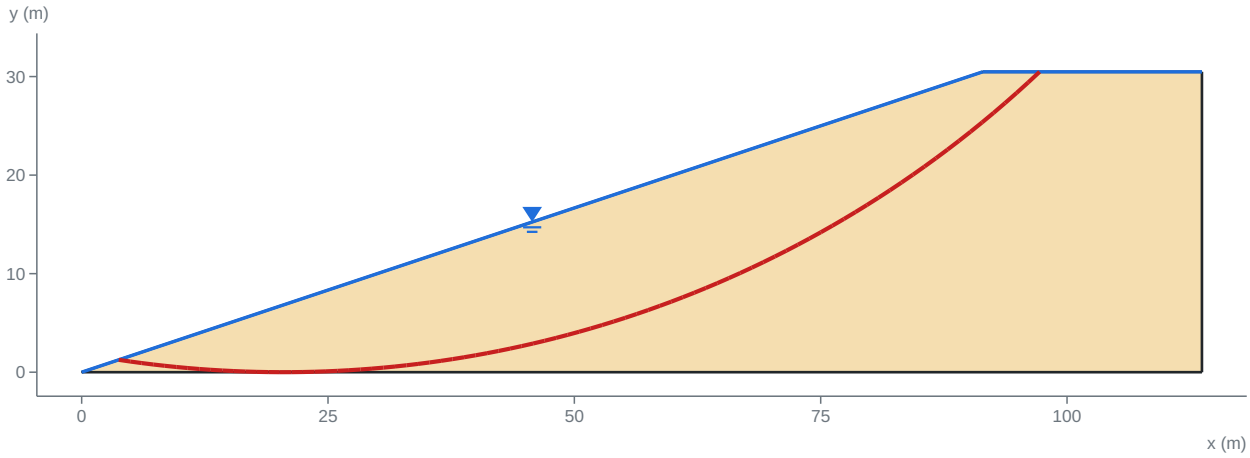



*Figure 121. V069a: critical circle (Bishop simplified, FOS = 1.200).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 19.60 | c' = 14.94 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.20 | stability chart | Morgenstern | 1.200 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | – | 1.200 |
| Spencer | – | 1.205 |

## 3.122 V069b — Morgenstern complete drawdown, the reference circle

*Morgenstern, N. (1963). Stability charts for earth slopes during rapid drawdown. Géotechnique 13(2). Verification manual of a commercial slope stability program.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V069a slope after complete drawdown on the reference critical circle (Fig. 100.2); pore pressure from the vertical head. Converted to SI.

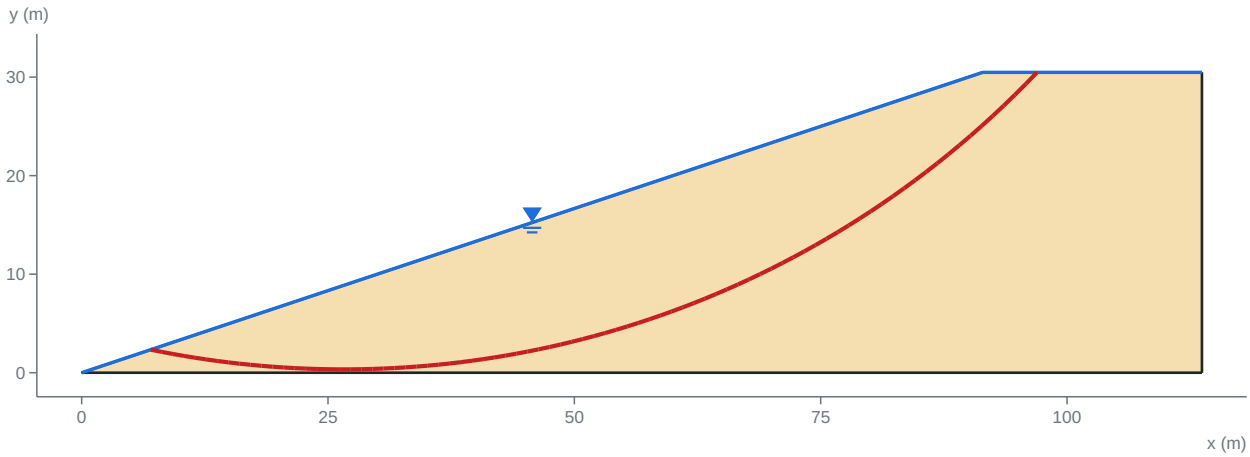



*Figure 122. V069b: analysed circle (Bishop simplified, FOS = 1.209).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 19.60 | c' = 14.94 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.20 | stability chart | Morgenstern (1963) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.212 | 1.209 |

*Morgenstern's 1.2 is read from a stability chart.*

## 3.123 V070 — Partial drawdown, $\bar{B} = 1$ (Morgenstern chart)

*Morgenstern (1963).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Homogeneous embankment after a partial drawdown from 100 to 50 ft with $\bar{B} = 1$: the piezometric line follows the face above the pool and stays at the pool below it, which loads the submerged face. Converted to SI.

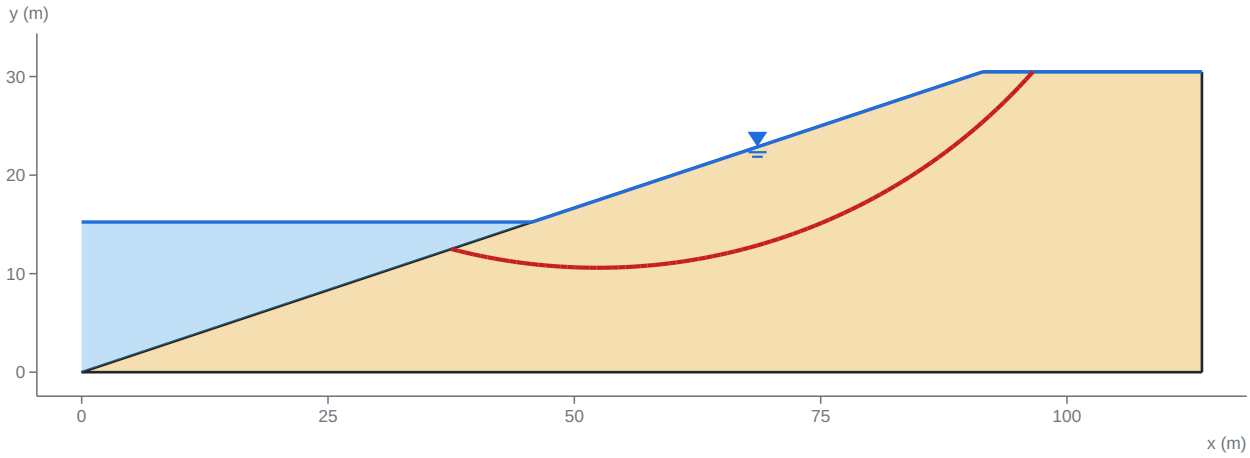



*Figure 123. V070: critical circle (Bishop simplified, FOS = 1.415).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Morgenstern soil | Mohr-Coulomb | 19.60 | c' = 14.94 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.41 | stability chart | Morgenstern | 1.415 |
| 1.52 | stability chart, circles tangent to the base | Morgenstern (1963) | – |
| 1.48 | stability chart, circles tangent to H/2 | Morgenstern (1963) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.417 | 1.415 |
| Spencer | 1.422 | 1.421 |

*Morgenstern's 1.41 is the lowest of three chart readings for circles tangent to the base, to H/2 and to H/4 (1.52, 1.48 and 1.41, the last interpolated), with his remark that a slightly lower value could perhaps be found; a free search may therefore come in below it.*

## 3.124 V071a — Pockoski & Duncan (2000) slope 1

*Pockoski & Duncan (2000), slope 1.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

2:1 slope with a water table, on the circle published by a commercial program; pore pressure from the vertical head below the piezometric line. Converted to SI.

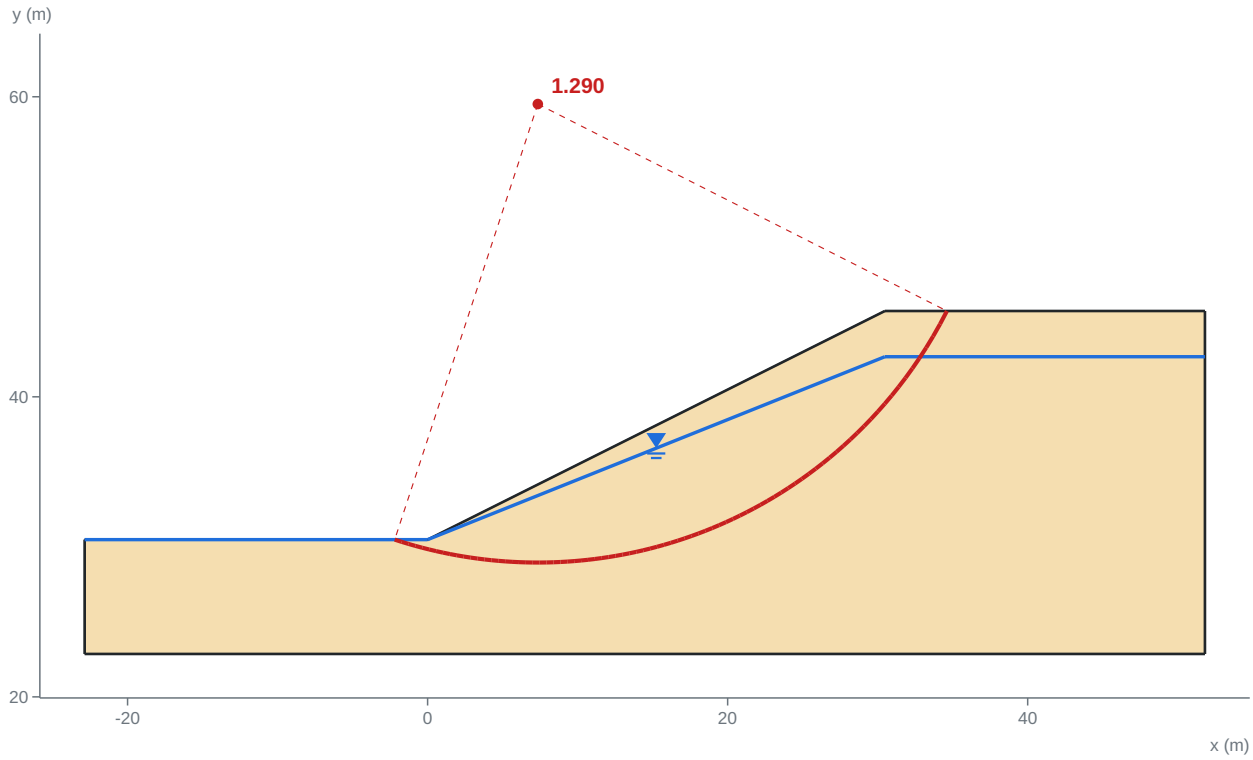



*Figure 124. V071a: analysed circle (Bishop simplified, FOS = 1.290).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sandy clay | Mohr-Coulomb | 18.85 | c' = 14.36 | 30.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.293 | 1.290 |
| Spencer | 1.300 | 1.296 |

*Pockoski & Duncan compare computer programs and publish no independent value.*

## 3.125 V071b — Pockoski & Duncan slope 1, circular search

*Pockoski & Duncan (2000), slope 1. Verification manual of a commercial slope stability program.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V071a slope (Pockoski & Duncan slope 1) with a circular search instead of the given circle; pore pressure from the vertical head. Converted to SI.

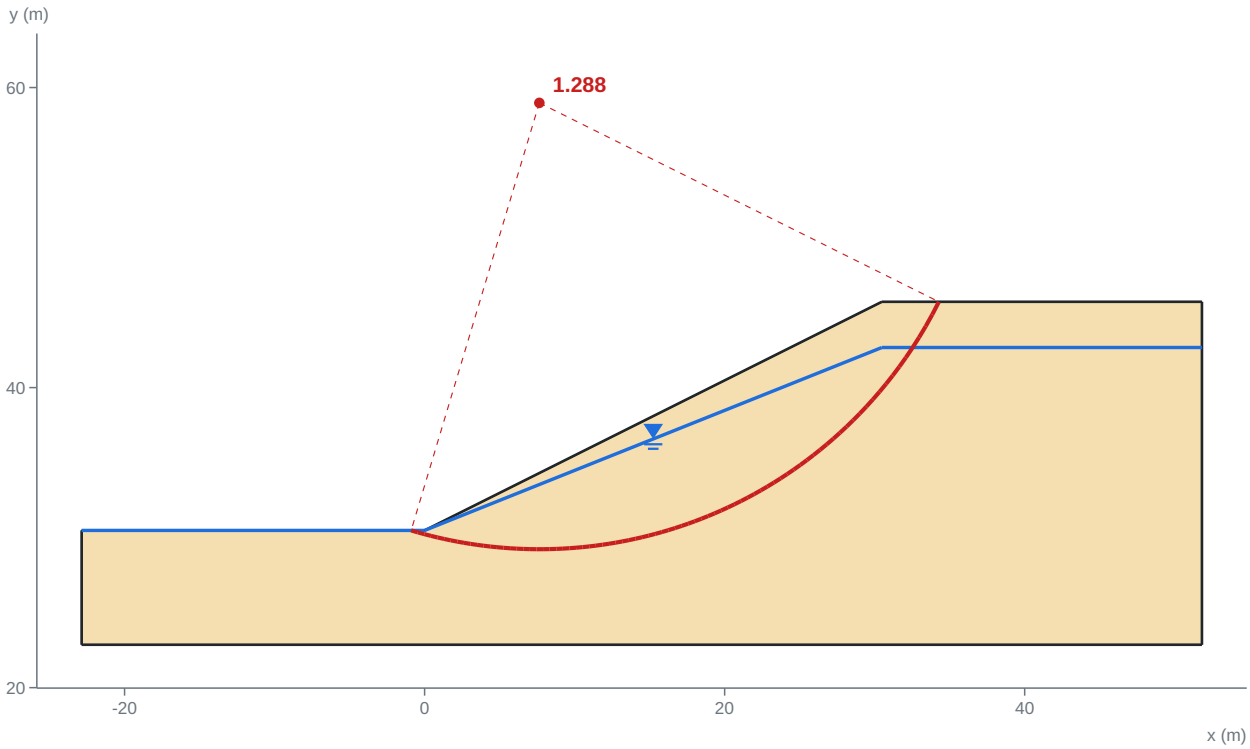



*Figure 125. V071b: critical circle (Bishop simplified, FOS = 1.288).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sandy clay | Mohr-Coulomb | 18.85 | c' = 14.36 | 30.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.293 | 1.288 |
| Spencer | 1.300 | 1.294 |
| Janbu simplified | 1.151 | 1.148 |

*Pockoski & Duncan compare computer programs and publish no independent value.*

## 3.126 V072a — Pockoski & Duncan test slope 2, dry tension crack

*Pockoski & Duncan (2000).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V071a slope with a dry 5.5 ft tension crack, on the reference critical circle. Converted to SI.

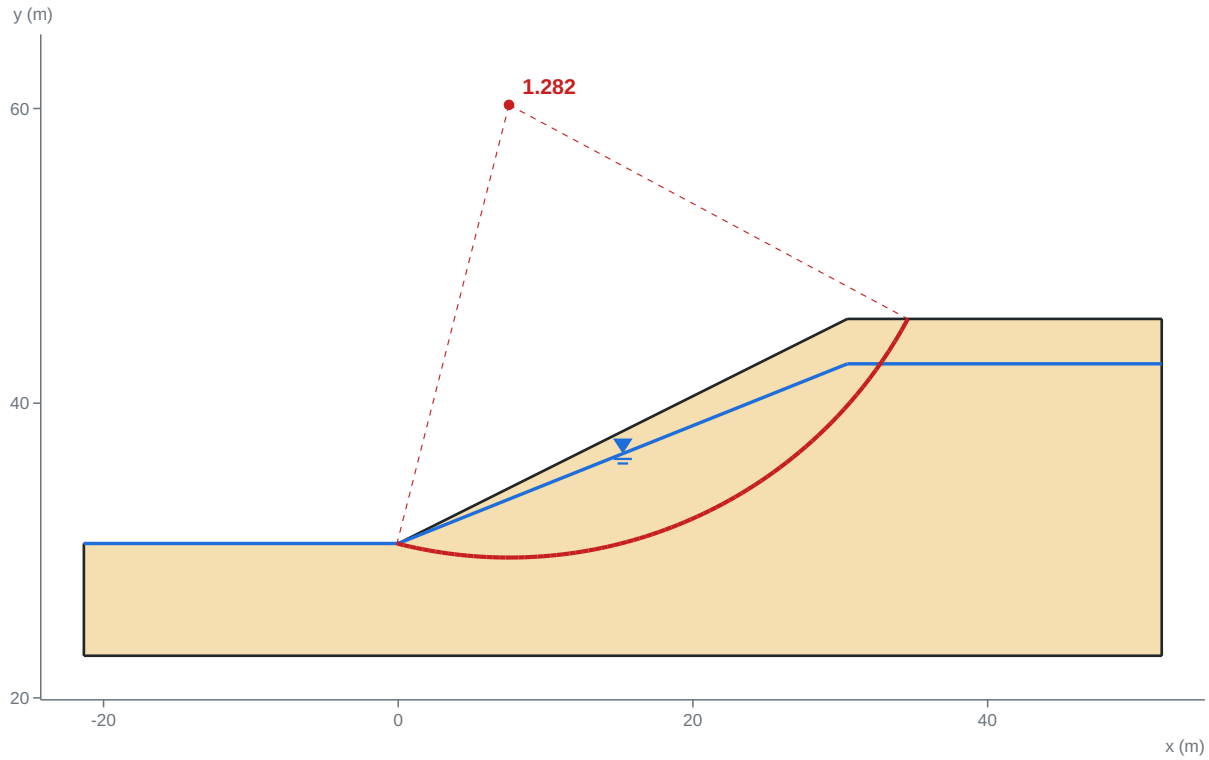



*Figure 126. V072a: analysed circle (Bishop simplified, FOS = 1.282).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sandy clay | Mohr-Coulomb | 18.85 | c' = 14.36 | 30.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.285 | 1.282 |
| Spencer | 1.290 | 1.288 |

*Pockoski & Duncan compare computer programs and publish no independent value.*

## 3.127 V072b — Pockoski & Duncan slope 2, circular search

*Pockoski & Duncan (2000). Verification manual of a commercial slope stability program.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V072a slope (dry 5.5 ft tension crack) with a circular search. Converted to SI.

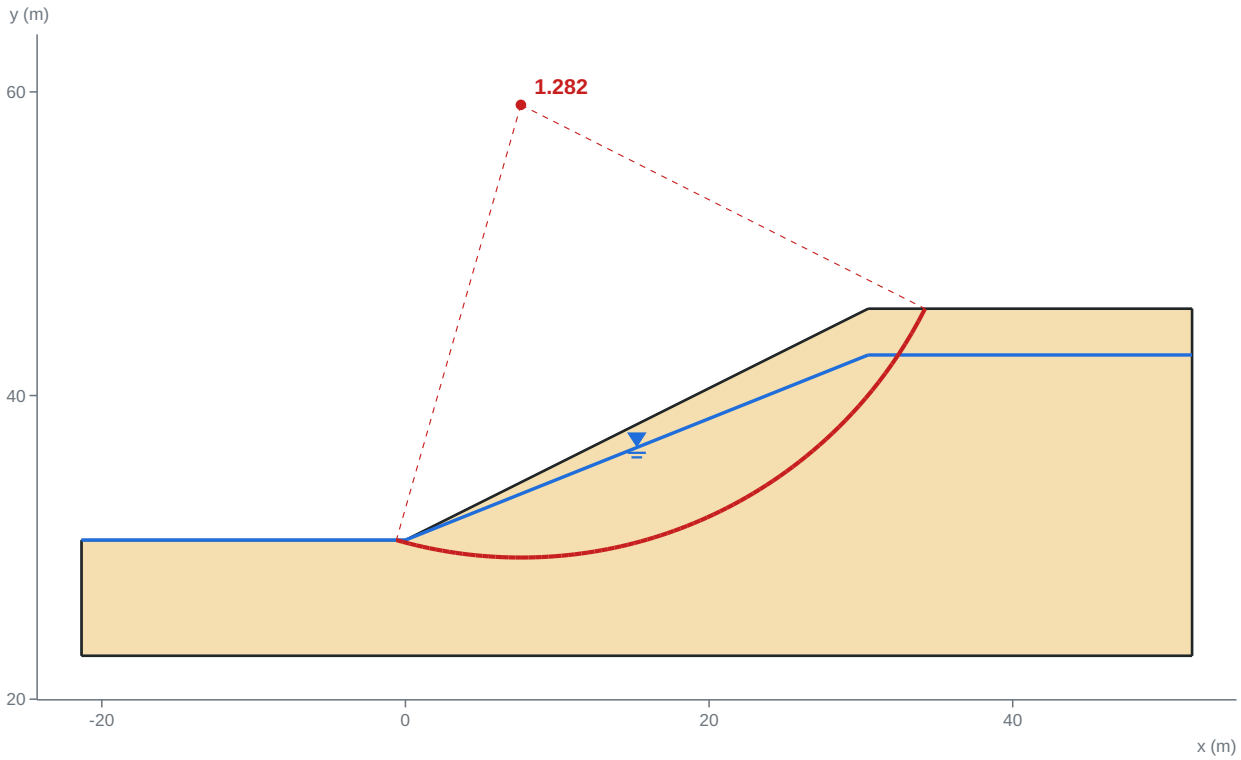



*Figure 127. V072b: critical circle (Bishop simplified, FOS = 1.282).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sandy clay | Mohr-Coulomb | 18.85 | c' = 14.36 | 30.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.285 | 1.282 |
| Spencer | 1.290 | 1.287 |
| Janbu simplified | 1.141 | 1.135 |

*Pockoski & Duncan compare computer programs and publish no independent value.*

## 3.128 V073 — Pockoski & Duncan test slope 3, clay seam and dry crack

*Pockoski & Duncan (2000).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Sandy clay over a 5 ft highly plastic clay seam on the model base, water table and a dry 6 ft tension crack; circular search. Converted to SI.

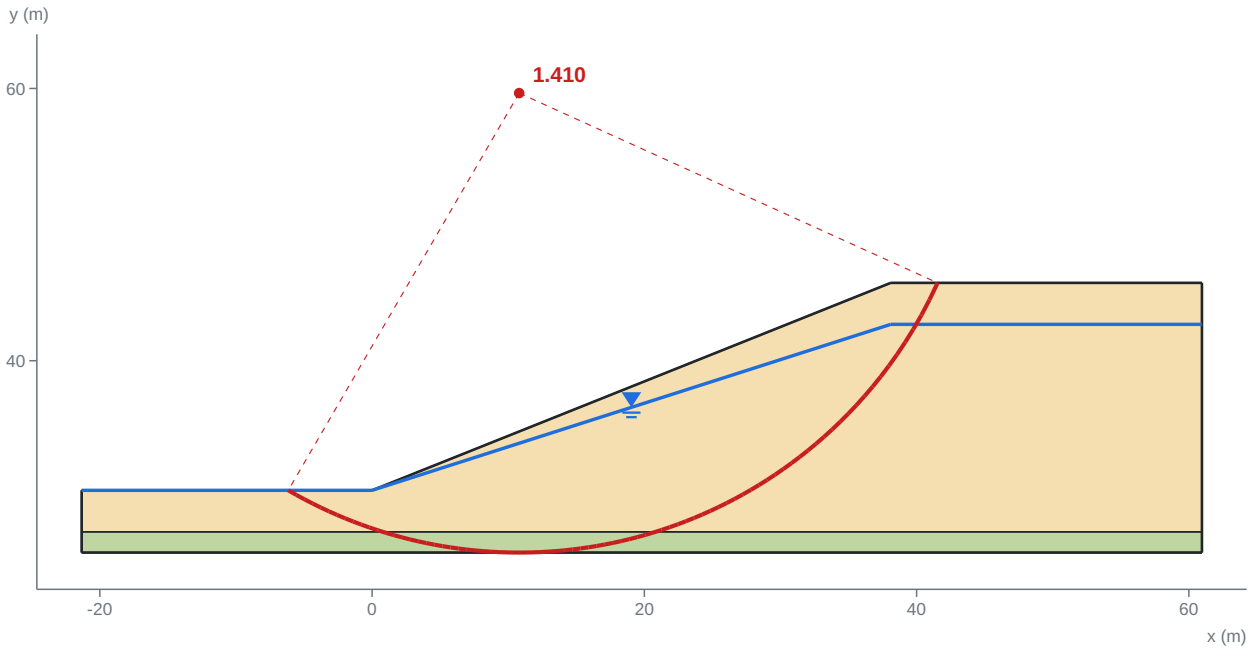



*Figure 128. V073: critical circle (Bishop simplified, circles only, FOS = 1.410).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sandy clay | Mohr-Coulomb | 20.42 | c' = 14.36 | 35.00° |
| Plastic clay seam | Mohr-Coulomb | 20.42 | c' = 0.00 | 25.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified, circles only | 1.417 | 1.410 |
| Spencer, circles only | 1.422 | 1.415 |
| Bishop simplified | 1.392 | 1.387 |
| Spencer | 1.400 | 1.395 |

*Pockoski & Duncan compare computer programs and publish no independent value.*

## 3.129 V074 — Soil-nailed wall with tension crack and surcharges

*Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

25 ft nailed wall in undrained clay (800 psf) with crest surcharges, a dry 7 ft crack and five passive nail rows at 15°; the reference circle truncated by the crack. Converted to SI.

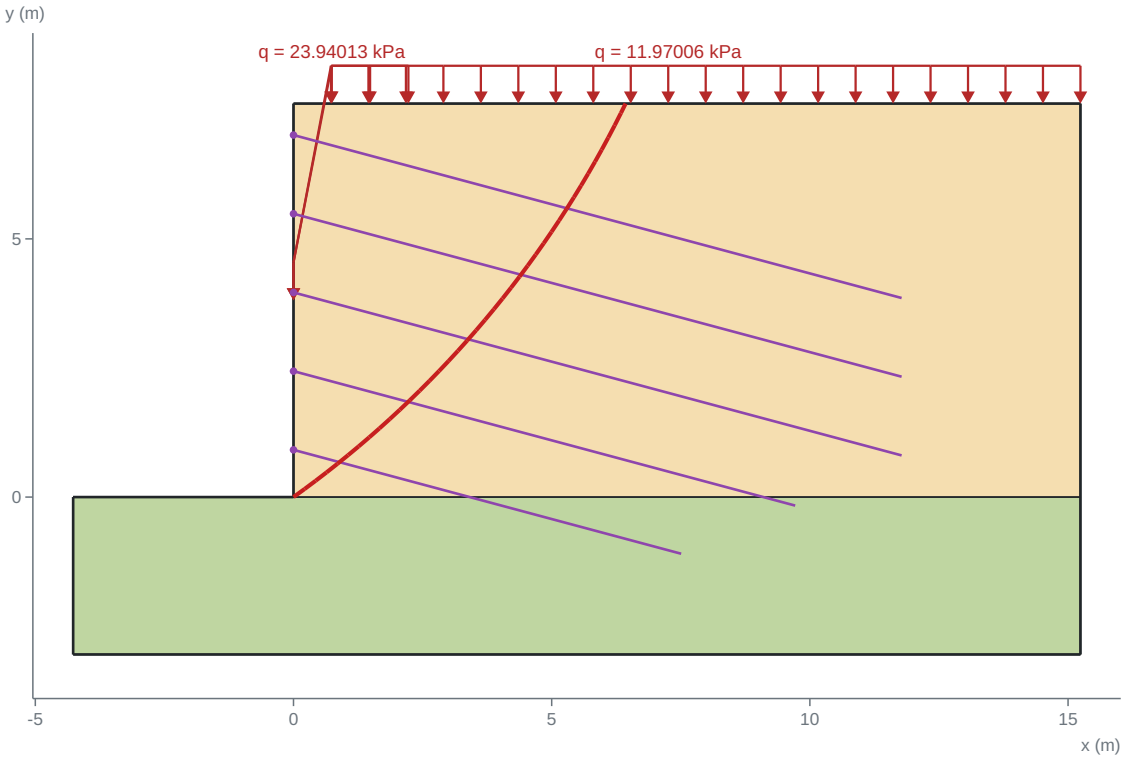



*Figure 129. V074: analysed surface (Spencer, FOS = 1.010).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sandy clay | Undrained (φ = 0) | 18.85 | Su = 38.30 | 0° |
| Firm soil | Mohr-Coulomb | 20.42 | c' = 239.4 | 35.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.009 | 1.010 |
| Janbu corrected | 1.073 | 1.075 |

*Pockoski & Duncan compare computer programs and publish no independent value.*

## 3.130 V075 — Tieback wall in sand, drawdown water table

*Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Single-row tieback wall in sand (φ′ = 30°; the tieback from 12 ft up the wall, 15° down, 37 ft long of which 22 ft bonded) with a water table drawn down to the wall toe (digitised from the reference figure) and pore pressure from the vertical head; the printed surface from the wall toe. Converted to SI.

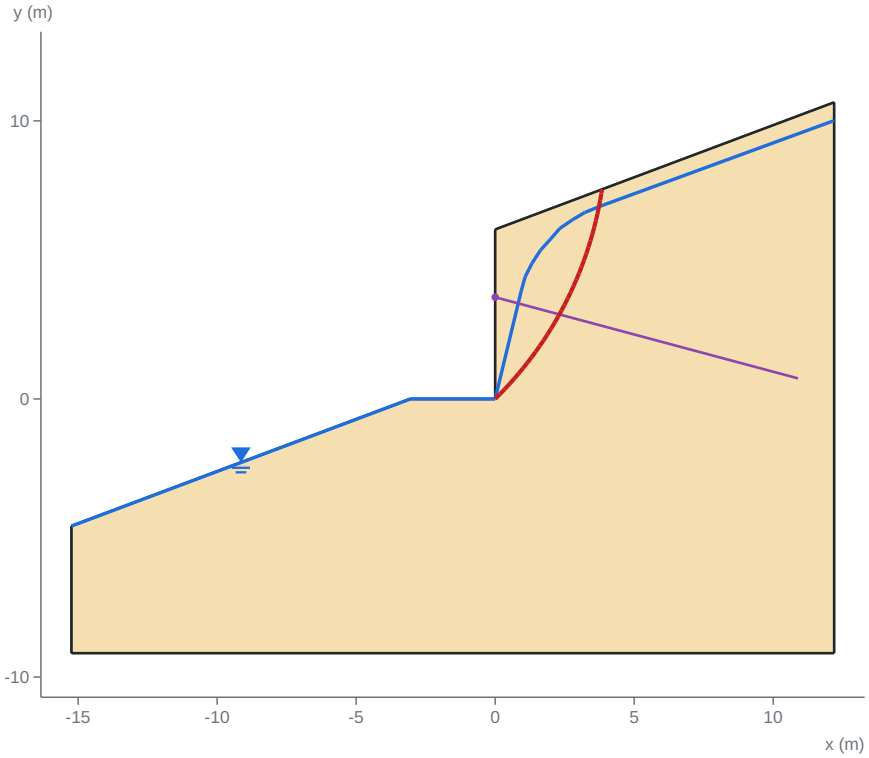



*Figure 130. V075: analysed surface (Spencer, FOS = 0.578).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sand | Mohr-Coulomb | 18.85 | c' = 0.00 | 30.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 0.570 | 0.578 |

*Pockoski & Duncan compare computer programs and publish no independent value.*

## 3.131 V076 — Tied-back wall in layered soil

*Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

44 ft tied-back excavation in eight layers, water table at grade in front of the wall and at el. 102.5 behind it (a vertical step at the wall), three tieback rows at 20° (88 ft, 40 ft bond). a commercial program's critical circle. Converted to SI.

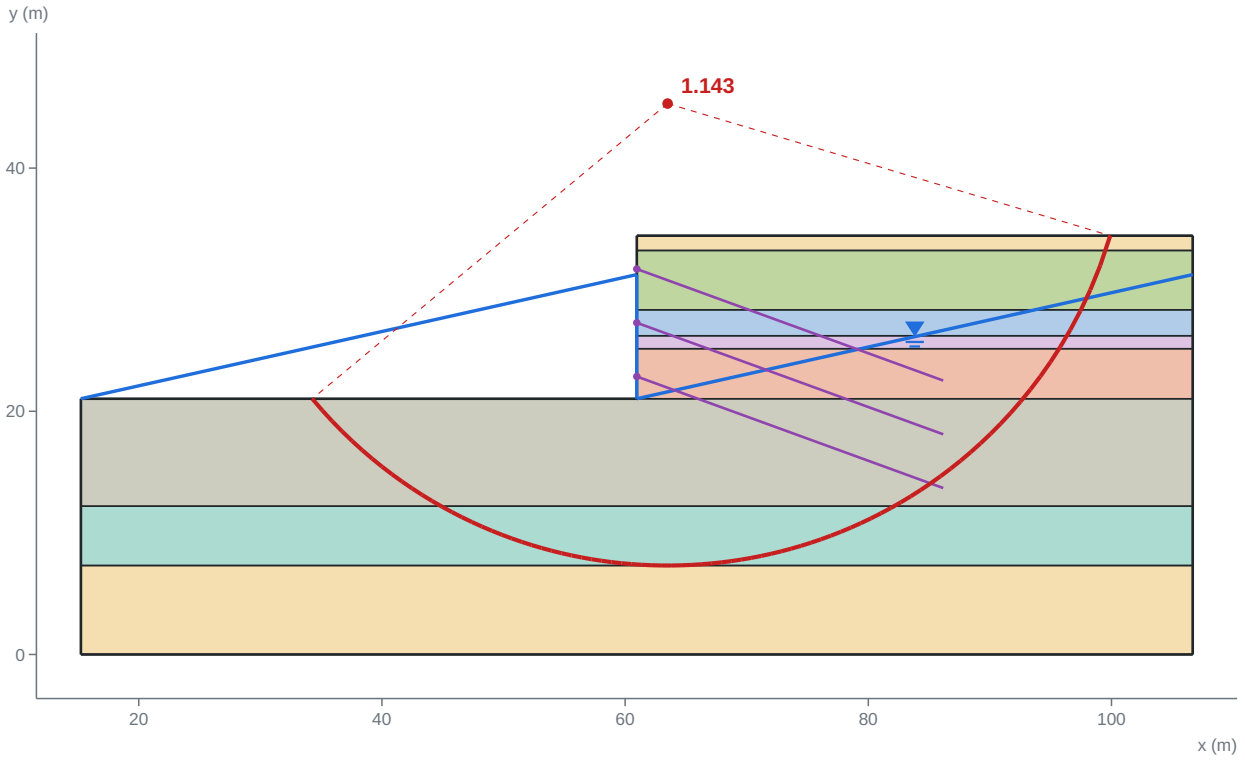



*Figure 131. V076: analysed circle (Bishop simplified, FOS = 1.143).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Granular Fill | Mohr-Coulomb | 18.91 | c' = 0.00 | 30.00° |
| Cohesive Fill | Mohr-Coulomb | 18.02 | c' = 0.00 | 30.00° |
| Organic Silt | Undrained (φ = 0) | 17.31 | Su = 43.09 | 0° |
| OC Crust | Undrained (φ = 0) | 18.50 | Su = 119.0 | 0° |
| Upper Marine Clay | Undrained (φ = 0) | 18.50 | Su = 79.96 | 0° |
| Middle Marine Clay | Undrained (φ = 0) | 18.50 | Su = 45.97 | 0° |
| Lower Marine Clay | Undrained (φ = 0) | 18.50 | Su = 51.95 | 0° |
| Glaciomarine Deposits | Undrained (φ = 0) | 23.11 | Su = 71.82 | 0° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.147 | 1.143 |
| Spencer | 1.145 | 1.141 |

*Pockoski & Duncan compare computer programs and publish no independent value.*

## 3.132 V077 — Prandtl bearing capacity on level ground

*Prandtl (1920). Über die Härte plastischer Körper. Nachr. Ges. Wiss. Göttingen, Math.-phys. Kl., 74–85.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Weightless (γ = 1e-6 kN/m³) c = 20 kPa soil (φ = 0) loaded by a strip load of 102.83 kPa = c·(2 + π) over 10 m, the exact bearing capacity. Analysed on the Prandtl mechanism of Fig. 26.2: active wedge under the load, a 90° fan arc of radius 7.071 m about the load's edge (5, 10), lowest point 2.929, and the passive wedge.

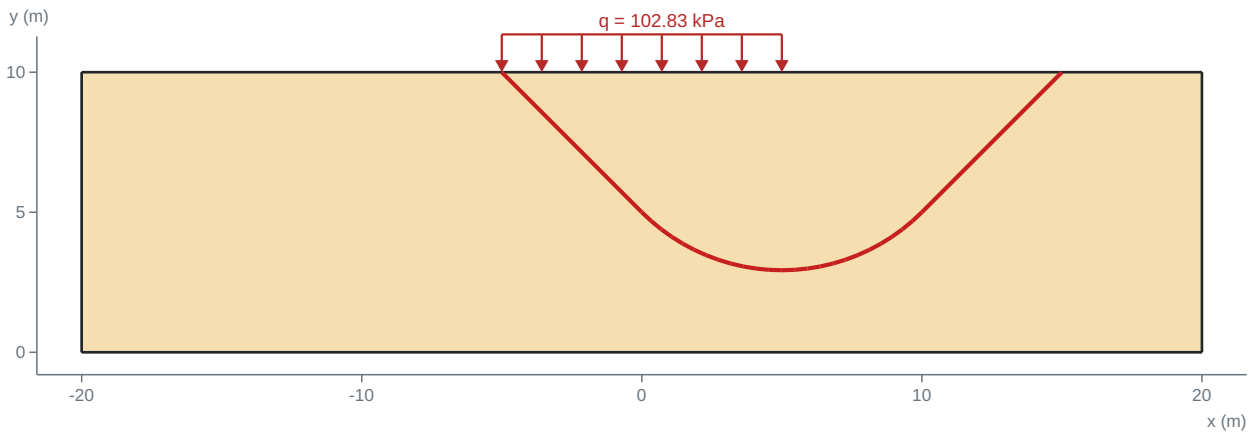



*Figure 132. V077: analysed surface (Spencer, FOS = 0.943).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| soil | Undrained (φ = 0) | 0.00 | Su = 20.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | plasticity theory | Prandtl (1920) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 0.940 | 0.943 |

*The theoretical factor of safety is 1.0; limit equilibrium with slices gives about 6 % less on this mechanism.*

## 3.133 V078 — Prandtl bearing capacity, fully specified Prandtl surface (reference values)

*Prandtl (1920). Über die Härte plastischer Körper. Nachr. Ges. Wiss. Göttingen, Math.-phys. Kl., 74–85.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Prandtl bearing capacity as a commercial program specifies it: weightless c = 20 kPa soil, 102.83 kPa over 10 m, the fully specified Prandtl surface (active wedge, 90° fan about the load edge, passive wedge).

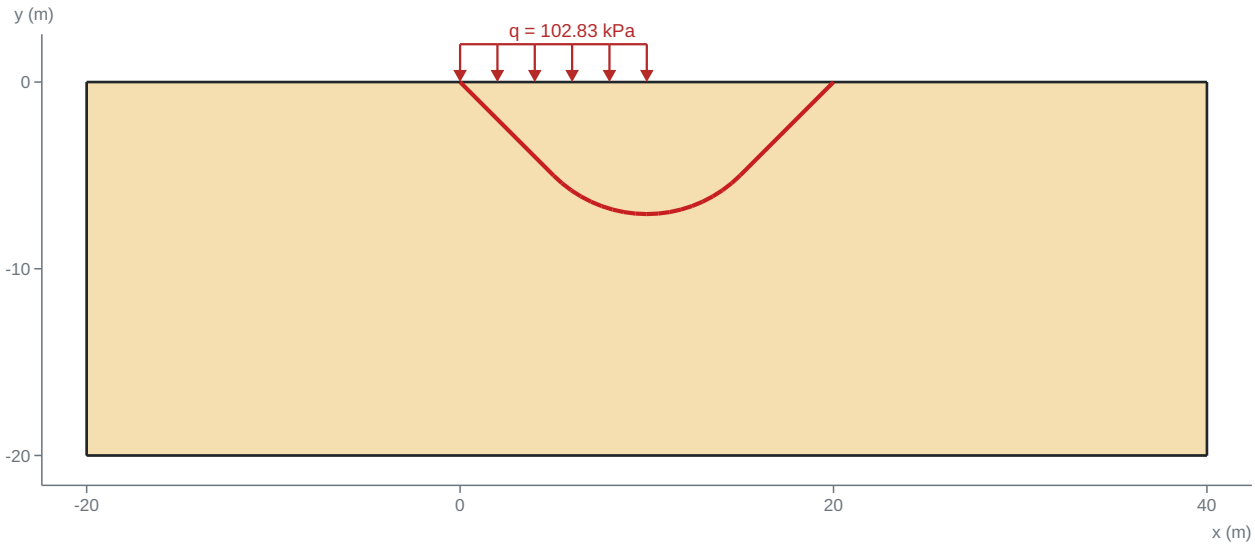



*Figure 133. V078: analysed surface (Morgenstern-Price (half-sine), FOS = 0.958).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Undrained (φ = 0) | 0.00 | Su = 20.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.00 | plasticity theory | Prandtl (1920) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 0.960 | 0.958 |

*The theoretical factor of safety is 1.0; limit equilibrium with slices gives about 4 % less on this mechanism.*

## 3.134 V079 — Priest (1993) rigid block on a plane, water-filled crack

*Priest, S.D. (1993). Discontinuity Analysis for Rock Engineering. Chapman & Hall, London; example 8.1, pp. 224–225.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Priest's example 8.1 as two commercial programs model it: a 30 m face with the crest at x = 17.0 (60.46°, Priest's is 60° with the crest at 17.32, see V080), a 30° plane from the toe, a vertical 15 m tension crack holding 3.75 m of water, and a water table rising from the toe to 3.75 m above the plane at the crack, which gives Priest's linear pressure along the plane.

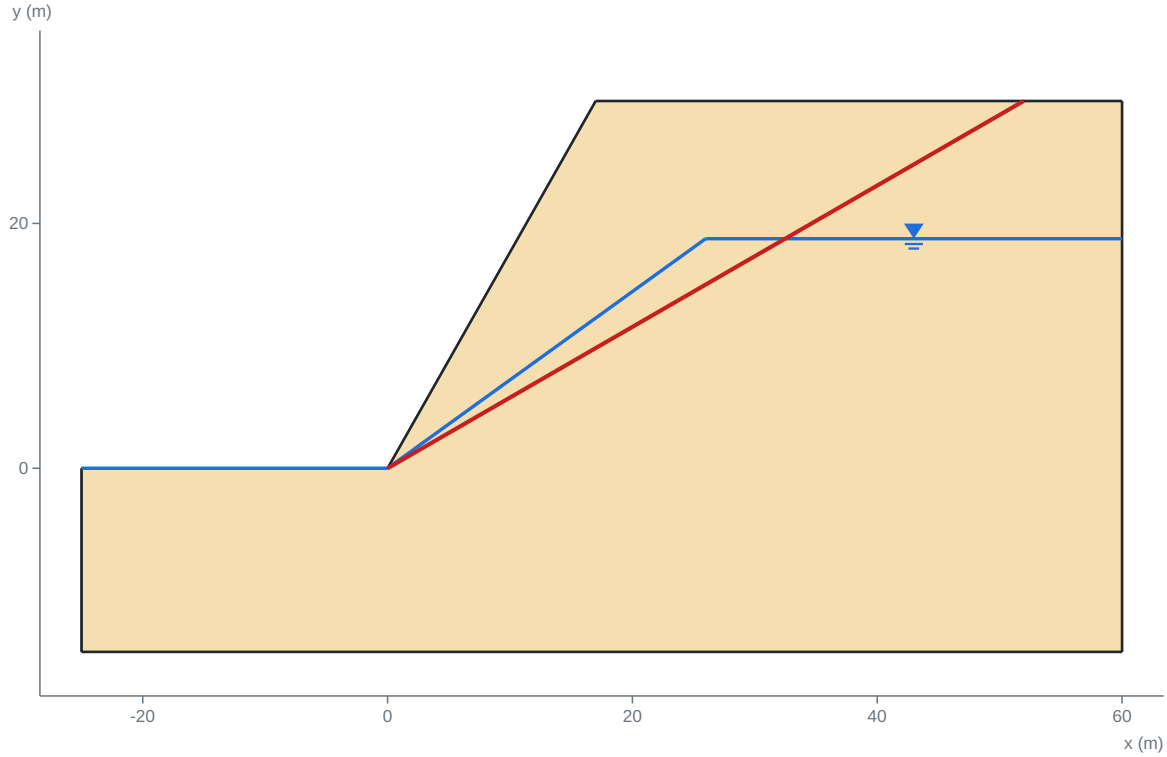



*Figure 134. V079: analysed surface (Janbu simplified, FOS = 1.048).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Block | Mohr-Coulomb | 25.00 | c' = 20.00 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.049 | rigid block, force equilibrium | Priest (1993) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu simplified | 1.049 | 1.048 |
| Spencer | 1.049 | 1.048 |

## 3.135 V080 — Priest (1993) rigid block, reference geometry (60° face)

*Priest, S.D. (1993). Discontinuity Analysis for Rock Engineering. Chapman & Hall, London; example 8.1, pp. 224–225.*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

Priest's example 8.1 with his geometry: a 60° face 30 m high (crest at x = 17.321), a 30° plane from the toe, a vertical 15 m tension crack 8.66 m behind the crest holding 3.75 m of water, and a water table rising from the toe to 3.75 m above the plane at the crack.

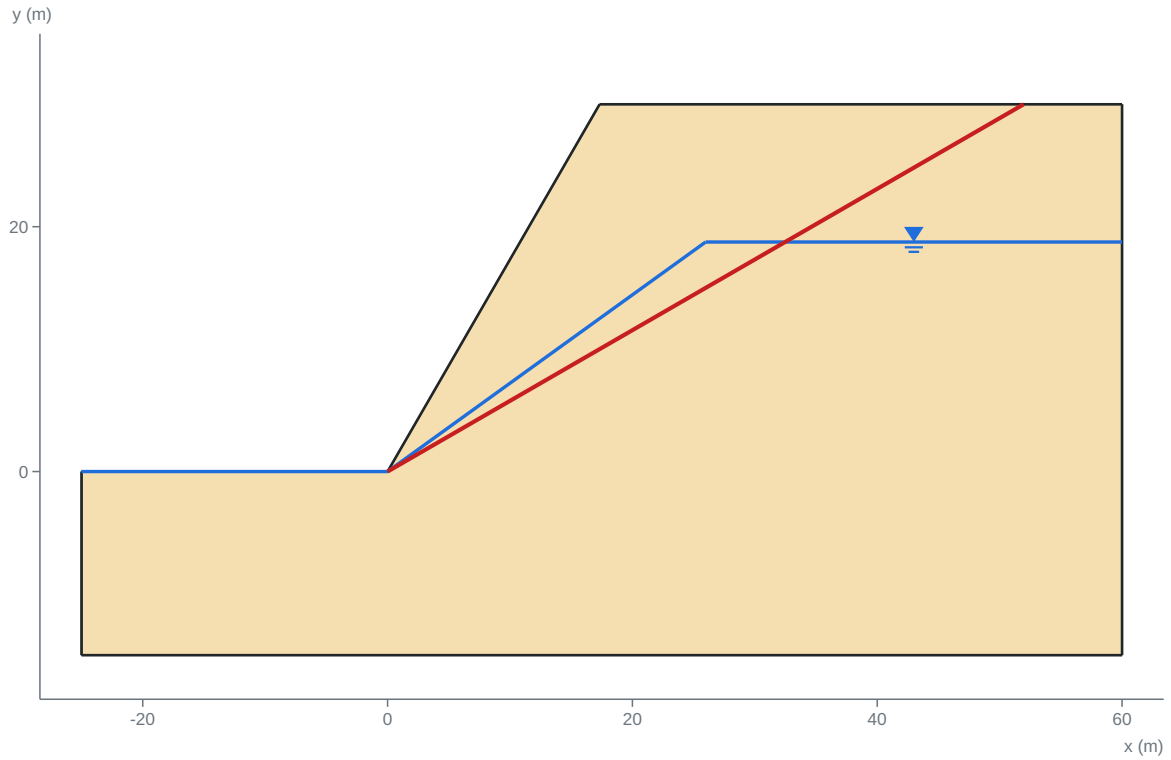



*Figure 135. V080: analysed surface (Morgenstern-Price (half-sine), FOS = 1.049).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Block | Mohr-Coulomb | 25.00 | c' = 20.00 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.049 | Spencer | Priest | 1.049 |
| 1.049 | rigid block, force equilibrium | Priest (1993) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Morgenstern-Price (half-sine) | 1.049 | 1.049 |
| Spencer | – | 1.049 |

## 3.136 V081a — Sharma (1996) reinforcement example, unreinforced

*Sharma (1996).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

45° cohesionless sand slope with a 40 kPa surcharge behind the crest, on the published toe circle.

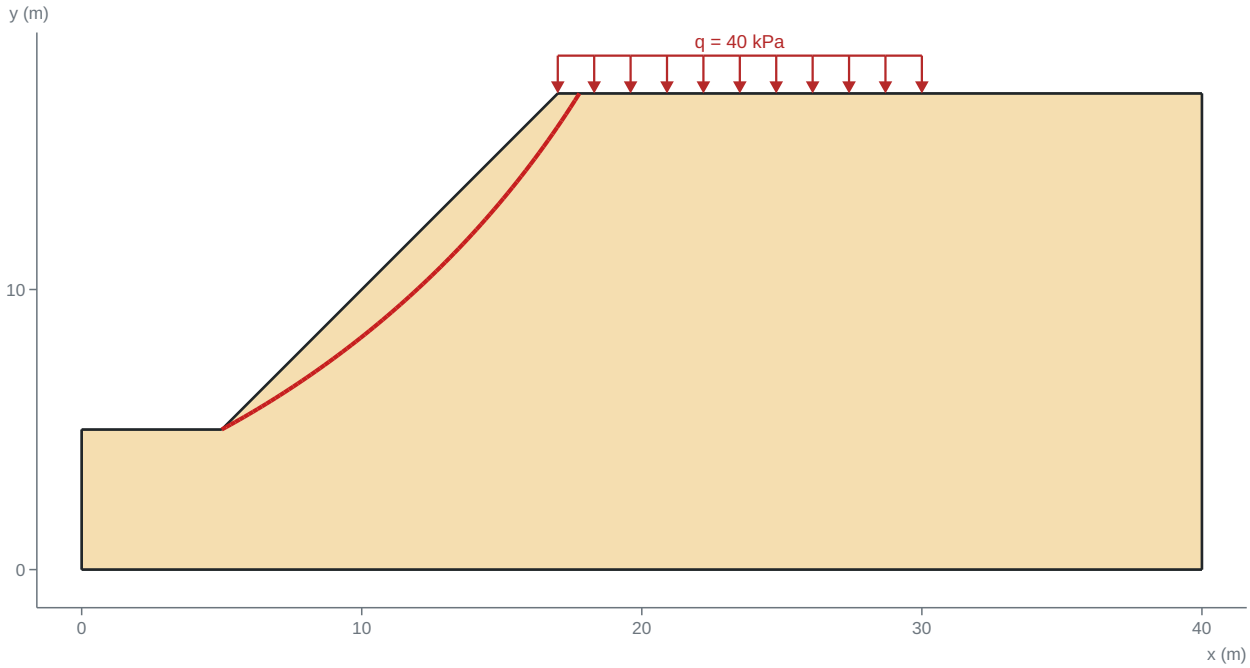



*Figure 136. V081a: analysed circle (Bishop simplified, FOS = 0.764).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Sand | Mohr-Coulomb | 20.00 | c' = 0.00 | 36.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 0.764 | 0.764 |

*The source compares programs and publishes no independent value.*

## 3.137 V081b — Sharma (1996) slope with a reinforced zone (φ = 54.93°)

*Sharma (1996). Verification manual of a commercial slope stability program.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V081a sand slope (φ′ = 36°, 40 kPa surcharge) with a 7.6 m reinforced zone of φ = 54.93° (tan φ = 1.96 tan 36°) behind the face, on the reference critical toe circle (8.707, 18.307, R = 13.813).

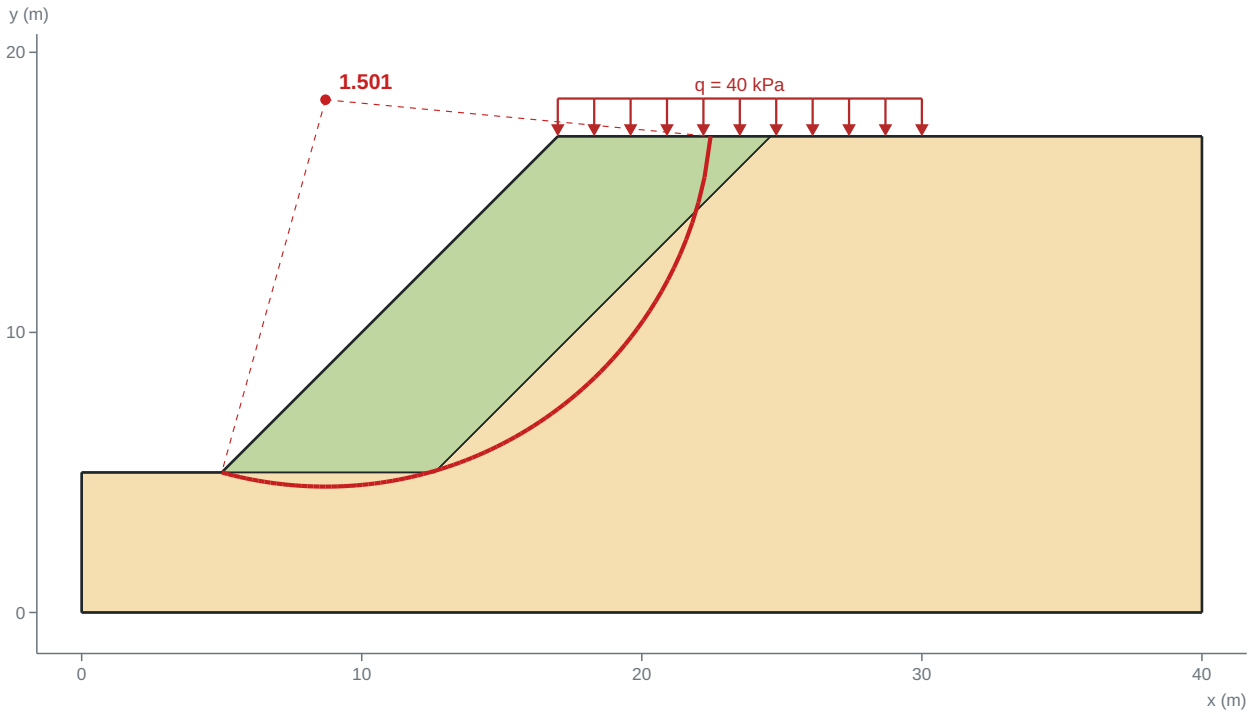



*Figure 137. V081b: analysed circle (Bishop simplified, FOS = 1.501).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Material 1 | Mohr-Coulomb | 20.00 | c' = 0.00 | 36.00° |
| Reinforced zone | Mohr-Coulomb | 20.00 | c' = 0.00 | 54.93° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.504 | 1.501 |

*The source compares programs and publishes no independent value.*

## 3.138 V082a — Sharma (1996) slope, 11 ft dry tension crack instead of the zero-strength cap

*Sharma (1996); Malkawi et al. (2001). Verification manual of a commercial slope stability program.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V082c slope with the zero-strength cap replaced by an 11 ft dry tension crack (Table 27.4, analysis 3), on the V082c circle; phreatic correction. Converted to SI.

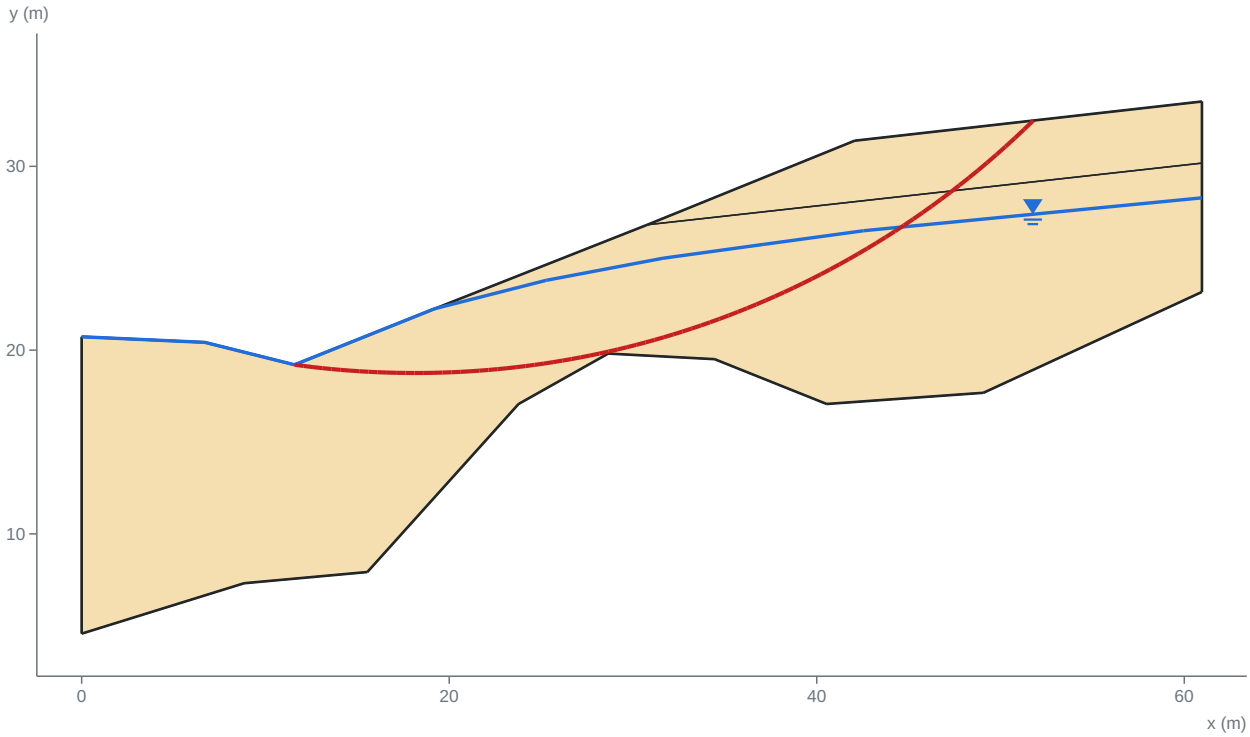



*Figure 138. V082a: analysed circle (Bishop simplified, FOS = 1.534).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil 1 | Mohr-Coulomb | 18.28 | c' = 23.94 | 14.00° |
| Soil 2 (zero strength) | Undrained (φ = 0) | 18.28 | Su = 0.00 | 0° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.532 | 1.534 |
| Spencer | 1.532 | 1.533 |
| Morgenstern-Price (half-sine) | 1.532 | 1.533 |
| Janbu corrected | 1.544 | 1.546 |

*The source compares programs and publishes no independent value.*

## 3.139 V082b — Sharma (1996) slope, 11 ft tension crack with 6 ft of water

*Sharma (1996); Malkawi et al. (2001). Verification manual of a commercial slope stability program.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V082a with 6 ft of water in the 11 ft crack (Table 27.4, analysis 4).

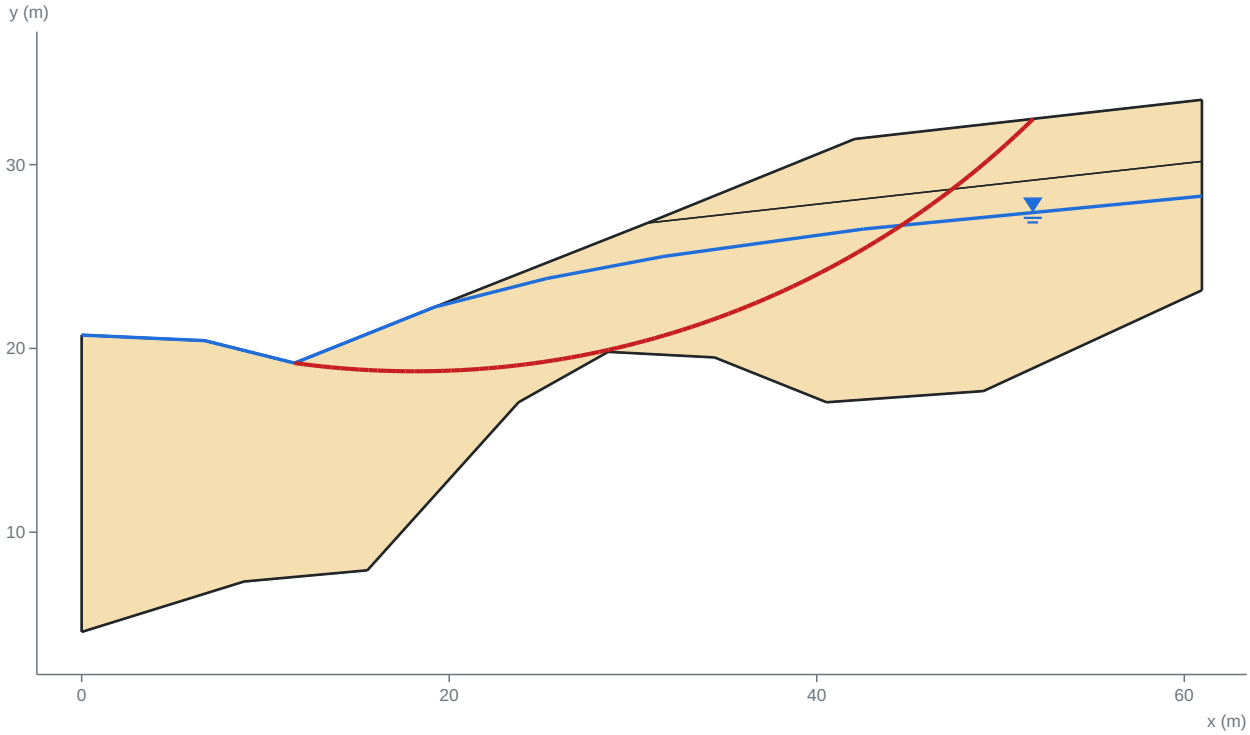



*Figure 139. V082b: analysed circle (Bishop simplified, FOS = 1.513).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil 1 | Mohr-Coulomb | 18.28 | c' = 23.94 | 14.00° |
| Soil 2 (zero strength) | Undrained (φ = 0) | 18.28 | Su = 0.00 | 0° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.511 | 1.513 |
| Spencer | 1.510 | 1.512 |
| Morgenstern-Price (half-sine) | 1.510 | 1.512 |
| Janbu corrected | 1.520 | 1.521 |

*The source compares programs and publishes no independent value.*

## 3.140 V082c — Sharma (1996) slope over undulating bedrock, zero-strength cap

*Sharma (1996); Malkawi et al. (2001).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Two-material slope over undulating bedrock with a zero-strength cap layer and a water table (phreatic correction); soil 1 has moist/saturated unit weights 116.4/124.2 pcf. Specified circle (59.52, 219.21, R = 157.68). Converted to SI.

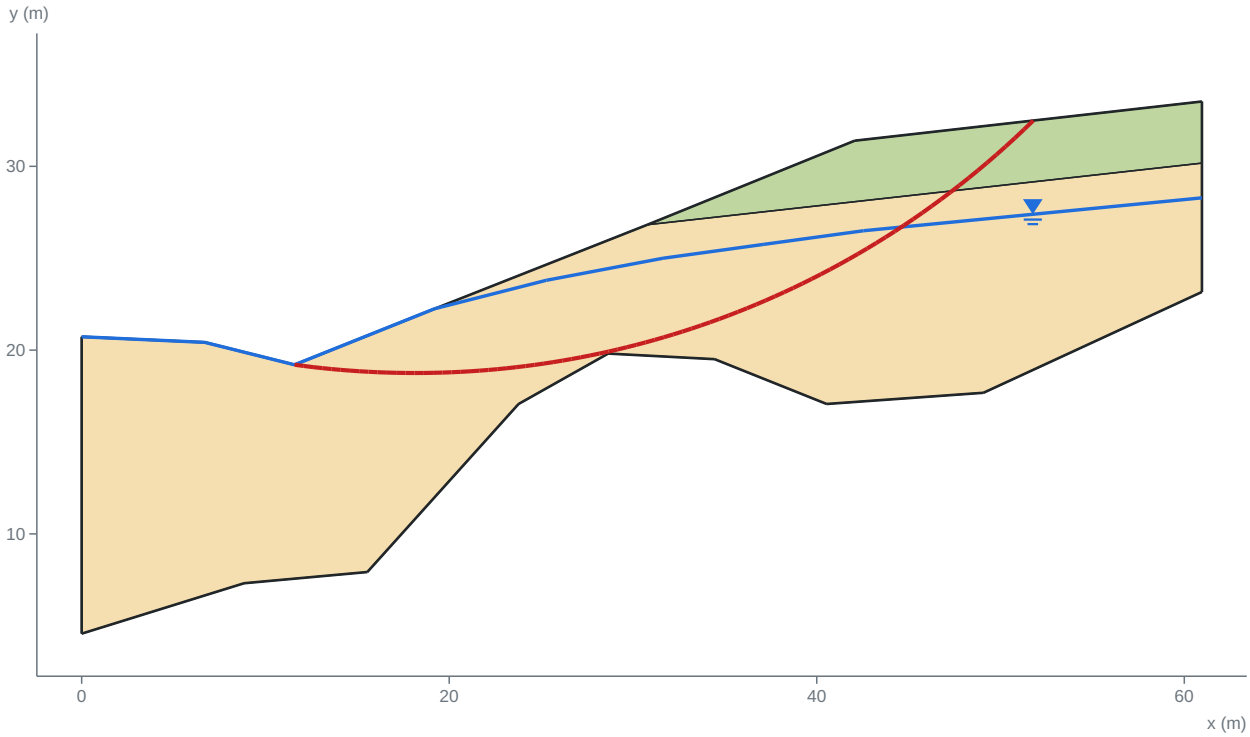



*Figure 140. V082c: analysed circle (Bishop simplified, FOS = 1.396).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil 1 | Mohr-Coulomb | 18.28 | c' = 23.94 | 14.00° |
| Soil 2 (zero strength) | Undrained (φ = 0) | 18.28 | Su = 0.00 | 0° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.396 | 1.396 |
| Spencer | 1.402 | 1.402 |
| Janbu corrected | 1.391 | 1.392 |
| Morgenstern-Price (half-sine) | 1.398 | 1.398 |

*The source compares programs and publishes no independent value.*

## 3.141 V082d — Sharma (1996) slope over undulating bedrock, circular search with exit limits

*Sharma (1996); Malkawi et al. (2001).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V082c slope (zero-strength cap, water table with phreatic correction) with a circular search whose circles must exit between x = 38 and 70 ft at the toe and 120 and 180 ft at the crest (Table 27.3). Converted to SI.

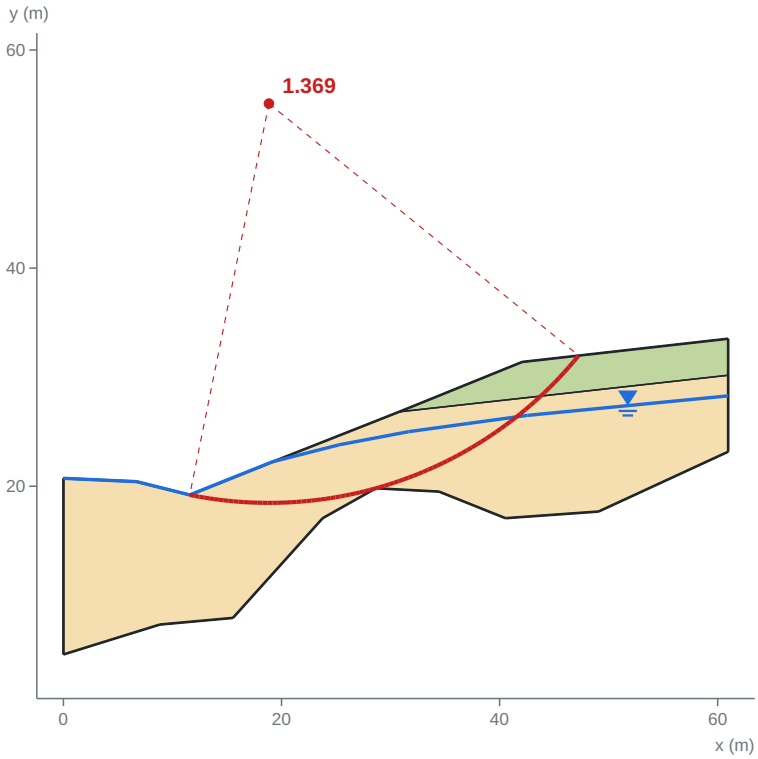



*Figure 141. V082d: critical circle (Bishop simplified, FOS = 1.369).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil 1 | Mohr-Coulomb | 18.28 | c' = 23.94 | 14.00° |
| Soil 2 (zero strength) | Undrained (φ = 0) | 18.28 | Su = 0.00 | 0° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.376 | 1.369 |
| Spencer | 1.382 | 1.376 |
| Morgenstern-Price (half-sine) | 1.378 | 1.371 |
| Janbu corrected | 1.345 | 1.343 |

*The source compares programs and publishes no independent value.*

## 3.142 V083 — Soil-nailed wall in clay (Amherst test wall)

*Sheahan & Ho (2003).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

6.1 m vertical cut in undrained clay (Su = 25 kPa) with two nail rows at 20° (tensile 118 kN, plate 86 kN, bond 15 kN/m, spacing 1.5 m; axial, passive) and the facing weight as a line load; the critical plane printed by a commercial program, from the toe (0, 0) to the crest (6.279, 6.1), 44.2°.

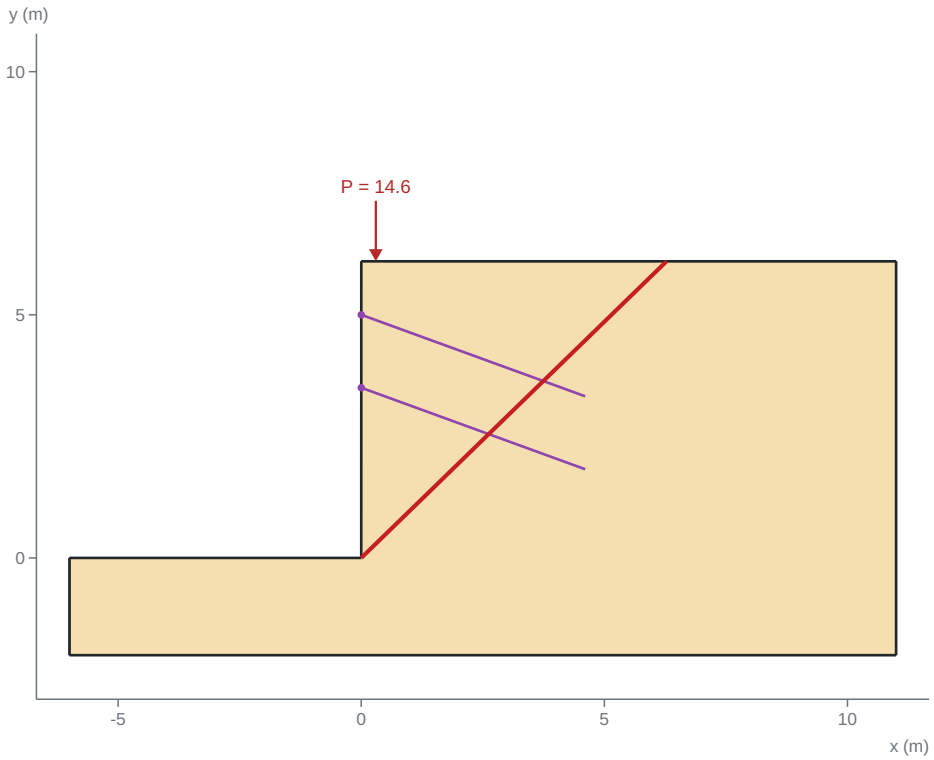



*Figure 142. V083: analysed surface (Janbu corrected, FOS = 0.885).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Amherst Clay | Undrained (φ = 0) | 18.90 | Su = 25.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.887 | force-equilibrium trial wedge | Sheahan & Ho (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 0.890 | 0.885 |

*Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.*

## 3.143 V084a — Soil-nailed wall in sand (Clouterre test wall)

*Sheahan & Ho (2003).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

7 m cut in sand (c′ = 3 kPa, φ′ = 38°) with seven passive nail rows at 10° (tensile 15 kN, plate 59 kN, bond 7.5 kN/m, i.e. the tensile capacity develops over 2 m; spacing 1.15 m) and the facing weight; 55° plane through the toe. V084b … V084f are the other planes of Table 48.3.

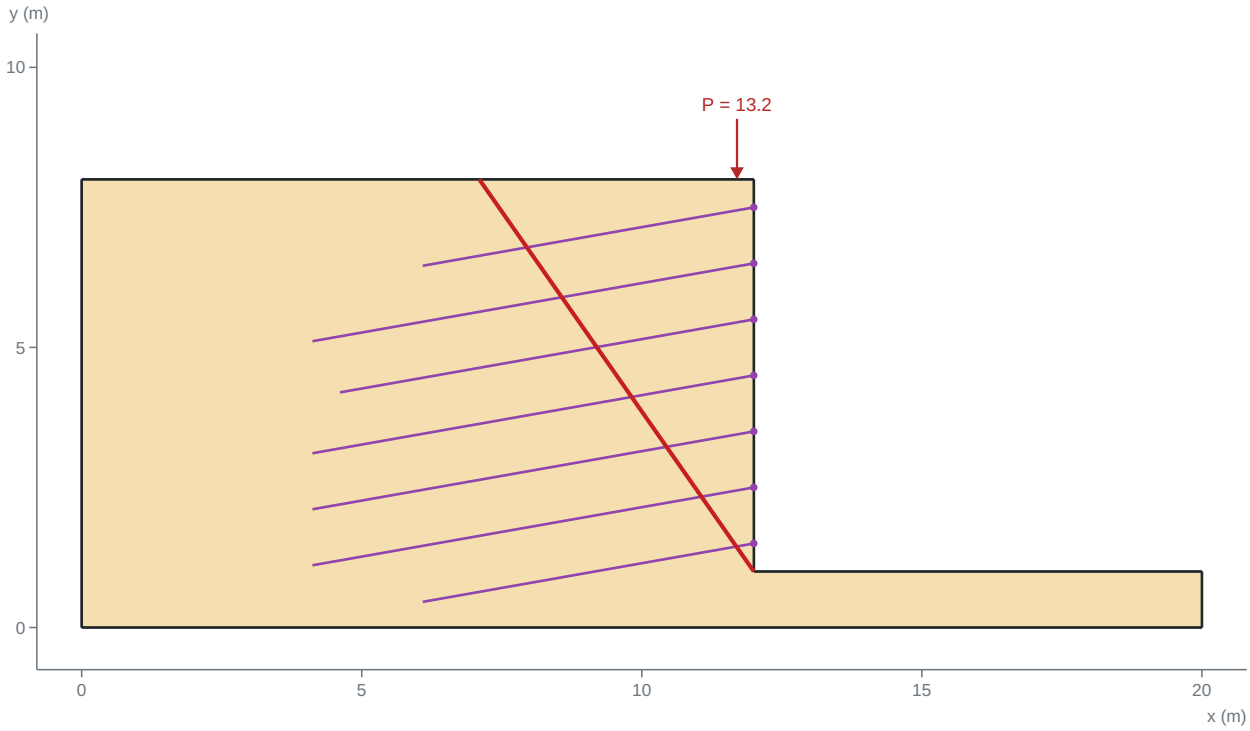



*Figure 143. V084a: analysed surface (Janbu corrected, FOS = 0.988).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fontainebleau Sand | Mohr-Coulomb | 20.00 | c' = 3.00 | 38.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.989 | force-equilibrium trial wedge | Sheahan & Ho (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 0.989 | 0.988 |

*Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.*

## 3.144 V084b — Soil-nailed wall in sand (Clouterre test wall), 45° plane

*Sheahan & Ho (2003).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

The V084a wall on the 45° plane through the toe.

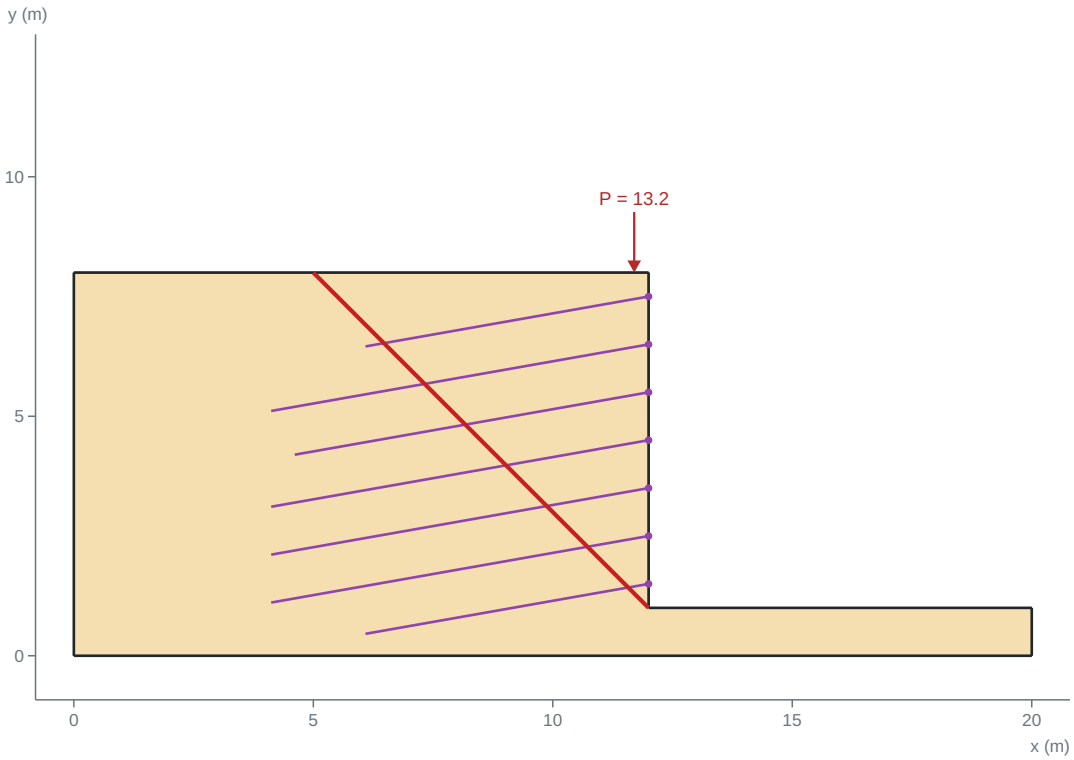



*Figure 144. V084b: analysed surface (Janbu corrected, FOS = 1.124).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fontainebleau Sand | Mohr-Coulomb | 20.00 | c' = 3.00 | 38.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.176 | force-equilibrium trial wedge | Sheahan & Ho (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 1.123 | 1.124 |

*Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 5 %.*

## 3.145 V084c — Soil-nailed wall in sand (Clouterre test wall), 50° plane

*Sheahan & Ho (2003).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

The V084a wall on the 50° plane through the toe.

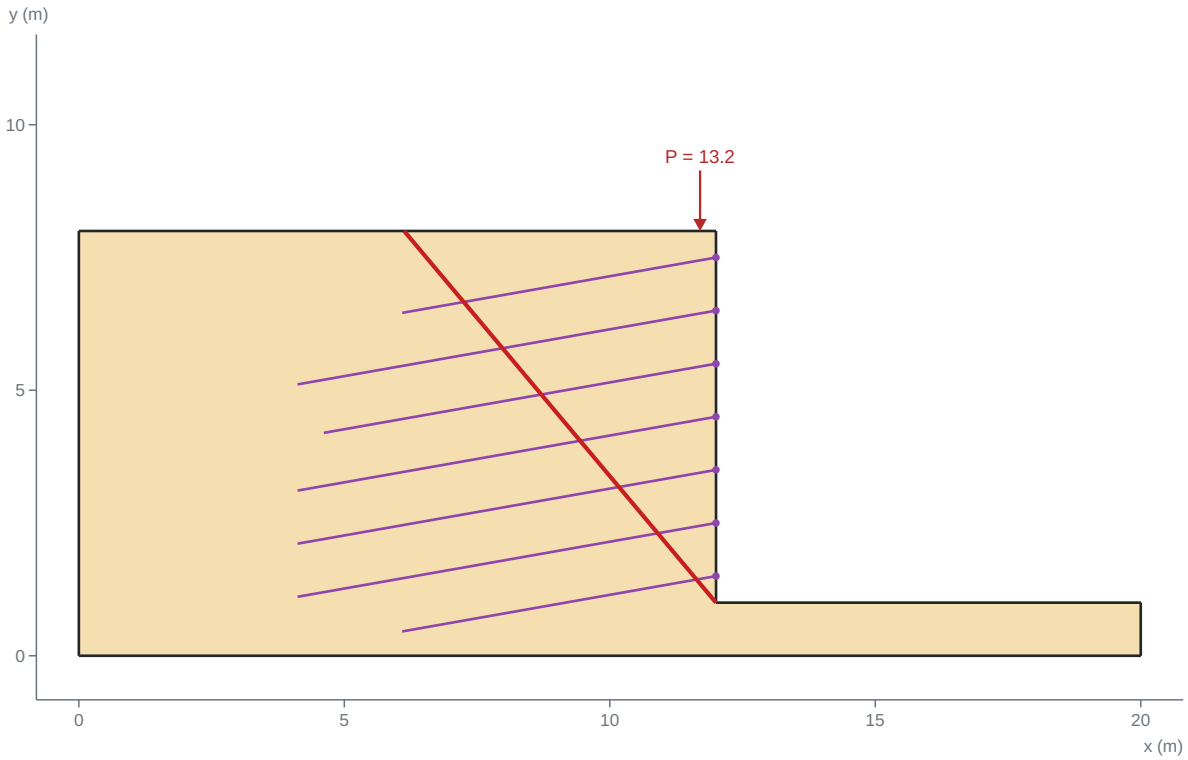



*Figure 145. V084c: analysed surface (Janbu corrected, FOS = 1.043).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fontainebleau Sand | Mohr-Coulomb | 20.00 | c' = 3.00 | 38.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.07 | force-equilibrium trial wedge | Sheahan & Ho (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 1.043 | 1.043 |

*Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.*

## 3.146 V084d — Soil-nailed wall in sand (Clouterre test wall), 60° plane

*Sheahan & Ho (2003).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

The V084a wall on the 60° plane through the toe.

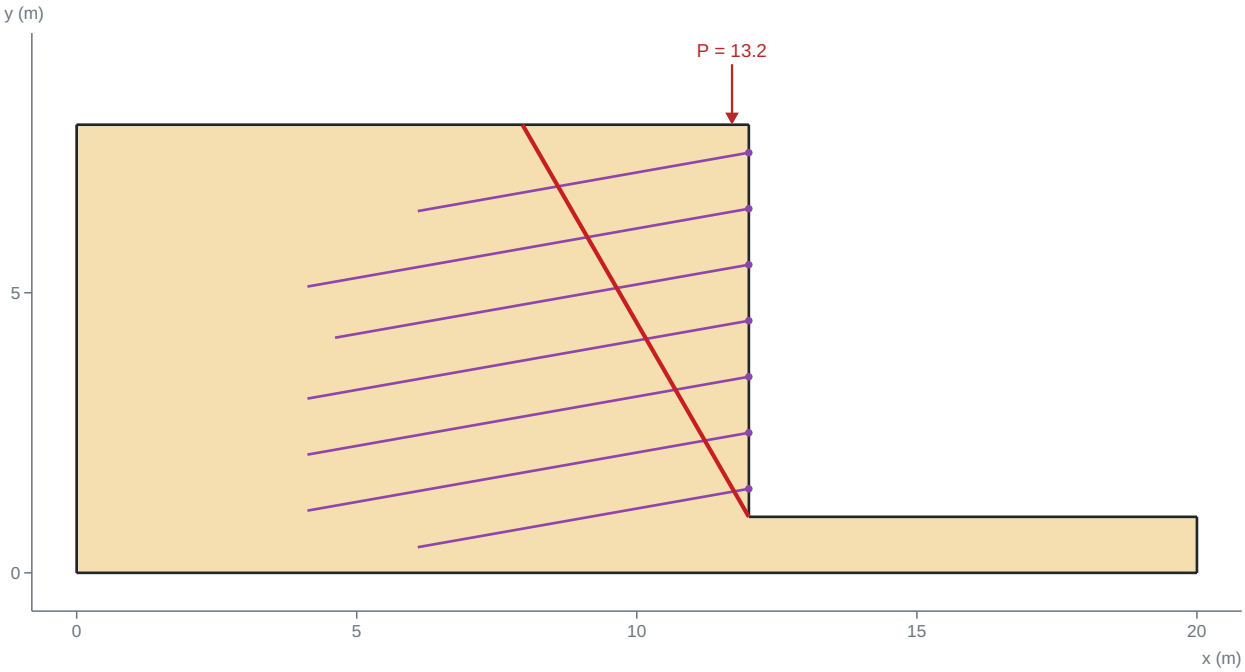



*Figure 146. V084d: analysed surface (Janbu corrected, FOS = 0.944).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fontainebleau Sand | Mohr-Coulomb | 20.00 | c' = 3.00 | 38.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.929 | force-equilibrium trial wedge | Sheahan & Ho (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 0.945 | 0.944 |

*Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.*

## 3.147 V084e — Soil-nailed wall in sand (Clouterre test wall), 65° plane

*Sheahan & Ho (2003).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

The V084a wall on the 65° plane through the toe.

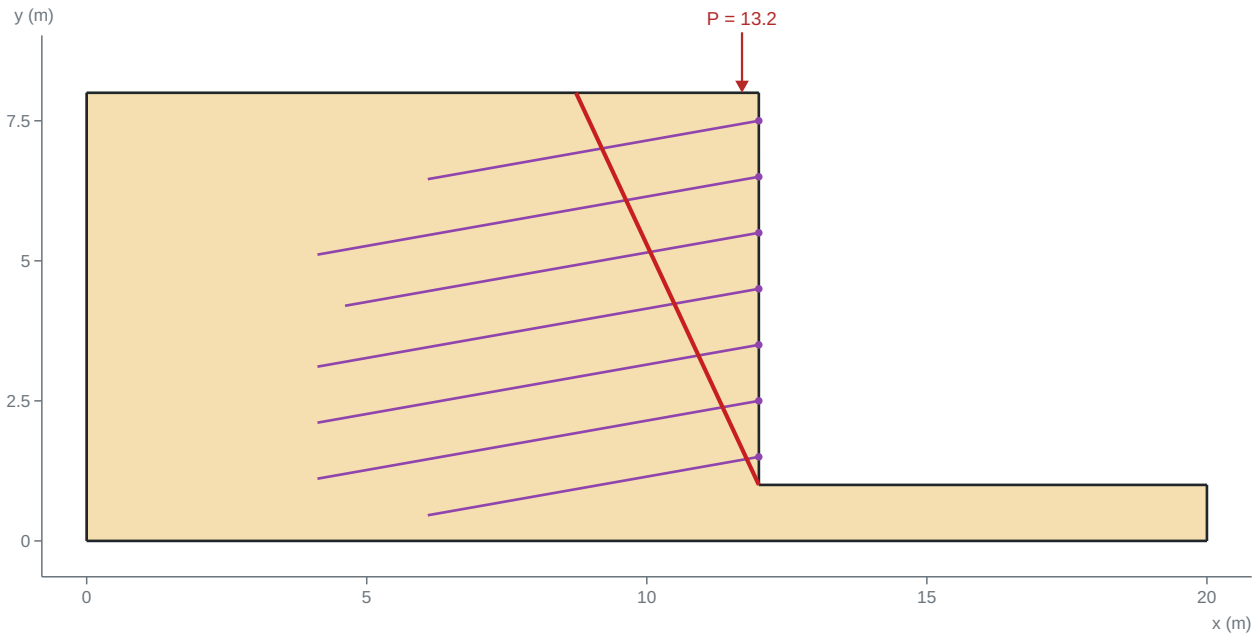



*Figure 147. V084e: analysed surface (Janbu corrected, FOS = 0.920).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fontainebleau Sand | Mohr-Coulomb | 20.00 | c' = 3.00 | 38.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.893 | force-equilibrium trial wedge | Sheahan & Ho (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 0.922 | 0.920 |

*Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.*

## 3.148 V084f — Soil-nailed wall in sand (Clouterre test wall), 70° plane

*Sheahan & Ho (2003).*

**Specified non-circular surface · Units: SI (m, kPa, kN/m³)**

The V084a wall on the 70° plane through the toe.

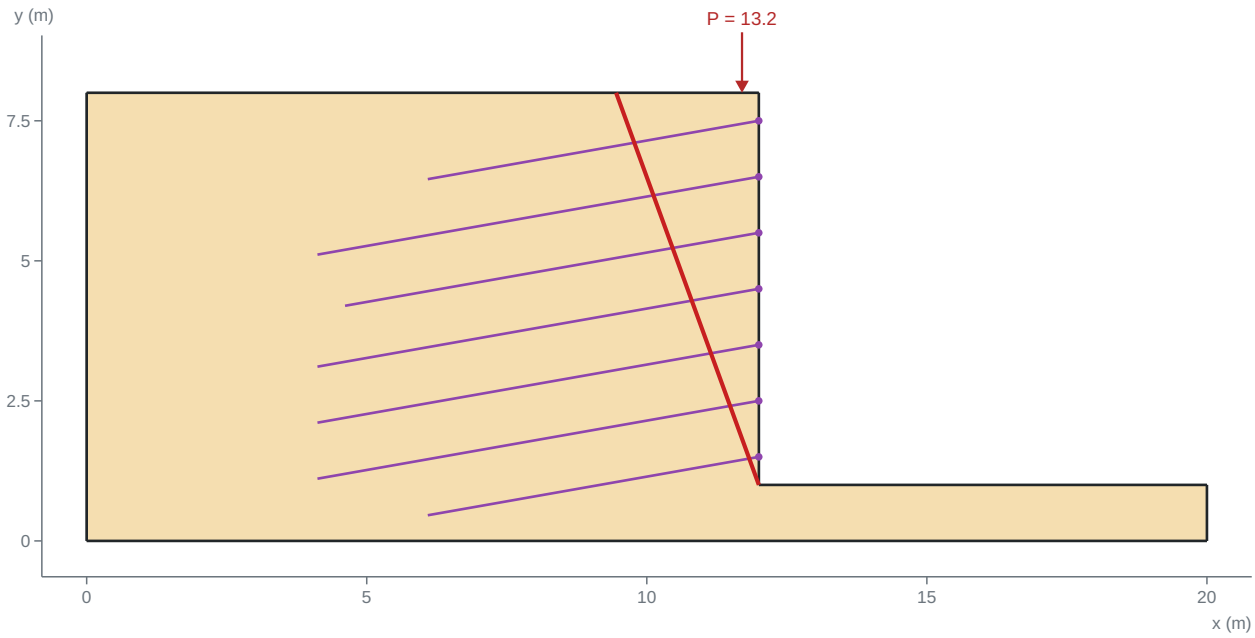



*Figure 148. V084f: analysed surface (Janbu corrected, FOS = 0.920).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fontainebleau Sand | Mohr-Coulomb | 20.00 | c' = 3.00 | 38.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.887 | force-equilibrium trial wedge | Sheahan & Ho (2003) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 0.923 | 0.920 |

*Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.*

## 3.149 V085 — Bradwell excavated slope, stiff-fissured London clay

*Skempton, A.W. & La Rochelle, P. (1965). The Bradwell slip: a short-term failure in London clay. Géotechnique 15(3), 221–242; Duncan, Wright & Brandon (2014), Soil Strength and Slope Stability, 2nd ed., §7.7.3.*

**Critical circle search · Units: imperial (ft, psf, pcf)**

Excavation in London clay (six sublayers with Su increasing with depth, S = f(datum)) under Marsh clay and a clay fill cracked to its full depth (11.4 ft); search for the deep failure, entering on the crest and leaving at the excavation toe.

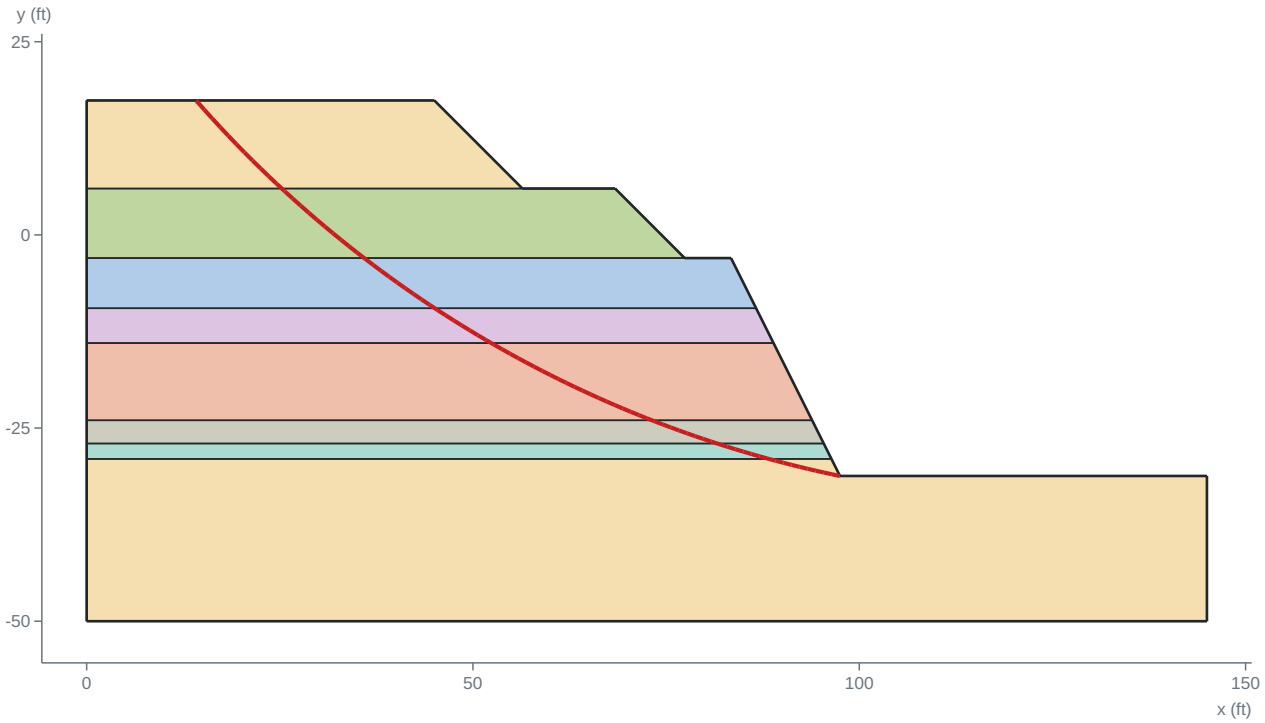



*Figure 149. V085: critical circle (Bishop simplified, FOS = 1.759).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Clay fill | Mohr-Coulomb | 110.0 | c' = 1.00 | 35.00° |
| Marsh clay | Mohr-Coulomb | 105.0 | c' = 300.0 | 0.00° |
| Brown London clay 1 | S = f(datum) | 120.0 | Su = 750.0 at y = -3.00, 90.00 psf/ft below | 0° |
| Brown London clay 2 | S = f(datum) | 120.0 | Su = 1335.0 at y = -9.50, 82.00 psf/ft below | 0° |
| Brown London clay 3 | S = f(datum) | 120.0 | Su = 1704.0 at y = -14.00, 53.00 psf/ft below | 0° |
| Brown London clay 4 | S = f(datum) | 120.0 | Su = 2234.0 at y = -24.00, 47.00 psf/ft below | 0° |
| Blue London clay 1 | S = f(datum) | 120.0 | Su = 2375.0 at y = -27.00, 47.00 psf/ft below | 0° |
| Blue London clay 2 | S = f(datum) | 120.0 | Su = 2469.0 at y = -29.00, 39.00 psf/ft below | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.76 | Bishop simplified | Duncan & Wright | 1.759 |
| 1.63 | Janbu simplified | Duncan & Wright | 1.627 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.762 | 1.759 |
| Spencer | 1.758 | 1.759 |
| Janbu corrected | 1.736 | 1.735 |
| Janbu simplified | 1.628 | 1.627 |

## 3.150 V086a — Tandjiria embankment, clay fill, geosynthetic T = 169 kN/m

*Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V086b clay-fill embankment with a 169 kN/m geosynthetic at its base (axial, active, fully anchored) and the water-filled 2 m crack, on the critical circle.

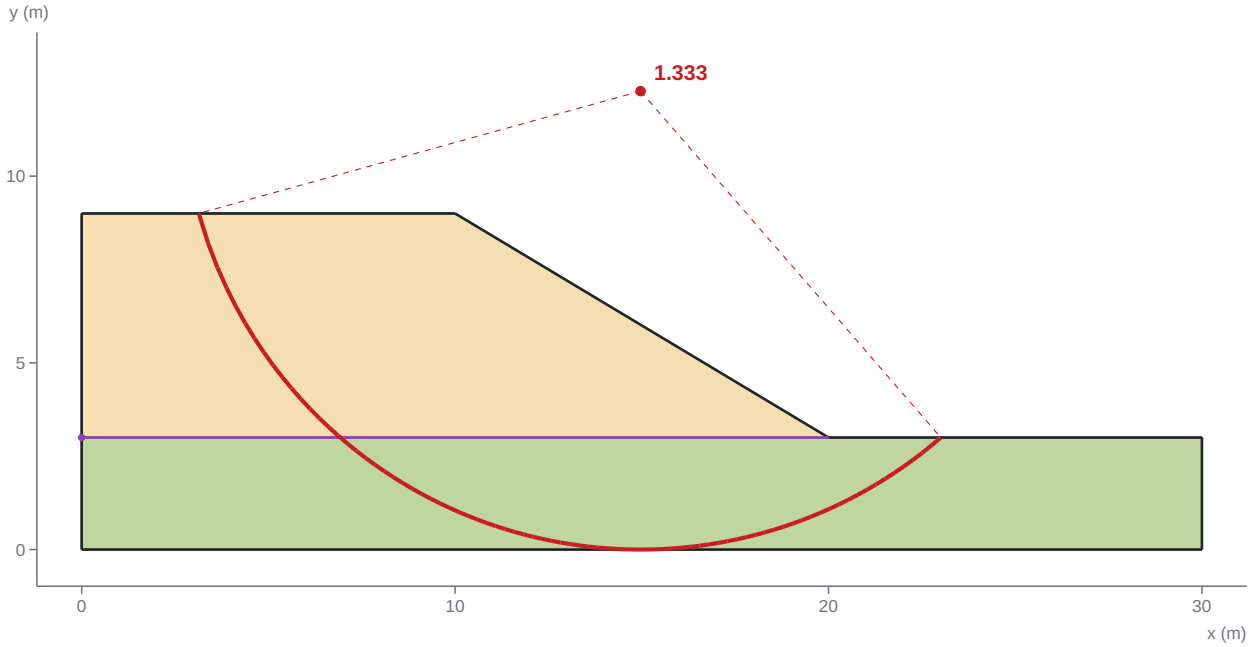



*Figure 150. V086a: analysed circle (Spencer, FOS = 1.333).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fill | Undrained (φ = 0) | 19.40 | Su = 20.00 | 0° |
| Soft Clay | Undrained (φ = 0) | 19.40 | Su = 20.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.35 | generalized method of slices, at T = 170.5 kN/m | Tandjiria, Low & Teh (2002) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.332–1.350 | 1.333 |

## 3.151 V086b — Tandjiria embankment, clay fill, water-filled crack

*Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Half-embankment (9 m crest, 10 m wide crest, 1V:1.67H face) of clay fill (Su = 20 kPa) on soft clay (Su = 20 kPa), γ = 19.4; 2 m water-filled tension crack, unreinforced; circular search. Also run with the automatic (Rankine) crack depth.

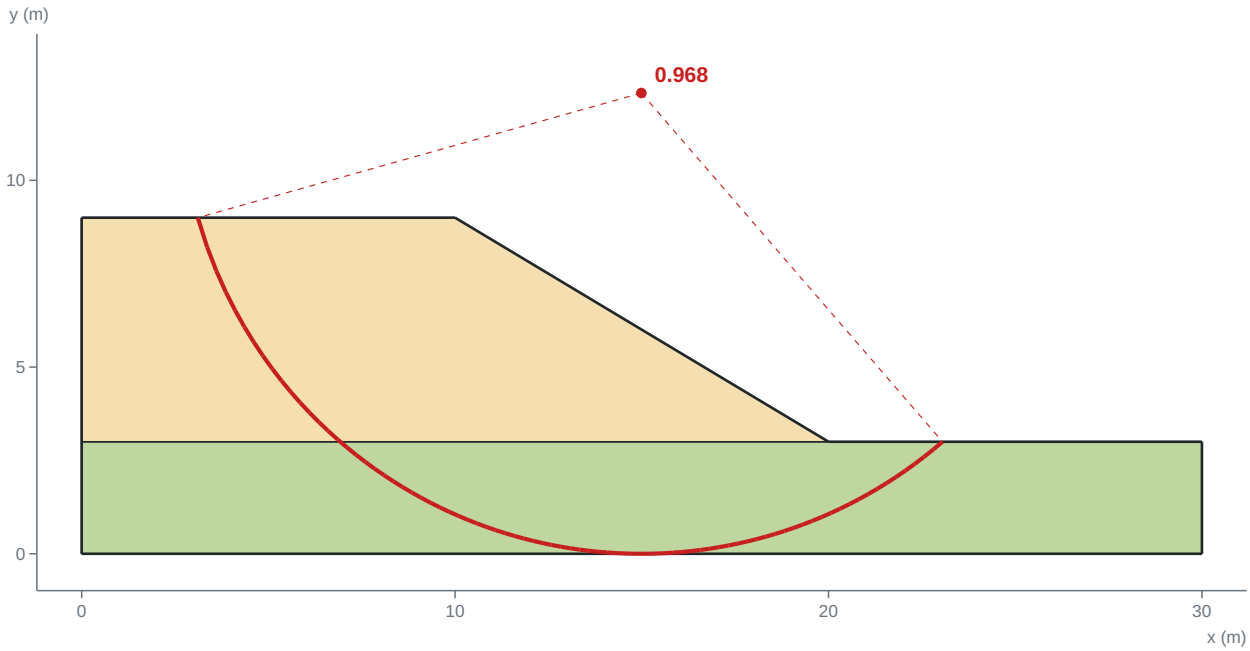



*Figure 151. V086b: critical circle (Spencer (2 m tension crack), FOS = 0.968).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fill | Undrained (φ = 0) | 19.40 | Su = 20.00 | 0° |
| Soft clay | Undrained (φ = 0) | 19.40 | Su = 20.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.981 | Spencer | Tandjiria, Low & Teh (2002) | 0.968 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer (2 m tension crack) | 0.975 | 0.968 |
| Spencer (tension crack of automatic depth) | 0.975 | 0.966 |
| Morgenstern-Price (half-sine) | 0.975 | 0.968 |

## 3.152 V086c — Tandjiria embankment, sand fill, dry crack

*Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V086 embankment as a sand fill (φ′ = 37°, γ = 17) on soft clay (Su = 20 kPa, γ = 20) with a dry 2 m tension crack, unreinforced; circular search.

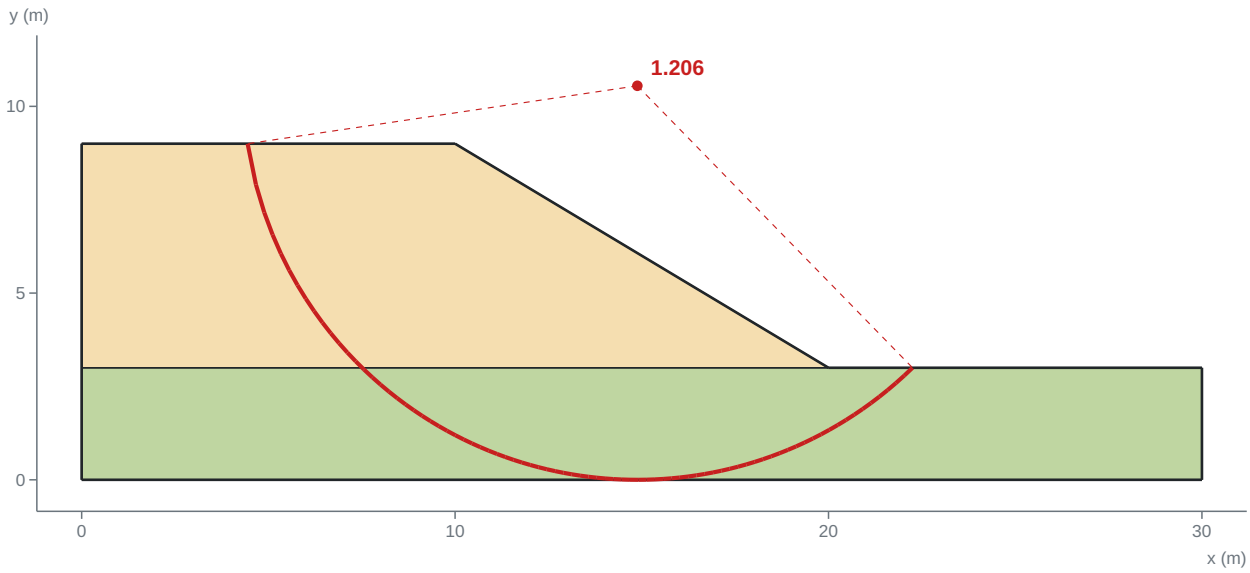



*Figure 152. V086c: critical circle (Spencer, FOS = 1.206).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fill | Mohr-Coulomb | 17.00 | c' = 0.00 | 37.00° |
| Soft clay | Undrained (φ = 0) | 20.00 | Su = 20.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.219 | Spencer | Tandjiria, Low & Teh (2002) | 1.206 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.209 | 1.206 |
| Morgenstern-Price (half-sine) | 1.218 | 1.215 |

## 3.153 V086d — Tandjiria embankment, clay fill, the paper's geometry

*Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443; problem 1, Table 2.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V086b clay-fill embankment as the paper gives it: a 30° face (p. 431, Fig. 4) and a water-filled tension crack 2c/γ = 2.06 m deep (Table 2 note), unreinforced; circular search.

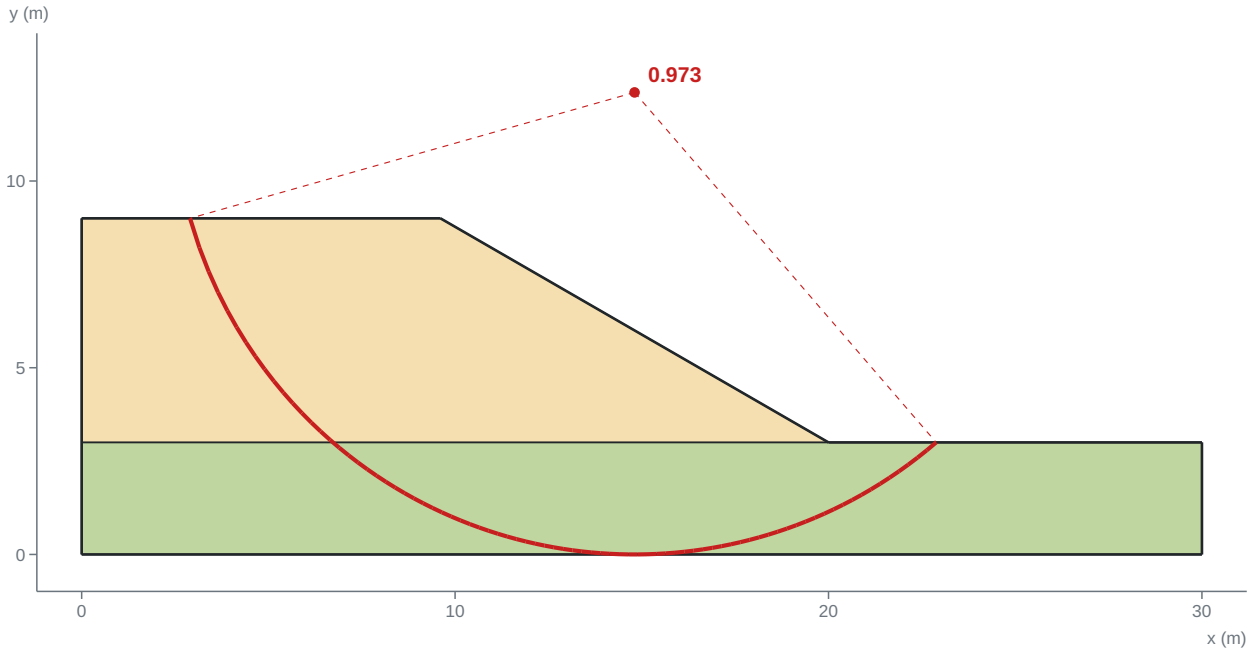



*Figure 153. V086d: critical circle (Spencer, FOS = 0.973).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fill | Undrained (φ = 0) | 19.40 | Su = 20.00 | 0° |
| Soft clay | Undrained (φ = 0) | 19.40 | Su = 20.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 0.981 | Spencer | Tandjiria, Low & Teh | 0.973 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | – | 0.973 |

*The paper uses the generalized method of slices without naming its side-force function; Spencer's method is cSlope's nearest. The soft clay's unit weight is not printed and is taken as the fill's (19.4).*

## 3.154 V086e — Tandjiria embankment, sand fill, the paper's geometry

*Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443; problem 2.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

The V086c sand-fill embankment as the paper gives it: a 30° face and no tension crack (the paper mentions one only for the clay fill), unreinforced; circular search.

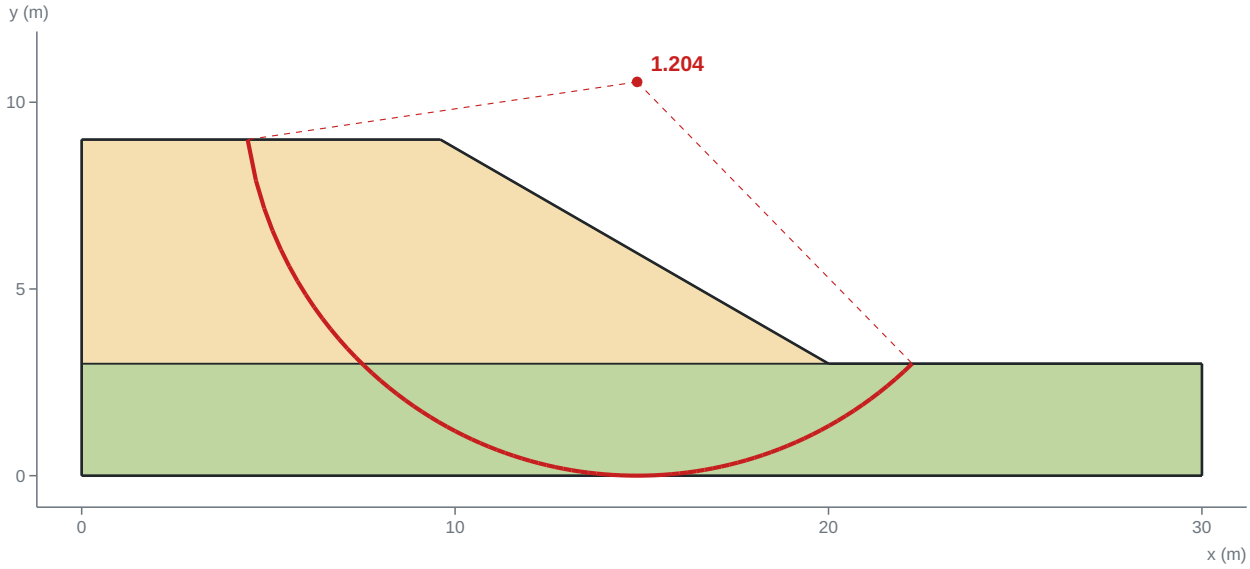



*Figure 154. V086e: critical circle (Spencer, FOS = 1.204).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Fill | Mohr-Coulomb | 17.00 | c' = 0.00 | 37.00° |
| Soft clay | Undrained (φ = 0) | 20.00 | Su = 20.00 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.219 | Spencer | Tandjiria, Low & Teh | 1.204 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | – | 1.204 |

*The paper uses the generalized method of slices without naming its side-force function; Spencer's method is cSlope's nearest. The soft clay's unit weight is not printed; 20 kN/m³ is assumed (17 gives 0.1 % less).*

## 3.155 V087 — USACE dam, chart-check properties (EM 1110-2-1902 fig. 4-3)

*USACE EM 1110-2-1902 (2003), fig. 4-3.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V088 dam family with single unit weights (embankment c = 200 psf, φ = 25°, γ = 115; sand 0/35/130; clay 0/27/115), slightly flatter faces and ponded water at el. 20 on both faces. the reference circle (−135, 169, R = 169.0), ending at the crest (−15, 50). Converted to SI.

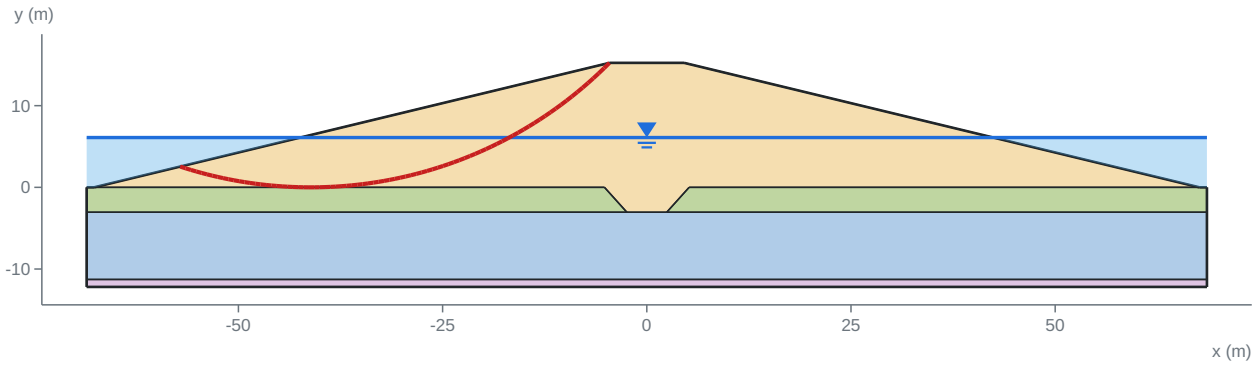



*Figure 155. V087: analysed circle (Janbu corrected, FOS = 2.248).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 18.07 | c' = 9.58 | 25.00° |
| Sand | Mohr-Coulomb | 20.42 | c' = 0.00 | 35.00° |
| Foundation clay | Mohr-Coulomb | 18.07 | c' = 0.00 | 27.00° |
| Rock | Mohr-Coulomb | 25.13 | c' = 0.00 | 45.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.30 | not stated | USACE EM 1110-2-1902 | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Janbu corrected | 2.290 | 2.248 |

## 3.156 V088 — USACE dam, end of construction (EM 1110-2-1902 fig. 4-1)

*USACE EM 1110-2-1902 (2003), fig. 4-1.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Symmetric 50 ft embankment (crest ±12 ft, toes ±206.18 ft) with c = 1000 psf, φ = 5°, a core trench through a 10 ft sand blanket over foundation clay (Su = 3000 psf) and rock; groundwater at the sand top, 7 ft dry crest crack; moist/saturated unit weights. Specified circle (102, 163, R = 163). Converted to SI.

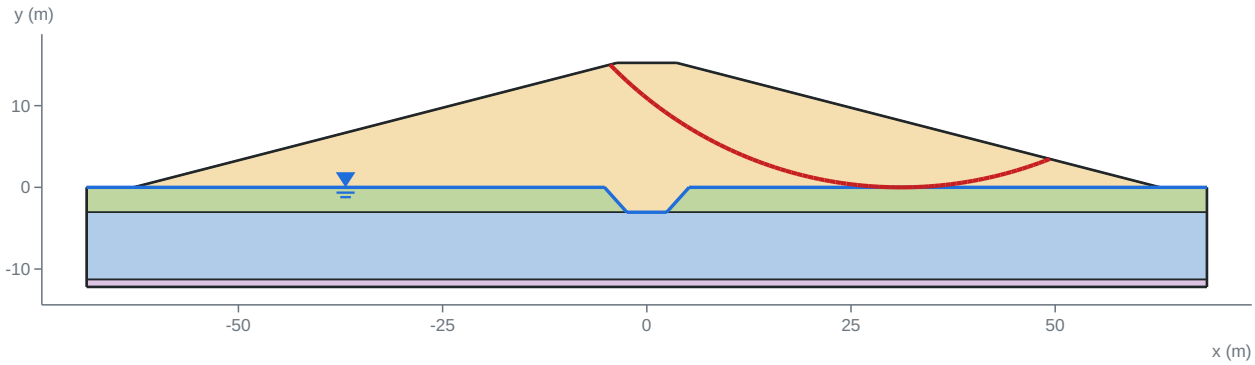



*Figure 156. V088: analysed circle (Spencer, FOS = 2.442).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 18.07 | c' = 47.88 | 5.00° |
| Sand | Mohr-Coulomb | 19.64 | c' = 0.00 | 35.00° |
| Foundation clay | Undrained (φ = 0) | 17.28 | Su = 143.6 | 0° |
| Rock | Mohr-Coulomb | 25.13 | c' = 0.00 | 45.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.44 | Spencer | USACE EM 1110-2-1902 | 2.442 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 2.445 | 2.442 |
| Bishop simplified | 2.447 | 2.442 |
| Janbu corrected | 2.430 | 2.444 |
| Morgenstern-Price (half-sine) | 2.447 | 2.442 |

## 3.157 V089 — USACE dam, upstream low pool (EM 1110-2-1902 fig. 4-2)

*USACE EM 1110-2-1902 (2003), fig. 4-2.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V088 dam family, 45 ft high (crest ±15 ft, toes ±200 ft), with drained strengths (embankment c = 100 psf, φ = 25°; sand 0/35; clay 0/28; rock 0/45; moist/saturated unit weights) and the pool at el. 20 standing on the upstream face. USACE's circle (−102, 163, R = 173), tangent to the clay top. Converted to SI.

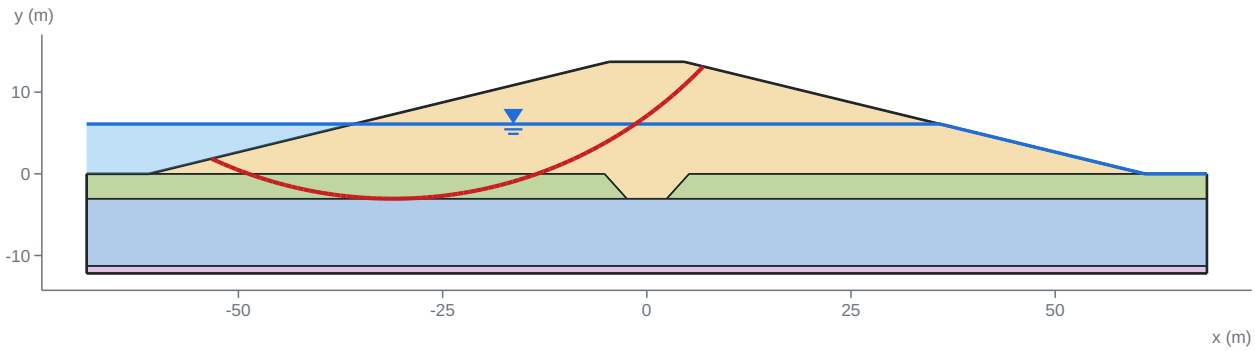



*Figure 157. V089: analysed circle (Bishop simplified, FOS = 2.698).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 18.07 | c' = 4.79 | 25.00° |
| Sand | Mohr-Coulomb | 19.64 | c' = 0.00 | 35.00° |
| Foundation clay | Mohr-Coulomb | 17.28 | c' = 0.00 | 28.00° |
| Rock | Mohr-Coulomb | 25.13 | c' = 0.00 | 45.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.71 | Bishop simplified | USACE EM 1110-2-1902 | 2.698 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.716 | 2.698 |
| Spencer | 2.736 | 2.721 |
| Janbu corrected | 2.650 | 2.647 |
| Morgenstern-Price (half-sine) | 2.744 | 2.727 |

## 3.158 V090 — USACE EM 1110-2-1902, example F-5

*US Army Corps of Engineers (2003). Slope Stability, EM 1110-2-1902, Appendix F.*

**Fixed circle · Units: imperial (ft, psf, pcf)**

Embankment (c = 1780 psf, φ = 5°) on a foundation (c = 1600 psf, φ = 2°), analysed with the circle given in the Corps of Engineers example.

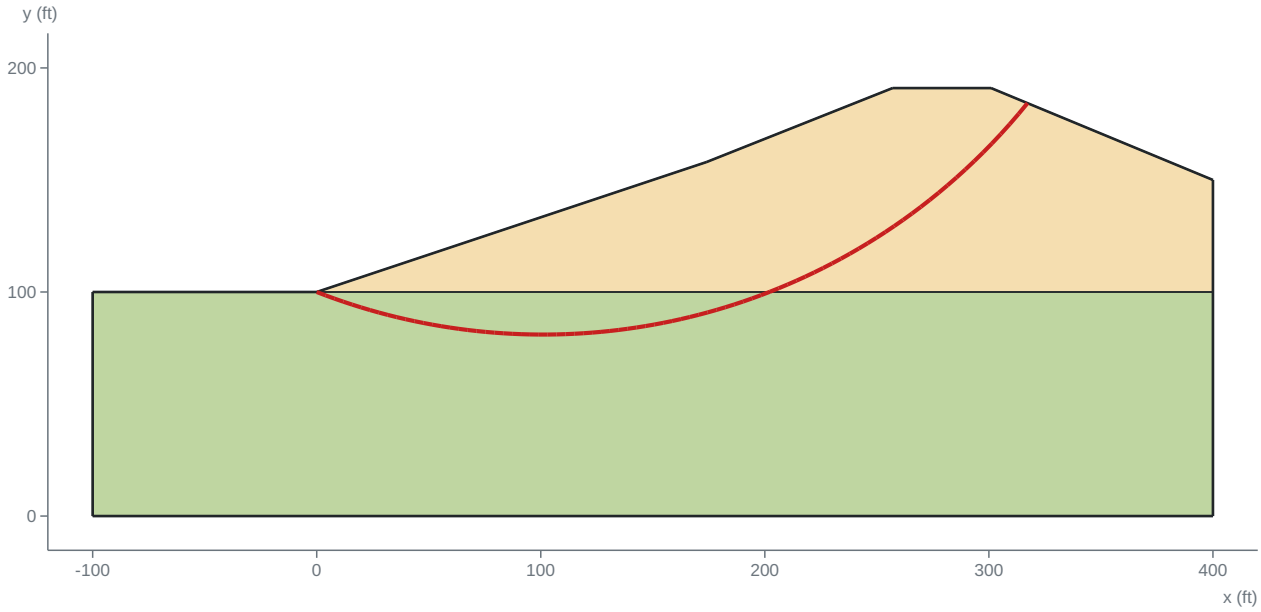



*Figure 158. V090: analysed circle (Bishop simplified, FOS = 1.320).*

| Material | Model | γ (pcf) | Strength (psf) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 135.0 | c' = 1780.0 | 5.00° |
| Foundation | Mohr-Coulomb | 127.0 | c' = 1600.0 | 2.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.33 | not stated | USACE EM 1110-2-1902 | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.332 | 1.320 |
| Spencer | 1.328 | 1.317 |
| Janbu corrected | 1.345 | 1.340 |
| Morgenstern-Price (half-sine) | 1.327 | 1.316 |

## 3.159 V091 — USACE example E-2 (fig. E-10), ponded water

*USACE EM 1110-2-1902 (2003), appendix E, example E-2, fig. E-10.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Undrained three-layer slope with 8 ft of water ponded against its toe (pool el. 0), on a commercial program's circle (48.262, 28, R = 47.683 ft; Fig. 68.2). Converted to SI.

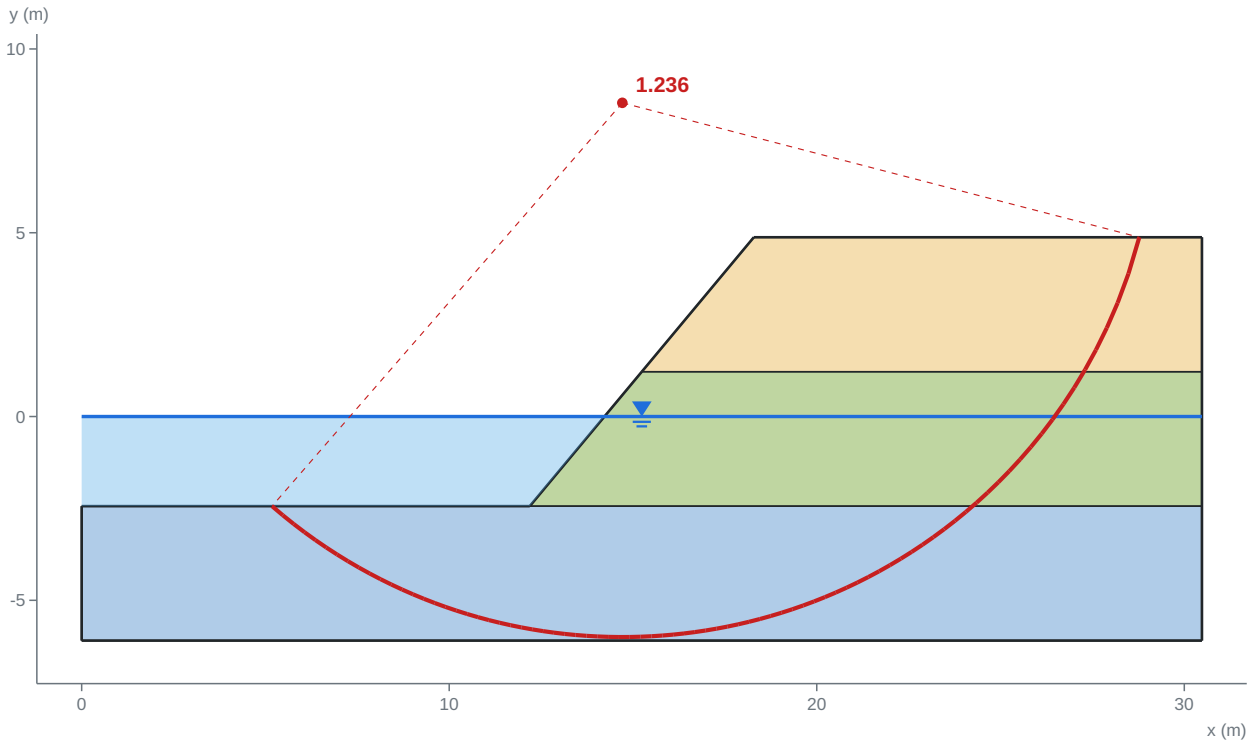



*Figure 159. V091: analysed circle (Bishop simplified, FOS = 1.236).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil 1 | Undrained (φ = 0) | 18.85 | Su = 28.73 | 0° |
| Soil 2 | Undrained (φ = 0) | 15.71 | Su = 19.15 | 0° |
| Soil 3 | Undrained (φ = 0) | 16.49 | Su = 23.94 | 0° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.24 | stability chart (example E-2) | USACE EM 1110-2-1902 | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.241 | 1.236 |
| Spencer | 1.241 | 1.236 |
| Morgenstern-Price (half-sine) | 1.244 | 1.236 |
| Janbu corrected | 1.385 | 1.390 |

*USACE's chart solution for this circle (example E-2, p. E-20) gives 1.24; the programs and cSlope give 1.235–1.244 on it. The manual's shallower circle tangent to el. −8 ft (example E-1) gives 1.14 and is the more critical of the two.*

## 3.160 V092 — USACE example F-6, steady-seepage dam

*USACE EM 1110-2-1902, example F-6.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

112 ft embankment on a granular foundation under steady seepage: piezometric line from the pool through the chimney drain to the tailwater, which stands on the downstream toe (free water). The upstream face rises from (30.2, 100) ft to the crest (60.8, 112). USACE's circle. Converted to SI.

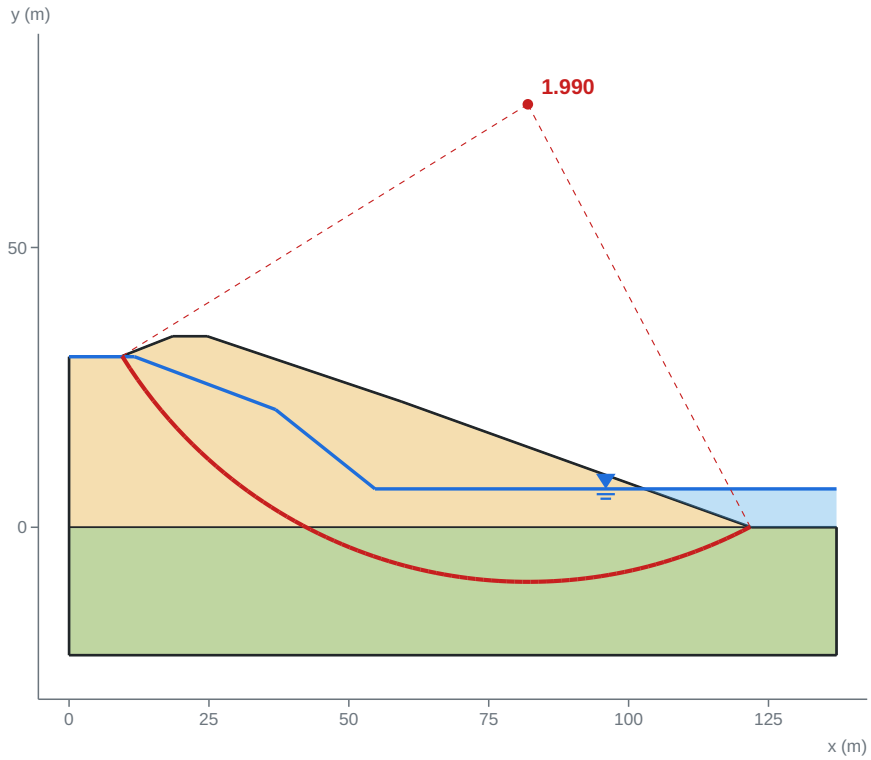



*Figure 160. V092: analysed circle (Bishop simplified, FOS = 1.990).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Embankment | Mohr-Coulomb | 20.42 | c' = 0.00 | 34.00° |
| Foundation | Mohr-Coulomb | 19.64 | c' = 0.00 | 35.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 2.01 | Simplified Bishop, nine-slice hand calculation on the manual's own piezometric line (example F-6) | USACE EM 1110-2-1902 | – |
| 2.07 | Modified Swedish (θ = 19.7°) on the manual's own piezometric line (example F-6) | USACE EM 1110-2-1902 | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 2.011 | 1.990 |
| Spencer | 2.026 | 2.005 |
| Janbu corrected | 1.830 | 1.823 |
| Morgenstern-Price (half-sine) | 2.027 | 2.005 |

*USACE's values are for the manual's own piezometric line, which its slice table (fig. F-17a) places lower than the line of the program version analysed here: the kink at about el. 53–59 ft instead of 69 ft, and the tailwater level reached about 18 ft further upstream. On the manual's line cSlope gives Bishop 2.05–2.06 and Spencer 2.06–2.08, depending on where the table's slices are placed: within 0.5 % of USACE's Modified Swedish value 2.07, and 2–2.6 % above its nine-slice Bishop hand calculation. The*

*manual's values are shown for reference and not compared with cSlope's analyses on the program's line.*

## 3.161 V093 — Yamagami & Ueta (1988)

*Yamagami & Ueta (1988).*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Homogeneous 2:1 slope, 5 m high; critical circle search.

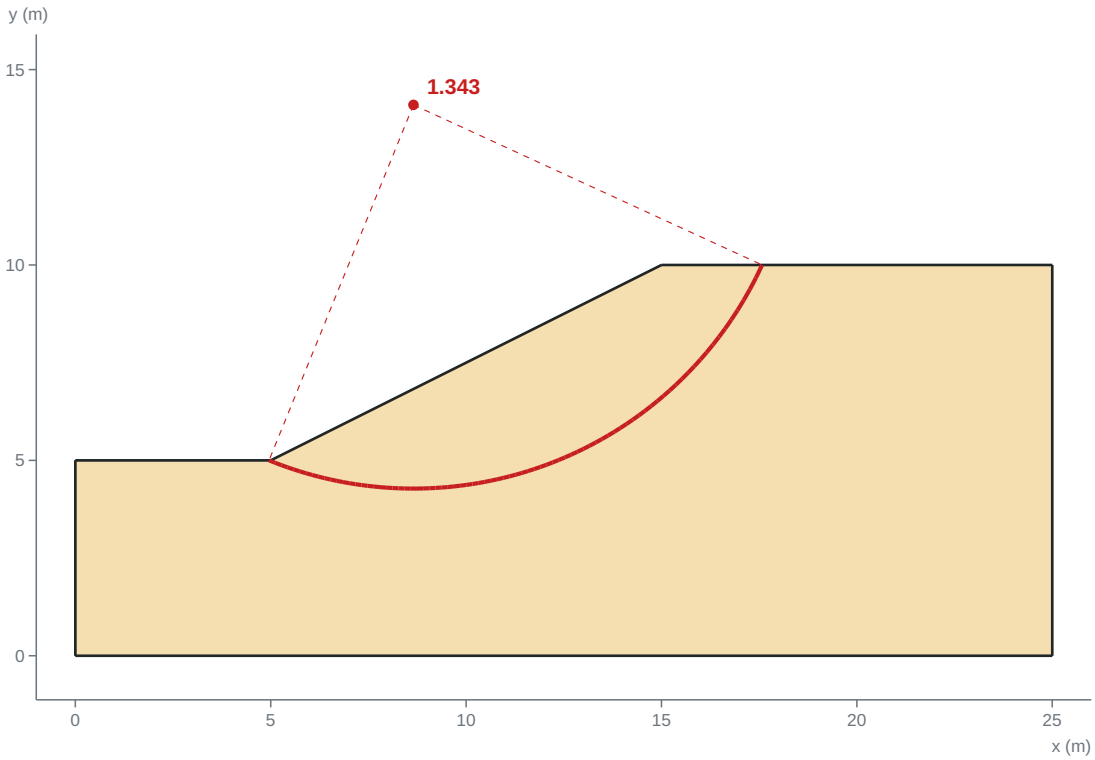



*Figure 161. V093: critical circle (Bishop simplified, FOS = 1.343).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 17.64 | c' = 9.80 | 10.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.348 | Bishop simplified | Yamagami & Ueta (1988) | 1.343 |
| 1.282 | Ordinary | Yamagami & Ueta (1988) | – |
| 1.185 | Janbu simplified, non-circular | Yamagami & Ueta (1988) | 1.176 |
| 1.339 | Spencer, non-circular | Yamagami & Ueta (1988) | 1.322 |
| 1.33 | Spencer, non-circular | Greco (1996) | 1.322 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.344 | 1.343 |
| Spencer, non-circular | 1.325 | 1.322 |
| Janbu simplified, non-circular | 1.178 | 1.176 |

## 3.162 V094 — Yamagami & Ueta (1988) four-layer slope, non-circular

*Greco (1996) example 4; Yamagami & Ueta (1988).*

**Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)**

Four-layer slope without water whose critical surface is non-circular; circular search followed by non-circular optimisation.

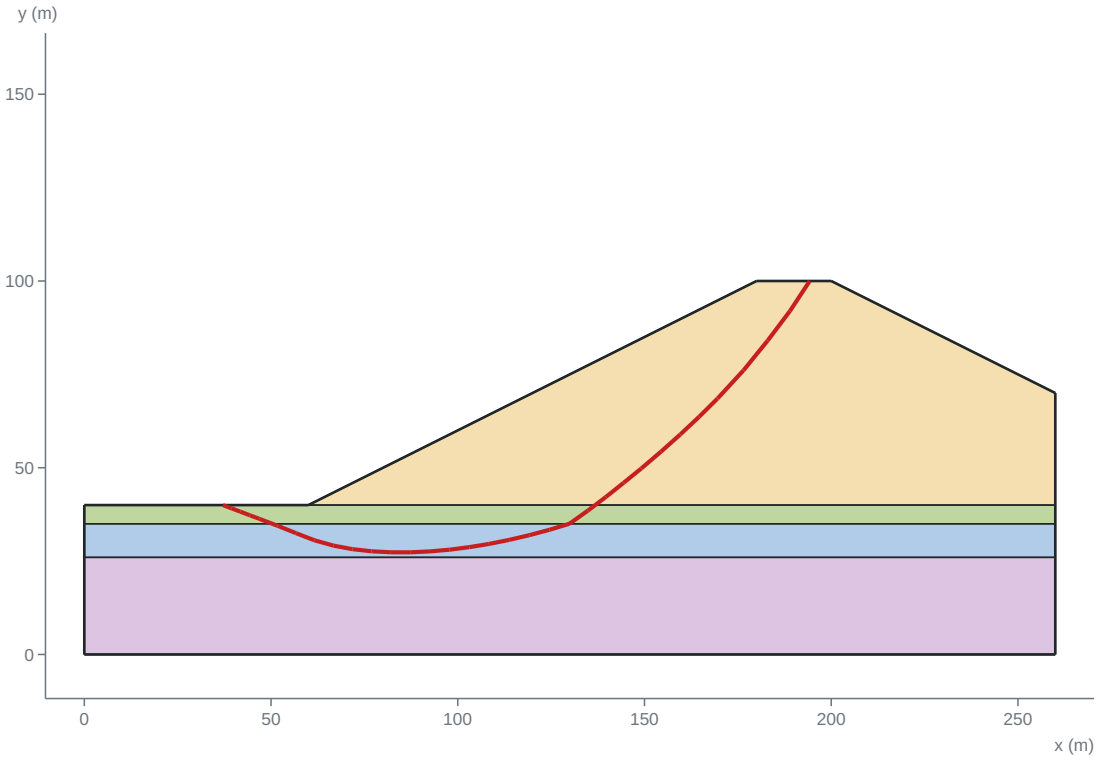



*Figure 162. V094: critical surface (Spencer, non-circular, FOS = 1.400).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Upper layer | Mohr-Coulomb | 20.38 | c' = 49.00 | 29.00° |
| Layer 2 | Mohr-Coulomb | 17.64 | c' = 0.00 | 30.00° |
| Layer 3 | Mohr-Coulomb | 20.38 | c' = 7.84 | 20.00° |
| Bottom layer | Mohr-Coulomb | 17.64 | c' = 0.00 | 30.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.40–1.42 | Spencer, non-circular | Greco (1996) | 1.400 |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, non-circular | 1.398 | 1.400 |
| Morgenstern-Price (half-sine), non-circular | 1.389 | 1.389 |

## 3.163 V095a — Yamagami micro-pile slope, no pile

*Yamagami et al. (2000).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Homogeneous 2:1 slope, 4 m high, without piles, on the published circle.

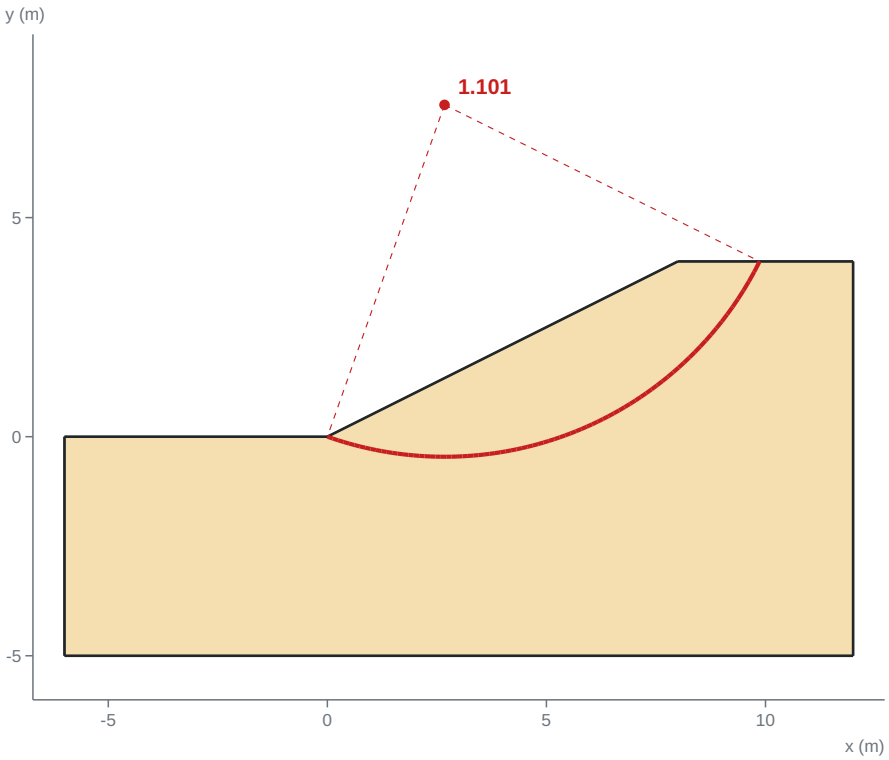



*Figure 163. V095a: analysed circle (Bishop simplified, FOS = 1.101).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Soil | Mohr-Coulomb | 15.68 | c' = 4.90 | 10.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.10 | not stated | Yamagami et al. (2000) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.102 | 1.101 |

## 3.164 V095c — Homogeneous slope with a micro-pile row

*Yamagami et al. (2000).*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V095b with a micro-pile row (1 m spacing) whose shear strength of 10.7 kN acts as a passive force tangent to the slip surface, on the reference with-pile circle (2.674, 7.376, R = 8.102).

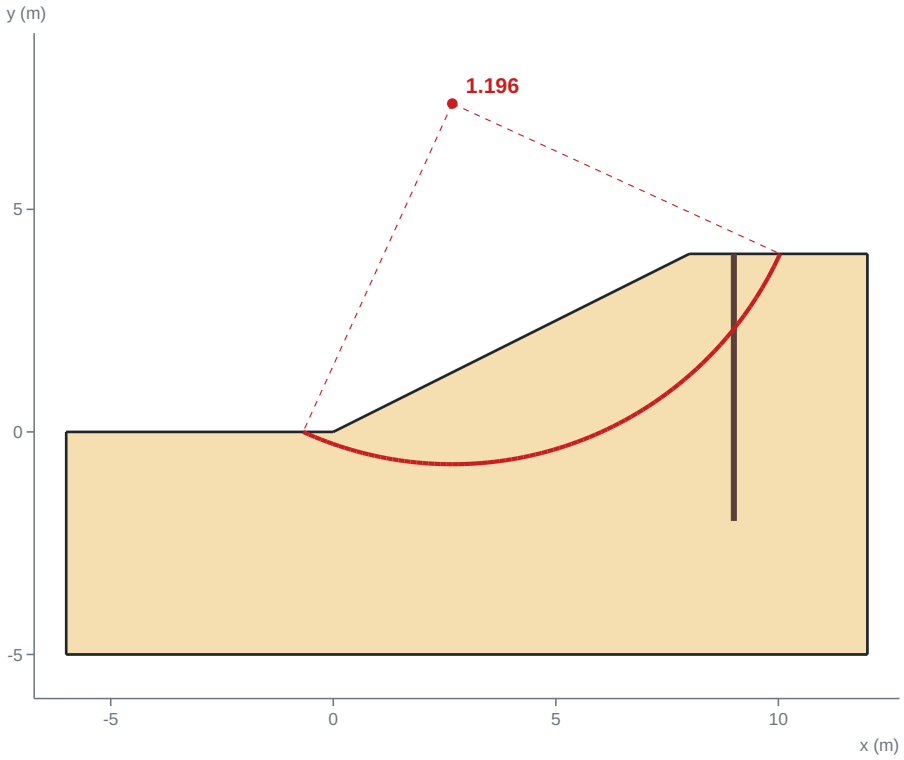



*Figure 164. V095c: analysed circle (Bishop simplified, FOS = 1.196).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Material 1 | Mohr-Coulomb | 15.68 | c' = 4.90 | 10.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.20 | not stated | Yamagami et al. (2000) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.193 | 1.196 |

## 3.165 V096a — Zhu & Lee heterogeneous slope, dry, deep circle

*Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

Four horizontal layers under a two-bench slope with a dry tension crack 1.015 m deep at the crest (Fig. 52.1; the depth of a commercial program's surface end below the crest), dry; the circular search lands in the deep family (surface 3).

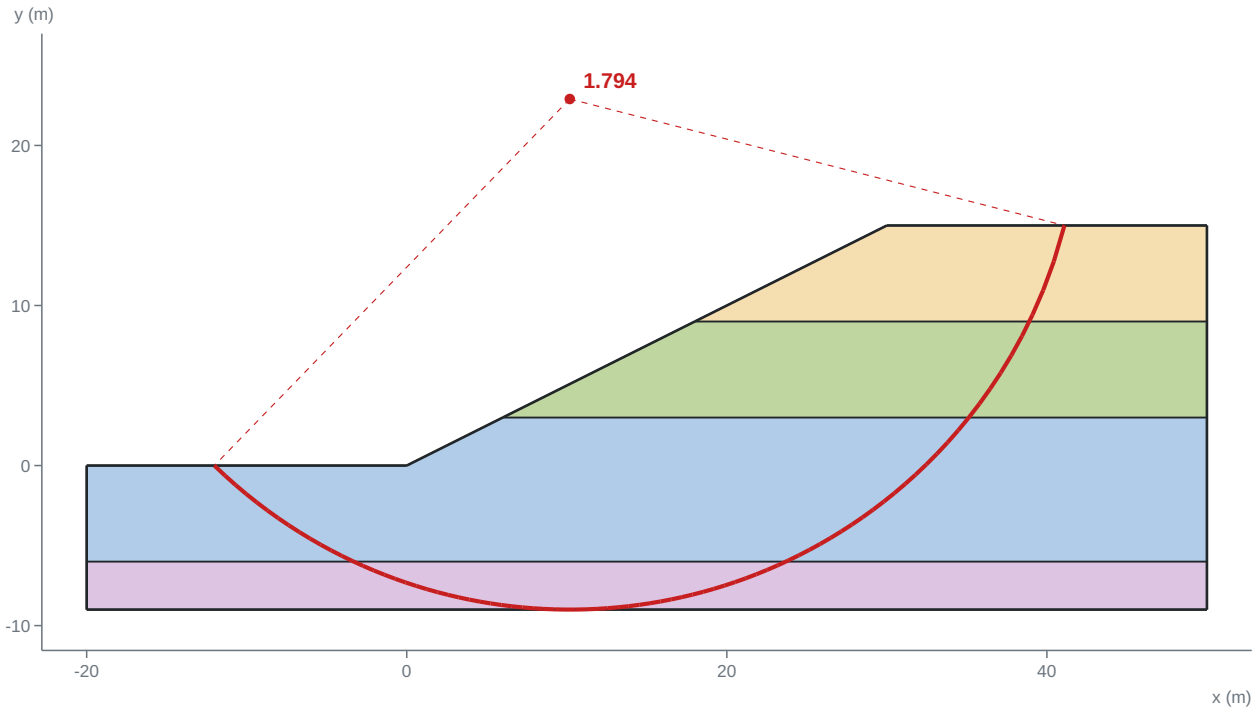



*Figure 165. V096a: critical circle (Bishop simplified, FOS = 1.794).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 (top) | Mohr-Coulomb | 18.80 | c' = 20.00 | 18.00° |
| Layer 2 | Mohr-Coulomb | 18.50 | c' = 40.00 | 22.00° |
| Layer 3 | Mohr-Coulomb | 18.40 | c' = 25.00 | 26.00° |
| Layer 4 (bottom) | Mohr-Coulomb | 18.00 | c' = 10.00 | 12.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.804 | 1.794 |
| Spencer | 1.804 | 1.794 |
| Morgenstern-Price (half-sine) | 1.790 | 1.780 |

*The circle and the 1.015 m crack follow a program's version of the problem. The paper gives its surface 3 only in a figure, and it enters the crest about 3 m nearer the slope than the program's circle; the paper's values for it are checked on the paper's own circle in V096d (dry) and V096e (with the water table).*

## 3.166 V096b — Zhu & Lee heterogeneous slope with a water table, deep circle

*Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.*

**Critical circle search · Units: SI (m, kPa, kN/m³)**

As V096a (dry 1.015 m crack) with the water table of Table 52.2.

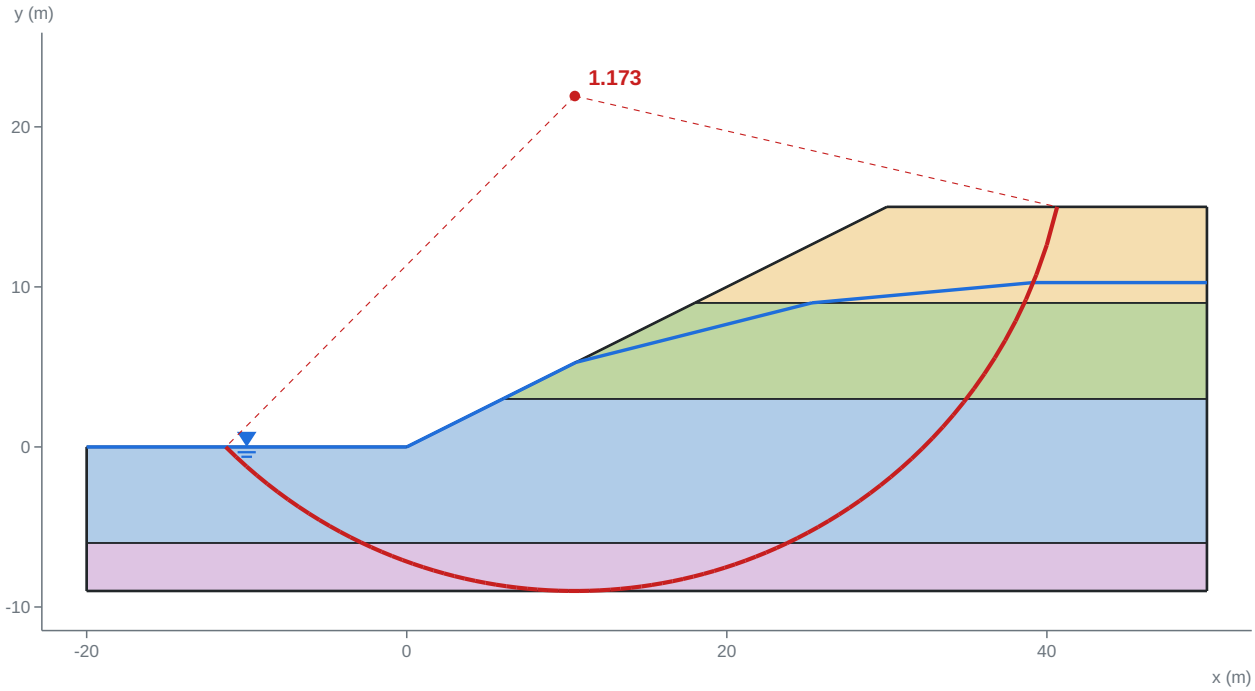



*Figure 166. V096b: critical circle (Bishop simplified, FOS = 1.173).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 (top) | Mohr-Coulomb | 18.80 | c' = 20.00 | 18.00° |
| Layer 2 | Mohr-Coulomb | 18.50 | c' = 40.00 | 22.00° |
| Layer 3 | Mohr-Coulomb | 18.40 | c' = 25.00 | 26.00° |
| Layer 4 (bottom) | Mohr-Coulomb | 18.00 | c' = 10.00 | 12.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.176 | 1.173 |
| Spencer | 1.189 | 1.187 |
| Morgenstern-Price (half-sine) | 1.174 | 1.172 |

*As V096a: the paper's values for its surface 3 with the water table are checked on the paper's own circle in V096e.*

## 3.167 V096c — Zhu & Lee slope, surface 3 (the reference circle), dry, 1 m crack

*Zhu & Lee (2002), Int. J. Numer. Anal. Meth. Geomech. 26(15). Verification manual of a commercial slope stability program.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V096a slope with the dry 1.015 m crack on the reference surface 3 circle (10.911, 23.208, R = 32.106; Fig. 52.4).

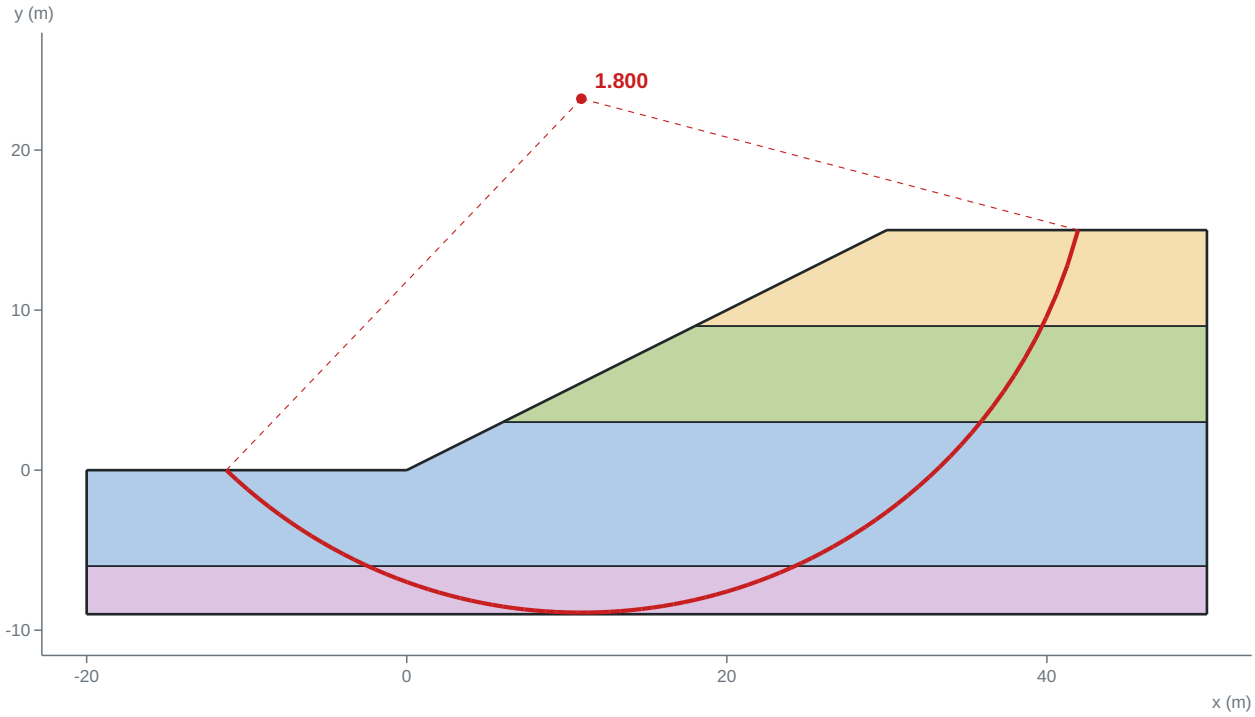



*Figure 167. V096c: analysed circle (Spencer, FOS = 1.800).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 | Mohr-Coulomb | 18.80 | c' = 20.00 | 18.00° |
| Layer 2 | Mohr-Coulomb | 18.50 | c' = 40.00 | 22.00° |
| Layer 3 | Mohr-Coulomb | 18.40 | c' = 25.00 | 26.00° |
| Layer 4 | Mohr-Coulomb | 18.00 | c' = 10.00 | 12.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | 1.804 | 1.800 |
| Bishop simplified | 1.804 | 1.799 |

*The program's reference circle enters the crest at x = 41.9 m, about 3 m further from the slope than the paper's surface 3 as drawn in its fig. 3; the paper's values are checked on the paper's circle in V096d and V096e.*

## 3.168 V096d — Zhu & Lee slope, the paper's surface 3, dry

*Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

The V096a slope, dry, no tension crack, on the paper's surface 3 digitised from its fig. 3: least-squares circle (9.7, 21.3, R = 29.8) through the drawn surface, leaving the lower bench at x = −11.1 and entering the crest at x = 38.8 (±0.3 m).

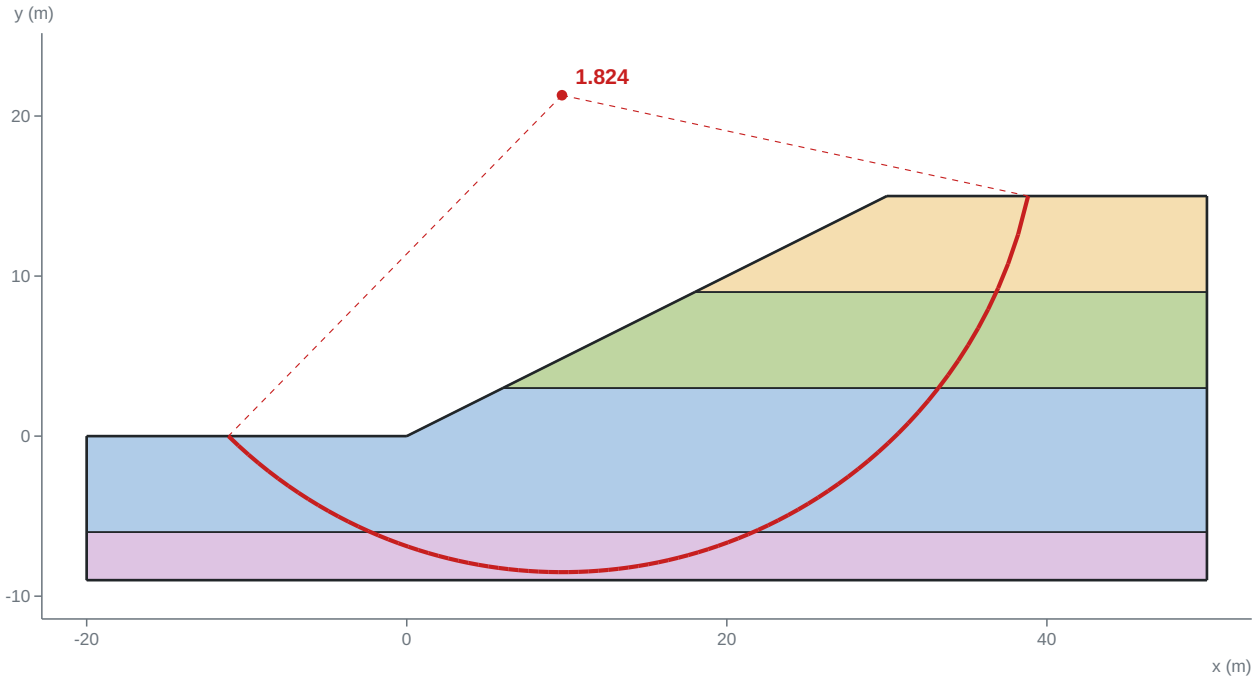



*Figure 168. V096d: analysed circle (Spencer, FOS = 1.824).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 | Mohr-Coulomb | 18.80 | c' = 20.00 | 18.00° |
| Layer 2 | Mohr-Coulomb | 18.50 | c' = 40.00 | 22.00° |
| Layer 3 | Mohr-Coulomb | 18.40 | c' = 25.00 | 26.00° |
| Layer 4 | Mohr-Coulomb | 18.00 | c' = 10.00 | 12.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.836 | Spencer | Zhu & Lee | 1.824 |
| 1.823 | Morgenstern-Price (half-sine) | Zhu & Lee | 1.808 |
| 1.744 | explicit solution (the paper's method) | Zhu & Lee (2002) | – |
| 1.429 | Bishop simplified | Zhu & Lee (2002) (disputed: see the note) | 1.823 |
| 1.229 | Ordinary | Zhu & Lee (2002) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | – | 1.824 |
| Morgenstern-Price (half-sine) | – | 1.808 |
| Bishop simplified | – | 1.823 |

*The paper gives surface 3 only in a figure. On the circle fitted to it cSlope comes within 1 % of the paper's Spencer and Morgenstern-Price values (1.836, 1.823), against 2 % on the program's circle of V096c; moving the fitted centre or radius by 0.3 m*

*changes the result by up to 1 %. The paper's simplified Bishop value 1.429 is 22 % below its own rigorous values for the same circle, where cSlope's Bishop gives 1.823 as expected on a circle; it is shown but not counted in the error statistics.*

## 3.169 V096e — Zhu & Lee slope, the paper's surface 3, with the water table

*Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V096d with the water table of the paper's fig. 3 (the points of V096b, which match the drawn line within 0.3 m).

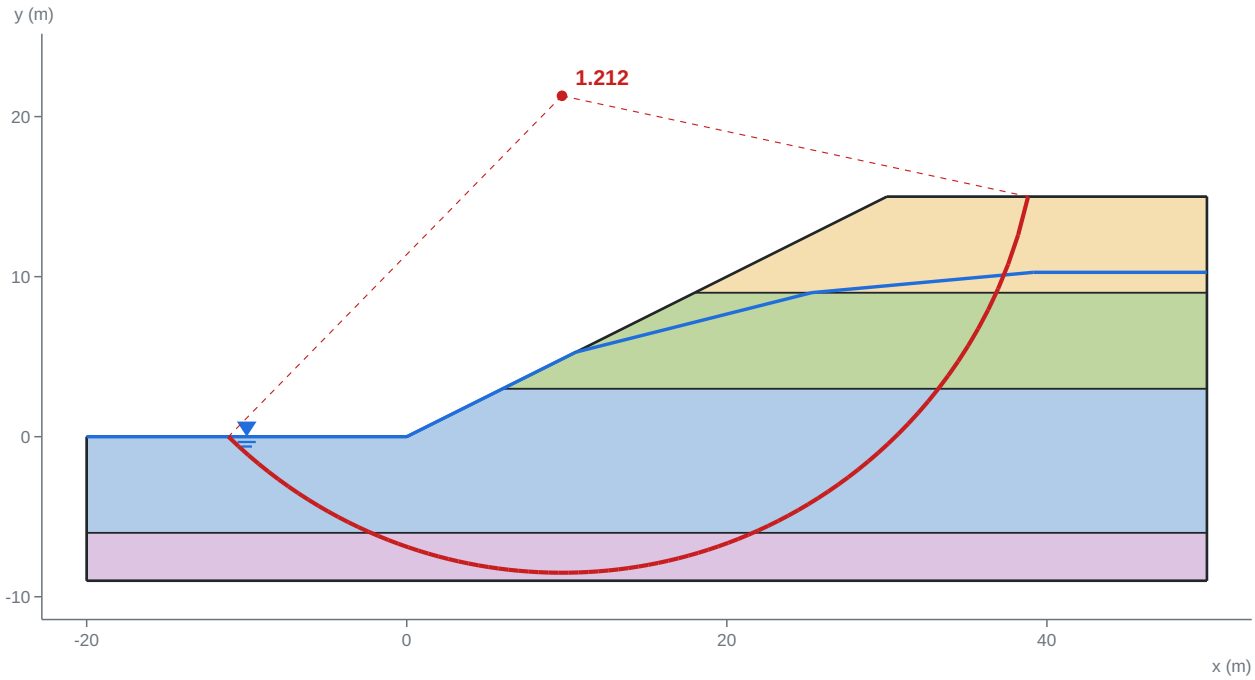



*Figure 169. V096e: analysed circle (Spencer, FOS = 1.212).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Layer 1 | Mohr-Coulomb | 18.80 | c' = 20.00 | 18.00° |
| Layer 2 | Mohr-Coulomb | 18.50 | c' = 40.00 | 22.00° |
| Layer 3 | Mohr-Coulomb | 18.40 | c' = 25.00 | 26.00° |
| Layer 4 | Mohr-Coulomb | 18.00 | c' = 10.00 | 12.00° |

| Published value | Method | Source | cSlope |
|---|---|---|---|
| 1.211 | Spencer | Zhu & Lee | 1.212 |
| 1.197 | Morgenstern-Price (half-sine) | Zhu & Lee | 1.197 |
| 1.166 | explicit solution (the paper's method) | Zhu & Lee (2002) | – |
| 1.079 | Bishop simplified | Zhu & Lee (2002) (disputed: see the note) | 1.199 |
| 0.922 | Ordinary | Zhu & Lee (2002) | – |

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer | – | 1.212 |
| Morgenstern-Price (half-sine) | – | 1.197 |
| Bishop simplified | – | 1.199 |

*On the circle fitted to the paper's figure cSlope reproduces the paper's Spencer and Morgenstern-Price values (1.211, 1.197) within 0.1 %. The paper's simplified Bishop value 1.079 lies 11 % below its own rigorous values for the same circle, as its dry value does; cSlope's Bishop gives 1.199. It is shown but not counted in the error statistics.*

## 3.170 V097a — Eurocode 7 cutting with surcharge

*Verification manual of a commercial slope stability program.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

Cutting in clay with an inclined piezometric line and a 35 kPa surcharge on the crest, analysed on the published circle.

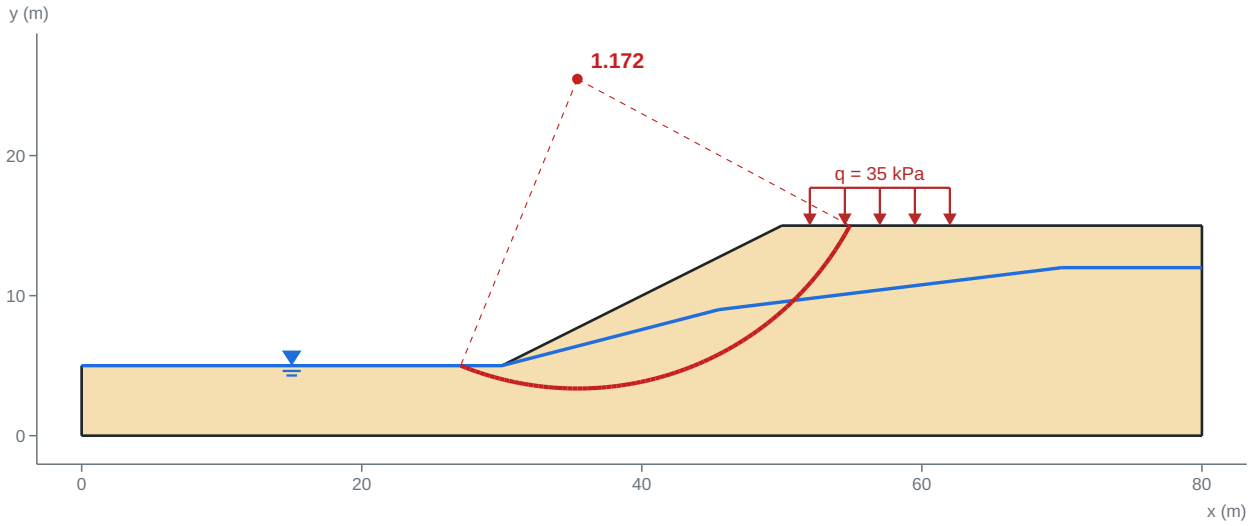



*Figure 170. V097a: analysed circle (Bishop simplified, FOS = 1.172).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Clay | Mohr-Coulomb | 20.00 | c' = 8.00 | 23.04° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.173 | 1.172 |

*The problem comes from a program manual only. The manual cites 1.193 (Bishop) from the textbook example it is based on, but its water table is approximate because the book gives no coordinates, so that value is not used as a check.*

## 3.171 V097b — Eurocode 7 cutting, partial factors applied by cSlope

*Verification manual of a commercial slope stability program.*

**Fixed circle · Units: SI (m, kPa, kN/m³)**

As V097a, but with the characteristic strength c' = 10 kPa, φ' = 28° and Design Approach 3 material factors (set M2: γc' = γφ' = 1.25) applied by the analysis; the surcharge is permanent (factor 1.0). The FOS is the over-design factor on design values.

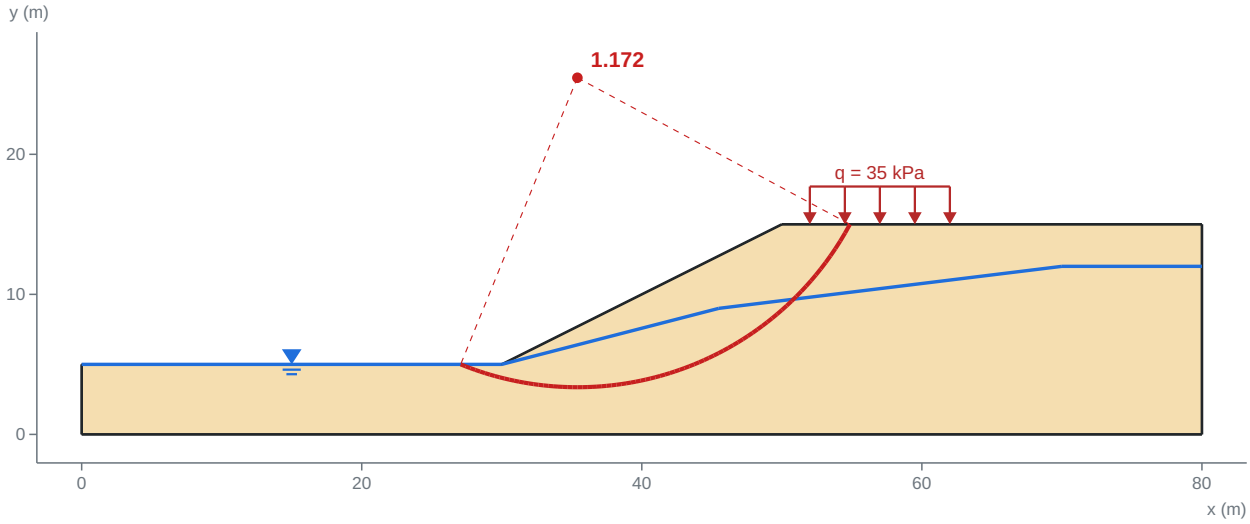



*Figure 171. V097b: analysed circle (Bishop simplified, FOS = 1.172).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| Clay | Mohr-Coulomb | 20.00 | c' = 10.00 | 28.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Bishop simplified | 1.173 | 1.172 |
| Spencer | 1.174 | 1.174 |

*The problem comes from a program manual only. The manual cites 1.193 (Bishop) from the textbook example it is based on, but its water table is approximate because the book gives no coordinates, so that value is not used as a check.*

## 3.172 V098 — Example slope without seismic load

*Verification manual of a commercial slope stability program.*

**Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)**

10 m, 2:1 slope in three layers over a horizontal base, dry, no seismic load; Spencer search, circular and non-circular.

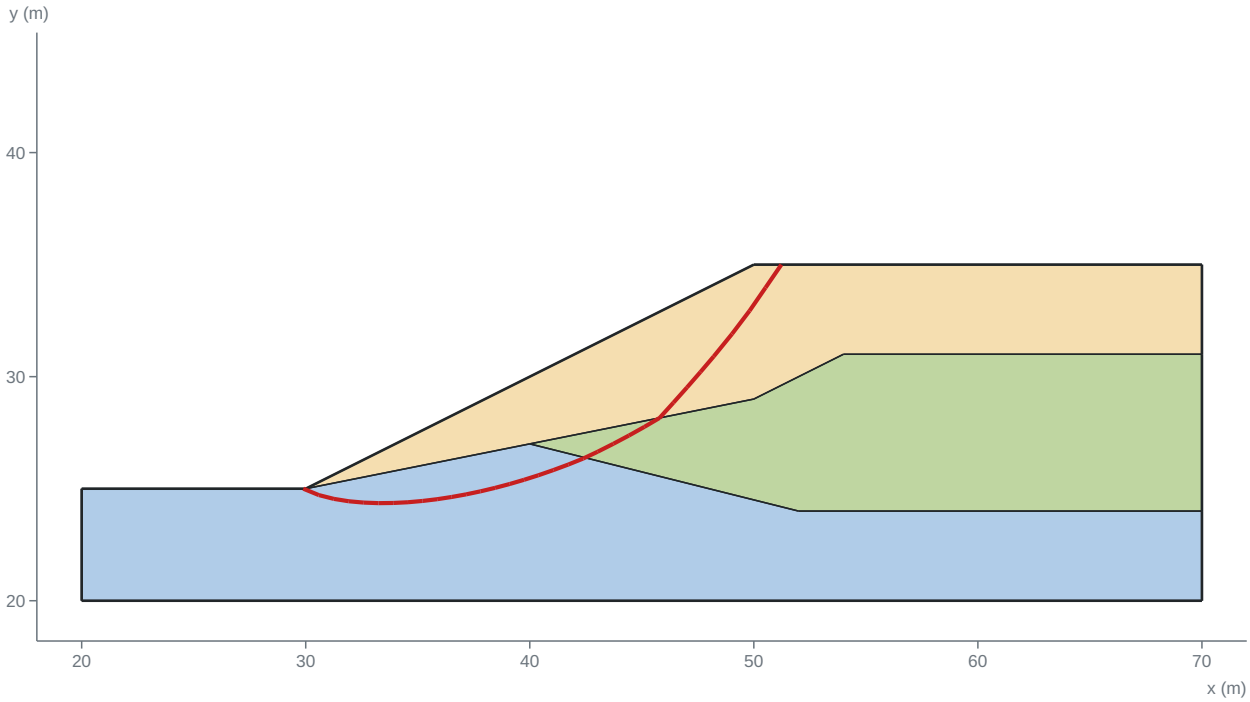



*Figure 172. V098: critical surface (Spencer, non-circular, FOS = 1.357).*

| Material | Model | γ (kN/m³) | Strength (kPa) | φ |
|---|---|---|---|---|
| soil1 | Mohr-Coulomb | 19.50 | c' = 0.00 | 38.00° |
| soil2 | Mohr-Coulomb | 19.50 | c' = 5.30 | 23.00° |
| soil3 | Mohr-Coulomb | 19.50 | c' = 7.20 | 20.00° |

*The source publishes no factor of safety for this problem.*

| cSlope analysis | Other programs | cSlope |
|---|---|---|
| Spencer, non-circular | 1.360 | 1.357 |
| Spencer | 1.372 | 1.372 |

*The problem comes from a program manual only; no independent value is published.*

# 4 Strength reduction by finite elements

cSlope's finite element analysis reduces c and tan φ by a factor F until the elastic–perfectly plastic slope can no longer be brought into equilibrium. The mesh is of six-node triangles, refined adaptively where the plastic dissipation at failure is largest until F changes by less than 0.01 or twice. Each problem is run with its source's flow rule (Davis B, associated, or non-associated with ψ = 0); the source's elastic parameters do not change F appreciably. The finite element factor is compared with the source's finite element, finite difference, limit analysis or theoretical value: within 2 % of a theoretical value or a pair of bounds, and within 5 % of a published numerical value, which depends on the mesh, the elements and the failure criterion.

| Problem | Published by | Published | Flow rule | Elements: F | cSlope |
|---|---|---|---|---|---|
| V001 | finite-element strength reduction (verification manual, problem 10), another program's manual | 1.4 | non-associated | 2811: 1.406, 4033: 1.402 | 1.402 |
| V002 | limit analysis, lower and upper bound, Kim et al. (2002) | 0.40–0.45 | associated | 2491: 0.425, 2573: 0.425 | 0.425 |
| V003 | finite-element strength reduction (verification manual, problem 12), another program's manual | 1.09 | non-associated | 1403: 1.107, 1846: 1.107 | 1.107 |
| V012 | finite-element strength reduction (verification manual, problem 24), another program's manual | 0.95 | non-associated | 6162: 0.910, 7014: 0.928, 8724: 0.928 | 0.928 |
| V013a | finite-element strength reduction (3D, without piles), Cai & Ugai (2000) | 1.14 | non-associated | 1366: 1.141, 2149: 1.130, 3035: 1.133 | 1.133 |
| V018 | plasticity theory (Prandtl mechanism), Chen & Shao (1988) | 1.0 | davis-b | 1543: 1.000, 1751: 1.000 | 1.000 |
| V019 | finite-element strength reduction, Cheng et al. (2007) | 1.47 | associated | 1832: 1.426, 1946: 1.408, 2157: 1.404 | 1.404 |
| V020 | finite-element strength reduction, Cheng et al. (2007) | 1.327 | associated | 1476: 1.377, 1532: 1.356, 1643: 1.353 | 1.353 |
| V026 | finite-element strength reduction, Duncan, Wright & Brandon (2014) | 1.19 | non-associated | 3026: 1.189, 3566: 1.186 | 1.186 |
| V038b | finite-element strength reduction, piezometric line, Duncan, Wright & Brandon (2014) | 1.07 | non-associated | 2978: 1.084, 3418: 1.074, 4174: 1.065 | 1.065 |
| V039 | finite-element strength reduction, Duncan, Wright & Brandon (2014) | 1.26 | non-associated | 5324: 1.240, 6292: 1.240 | 1.240 |
| V044 | finite-element strength reduction (verification manual, problem 17), another program's manual | 2.0 | non-associated | 2133: 1.985, 3190: 1.980 | 1.980 |
| V045 | finite-element strength reduction (verification manual, problem 17), another program's manual | 1.78 | non-associated | 2140: 1.785, 3066: 1.785 | 1.785 |
| V046a | finite-element strength reduction (verification manual, problem 18), another program's manual | 1.34 | non-associated | 2802: 1.340, 3113: 1.337 | 1.337 |
| V046b | finite-element strength reduction (verification manual, problem 18), another program's manual | 1.13 | non-associated | 2922: 1.183, 3425: 1.186 | 1.186 |
| V047a | finite-element strength reduction (verification manual, problem 1), another program's manual | 0.99 | non-associated | 2555: 0.971, 3807: 0.951, 5401: 0.956 | 0.956 |
| V049 | finite-element strength reduction (verification manual, problem 2), another program's manual | 1.36 | non-associated | 4522: 1.335, 6232: 1.335 | 1.335 |
| V050b | finite-element strength reduction (verification manual, problem 5), another program's manual | 1.26 | non-associated | 2530: 1.283, 2855: 1.283 | 1.283 |
| V052a | finite-element strength reduction (verification manual, problem 4), another program's manual | 1.88 | non-associated | 3092: 1.780, 3417: 1.789 | 1.789 |
| V099 | finite-element strength reduction with finite element seepage (verification manual, problem 7), another program's manual | 1.48 | non-associated | 4354: 1.551, 4935: 1.543 | 1.543 |
| V053 | finite-element strength reduction (verification manual, problem 16), another program's manual | 1.02 | non-associated | 3380: 0.979, 3688: 0.969 | 0.969 |

| Problem | Published by | Published | Flow rule | Elements: F | cSlope |
|---|---|---|---|---|---|
| V055 | finite elements, Griffiths & Lane (1999) | 1.4 | non-associated | 2200: 1.356, 3288: 1.346, 5077: 1.342 | 1.342 |
| V056 | finite elements (read from fig. 7), Griffiths & Lane (1999) | 0.6 | non-associated | 2663: 0.538, 2845: 0.517, 3293: 0.509 | 0.509 |
| V057a | finite elements, Griffiths & Lane (1999) | 1.45 | non-associated | 2296: 1.455, 2836: 1.451 | 1.451 |
| V057b | finite elements, Griffiths & Lane (1999) | 2.03 | non-associated | 2296: 2.094, 2444: 2.071, 2912: 2.042 | 2.042 |
| V058 | finite elements, Griffiths & Lane (1999) | 1.9 | non-associated | 3566: 1.863, 4066: 1.854 | 1.854 |
| V059a | finite elements (read from fig. 2), Guo & Griffiths (2020) | 1.28 | non-associated | 2046: 1.284, 2603: 1.281 | 1.281 |
| V059b | finite elements, Guo & Griffiths (2020) | 1.35 | non-associated | 2046: 1.353, 2695: 1.353 | 1.353 |
| V059c | finite elements (read from fig. 2), Guo & Griffiths (2020) | 1.36 | non-associated | 2046: 1.396, 2260: 1.369, 2713: 1.362 | 1.362 |
| V063 | finite-element strength reduction, Huang & Jia (2009) | 2.5 | non-associated | 3530: 2.427, 4872: 2.403, 6770: 2.403 | 2.403 |
| V067 | finite-element strength reduction (verification manual, problem 19), another program's manual | 1.41 | non-associated | 3620: 1.465, 3896: 1.447, 4469: 1.439 | 1.439 |
| V068 | limit analysis, lower and upper bound, Kim et al. (2002) | 1.13–1.27 | associated | 2024: 1.180, 2445: 1.177 | 1.177 |
| V077 | plasticity theory, Prandtl (1920) | 1.0 | davis-b | 4162: 1.004, 5303: 1.001 | 1.001 |
| V093 | finite-element strength reduction (verification manual, problem 13), another program's manual | 1.33 | non-associated | 2115: 1.339, 3399: 1.339 | 1.339 |
| V094 | finite-element strength reduction (verification manual, problem 15), another program's manual | 1.39 | non-associated | 2134: 1.371, 2626: 1.354, 3700: 1.351 | 1.351 |